Semisimple Deformations of Contraction Algebras for Smooth Threefold Flops

Shutao Bi*

October 2026

Abstract

Building on the work of Bryan–Katz–Leung, Hua–Toda and Kawamata, we study deformations of contraction algebras and the local geometry of the scheme of associative algebra laws. Let A be the basic contraction algebra of a completed projective flopping contraction of a smooth complex threefold. We construct a finite free polynomial family that becomes a constant semisimple algebra away from the origin, with the original curve projectors fixed. For each restricted positive root β, the split generic fibre of this family has n_β matrix blocks of size $\sum_i \beta_i$, and the labelled projector e_i has rank β_i on each block. Here n_β is the corresponding genus-zero Gopakumar–Vafa invariant. For every finite free associative formal deformation of A with geometrically semisimple generic fibre, that geometric generic fibre has the same matrix factors and the same ranks for any lifts of the original projectors. The scheme of associative algebra laws is smooth at the point defined by A, and its unique irreducible component through that point is the closure of the corresponding semisimple orbit.

The proof uses the curve divisors described by Katz–Morrison and Nabijou–Wemyss, universal algebra and cocenter presentations, and the orbit method of Jelisiejew–Smoktunowicz. The presentations also prove the three geometric Hua–Toda flatness assertions over arbitrary complete Noetherian local complex bases.

Near A, the centre and cocenter form dual vector bundles compatible with arbitrary base change. For the type D algebras $D_{n,2}$, $n \geq 2$, the centre morphism is not smooth. These are counterexamples to the smoothness assertion in Conjecture 1.4(2) of Jelisiejew–Smoktunowicz.

*Email: shutaobi@mail.sdu.edu.cn. Homepage: https://shutaobi.github.io/.

Contents

1 Introduction **3**

2 Constructing a semisimple deformation for one curve **8**
2.1 Root divisors and a deformation into ordinary double points . . . 12
2.2 Tilting generators and the local matrix blocks . . . 28
2.3 Assembling a finite free deformation . . . 34

3 The cocenter and uniqueness of the semisimple generic fibre **37**
3.1 Statement and proof strategy . . . 37
3.2 The orbit criterion . . . 38
3.3 Symmetry, periodicity, and Tate duality . . . 41
3.4 Reduction to the cocenter dimension . . . 44
3.5 Universal marked cocenter on the Cartan cover . . . 44
3.6 The coefficient-five E_8 computation . . . 47
3.7 Constructing the cocenter presentation . . . 49
3.8 Descent to the marked base . . . 50
3.9 Restriction to the central arc . . . 53
3.10 Completion of the uniqueness proof . . . 55

4 Universal presentations and flat families **56**
4.1 Freeness and traces over local bases . . . 57
4.2 One curve: presentations and pullback along classifying maps . . . 59
4.3 Idempotent subalgebras associated to several curves . . . 64

5 Semisimple generic fibres with the curve projectors retained **89**
5.1 Constructing a semisimple generic fibre with fixed projectors . . . 89
5.2 Fixed-idempotent laws and the orbit criterion . . . 95
5.3 Periodicity and general uniqueness . . . 97
5.4 Projector ranks in examples . . . 100

6 Consequences for centres and their deformations **101**
6.1 Unobstructedness . . . 102
6.2 Centres and cocenters in associative families . . . 102
6.3 The centre morphism . . . 104
6.4 The centre tangent map and the smooth locus . . . 106
6.5 Type D counterexamples to Conjecture 1.4(2) . . . 108
6.6 Exact tangent data for $D_{2,2}$ . . . 113

Acknowledgements **115**

A Root enumeration for the primitive marked pairs **115**

B The coefficient-five E_8 calculation **116**
B.1 The ρ-line relations . . . 116
B.2 Relation matrices and modular rank bounds . . . 119

C Exact linear algebra for $D_{2,2}$ **120**

References **122**

1 Introduction

Does a contraction algebra determine the semisimple geometric generic fibres of its associative deformations? For several exceptional curves, this asks for both the matrix factors and the ranks of the original curve projectors. The geometric families of Hua–Toda [17, Theorem 4.2] and Kawamata [23, Theorem 3.1(2)–(3)] provide candidate semisimple algebras against which to compare the geometric generic fibres of arbitrary finite free associative formal deformations.

The cocenter $HH_0(A) = A/[A, A]$ supplies the numerical comparison: when A is semisimple, $\dim HH_0(A)$ is the number of matrix factors of A. Together with the orbit method of Jelisiejew–Smoktunowicz [20, Section 1 and Theorem 2.14], the cocenter calculation proves smoothness of the scheme of associative algebra laws at A and determines which semisimple geometric generic fibres can occur. The resulting local geometry also controls how centres vary in associative families.

Let $f : X \to \operatorname{Spec} R$ be the completion at the exceptional image point of a simple flopping contraction of a smooth complex threefold, with exceptional curve $C \cong \mathbb{P}^1$. The contraction algebra of Donovan–Wemyss [8] records noncommutative deformations of $\mathcal{O}_C(-1)$. For the basic Van den Bergh generator $R \oplus N$, put

$$\Lambda = \operatorname{End}_R(R \oplus N)^{\mathrm{op}}, \qquad A = A_{\mathrm{con}} = \Lambda/\Lambda e_0 \Lambda,$$

where e_0 projects onto the free summand. Thus the quotient removes endomorphisms that factor through finite sums of that free summand. Bryan, Katz and Leung deform the neighbourhood of C to a family with isolated $(-1,-1)$-curves and identify the number of such curves in each homology class with the corresponding Hilbert-scheme multiplicity [6, Section 2, Proposition 2.3 and Lemma 2.8]. For a contraction of length ℓ, these multiplicities are the genus-zero GV invariants $n_1, \dots, n_\ell$; n_j counts curves of class $j[C]$ [6, 36]. Katz's local DT–GV construction recovers the same multiplicities for contractible curves [38, Proposition 3.3 and Corollary 3.4]. The invariants n_j define the semisimple algebra

$$S_{\mathrm{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \tag{1.1}$$

Toda's width formula gives $\dim_{\mathbb{C}} A = \sum_j j^2 n_j = \dim_{\mathbb{C}} S_{\mathrm{GV}}$ [36, Theorem 1.1]. A semisimple deformation realizes this numerical equality at the level of multiplication.

For finite-dimensional algebras over an algebraically closed field k of characteristic zero, we say that A *deforms to* S if a finite free unital $k[[s]]$-algebra has special fibre A and geometric generic fibre $S \otimes_k \overline{k((s))}$. Jelisiejew–Smoktunowicz prove that this deformation notion agrees with the polynomially split and orbit-closure formulations [20, Theorem 2.20]. In the polynomially split formulation, the family is defined over $k[u]$ and becomes the constant algebra S after inverting u. Thus, if the geometric generic fibre is semisimple, S is a product of matrix algebras over k, and its block sizes and multiplicities agree with those of that fibre.

Conjecture 1.1 (Wemyss; cf. [20, Questions 1.1–1.2])**.** *The algebra A deforms to S_{GV}, and every complex semisimple algebra to which A deforms is isomorphic to S_{GV}.*

Semisimple generic fibres and smoothness of algebra laws

For the general statements, let $f_0 : X_0 \to Y_0$ be the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold. Fix the normalized basic

Van den Bergh generator and its labelled nonfree projectors $(e_i)_{i\in J}$. A general surface section gives an ADE diagram with retained subset J. We use the simultaneous ADE resolutions of Katz–Morrison [27, Theorem 1], together with the restricted-root formulation and perturbation construction of Nabijou–Wemyss [29, Theorem 2.6 and Section 2.6]. A nonzero restricted positive root $\beta = (\beta_i)_{i\in J}$ is the vector of retained coefficients of a positive root; distinct proportional vectors remain distinct. Let n_β be the corresponding genus-zero GV invariant, using the curve-counting description in [29, Section 3.1], and set

$$s_\beta = \sum_i \beta_i, \qquad N = \sum_\beta n_\beta, \qquad E = \bigoplus_{i\in J} \mathbb{C}e_i.$$

A marking is the fixed embedding $E \hookrightarrow A$. The embedding records which projector belongs to which curve. The associated Peirce spaces are the vector spaces e_iAe_j; together they decompose the underlying vector space of A. On a matrix factor, the marking is described by the ranks of the orthogonal projectors e_i, whose sum is the identity of that factor.

Write X_d for the scheme of associative laws on a fixed d-dimensional vector space with fixed identity, and let $G = \mathrm{Stab}_{\mathrm{GL}_d}(1)$ be the group of allowed changes of basis. The fixed-idempotent scheme X_E also fixes the Peirce spaces e_iAe_j and the labelled vectors e_i; denote its change-of-basis group by G_E. Orbits in these schemes represent isomorphism classes with the indicated data fixed.

Theorem 1.2 (Semisimple generic fibres and smoothness)**.** *For the basic contraction algebra A of f_0, put*

$$S_E = \prod_\beta \mathrm{End}_{\mathbb{C}} \left(\bigoplus_{i\in J} \mathbb{C}^{\beta_i} \right)^{\times n_\beta}.$$

(i) *There is a finite free marked $\mathbb{C}[u]$-algebra $\mathscr{P}$ with $\mathscr{P}/(u) \cong A$ and $\mathscr{P}[u^{-1}] \cong S_E \otimes_{\mathbb{C}} \mathbb{C}[u, u^{-1}]$. On each β-factor, e_i has rank β_i.*

(ii) *Every finite free formal associative deformation of A over $\mathbb{C}[[t]]$ with geometrically semisimple generic fibre has, on that geometric generic fibre and with respect to any lifts of the original projectors, the matrix factors and ranks prescribed by S_E. Forgetting the projectors, the corresponding complex semisimple algebra is $\prod_m M_m(\mathbb{C})^{\times N_m}$, where $N_m = \sum_{s_\beta = m} n_\beta$.*

(iii) $\dim HH_0(A) = \dim Z(A) = N$. *Both X_d and X_E are smooth at A. In each scheme, the unique irreducible component through A is the closure of the corresponding semisimple orbit.*

For one curve, β consists of the single coefficient j, so the theorem proves Conjecture 1.1. Theorem 5.1 constructs the family with split generic fibre S_E, and Theorem 5.12 proves that all geometrically semisimple generic fibres have these matrix factors and projector ranks. The vector of projector ranks retains information lost when one records only the matrix size: its entries specify how each original curve projector acts on the corresponding factor. The theorem determines the geometric generic fibre with these projectors; the integral families can still differ.

Relation to earlier work

For a single curve, the recovery of GV invariants from the abstract contraction algebra is already a theorem of Hua–Toda: their noncommutative Donaldson–Thomas formula determines

these invariants [17, Theorem 4.4 and Corollary 4.6]. We ask in addition which semisimple algebras can occur as geometric generic fibres of associative deformations. Hua–Toda's flatness conjecture [17, Conjecture 4.3] concerns the relative algebra, centre and cocenter in geometric families. We treat these three objects over complete Noetherian local bases in Section 4.

Kawamata's proof records the individual summand multiplicities [23, Theorem 3.1(3) and its proof], which give the projector ranks in the generic matrix factors. Our construction through the universal quotient makes the comparisons over formal classifying arcs explicit and produces a polynomially split family with fixed projectors. These steps are proved in Section 5.

Explicit algebraic approaches include the deformation of a nine-dimensional contraction algebra to $M_2(\mathbb{C}) \times \mathbb{C}^5$ by Bowman–Puljić–Smoktunowicz [3, Section 4], and Smoktunowicz's study of strongly flat deformations [34]. Using the local normal forms of Brown–Wemyss [5], Jelisiejew–Smoktunowicz prove semisimple existence and uniqueness for Types A and D [20, Theorem 1.3]. We use their comparison of tangent dimensions with a known semisimple orbit. The cocenter calculation supplies the missing numerical equality for the exceptional cases, and relative Hochschild cochains give the version with every projector fixed. Thus the proof of uniqueness applies to all the finite free associative formal deformations in the main theorem.

Flatness for several exceptional curves

For several curves, the universal retained algebra and its cocenter have square presentations. Their determinants convert intersection orders with the curve divisors into ranks of the resulting flat family. A geometric family pulls these matrices back along classifying maps to power-series rings $B[[w]]$, where B is the deformation base and w is the surface-section parameter. The resulting cokernels are free over every complete Noetherian local deformation base, including bases with nilpotents. This proves geometric flatness even when the generic fibre is not semisimple.

Theorem 1.3 (Geometric Hua–Toda flatness)**.** *Let f_B be a flat deformation of f_0 over a complete Noetherian local complex algebra $(B, \mathfrak{m}, \mathbb{C})$. Form the relative contraction quotient $\mathscr{A}_B$ using the lift of the fixed marked basic generator and the full finite moving support. Then the algebra, cocenter, and centre are finite free, with ranks*

$$\operatorname{rank}_B \mathscr{A}_B = \sum_{\beta} n_\beta s_\beta^2, \qquad \operatorname{rank}_B HH_0(\mathscr{A}_B/B) = \operatorname{rank}_B Z(\mathscr{A}_B) = N.$$

The centre and cocenter commute with every commutative scalar extension $B \to B'$, and every symmetrizing trace of the central algebra lifts to a perfect relative trace. The corresponding local statement holds over Noetherian algebraic or analytic bases when the relative quotient is finite over the base.

We obtain the algebra presentation from retained corners of the Etingof–Rains central extension [10, Theorem 3.2]. For the cocenter, we combine the full-algebra trace results of Etingof–Latour–Rains [11, Theorem 3.1 and Corollary 3.5] with a vertex-deletion argument. The resulting Theorem 4.15 relates the corner, the full central extension and the deleted subdiagram. Pulling back the algebra and cocenter presentations gives the ranks for the relative quotient defined in Remark 4.1. Flatness and local base change are proved in Theorem 4.26 and Corollary 4.27.

Consequences for centres

The main theorem gives a smooth open neighbourhood of A in the scheme of associative algebra laws. On this neighbourhood, centres and cocenters vary in vector bundles of constant rank. The vector-bundle statement applies to arbitrary finite free associative families through A. There is also a converse problem. Given a deformation of the commutative algebra $Z(A)$, can it be obtained as the centre of a deformation of A? The tangent map of the centre morphism detects first-order obstructions to this lifting problem. For a commutative algebra Z, let $T^1_{\mathrm{com}}(Z)$ be its Harrison space of first-order commutative deformations [15].

Theorem 1.4 (The centre deformation problem)**.** *For A as above, with $d = \dim A$, the following hold.*

(i) $\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+N}]]$*; in particular A is unobstructed.*

(ii) *Near A in X_d, the centre and cocenter of the tautological algebra are dual vector bundles of rank N, compatible with arbitrary base change. Every finite free associative deformation of A over a Noetherian local complex base inherits these properties and a lift of any chosen symmetrizing trace of the closed fibre.*

(iii) *The centre morphism is smooth at an algebra law B in this neighbourhood if and only if $\chi_B : HH^2(B) \to T^1_{\mathrm{com}}(Z(B))$ is surjective. This holds on a dense open subset containing the semisimple laws.*

(iv) *For $D_{n,2}$, $n \geq 2$, a polynomial family of centres inside the smoothable component has first-order class outside the image of χ. Thus the centre morphism is not smooth at $D_{n,2}$. For $D_{2,2}$, nonlifting also holds without a chosen identification of the special centre.*

These consequences establish unobstructedness and local constancy of the centre dimension in [20, Conjecture 1.4] for the contractions considered here. The type D algebras $D_{n,2}$, $n \geq 2$, give counterexamples to the smoothness assertion in part (2). The centre morphism remains non-smooth even after restricting its target to the smoothable component of commutative algebra laws.

Outline of the proof

The proof uses root divisors to connect the geometric construction to the dimensions needed in the orbit argument. Write C_J for the universal contraction quotient with the projectors indexed by J retained, and R_J for its parameter ring. Let $P_\beta = 0$ be the divisor along which a curve of restricted class β appears. The zeroth Fitting ideals of the following modules are generated by products of these divisor equations. Their exponents record three different measurements:

module	exponent of P_β	information recorded
$e_i C_J e_j$	$\beta_i \beta_j$	individual Peirce dimensions
C_J	s_β^2	dimension of the matrix algebra
$HH_0(C_J/R_J)$	1	number of simple blocks

Restricting to a classifying arc turns these exponents into ranks by multiplying them by the divisor intersection orders n_β. The one-curve exponents j^2 and 1 are the scalar version of this description. For one curve, the comparison with the original GV invariant is a local

Hilbert-length calculation: the complete obstruction space and tangent space have the same dimension, making the relative Hilbert ring finite flat; its reduced generic points are the isolated $(-1,-1)$-curves (Proposition 2.19). For a marked configuration, the same local calculation compares formal classifying arcs with the analytic arc used in the Nabijou–Wemyss counting convention (Corollary 4.25). The rank formulas provide inputs for uniqueness; the marked product itself comes from the geometric construction. The perturbation of the classifying arc uses the Katz–Morrison and Bryan–Katz–Leung picture [27, 6]. The subsequent identification of matrix factors uses determinant degrees of the transported generator, following Hua–Toda and Kawamata [17, Theorem 4.2 and its proof] [23, Theorem 3.1(3) and its proof].

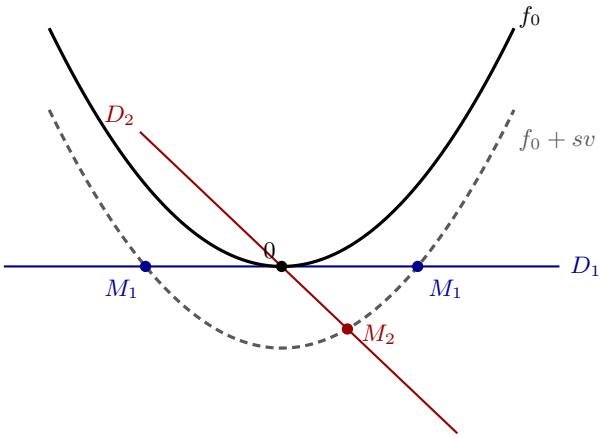


Figure 1: A schematic slice of the marked base with two curve divisors. The central classifying arc is tangent to D_1; a small translation separates that tangency into transverse intersections. An intersection with D_j gives the local factor M_j, while the conserved intersection number determines how many such factors occur in the generic product of matrix algebras. This figure depicts the one-curve divisor-count construction of Bryan–Katz–Leung [6, Section 2.1, Proposition 2.3, and Lemma 2.8].

August's symmetry theorem [2, Theorem 2.6], the bimodule sequence of Jasso–Keller–Muro [19, Theorem 3.2.5], and Linckelmann's Tate duality [28, equation (2.11)] give $\dim HH^1(A) = \dim HH^2(A)$. The cocenter rank then makes the tangent dimension equal to the dimension of the constructed semisimple orbit. This equality forces smoothness and a unique component through A, and hence determines the possible semisimple geometric generic fibres. Relative Hochschild cochains give the corresponding tangent formula with every projector fixed, which proves the full marked statement.

In the one-curve argument, the coefficient-five E_8 corner is computed from the exact relations and rank certificate in Subsection 3.6 and Appendix B, using the results of Etingof–Rains and Etingof–Latour–Rains [10, 11]. The exact sequence in Theorem 4.15 subsequently computes the cocenter dimension for arbitrary retained subsets. The appendices record the exact root enumeration, the E_8 calculation, and the $D_{2,2}$ tangent data; the source distribution includes their verification programs.

Conventions and organization. All algebras are associative and unital. All parameter rings and deformation bases in the geometric constructions, including local Hilbert algebras,

are commutative. The geometry is over $\mathbb{C}$ unless a field extension or a split local model over another field is specified; the general orbit criterion works over an algebraically closed field of characteristic zero. We write $A^e = A \otimes A^{\mathrm{op}}$, $HH^i(A) = \mathrm{Ext}^i_{A^e}(A, A)$, and $D = \mathrm{Hom}_{\mathbb{C}}(-, \mathbb{C})$. The symbol $[A, A]$ denotes the linear span of commutators, and $HH_0(A) = A/[A, A]$. The parameters w, s, z denote respectively the surface slice, the external deformation, and the central-extension element. In the multi-curve construction $\mathscr{E}$ denotes the central extension; $E = \bigoplus_i \mathbb{C}e_i$ denotes the separable algebra of labelled idempotents.

Sections 2 and 3 construct a family with semisimple generic fibre for one curve. They also prove that every finite free associative formal deformation of A with geometrically semisimple generic fibre has the same matrix factors as this family. Section 4 begins with a criterion for freeness after pullback and the lifting of symmetrizing traces. It then constructs the universal presentations and identifies their geometric pullbacks, first for one curve and then for several curves. Section 5 determines the semisimple geometric generic fibres with the curve projectors fixed and proves smoothness of the corresponding schemes of algebra laws. Section 6 derives the consequences for centres in associative families and studies the centre-lifting obstruction.

2 Constructing a semisimple deformation for one curve

For a simple one-curve flop, the construction combines three geometric inputs. Katz–Morrison's length classification and construction are recalled in [21, Theorems 1.1–1.2]; Bryan–Katz–Leung describe the curve-divisor count in a one-parameter analytic deformation [6, Section 2.1, Proposition 2.3, and Lemma 2.8]; and Toda relates the resulting GV multiplicities to the contraction-algebra dimension [36, Section 2.4 and Theorem 1.1]. We combine them with a fixed tilting generator to construct a finite free algebra family. Root-divisor intersections determine the number of local factors, and the determinant degree of the generator determines their matrix sizes. Throughout this section the ground field is $\mathbb{C}$. Let

$$f\colon X \longrightarrow Y = \operatorname{Spec} R$$

be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold. Thus $\mathrm{Exc}(f) = C \cong \mathbb{P}^1$, the scheme-theoretic length ℓ belongs to $\{1, \dots, 6\}$, and the contraction algebra is denoted by A_{con}. We write n_j for the genus-zero Gopakumar–Vafa invariant in class $j[C]$.

Theorem 2.1 (Semisimple existence)**.** *Let f be any completed simple one-curve flop of a smooth complex threefold, of length ℓ, with contraction algebra A_{con} and genus-zero GV invariants n_j. There exists a finite free $\mathbb{C}[[s]]$-algebra $\mathscr{A}$ such that*

$$\mathscr{A}/(s) \cong A_{\mathrm{con}}.$$

After a finite splitting extension $K'/\mathbb{C}((s))$,

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \tag{2.1}$$

Definition 2.2 (Marked partial-resolution data)**.** Let $(\Phi, *)$ be the primitive marked ADE pair associated to a general hyperplane section, with retained vertex $*$. The root system Φ

has complex Cartan space $\mathfrak{h}$, Weyl group W, and simple roots $\Delta = \{\alpha_i\}$, which determine the positive roots Φ^+. The marking fixes the retained simple root α_* and the corresponding summand projector in the endomorphism algebra. For a root $\alpha = \sum_i c_i(\alpha)\alpha_i$, the marked coefficient is $c_*(\alpha)$; the Levi roots are those with $c_*(\alpha) = 0$. Put

$$W_0 = \langle s_i : i \neq * \rangle, \qquad H = \mathfrak{h}/W_0,$$

and let $\pi_W : H \to \mathfrak{h}/W$ be the induced finite map to the semiuniversal ADE base. Choose affine coordinates on H centred at the origin.

If θ is the highest root, the length classification gives

$$\ell = c_*(\theta)$$

[27]. The curve divisors and symbolic thickenings are indexed by $1 \leq j \leq \ell$ [6, Proposition 2.3 and Lemma 2.4]. Modulo the Levi root lattice, $[\alpha] = c_*(\alpha)[\alpha_*]$. Degrees of extended line bundles express the comparison with curve classes; see Lemma 2.10.

The retained root coefficient labels a curve class, whereas the quotient by the Levi Weyl group parametrizes the marked partial resolution. The six primitive cases are shown in Figure 2.

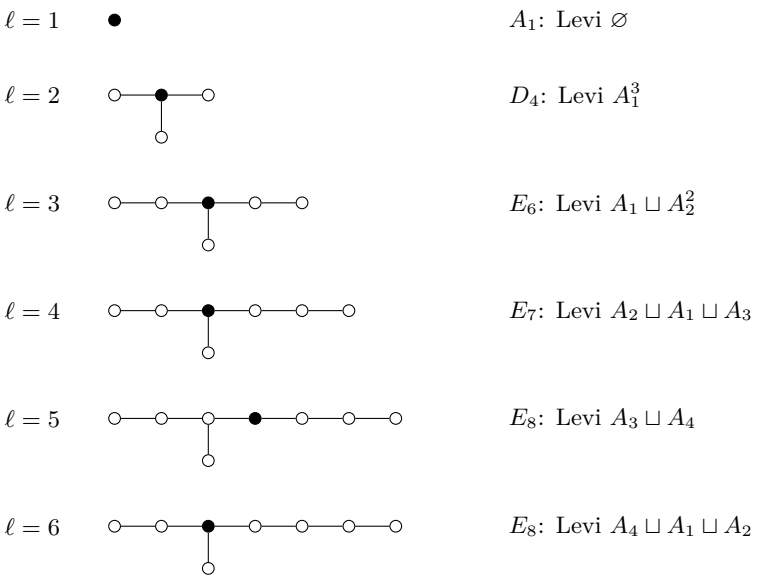


Figure 2: The six primitive marked ADE pairs. The filled vertex is the retained vertex α_*; the highest root has coefficient $\ell = c_*(\theta)$ at this vertex, and the Levi type is the diagram left after deleting it. This is the primitive one-curve situation in the Bryan–Katz–Leung divisor picture [6, Section 2.1 and Proposition 2.3]. For lengths $2, 3, 4, 6$, the retained vertex is the branching vertex used in the corner computations of Etingof–Latour–Rains. The coefficient-five corner is computed in Proposition 3.15.

Lemma 2.3 (The marked analytic classifying arc)**.** *After replacing the source and target of the contraction by sufficiently small analytic neighbourhoods, there is a holomorphic map*

$$f_0 \colon \mathbb{D}_w \longrightarrow H, \qquad f_0(0) = 0,$$

for which the given analytic contraction is isomorphic, as a marked contraction, to the pullback of the simultaneous partial resolution over H. The completion of f_0 at $w = 0$ is a formal classifying arc

$$\nu_0 \colon \operatorname{Spf} \mathbb{C}[[w]] \longrightarrow H,$$

whose pullback is isomorphic to the original completed marked contraction.

Proof. Choose a general analytic hyperplane parameter w through the contraction point. The central fibre of the resulting hyperplane-section family is the marked partial resolution of the ADE surface section. The semiuniversal deformation spaces of the full resolution, the marked partial resolution, and the ADE surface are, respectively,

$$\mathfrak{h}, \qquad H = \mathfrak{h}/W_0, \qquad \mathfrak{h}/W.$$

Bryan–Katz–Leung set up the compatible analytic classifying maps and their curve-divisor geometry [6, Section 2.1, especially Propositions 2.2–2.3]. The local universal-flop formulation used here is also stated in [21, Theorem 1.2]. These results identify the resolved threefold neighbourhood with the Cartesian pullback of the simultaneous marked partial resolution. In the formal assertion, the pullback is completed at the full maximal ideal of the central target point. More explicitly, a local presentation of the universal target as $\mathbb{H}_\ell[z_1, \dots, z_N]/(G_1, \dots, G_q)$ gives the completed pulled-back target

$$\mathbb{C}[[w, z_1, \dots, z_N]]/(G_1(\nu_0(w), z), \dots, G_q(\nu_0(w), z)).$$

This is the completion of the analytic Cartesian target at its contraction point; base change of the projective source gives the original completed marked contraction. The later subsection compares the contraction quotient after transporting the tilting generator. □

Let S_0 be the central partial-resolution surface, and write $\mathcal{I}_C \subset \mathcal{O}_{S_0}$ for the ideal of C in that surface.

Proposition 2.4 (The marked family and GV multiplicities)**.** *For the classifying arc of Lemma 2.3, the simultaneous partial resolution has the following properties.*

(i) *The base H is smooth affine space, the quotient $\pi_W \colon H \to \mathfrak{h}/W$ is finite, and the universal simultaneous partial resolution*

$$\pi_\ell \colon \mathcal{Y}_\ell \longrightarrow \mathcal{X}_\ell = \operatorname{Spec} \mathcal{R}_\ell$$

is projective with source and target flat over H. Its fibres have dimension at most one and

$$R\pi_{\ell *}\mathcal{O}_{\mathcal{Y}_\ell} \cong \mathcal{O}_{\mathcal{X}_\ell}.$$

The universal flopping algebra is finite over its centre $\mathcal{R}_\ell$.

(ii) *The curve discriminant is a union of precisely ℓ reduced irreducible divisors*

$$D^{\mathrm{curv}} = D_1 \cup \cdots \cup D_\ell.$$

At a general point of D_j, the partial-resolution surface contains one smooth (-2)-curve in class $j[C]$. To specialize a chosen curve, retain a positive root $\alpha = \sum_i c_i \alpha_i$, with $c_* = j$,

and its hyperplane $H_\alpha \subset \mathfrak{h}$ on the Cartan cover. If E_i are the simple exceptional curves on the full central resolution, the effective root divisor $E_\alpha = \sum_i c_i E_i$ has a projective flat deformation over this chosen H_α, with central fibre E_α. At a single-root point its fibre maps isomorphically to the corresponding curve in the partial resolution. This parameterization keeps the root choice. Independently, write $C_j \subset S_0$ for the symbolic thickening defined by $\mathcal{I}_C^{(j)}$.

(iii) *A sufficiently small boundary-proper generic perturbation of f_0 meets D_j in k_j points, counted with multiplicity, where k_j is the length of the local Hilbert scheme at C_j.*

(iv) *Toda's pure thickening is*

$$\mathcal{O}_{C^{(j)}} = (\mathcal{O}_{S_0}/\mathcal{I}_C^j)/Q,$$

where Q is the maximal zero-dimensional subsheaf, and n_j is the length of its local Hilbert scheme. A generic geometric deformation has exactly n_j disjoint $(-1,-1)$-curves in class $j[C]$.

(v) *The contraction algebra satisfies*

$$\dim_{\mathbb{C}} A_{\mathrm{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Proof. The universal-flop construction supplies the smooth marked base, finite quotient map, flat source and target, projectivity, fibre-dimension bound, and $R\pi_{\ell *}\mathcal{O} = \mathcal{O}$ [21, Sections 2.3 and 3.3]. The universal endomorphism algebra is finite over $\mathcal{R}_\ell$ by coherence of proper direct image. Its positive generator is globally generated and contains $\mathcal{O}$; commuting with the free projector and with all maps from $\mathcal{O}$ makes a central endomorphism scalar. Thus its centre is exactly $\mathcal{R}_\ell$, proving (i).

The restricted-root description gives the reduced curve divisors [29, Theorem 2.6]; in the primitive one-curve case the retained coefficients are $1, \dots, \ell$, with one orbit for each [6, Proposition 2.3 and Lemma 2.4]. Their reduced equations and orbit counts are checked in Lemmas 2.5 and 2.6. Lemma 2.8 constructs the specified effective root divisor over its chosen Cartan root hyperplane, with its scheme-theoretic central fibre and arbitrary-base-change compatibility. This supplies the projective flat family in (ii). At a single non-Levi root, no deleted root vanishes, so the full-to-partial contraction preserves the root curve. The degree calculation in Lemma 2.10 gives its class $j[C]$.

The symbolic and pure thickenings coincide by Lemma 2.13. Katz identifies their isolated Hilbert-scheme length with the genus-zero GV invariant through the stable-sheaf moduli component [38, Lemma 3.2 and Proposition 3.3], so $k_j = n_j$. Proposition 2.19 compares this original Hilbert length directly with the roots of the prescribed constant translation, giving $k_j = n_j = m_j$. A small boundary-proper transverse perturbation has the same intersection number m_j by the argument principle; the single-root local model gives one $(-1,-1)$-curve in class $j[C]$ at each intersection. This proves (iii) and (iv). Finally Toda's width formula gives (v) [36, Theorem 1.1 and Sections 2.2–2.4], with the same completed basic contraction algebra and the same GV conventions. □

2.1 Root divisors and a deformation into ordinary double points

The marked root coefficient defines a divisor on the deformation base. Bryan–Katz–Leung's simultaneous-partial-resolution construction gives the analytic divisor picture [6, Section 2.1 and Proposition 2.3]. That construction builds on Katz–Morrison's root-hyperplane description of simultaneous resolutions [27, Theorem 1]. This divisor geometry permits a formal constant translation while retaining its finite moving support. Let $q : \mathfrak{h} \to H$ be the quotient map, and for a root α let $H_\alpha \subset \mathfrak{h}$ be its root hyperplane. The group W_0 preserves the marked coefficient and permutes the non-Levi roots with a fixed coefficient. If $\mathcal{O}$ is one such orbit, the square-free product

$$p_{\mathcal{O}} = \prod_{\alpha \in \mathcal{O}} \alpha$$

is W_0-invariant and descends to an equation on H. The roots of marked coefficient j form a single W_0-orbit $\mathcal{O}_j$ (Lemma 2.5), so $p_{\mathcal{O}_j}$ descends to a polynomial $P_j \in \mathcal{O}(H)$ with $q^*P_j = p_{\mathcal{O}_j}$. This polynomial cuts out the reduced irreducible divisor D_j of Proposition 2.4(ii). Put

$$D^{\mathrm{curv}} = \bigcup_{j=1}^{\ell} D_j;$$

This is the curve discriminant, the locus where the marked partial resolution has a complete exceptional curve. Its good points have one vanishing root; the constant translation below separates the corresponding root intersections away from the higher-codimension locus.

2.1.1 Root divisors and transverse local models

Lemma 2.5 (The coefficient strata are single W_0-orbits)**.** *For each of the six primitive marked ADE pairs and each $1 \le j \le \ell$, the set*

$$\mathcal{O}_j := \{\alpha \in \Phi^+ : c_*(\alpha) = j\}$$

is a single W_0-orbit. Its cardinalities are:

length ℓ	*marked pair*	$\lvert\mathcal{O}_1\rvert, \ldots, \lvert\mathcal{O}_\ell\rvert$
1	A_1	1
2	D_4	8, 1
3	E_6	18, 9, 2
4	E_7	24, 18, 8, 3
5	E_8, $c_*(\theta) = 5$	40, 30, 20, 10, 4
6	E_8, $c_*(\theta) = 6$	30, 30, 20, 15, 6, 5

In particular $\sum_j \lvert\mathcal{O}_j\rvert$ is the number of positive roots with $c_*(\alpha) > 0$, written N_p in Subsection 3.5, and each D_j is irreducible.

Proof. Bryan–Katz–Leung identify a marked coefficient with a curve-divisor component [6, Proposition 2.3]. The group W_0 preserves c_*, hence acts on every $\mathcal{O}_j$. The reflection calculation in Appendix A enumerates these roots and their W_0-orbits, giving the displayed cardinalities and transitivity. A proper factor of the descended product $p_{\mathcal{O}_j}$ would select a proper nonempty W_0-stable subset of $\mathcal{O}_j$, contradicting transitivity. Thus D_j is irreducible. Lemma 2.6 supplies its reducedness. □

Lemma 2.6 (Reduced root divisors and the higher-root locus)**.** *Let α be a non-Levi root. The quotient $\mathfrak{h} \to H = \mathfrak{h}/W_0$ is étale at a general point of H_α. Consequently the divisor descended from every W_0-orbit of non-Levi root hyperplanes is generically reduced.*

There is a closed subset $\Sigma^{\mathrm{bad}} \subset D^{\mathrm{curv}}$ of codimension at least two in H, containing:

(i) *the singular loci of all curve-discriminant components;*

(ii) *intersections of distinct curve-discriminant components;*

(iii) *the images of intersections $H_\alpha \cap H_\beta$, where α is non-Levi and $\beta \neq \pm\alpha$.*

We call the points of $D^{\mathrm{curv}} \setminus \Sigma^{\mathrm{bad}}$ good. At each Cartan lift of a good point of D_j, exactly one root pair $\{\pm\alpha\}$ vanishes.

Proof. The root-hyperplane input from Katz–Morrison makes the first assertion a stabilizer calculation upstairs. Choose

$$x \in H_\alpha \setminus \bigcup_{\beta \neq \pm\alpha} H_\beta.$$

Its stabilizer in W is $\{1, s_\alpha\}$. Since α is not a Levi root, $s_\alpha \notin W_0$, and the W_0-stabilizer of x is trivial. A finite quotient in characteristic zero is étale at a point with trivial stabilizer. Therefore the product of the distinct linear root equations descends with generic multiplicity one.

For the bad locus, distinct root hyperplanes meet in codimension two. A finite morphism preserves the dimension of an irreducible closed subset, so all images in (iii) have codimension at least two in H. Distinct reduced curve divisors have no common codimension-one component, and the singular locus of a reduced divisor in the smooth space H also has ambient codimension at least two. Their finite union has the required codimension. Outside it, every Cartan lift lies on exactly one root hyperplane. For A_1, there are no intersections of distinct root hyperplanes and the bad locus is empty. □

Lemma 2.7 (Quadratic reflection coordinate)**.** *Let $\bar{x} \in D_j \setminus \Sigma^{\mathrm{bad}}$, and choose a Cartan lift $x \in H_\alpha$. There are completed local coordinates $(\zeta, z_2, \ldots, z_r)$ at $\bar{x}$ and $(t, z_2, \ldots, z_r)$ at its image in $\mathfrak{h}/W$ such that*

$$\widehat{\mathcal{O}}_{H,\bar{x}} = \mathbb{C}[[\zeta, z_2, \ldots, z_r]], \qquad \widehat{\mathcal{O}}_{\mathfrak{h}/W, \pi_W(\bar{x})} = \mathbb{C}[[t, z_2, \ldots, z_r]],$$

and $\pi_W^* t = \zeta^2$, while $D_j = (\zeta = 0)$.

Proof. At the chosen lift, the W_0-stabilizer is trivial and the full Weyl stabilizer is the order-two group generated by s_α. This proposition also holds at every other single-root point. Indeed, write the Cartan point as $a + ib$ with a, b real and choose a generic real c. The vector $a + cb$ has the same vanishing roots and the same stabilizer, since only finitely many linear conditions must be avoided. For a real vector in the closed dominant chamber, a reduced-word expansion of $v - wv$ is a sum of positive roots with coefficients given by its nonnegative simple-root pairings. If $wv = v$, all coefficients vanish, so its stabilizer is generated by the walls containing it. Here the only such root pair is $\{\pm\alpha\}$. Moreover $s_\alpha \notin W_0$, since the retained fundamental weight is fixed by W_0 but not by this non-Levi reflection. The reflection acts by $\zeta \mapsto -\zeta$ on the normal line to H_α and trivially along the hyperplane. The resulting invariant completed local ring is therefore $\mathbb{C}[[\zeta^2, z_2, \ldots, z_r]]$, which gives the asserted quadratic parameter t. □

Lemma 2.8 (A scheme-theoretic family for a chosen root)**.** *Let $\mathcal{Z} \to \mathfrak{h}$ be the standard full simultaneous resolution of an ADE surface, with the central exceptional curves C_i identified with the simple roots α_i. For a positive root $\alpha = \sum_i c_i \alpha_i$, put $H_\alpha = \{\alpha = 0\} \subset \mathfrak{h}$. There is a closed subscheme*

$$\mathcal{E}_\alpha \subset \mathcal{Z} \times_{\mathfrak{h}} H_\alpha$$

which is projective and flat over H_α, whose central fibre is exactly the effective Cartier divisor $\sum_i c_i C_i$, with its scheme structure. At a point where the only vanishing roots are $\{\pm\alpha\}$, its fibre is the reduced exceptional (-2)-curve. Formation of the displayed family by base change preserves projectivity, flatness, and this central fibre. In particular, for an arc $\operatorname{Spec} R \to H_\alpha$ from a complete DVR, taking the closed point to the Cartan origin and the generic point to the single-root locus, the pullback is a projective flat curve over R with these special and generic fibres.

Proof. We use the fixed marking and standard resolution of [27, Theorem 1 and Sections 4–5]; all page numbers in this proof refer to the supplied arXiv version `alg-geom/9202002v1`. The compatible standard partial models are those of [29, Section 2.4, diagram (2.E)]. The word "lifts" in Katz–Morrison's Theorem 1(3) is used below only for its fixed-marking fibrewise effectivity assertion. We do not read that statement as already asserting the global scheme family claimed here. In particular, the following argument for types A, D does not use their displayed equations for root families on page 24.

A closure criterion. Suppose S is a smooth integral complex variety, $q : T \to S$ is smooth with geometrically integral, nonproper surface fibres, and T is integral. Suppose further that $\pi : T \to X$ is projective and that $\sigma : S \to X$ is a section over S. Let C_η be a geometrically integral curve in $\pi^{-1}(\sigma(\eta))$ over the generic point η of S. Define E to be its reduced scheme-theoretic closure in T. This is an integral codimension-one closed subscheme of the regular scheme T, so it is an effective Cartier divisor. It lies scheme-theoretically in $\pi^{-1}(\sigma(S))$: the latter is closed and contains its generic curve, so its ideal vanishes on the reduced integral scheme E. Consequently $E \to S$ is projective. No geometric fibre E_b can be the entire surface T_b, since E_b is proper whereas T_b is not. A local equation for E therefore restricts to a nonzero function on the integral surface T_b, and is a nonzerodivisor there. The fibrewise Cartier criterion [35, Tag 062Y, part (3)] makes E a relative effective Cartier divisor, flat over S. Its image is closed and contains η, so all its fibres are nonempty. Every fibre is a Cartier divisor on a smooth surface, and hence has no embedded points.

We will also use the following elementary degree computation, independently of any degree-specialization statement later in the thesis. For a projective flat curve family $E \to S$ and a line bundle L on it, both $\chi(E_b, L_b)$ and $\chi(E_b, \mathcal{O}_{E_b})$ are locally constant. This follows, for example, by representing their derived pushforwards locally by perfect complexes and taking alternating ranks [35, Tags 0DI4 and 0DJT]. If a fibre is the Cartier divisor $D = \sum m_i C_i$ on a smooth surface, then

$$\chi(D, L|_D) - \chi(D, \mathcal{O}_D) = \sum_i m_i \deg(L|_{C_i}). \tag{$*$}$$

Indeed, adding one component at a time gives $0 \to \mathcal{O}_{C_i}(-D') \to \mathcal{O}_{D'+C_i} \to \mathcal{O}_{D'} \to 0$. Tensoring by L, subtracting Euler characteristics, and using Riemann–Roch on C_i proves $(*)$ by induction. This argument allows repeated components.

Type A_{n-1}. Use the graph-closure construction on Katz–Morrison page 20 for $XY = \prod_{a=1}^{n}(Z + t_a)$, with projective coordinates $[u_l : v_l]$, $1 \le l \le n-1$. We use the graph closure itself, not a

claim that the abbreviated displayed relations generate its whole ideal. Its projection π to the affine surface family is projective. The standard simultaneous resolution is a smooth family of integral surfaces: equivalently, its total space is smooth, its fibres are the smooth minimal resolutions of dimension two, and miracle flatness followed by the smooth-fibre criterion proves smoothness of the family. These facts persist after restriction to $H = \{t_i = t_j\}$, $i < j$. Each fibre is nonproper, since it maps surjectively to its two-dimensional affine surface.

The polynomial section

$$\sigma : H \longrightarrow X_H, \qquad (X, Y, Z) = (0, 0, -t_i)$$

is singular in every fibre. At the generic point of H, it is the single ordinary double point. Let E be the closure of its unique reduced exceptional $\mathbb{P}^1$. The closure criterion proves projectivity, flatness, and the fibrewise Cartier assertion without any flatness claim based only on constancy of genus.

Let L_l be the pullback of $\mathcal{O}_{\mathbb{P}^1}(1)$ from the l-th factor of the graph. On the central exceptional configuration, the coordinate description on page 20 gives $\deg(L_l|_{C_a}) = \delta_{la}$: only the a-th coordinate varies on C_a, and it parametrizes that curve isomorphically. On the generic exceptional curve above $t_i = t_j$, the factors $l < i$ and $l \geq j$ are constant. The factors $i \leq l < j$ are related by multiplication by nonzero products of $t_a - t_i$, so each parametrizes that $\mathbb{P}^1$ isomorphically. Thus the generic L_l-degree is one exactly for $i \leq l < j$, and is zero otherwise. Since E_0 is supported on the central exceptional configuration, write $E_0 = \sum m_a C_a$. Applying $(*)$ gives $m_a = 1$ for $i \leq a < j$ and zero otherwise. This is exactly the positive-root divisor. The rank-one case, where H is a point, has the same calculation with $E = C_1$.

Type D_n. We use the actual two-stage construction on Katz–Morrison pages 21–22: first blow up $(X - P, Y - S)$, with coordinate $[u : v]$, and then apply the A-type graph resolution on the chart $v \neq 0$. Here P, Q, S, G have precisely the definitions in their equations (4)–(7), (12); in particular

$$f(U) = \prod_a (U + t_a) = UP(-U^2) + Q(-U^2), \quad g(Z) = \prod_a (Z + t_a^2), \quad \gamma = \prod_a t_a,$$

and the affine surface equation is

$$X^2 + Y^2 Z - \frac{g(Z) - \gamma^2}{Z} + 2\gamma Y = 0,$$

where the quotient is polynomial. The construction is projective and is the smooth full simultaneous resolution, for the same reasons as above. The following homogeneous coordinate ratios are read from the ambient construction on page 22, and also follow directly from its A-type graph construction:

$$[u_l : v_l] = [v^{n-l-2}(v^2 Z + u^2) : \prod_{a=l+1}^{n} (u + t_a v)] \quad (1 \leq l \leq n-2), \tag{D1}$$

$$[u_{n-1} : v_{n-1}] = [v^2 Z + u^2 : v(u + t_n v)]. \tag{D2}$$

These are equalities wherever the indicated pair does not vanish simultaneously; on $v \neq 0$ the graph resolution resolves their indeterminacies. They define global maps to $\mathbb{P}^1$: near $v = 0$, $u \neq 0$, the second entry of (D1) is a unit and the first entry of (D2) is a unit. Hence they

extend the graph maps regularly across that complement and agree on the overlap. Denote the pullbacks of $\mathcal{O}(1)$ by L_l, and denote the line bundle from the first $[u : v]$ by L_0. No coordinate indexed by zero is required in (D1)–(D2).

For a root hyperplane $H = \{t_j = \varepsilon t_k\}$, where $j < k$ and $\varepsilon \in \{1, -1\}$, put

$$\sigma(t) = \left(0,\ \varepsilon \prod_{a \neq j,k} t_a,\ -t_k^2\right). \tag{D3}$$

Substitution in the surface equation and its first derivatives shows that this is a polynomial singular section. One may check where $t_k \neq 0$, using that $-t_k^2$ is a double root of g, and then extend the polynomial identities to all of H. For a generic point of H this is the unique ordinary double point. Define E by the closure criterion using its reduced exceptional curve. In particular, the problematic family equations on page 24 play no role in the definition of E or in its flatness.

Here is the full central degree matrix. The descriptions of the central curves on page 23 give, for $1 \leq a < n$,

$$\deg(L_0|_{C_a}) = 0, \qquad \deg(L_l|_{C_a}) = \delta_{la} \quad (1 \leq l < n).$$

The curve C_n is parametrized by $[u : v]$. Substituting the central parameters into (D1)–(D2) and cancelling on its dense torus shows that its other maps are

$$[u_l : v_l] = [v^{n-l-2} : u^{n-l-2}]\ (l \leq n-2), \qquad [u_{n-1} : v_{n-1}] = [u : v].$$

Thus

$$\deg(L_0|_{C_n}) = 1, \quad \deg(L_l|_{C_n}) = n - l - 2\ (l \leq n-2), \quad \deg(L_{n-1}|_{C_n}) = 1. \tag{D4}$$

In particular this matrix determines every coefficient of an exceptional divisor from its $L_0, L_1, \ldots, L_{n-1}$ degrees.

For $\varepsilon = 1$, the generic first coordinate is $[u : v] = [-t_j : 1]$, and the remaining construction is the ordinary A-resolution at the repeated factors $U + t_j, U + t_k$. Its generic L_0-degree is zero and its L_l-degrees are one for $j \leq l < k$ and zero otherwise. Equations $(*)$ and (D4) therefore give $E_0 = C_j + \cdots + C_{k-1}$.

For $\varepsilon = -1$, take generic parameters with $t_k \neq 0$ and with no other coinciding squares. There are no repeated factors of f, so the second-stage A-resolution is an isomorphism on this fibre. The first blowup resolves its ordinary double point; its exceptional curve maps isomorphically to the first $\mathbb{P}^1$. For example $f(t_k) = f(-t_k) = 0$ implies $P(-t_k^2) = Q(-t_k^2) = 0$, and then $Y = S$ on (D3), so both defining linear relations of that blowup vanish on this whole $\mathbb{P}^1$. Its reduced exceptional curve is consequently that $\mathbb{P}^1$, with L_0-degree one. In (D1), substitute $Z = -t_k^2$ and cancel precisely those of $(u + t_j v), (u + t_k v)$ which occur in the denominator. The remaining homogeneous numerator and denominator have no common zero. The resulting degrees are

$$d_l = \deg(L_l|_{E_\eta}) = \begin{cases} n - l - 2, & l < j, \\ n - l - 1, & j \leq l < k, \\ n - l, & k \leq l \leq n - 2, \end{cases} \qquad (1 \leq l \leq n-2), \tag{D5}$$

where an empty range contributes nothing. Equation (D2) gives

$$d_{n-1} = \begin{cases} 1, & k = n, \\ 2, & k < n. \end{cases} \tag{D6}$$

There is no common zero at $v = 0$: the denominator of (D1) is nonzero there and the numerator of (D2) is nonzero there.

Write $E_0 = \sum m_a C_a$. Equations $(*)$, (D4)–(D6) give

$$m_n = 1, \qquad m_l = d_l - (n - l - 2)\ (l \le n - 2), \qquad m_{n-1} = d_{n-1} - 1. \tag{D7}$$

In the standard D_n convention $\alpha_l = e_l - e_{l+1}$ for $l < n$, $\alpha_n = e_{n-1} + e_n$, these are exactly the coefficients of $e_j + e_k$: if $k \le n - 2$, they are one from j through $k - 1$, two from k through $n - 2$, and one at both fork vertices; if $k = n - 1$, they are one from j through n; if $k = n$, they are one from j through $n - 2$ and at n, and zero at $n - 1$. For $(j, k) = (n - 1, n)$ this is just C_n. This includes every boundary case without introducing $[u_0 : v_0]$.

Since each central fibre is already an effective Cartier divisor on its smooth surface, determination of all its multiplicities determines its ideal $\mathcal{O}(-\sum c_a C_a)$, hence its full scheme structure. There are no additional embedded points to specify. At any single-root point the fibre is mC for the sole exceptional (-2)-curve C. Constancy of $\chi(\mathcal{O}_{E_b}) = 1$ and the filtration of $\mathcal{O}_{mC}$ give $\chi(\mathcal{O}_{mC}) = \sum_{r=0}^{m-1}(2r + 1) = m^2$. Consequently $m = 1$, including at nongeneric single-root points.

Types E_6, E_7, E_8. For E_n, $n = 6, 7, 8$, we supply the scheme-theoretic step. Katz–Morrison construct

$$p : \overline{\mathcal{Z}} \longrightarrow \mathfrak{h}, \qquad \mathcal{Z} = \overline{\mathcal{Z}} \setminus \mathcal{B},$$

by successively blowing up the ordered sections on the smooth locus of a fixed cuspidal cubic in $\mathbb{P}^2 \times \mathfrak{h}$ [27, Section 5, arXiv version pp. 25–26]. Here p is smooth and projective, and $\mathcal{B}_b$ is an integral anticanonical curve on each fibre. The anticanonical class is nef and has square $9 - n > 0$. The sections remain sections throughout this construction: blowing up a section on the relative cubic leaves its strict transform isomorphic to the cubic, since that section is a Cartier divisor on the cubic. Thus this description also gives global line bundles for the pullback of a line in $\mathbb{P}^2$ and for the total transforms of the ordered exceptional divisors.

Use these line bundles to extend the marked class $\sum_i c_i[C_i]$ to a line bundle L_α on $\overline{\mathcal{Z}}$, and restrict it to $p_\alpha : \overline{\mathcal{Z}}_{H_\alpha} \to H_\alpha$. This uses the fixed integral coefficients in the marked blowup basis; it is not an extension of a merely numerical class chosen separately on each fibre. Katz–Morrison's Theorem 1(3), with the marking fixed immediately before that theorem, says that this class is effective at every closed complex point of H_α; the ordered blowup basis identifies the same class throughout this family. This is the precise imported effectivity assertion. Upper semicontinuity for the fixed line bundle on the projective family shows that the locus $h^0 \ge 1$ is closed. It contains all closed complex points, and hence all of the finite-type base. Cohomology of a proper scheme commutes with extension of its ground field, so effectivity holds at every geometric point. Its square is -2, and its canonical pairing is zero.

We explicitly record a source correction: in the supplied arXiv version, the last row of Table 4 on page 26 prints the upper limit 7 twice, whereas the stated E_8 cubic configuration requires 8. With distinct indices, the printed class $q_7 = 3e_0 - 2e_{i_1} - \sum_{a=2}^{7} e_{i_a}$ has $q_7^2 = -1$ and $K \cdot q_7 = -1$, so is not a root. The corrected class and equation are

$$q_8 = 3e_0 - 2e_{i_1} - \sum_{a=2}^{8} e_{i_a}, \qquad 2t_{i_1} + \sum_{a=2}^{8} t_{i_a} = 0.$$

They satisfy $q_8^2 = -2$, $K \cdot q_8 = 0$, and the equation is exactly $q_8^{\perp}$, by the distinguished coordinates on page 11. This is an explicit correction justified by the lattice and the adjacent eight-point

description, not an author-issued erratum. The proof uses Theorem 1(3), not the erroneous printed row.

There is exactly one effective divisor in this class on every geometric fibre. Indeed, if its complete linear system had a nonzero moving part M, nefness of $-K$ would give $(-K)\cdot M = 0$. The intersection form in the blowup basis is $\operatorname{diag}(1,-1,\dots,-1)$, and $(-K)^2 = 9-n > 0$, so its restriction to $(-K)^{\perp}$ is negative definite. The numerical class of a nonzero effective divisor is nonzero, since it has positive intersection with an ample divisor. Thus $M^2 < 0$. Two general members of the moving part have no common irreducible component and therefore have nonnegative intersection, a contradiction. It follows that $h^0(L_{\alpha,b}) = 1$. Serre duality gives $h^2(L_{\alpha,b}) = 0$, because an effective divisor of class $K - L_{\alpha,b}$ would have negative intersection $-(9-n)$ with the nef class $-K$. Riemann–Roch on this rational surface gives

$$\chi(L_{\alpha,b}) = 1 + \tfrac{1}{2}((-2)-0) = 0, \qquad h^1(L_{\alpha,b}) = 1.$$

Cohomology and base change now imply that $M_\alpha = p_{\alpha*}L_\alpha$ is a line bundle and that its formation commutes with arbitrary base change on H_α. For completeness, the flat projective morphism and the line bundle give a perfect complex $Rp_{\alpha*}L_\alpha$, compatible with arbitrary derived base change [35, Tags 0DI4 and 0DJT]. Locally represent it by a bounded complex of finite free modules. The constant fibre cohomology dimensions just computed force constant ranks of its differentials. Since H_α is reduced, the larger minors vanish, and the constant-rank matrices split locally. Its cohomology modules are therefore locally free and commute with arbitrary base change; in degree zero the rank is one.

The evaluation map $p_\alpha^* M_\alpha \to L_\alpha$ defines a section of $L_\alpha \otimes p_\alpha^* M_\alpha^{-1}$ which is nonzero on every geometric fibre. Let $\mathcal{E}_\alpha$ be its zero scheme. Each fibre is an effective Cartier divisor on a smooth integral surface. The fibrewise Cartier criterion makes $\mathcal{E}_\alpha$ a relative effective Cartier divisor, hence flat over H_α [35, Tag 062Y]. It is projective as a closed subscheme of $\overline{\mathcal{Z}}_{H_\alpha}$.

This divisor lies in $\mathcal{Z}_{H_\alpha}$. Indeed, each component of its fibre has intersection zero with the nef class $-K$. The boundary $\mathcal{B}_b$, of anticanonical class and positive square, cannot be such a component. Its total intersection with the effective divisor is zero, so positivity of local intersection multiplicities shows that the supports are disjoint. Thus the closed divisor on the projective compactification is contained in its indicated open part; projectivity is retained.

On the central fibre, $\sum_i c_i C_i$ is already an effective divisor of the specified marked class. Uniqueness therefore identifies the fibre with this Cartier divisor, including all multiplicities, not merely with its support or fundamental cycle. Its ideal is $\mathcal{O}(-\sum_i c_i C_i)$; it has no embedded points. At a single-root point its support is the sole exceptional curve. The square -2 of the root class forces its multiplicity to be one. Finally, projectivity and flatness are stable under arbitrary base change, as are relative effective Cartier divisors [35, Tag 056Q]. The assertion about complete DVRs is ordinary algebraic base change. If the arc is initially given only as a compatible formal arc into the completion of H_α at the origin, its finitely many coordinate images are elements of $R = \varprojlim R/\mathfrak{m}^a$; they satisfy the hyperplane equation in R. Since H_α is affine, they define an actual morphism $\operatorname{Spec} R \to H_\alpha$. The base-changed projective flat divisor is therefore algebraic, and its formal completion is exactly the compatible system of its reductions modulo $\mathfrak{m}^a$. □

Remark 2.9 (The family remains on the full resolution). We do not take a scheme-theoretic image of $\mathcal{E}_\alpha$ in a partial resolution. A line bundle on the partial family is pulled back to the full family and then restricted to $\mathcal{E}_\alpha$. At a single non-Levi root the full-to-partial morphism is

an isomorphism on the unique root curve, while on the central fibre the components indexed by the deleted vertices are contracted. This is the compatible standard resolution diagram of [29, Section 2.4 and proof of Theorem 2.6]. Consequently no preservation of flatness under contraction, and no identification with a symbolic thickening on the partial surface, is needed here.

Lemma 2.10 (Specialization of the marked degree)**.** *Let $\mathscr{L}$ be a line bundle on the marked partial-resolution family, and let E_b be its exceptional curve over a good point $b \in D_j$. Then*

$$\deg_{E_b} \mathscr{L}_b = j \deg_C \mathscr{L}_0.$$

Proof. Choose a Cartan lift $x \in H_\alpha$ of b, with $\alpha = \sum_i c_i \alpha_i$ and $c_* = j$. Pull $\mathscr{L}$ back to the full simultaneous resolution over $\mathfrak{h}$. Lemma 2.8 provides a projective flat family on the chosen root hyperplane with central fibre $C_\alpha = \sum_i c_i C_i$, as an effective Cartier divisor. Restrict this projective flat root-cycle family along the radial line $\tau \mapsto \tau x$. Its central fibre is C_α; at $\tau = 1$ its fibre is the single (-2)-curve corresponding to α. We retain the chosen root on this Cartan family.

The degree of the pulled-back line bundle is constant in this family, as the difference of the Euler characteristics of the line bundle and of the structure sheaf. Its central degree is

$$\deg_{C_\alpha} \mathscr{L} = \sum_i c_i \deg_{C_i} \mathscr{L}.$$

For every deleted vertex $i \neq *$, the curve C_i maps to a point in the partial resolution, so the pulled-back degree is zero. The retained component maps isomorphically to C, giving central degree $j \deg_C \mathscr{L}_0$. At the good point x no Levi root vanishes, so the sole root curve maps isomorphically to E_b. This gives the claimed equality. For a split field-valued good point, base change the algebraic root-cycle family and its radial line to that field; the same Euler-characteristic calculation gives the identical degree formula. The argument takes place before the Cartan quotient and does not require a curve family over the whole reduced divisor D_j. □

Lemma 2.11 (The transverse threefold is an ordinary double point)**.** *At a point of $D_j \setminus \Sigma^{\mathrm{bad}}$, the transverse surface deformation is*

$$xy - z^2 = t.$$

After pullback to the marked partial-resolution base it is

$$xy - z^2 = \zeta^2,$$

and hence, after $\xi_+ = z + i\zeta$, $\xi_- = z - i\zeta$, it is the ordinary double point

$$xy - \xi_+ \xi_- = 0.$$

If a one-dimensional classifying curve meets D_j transversely, its partial resolution contains one $(-1,-1)$-curve in numerical class $j[C]$.

Proof. At a good point, the vanishing root subsystem is $\{\pm\alpha\} \cong A_1$. Katz–Morrison's explicit simultaneous resolution identifies this fibre configuration with a single A_1 singularity [27, Theorem 1 and Sections 4–5]. Their hypersurface deformation has an independent constant-term coefficient [27, Tables 2–3]; differentiation with respect to that coefficient shows that its total space is smooth. These statements apply at the chosen point and do not require it to be generic in its root hyperplane.

The invertible fibre Hessian gives a formal critical section over the base. Translate by that section and apply formal Morse elimination in the fibre variables. The actual local equation becomes $xy - z^2 = g(t, z_2, \dots, z_r)$. Its total space is smooth, so $dg \neq 0$ at the point. Its discriminant has support $t = 0$ by Lemma 2.7; hence $g = tu$, where $dg \neq 0$ makes u a unit. Replacing t by tu and ζ by $\zeta\sqrt{u(\zeta^2, z_2, \dots, z_r)}$ preserves the square relation and gives

$$xy - z^2 = t, \qquad t = \zeta^2.$$

No Levi root vanishes at the chosen lift, so the full-to-partial contraction changes no curve in this neighbourhood. Formal versality of the resolved A_1 surface, whose resolved parameter σ has smoothing parameter σ^2 [21, Section 2.3], identifies this local simultaneous resolution with one of the two standard A_1 resolutions; after the preceding normalization $\sigma = \pm\zeta$. Since $(z + i\zeta)(z - i\zeta) = z^2 + \zeta^2$, the pulled-back equation is exactly $xy - \xi_+\xi_- = 0$. On a transverse classifying curve, ζ is a local parameter. Using it as the curve coordinate absorbs the higher terms in the original parametrization and gives the same completed ODP equation. The partial resolution is one of the two small resolutions of this ODP, so its exceptional curve has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. The degree comparison in Lemma 2.10 identifies its class as $j[C]$. □

2.1.2 Perturbing the classifying arc

We perturb the classifying arc transversely to the curve divisors in order to realize the local factors simultaneously. Bryan–Katz–Leung use a boundary-proper analytic generic perturbation [6, Section 2.1 and Lemma 2.8]. The constant translation below stays inside a fixed disk, so the boundary contributes no new intersections and the total divisor multiplicities remain visible.

Lemma 2.12 (The central arc is not contained in the curve discriminant)**.** *The image of f_0 is contained in no D_j. After shrinking to a closed disk $\overline{\mathbb{D}}_\rho$,*

$$f_0^{-1}(D^{\mathrm{curv}}) \cap \overline{\mathbb{D}}_\rho = \{0\}, \qquad f_0(\partial\mathbb{D}_\rho) \cap D^{\mathrm{curv}} = \varnothing.$$

If P_j is a reduced equation of D_j, then

$$0 < m_j := \mathrm{ord}_{w=0}\, P_j(f_0(w)) < \infty.$$

Proof. If the classifying arc were contained in D_j, then for every nearby nonzero w its partial-resolution surface would contain a complete exceptional curve, by the full curve-discriminant description [29, Theorem 2.6]. The actual Cartesian classifying identification would give exceptional curves of the original threefold contraction at those nonzero values of w. Its only exceptional curve is C, over $w = 0$, which is a contradiction. Thus every reduced equation P_j pulls back to a nonzero holomorphic function whose zero at the origin is isolated. There are finitely many j, so one common shrinking gives the boundary statement and the finite, positive orders. □

Lemma 2.13 (Symbolic and pure thickenings coincide)**.** *The Bryan–Katz–Leung symbolic thickening C_j equals Toda's pure one-dimensional thickening $C^{(j)}$. Consequently their Hilbert multiplicities satisfy $k_j = n_j$.*

Proof. Work in a stalk $\mathcal{S}$ of the normal surface S_0, and let $\mathfrak{p} \subset \mathcal{S}$ be the height-one prime of C. By definition,

$$\mathfrak{p}^{(j)} = \mathfrak{p}^j \mathcal{S}_{\mathfrak{p}} \cap \mathcal{S}.$$

The quotient $\mathfrak{p}^{(j)}/\mathfrak{p}^j$ is supported at closed points and hence has finite length. The ideal $\mathfrak{p}^{(j)}$ is $\mathfrak{p}$-primary, so $\mathcal{S}/\mathfrak{p}^{(j)}$ has only the associated prime $\mathfrak{p}$ and therefore has no nonzero finite-length submodule. In the exact sequence

$$0 \longrightarrow \mathfrak{p}^{(j)}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^{(j)} \longrightarrow 0,$$

the first term is the maximal finite-length submodule of $\mathcal{S}/\mathfrak{p}^j$, namely Toda's Q. Thus $(\mathcal{O}/I^j)/Q = \mathcal{O}/I^{(j)}$, so the local Hilbert rings are identical. Katz's direct identification of this isolated Hilbert component with the stable-sheaf GV component gives $k_j = n_j$ [38, Lemma 3.2 and Proposition 3.3]; this step does not require a curve family over the full divisor D_j. □

Lemma 2.14 (Parametric transversality with boundary)**.** *Let M be a compact smooth manifold with boundary, let $\mathcal{U}$ and N be smooth manifolds, and let $S_1, \dots, S_m \subset N$ be smooth locally closed submanifolds. If a smooth map $\Xi\colon M \times \mathcal{U} \to N$ and its boundary restriction are transverse to every S_i, then the parameters for which both slice maps are simultaneously transverse to all S_i form a residual dense subset of $\mathcal{U}$. If $\dim_{\mathbb{R}} M = 2$, such a slice misses every stratum of real codimension at least four, and its boundary misses every stratum of real codimension at least two.*

Proof. For one stratum, put $Z = \Xi^{-1}(S)$. Transversality makes Z a smooth incidence manifold, with boundary incidence Z_∂. A parameter is a regular value of $Z \to \mathcal{U}$ exactly when its slice is transverse to S. Indeed, write the normal differential as $A\,dx + B\,du$. Total transversality says that the images of A and B span the normal space, while surjectivity of the incidence projection says that $\operatorname{im} B \subseteq \operatorname{im} A$. Together these conditions are equivalent to surjectivity of A, which is slice transversality. The same argument applies on the boundary. When an incidence manifold has dimension smaller than the parameter space, its whole image is null by the local smooth-map dimension estimate. In the remaining dimensions, Sard's theorem [33, Theorems 4.1 and 7.2] makes the critical-value sets null in parameter coordinates. Exhaust the closed critical loci in the incidence manifolds by countably many compact sets. Their images are closed null sets, hence nowhere dense; removing them gives a residual dense set of regular values. For boundary charts apply the theorem to a smooth extension across the boundary, and apply it separately to the boundary incidence projection. Intersect the resulting sets for the finitely many strata. The dimension formula for transverse inverse images gives the two avoidance assertions. □

Lemma 2.15 (A boundary-controlled constant translate)**.** *There is an arbitrarily small vector $v \in H \cong \mathbb{C}^{\dim H}$ such that $f_v(w) = f_0(w) + v$ is transverse to every smooth codimension-one curve-discriminant stratum, avoids Σ^{bad}, and has no curve-discriminant point on $\partial\mathbb{D}_\rho$. The vector can be chosen so that, for some $\epsilon > 0$,*

$$f_0(w) + sv \notin D^{\mathrm{curv}} \quad \text{for } w \in \partial\mathbb{D}_\rho, \quad |s| \le 1 + \epsilon.$$

Proof. Decompose the curve discriminant into finitely many smooth locally closed algebraic pieces adapted to Σ^{bad}. Such a decomposition is obtained by taking the smooth parts of the irreducible components away from their intersections and repeating on the omitted closed subsets. Their dimensions strictly decrease, so the process terminates. The good pieces have complex codimension one; all pieces in Σ^{bad} have complex codimension at least two. The translation map

$$\overline{\mathbb{D}}_\rho \times H \longrightarrow H, \qquad (w, \xi) \longmapsto f_0(w) + \xi$$

and its boundary restriction are submersions. Lemma 2.14 therefore gives translations for which the disk avoids the codimension-at-least-four bad strata and its boundary avoids every curve-discriminant stratum. The compact boundary of the central disk is disjoint from the closed curve discriminant, so reduced local equations have a positive minimum there. Uniform continuity supplies a ball of translations preserving boundary nonvanishing; choose the transverse parameter in a slightly smaller ball. □

2.1.3 Root counts and the formal generic fibre

The translation preserves each divisor intersection number. Weierstrass polynomials make the moving intersections into finite families over the translation parameter; their formal generic fibre supplies the points of the local matrix blocks.

Lemma 2.16 (Conservation of the root intersections)**.** *Put $F(s, w) = f_0(w) + sv$ and $G_j(s, w) = P_j(F(s, w))$. On a connected s-disk U containing 0 and 1, the zero scheme of $G_j(s, \cdot)$ is finite flat of degree m_j and is represented by a monic polynomial*

$$W_j(s, T) \in \mathcal{O}(U)[T], \qquad \deg_T W_j = m_j, \qquad W_j(0, T) = T^{m_j}.$$

Proof. Boundary nonvanishing makes the argument-principle integral

$$\frac{1}{2\pi i} \int_{\partial \mathbb{D}_\rho} \frac{\partial_w G_j(s, w)}{G_j(s, w)}\, dw$$

constant in s, with value m_j. For $r \geq 1$, the analogous integral with an extra factor w^r is holomorphic in s and gives the r-th power sum of the zeros, counted with multiplicity. Newton's identities then give holomorphic elementary symmetric functions on the whole parameter disk. Let $W_j(s, T)$ be the monic polynomial with these coefficients.

For each s, the roots of W_j, with multiplicities, are exactly the zeros of $G_j(s, \cdot)$ in the w-disk. Joint holomorphic division can be checked by a contour formula: for a holomorphic function $h(s, w)$, its remainder modulo W_j is

$$R_h(s, w) = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{h(s, z)}{W_j(s, z)} \frac{W_j(s, z) - W_j(s, w)}{z - w}\, dz.$$

This is a polynomial in w of degree less than m_j, and Cauchy's formula gives $h = W_j Q_h + R_h$ with Q_h holomorphic. For $h = G_j$, the remainder vanishes at every root with its multiplicity, so $R_{G_j} = 0$. The quotient Q_{G_j} is a unit, since an additional zero would change the counted multiplicity. Thus the two equations define the same analytic scheme, whose direct image is free with basis $1, w, \ldots, w^{m_j - 1}$. Boundary nonvanishing makes the support proper over the parameter disk. At $s = 0$, all m_j zeros lie at the origin, giving $W_j(0, T) = T^{m_j}$. □

Lemma 2.17 (Passage from one good fibre to the formal generic fibre)**.** *Put $B = \mathbb{C}[[s, w]]$ and $K = \mathbb{C}((s))$. The polynomials $W_j \in K[T]$ are separable and pairwise coprime. The classifying map on* $\operatorname{Spec} B[1/s]$ *avoids Σ^{bad}, and every root is transverse to the smooth part of its unique divisor D_j.*

Proof. The good fibre at $s = 1$ controls the formal generic fibre. At $s = 1$, the discriminant of each W_j and every pairwise resultant are nonzero. These are therefore nonzero holomorphic functions of s. Their Taylor series may be divisible by s, but become units after passing to K. This proves separability and coprimality.

For the bad locus, put $W = \prod_j W_j$, of degree M, and work in the free root-incidence algebra $\mathcal{O}(U)[w]/(W)$. Choose finitely many polynomial generators of the ideal of Σ^{bad}. Their compositions with F have polynomial remainders modulo W, by the contour division above. At $s = 1$ these remainders generate the unit ideal: the roots are simple and none maps into the bad locus. The matrix of their generated ideal in this free rank-M module therefore has a maximal minor $a(s)$ nonzero at $s = 1$. Its adjugate makes $a(s)$ annihilate the bad-incidence quotient. Its Taylor series is nonzero, so it becomes a unit in K, and that quotient vanishes after inverting s. Since $\Sigma^{\mathrm{bad}} \subset D^{\mathrm{curv}}$, every bad point would lie in this root incidence. This proves formal avoidance. Finally, at a smooth point of D_j, separability is the condition $\partial_w G_j \neq 0$, which is precisely transversality. □

The numerical comparison uses a local deformation argument. The following criterion turns a complete obstruction space of the expected dimension into conservation of the length of an isolated point.

Lemma 2.18 (A complete obstruction bound and conservation of length)**.** *Let $R = \mathbb{C}[[s]]$, and let $\mathscr{H}$ be a complete Noetherian commutative local R-algebra with residue field $\mathbb{C}$. Suppose*

$$k = \dim_{\mathbb{C}} \mathscr{H}/(s) < \infty, \qquad k > 0, \qquad a = \dim_{\mathbb{C}} \frac{\mathfrak{m}_{\mathscr{H}}}{\mathfrak{m}_{\mathscr{H}}^2 + s\mathscr{H}}.$$

Assume that the functor of local continuous R-homomorphisms $\mathscr{H} \to A$, on commutative local Artinian R-algebras with residue field $\mathbb{C}$, has a finite-dimensional complete obstruction space V with $\dim_{\mathbb{C}} V \leq a$. Explicitly, for each small extension $A' \twoheadrightarrow A$, whose kernel N is annihilated by the maximal ideal of A', the obstruction lies in $V \otimes_{\mathbb{C}} N$, commutes with linear pushouts of N, and vanishes exactly when the given object lifts. Then $\mathscr{H}$ is finite free of rank k over R. It is a one-dimensional complete intersection.

Proof. Lift a basis of the relative cotangent space to $x_1, \dots, x_a \in \mathfrak{m}_{\mathscr{H}}$. Together with s, these elements generate the maximal ideal by Nakayama. Successive approximation in the complete local ring gives a minimal relative presentation

$$B = R[[x_1, \dots, x_a]] \twoheadrightarrow \mathscr{H}, \qquad J = \ker(B \to \mathscr{H}), \quad \mathfrak{m} = (s, x_1, \dots, x_a).$$

Relative minimality says $J \bmod s \subset (x_1, \dots, x_a)^2$, so every $\partial f/\partial x_i$, for $f \in J$, lies in $\mathfrak{m}$. Artin–Rees gives q with $J \cap \mathfrak{m}^q \subset \mathfrak{m}J$ [35, Lemma 10.51.2, Tag 00IN]. Consider the small extension

$$B/(\mathfrak{m}J + \mathfrak{m}^q) \longrightarrow B/(J + \mathfrak{m}^q), \qquad \text{kernel } J/\mathfrak{m}J.$$

The coefficients of its obstruction in a basis of V span the kernel. Otherwise a nonzero one-dimensional quotient of the kernel annihilating those coefficients would have zero obstruction

and admit a lift. That lift changes each x_i only by a small-kernel element and keeps s fixed. Taylor expansion and the displayed derivative condition then leave every relation's image unchanged, contradicting a relation with nonzero image in that quotient. Thus $\dim J/\mathfrak{m}J \le a$.

Since $\mathscr{H}/(s)$ is nonzero Artinian, (J, s) is $\mathfrak{m}$-primary. If $r \le a$ is the minimal number of generators of J, the height bound gives $a+1 \le r+1$ [35, Lemma 10.60.12, Tag 0BBZ], so $r = a$. Choose generators $f_1, \dots, f_a$ of J. The sequence $f_1, \dots, f_a, s$ is a full system of parameters in the regular local ring B, and hence is a regular sequence [35, Lemmas 10.106.3 and 10.104.2, Tags 00NQ and 02JN]. Consequently $\mathscr{H} = B/J$ is Cohen–Macaulay of dimension one, and s is a nonzerodivisor. It is finite over R: lift a finite basis modulo s, express remainders successively as multiples of s, and sum using completeness. The s-adic and maximal-ideal topologies are cofinal because the closed quotient is Artinian. It follows that $\mathscr{H}$ is finite free over R, of rank k. □

Proposition 2.19 (The conserved multiplicities are the GV invariants)**.** *For each j,*

$$m_j = n_j. \tag{2.2}$$

Thus the formal constant-translation family has exactly n_j coefficient-j roots.

Proof. We compare the original Hilbert length with the curves of the prescribed constant translation. Write X_{orig} for the original smooth threefold representative before target completion. Put $R_0 = \mathbb{C}[[s]]$, $D = R_0[[w]]$, and $K = \mathbb{C}((s))$. Before completing the target, pull back the fixed algebraic simultaneous partial resolution along the original map $F(s, w) = f_0(w) + sv$, and write the resulting projective morphism as

$$\pi_D : X_D \longrightarrow Y_D = \operatorname{Spec} Q_D.$$

The algebra Q_D is of finite type over the Noetherian ring D. The source is flat over D, hence over R_0, by the universal flatness in Proposition 2.4(i). Its full central target completion is the geometric family under consideration. Write $X_c = X_D/(s)$, $F_j = \mathcal{O}_{C_j}$, and $I_j = \ker(\mathcal{O}_{X_c} \to F_j)$.

The actual local Hilbert functor. Take $M = \mathrm{Hilb}_{X_D/Y_D}$, regarded as an R_0-space. It is an algebraic space locally of finite presentation over Y_D, since π_D is separated and of finite presentation [35, Proposition 99.9.4, Tag 0D01]. An object over A consists of a target map $y_A : \operatorname{Spec} A \to Y_D$ and a proper flat closed subscheme of $X_D \times_{Y_D, y_A} \operatorname{Spec} A$ [35, Situation 99.9.1, Tag 0CZY, and the following paragraph in Section 99.9, Tag 0CZX]. Choose an étale scheme chart $U \to M$ and a $\mathbb{C}$-point u_j over $[C_j]$, and set

$$\mathscr{H}_j = \widehat{\mathcal{O}}_{U,u_j}.$$

This is a complete Noetherian local R_0-algebra. Formal étaleness shows that it prorepresents the local Artin functor with this chosen central point.

This functor also parametrizes all embedded proper deformations in $X_D \times_{R_0} \operatorname{Spec} A$; it does not fix their target point. Indeed Katz gives $H^0(F_j) = \mathbb{C}$ and $H^q(F_j) = 0$ for $q > 0$ [38, Lemma 3.2]. For a proper-support flat sheaf deformation F_A, induction over small Artin extensions gives $H^0(F_A) \cong A$ and $H^q(F_A) = 0$ for $q > 0$. Explicitly, a small kernel N gives the cohomology sequence of $0 \to F_j \otimes N \to F_A \to F_{A/N} \to 0$; lifting the central section 1 and comparing lengths makes $A \to H^0(F_A)$ an isomorphism. Evaluation $\mathcal{O}_{X_A} \to F_A$ is surjective by Nakayama. Changing its generator by a unit does not change its kernel ideal. Moreover

$\mathrm{End}(F_A) = H^0(F_A) = A$, so the action of Q_D determines the unique moving section y_A. These operations commute with Artin base change and invert the forgetful map from Hilbert quotients to sheaf deformations. The kernel ideal therefore records the sheaf deformation uniquely.

For $s = 0$, these are the deformations in the original *threefold*. Each is annihilated by a power of the central target ideal: its action is zero on the closed sheaf, and each further power lands in the next power of the Artin maximal ideal. The corresponding finite target neighbourhoods agree before and after full completion. The prescribed comparison with the original contraction therefore gives

$$\mathscr{H}_j/(s) \cong \widehat{\mathcal{O}}_{\mathrm{Hilb}(X_{\mathrm{orig}}),[C_j]}, \qquad \dim_{\mathbb{C}} \mathscr{H}_j/(s) = k_j = n_j.$$

Here the last equality is Lemma 2.13; the ring is Artinian by Katz's isolated-component result. *Complete obstructions for the prescribed ambient family.* Let $A' \twoheadrightarrow A$ be a small extension of local Artin R_0-algebras, with kernel N, and let $0 \to K_A \to \mathcal{O}_{X_A} \to F_A \to 0$ represent an object. Both $\mathcal{O}_{X_A}$ and F_A are A-flat, so K_A is A-flat. If $\widetilde{K}_A$ is the inverse image of K_A in $\mathcal{O}_{X_{A'}}$, ambient flatness gives the canonical extension

$$0 \longrightarrow F_A \otimes_A N \longrightarrow \widetilde{K}_A/(K_A \otimes_A N) \longrightarrow K_A \longrightarrow 0.$$

Its class is the quotient obstruction of [35, Remark 99.7.9, Tag 0CZU]. It vanishes exactly when a flat quotient lift exists: a splitting gives the lifted ideal by inverse image in $\widetilde{K}_A$, and its intersection with $\mathcal{O}_{X_A} \otimes_A N$ is $K_A \otimes_A N$. Consequently the quotient satisfies the nilpotent flatness criterion of [35, Lemma 91.11.2, Tag 08MQ]. Conversely a flat quotient lift gives precisely such a splitting. The construction commutes with linear pushouts of N. The lift is coherent and supported over a finite central target thickening, so its support is proper; hence it belongs to the Hilbert functor just identified. The quotient setup [35, Situation 99.7.1, Tag 082M] does not require X_D/R_0 to be of finite type.

Derived restriction sends the flat K_A to I_j, so the obstruction space is $V_j = \mathrm{Ext}^1_{X_c}(I_j, F_j)$. The differences between lifts give tangent space $T_j = \mathrm{Hom}_{X_c}(I_j, F_j)$ [35, Remark 99.7.10, Tag 0CZV]. In the long exact sequence obtained from $0 \to I_j \to \mathcal{O}_{X_c} \to F_j \to 0$, evaluation at 1 identifies $\mathrm{Hom}(F_j, F_j)$ with $H^0(F_j)$. Together with $H^1(F_j) = H^2(F_j) = 0$, this gives

$$T_j \cong \mathrm{Ext}^1(F_j, F_j), \qquad V_j \cong \mathrm{Ext}^2(F_j, F_j).$$

Proper-support duality makes these spaces dual. To apply it, take the original smooth quasi-projective threefold representative and any projective closure. The proper support of F_j is closed in that closure and misses its boundary. Extension by zero is therefore coherent and perfect, including at the boundary where it is zero. Proper duality for this perfect object applies even if the closure is singular [35, Lemmas 48.27.1(5) and 48.27.4, Tags 0FVV and 0FVY]. By open restriction of upper shriek [35, Lemma 48.17.2, Tag 0G4J], the dualizing complex restricts to the dualizing complex of the smooth open set. This quasi-projective threefold immerses into a projective space; the smooth case of the fundamental-class formula therefore identifies that restriction with $\omega_X[3]$ [35, Lemma 48.29.2, Tag 0E9Z]. The Gorenstein target's canonical line is trivial near its closed point, and crepancy trivializes ω_X near the support. Denote the zero extension by $\overline{F}_j$ and set $E = R\mathcal{H}om(\overline{F}_j, \overline{F}_j)$. Perfect duality and symmetry of tensor product identify $E^\vee$ with E. Since E is supported where ω_X is trivial, the proper-duality formula gives $\mathbb{H}^i(\overline{X}, E)^\vee \cong \mathbb{H}^{3-i}(\overline{X}, E)$. In particular, $\mathrm{Ext}^2(F_j, F_j) \cong \mathrm{Ext}^1(F_j, F_j)^\vee$. These self-Ext groups

are unchanged by the full target completion: their proper pushforwards have finite-length cohomology, flat completion commutes with the perfect self-Hom complex, and derived base change applies [35, Lemma 36.22.5, Tag 08IB]. On X_c, perfection near the support descends from the faithfully flat completed local rings, which are the identified regular local rings of the completed smooth threefold. The same self-Hom comparison therefore transfers the equality to X_c. Thus $\dim V_j = \dim T_j =: a$.

The preceding calculation supplies every hypothesis of Lemma 2.18: the relative tangent space is T_j, the natural complete obstruction space is V_j, their dimensions agree, and the closed algebra has length k_j. Hence $\mathscr{H}_j$ is finite free over R_0, of rank k_j.

Actual generic objects and their specialization. This finite hull is already the actual local chart ring. Indeed $\mathcal{O}_{U,u_j}$ injects into its completion by Krull intersection, and is an R_0-submodule of the finite module $\mathscr{H}_j$. It is therefore finite and closed for the s-adic topology. Its image is also dense for that topology, by cofinality with the maximal-ideal topology, so the two rings agree [35, Lemma 10.51.4, Tag 00IP, and Lemma 10.97.1, Tag 00MA]. Thus $\operatorname{Spec}\mathscr{H}_j$ carries the actual universal Hilbert family, and its generic local rings have the actual Hilbert tangent spaces. Its moving target map extends uniquely through full completion: if $\mathfrak{n}$ is the central target ideal, then $\mathfrak{n}^N\mathscr{H}_j \subset s\mathscr{H}_j$ for some N. Base change through this extension preserves the family.

Work over a common finite splitting extension $\mathbb{C}((u))/K$, with valuation ring $\mathbb{C}[[u]]$. The Weierstrass and transversality lemmas give exactly m_j distinct coefficient-j roots. Lemma 2.11 gives one split $(-1,-1)$-curve at each root, with reduced scheme fibre $\mathbb{P}^1$; these are all the complete curves. Form the generic base from $D[1/s]$ before completing at any moving point. The finite critical-target algebra of Lemma 4.22 contains each exceptional image. Its finiteness over R_0 makes all coordinates of that image integral over R_0, so they extend uniquely to a target section over $\mathbb{C}[[u]]$ specializing to the original point. This controls the target coordinates as well as the roots in w.

Let E be one such curve of class $j[C]$. In the projective inverse image of this target section, take its schematic closure $\overline{E}$. Its ideal is u-saturated, so its quotient has no u-torsion and is flat over the DVR. Saturation also proves uniqueness of this flat closure. Choose a relatively very ample line bundle L from the universal model and put $h = \deg_C L > 0$. The marked degree comparison gives $\deg_E L = jh$, so constancy of Hilbert polynomials gives $P_{\overline{E}_0}(t) = jh\,t + 1$. The special fibre $Z = \overline{E}_0$ is scheme-theoretically contained in the original fibre $C_\ell \subset S_0$ [38, Lemma 3.2(ii)]. Its pure quotient has cycle $j[C]$, hence equals $\mathcal{O}_{C_j}$: on the normal surface, the primary ideals at height one are determined by their multiplicities and are the symbolic powers. If T is the zero-dimensional kernel, Katz's $\chi(\mathcal{O}_{C_j}) = 1$ gives

$$1 = \chi(\mathcal{O}_Z) = 1 + \operatorname{length} T.$$

Therefore $Z = C_j$ as a closed subscheme. The resulting Hilbert map lifts through the chosen chart over the strictly henselian $\mathbb{C}[[u]]$ and gives a generic point of $\operatorname{Spec}\mathscr{H}_j$.

Conversely any geometric generic hull point is a subscheme of a contraction fibre with Hilbert polynomial $jh\,t + 1$. That fibre has a curve and is the reduced integral $\mathbb{P}^1$ at one of the roots. Its only one-dimensional closed subscheme is itself, so the object is exactly the corresponding exceptional curve. Two hull points representing the same curve give the same saturated closure and the same étale lift with prescribed closed point; hence they agree. The count consequently assigns one point to each embedded curve. Finally $\operatorname{Ext}^1(\mathcal{O}_E, \mathcal{O}_E) = 0$ on a split $(-1,-1)$-curve, by Lemma 4.23. Together with $H^0(\mathcal{O}_E) = \mathbb{C}((u))$ and $H^1(\mathcal{O}_E) = 0$, this

makes its embedded Hilbert tangent space zero. Every geometric generic local ring is therefore a field by Nakayama, so the generic fibre is reduced. We obtain

$$n_j = k_j = \operatorname{rank}_{R_0} \mathscr{H}_j = \#\{\text{generic exceptional curves of class } j[C]\} = m_j.$$

The count is for the original constant translation. For a sufficiently small boundary-proper transverse analytic perturbation, the argument principle gives the same reduced-divisor intersection number m_j. This also gives the original local Hilbert count for that perturbation. □

Proposition 2.20 (A formal marked deformation into ordinary double points)**.** *There is a finite extension K'/K with valuation ring $\mathbb{C}[[u]]$, where $K' = \mathbb{C}((u))$ and $s = u^e$ for some $e \geq 1$. The completed local models at the generic curve points are small resolutions of split ordinary double points. Moreover,*

$$W_j(u^e, w) = \prod_{a=1}^{n_j} (w - w_{j,a}(u)), \qquad w_{j,a}(u) \in u\mathbb{C}[[u]],$$

with all roots pairwise distinct, including roots belonging to different coefficients. The pullback of the universal partial resolution has exactly n_j pairwise disjoint $(-1,-1)$-curves in class $j[C]$, and no other complete curves.

Proof. Split the finitely many separable polynomials W_j and the Cartan lifts of their parameter points. The single-root and transverse conditions are algebraic and hold at these field-valued points. The explicit ADE hypersurface equations, invertible fibre Hessian, and unit-smoothing argument in Lemma 2.11 persist after scalar extension. They therefore identify the resulting curve points as geometrically ordinary double points. The formal coordinate recursions use only the inverse Hessian and characteristic-zero integer denominators; splitting their quadratic tangent cones requires only finitely many square roots. Enlarge the field by one finite extension that supplies these roots and the local rulings at all the finitely many points. The formal coordinate reduction can be made explicit. Write a completed hypersurface equation as $q(x) + r_d(x) +$ terms of degree at least $d+1$, where q is nondegenerate quadratic, $d \geq 3$, and r_d is homogeneous of degree d. If A is the Hessian of q, the substitution

$$x \longmapsto x - \frac{1}{d} A^{-1} \nabla r_d$$

removes r_d: Euler's identity gives $x^{\mathrm{t}} \nabla r_d = d r_d$, and all other new terms have degree at least $d+1$. Successive substitutions converge in the maximal-ideal topology to a formal automorphism with identity linear part. Thus the equation becomes its quadratic part. Diagonalizing that part and adjoining square roots of its nonzero coefficients and of -1 gives the split equation $a_1 a_2 - b_1 b_2 = 0$. The small resolutions correspond to its two rulings. Denote the resulting field by K'. The normalization of $\mathbb{C}[[s]]$ in K' is a finite complete local DVR with residue field $\mathbb{C}$ [35, Remark 15.113.6 and Lemmas 10.161.8, 10.153.4, Tags 09E8, 032L and 04GH]. Choose a uniformizer t. Successively taking residues gives a unique convergent expansion in powers of t, so this ring is $\mathbb{C}[[t]]$. Write $s = t^e a(t)$ with $a(t)$ a unit. An e-th root of its nonzero constant coefficient and the binomial series for its remaining unit part give $a(t)^{1/e}$. Then $u = t a(t)^{1/e}$ is a uniformizer and $s = u^e$. Each root is integral because W_j is monic, and $W_j(0,T) = T^{n_j}$ forces its residue to be zero. Thus $w_{j,a}(u) \in u\mathbb{C}[[u]]$. Separability and coprimality give all the asserted

distinctness. Lemma 2.11 supplies one $(-1,-1)$-curve of class $j[C]$ at each root; different roots lie in different w-fibres and are disjoint.

It remains to exclude further complete curves. The completed target is affine, so every complete curve in the source maps to a point and lies in the exceptional locus; the regular function w is constant on it. The curve discriminant D^{curv} is the full locus of complete exceptional curves. Every generic point of the formal disk on that locus is a good point of one unique D_j, where the unique curve has already been counted, and the disk avoids the codimension-two bad locus. Hence the list is exhaustive. □

Remark 2.21. The relative family retains all moving roots over $\mathbb{C}[[s,w]]$. Its generic fibre is formed over $\mathbb{C}[[s,w]][1/s]$ and finite extensions of that ring, as in Lemma 2.17. Over $\mathbb{C}((u))[[w]]$, the element $w - w_{j,a}(u)$ is a unit whenever $w_{j,a} \neq 0$. Thus one must localize the finite relative family before making any further completion; completing first at $w = 0$ would discard moving root points.

2.2 Tilting generators and the local matrix blocks

The deformed classifying arc locates the exceptional curves and records their classes. We now identify the algebra at each curve. The fixed universal generator survives the relevant classifying base changes; at a split ordinary double point, its determinant degree counts the nonfree projective summands, whose stable endomorphisms produce the matrix block.

2.2.1 The universal generator and base change

Karmazyn's universal-flop construction supplies the simultaneous geometry and its universal algebra [21, Sections 2.3 and 3.3]. Van den Bergh's perverse generators select a normalized basic generator on that family [37, Propositions 3.2.5–3.2.7]. The lemmas below establish how this generator pulls back along an arbitrary Noetherian classifying base. The universal length-ℓ partial resolution carries a basic zero-perverse generator and its positive dual

$$\mathcal{T} = \mathcal{O} \oplus \mathcal{T}', \qquad \mathcal{P} = \mathcal{T}^{\vee} = \mathcal{O} \oplus \mathcal{P}'.$$

Set

$$\mathcal{A}_\ell = \mathrm{End}_{\mathcal{Y}_\ell}(\mathcal{T})^{\mathrm{op}} \cong \mathrm{End}_{\mathcal{Y}_\ell}(\mathcal{P}), \qquad e_0 = [\mathcal{O}].$$

The universal source, target, centre, and algebra are flat over the marked base $H = \operatorname{Spec} \mathbb{H}_\ell$ by the universal-flop construction [21, Sections 2.3 and 3.3], and the universal target is affine of finite type over H. The following lemmas establish the Tor-independence and ordinary endomorphism base change needed for an arbitrary Noetherian classifying algebra. The positive nonfree summand is the normalized maximal extension

$$0 \longrightarrow \mathcal{O}^{\oplus(\ell-1)} \longrightarrow \mathcal{P}' \longrightarrow \mathcal{L} \longrightarrow 0, \qquad \deg_C \mathcal{L} = 1. \tag{2.3}$$

Thus $\operatorname{rk} \mathcal{P}' = \ell$ and $\deg_C \det \mathcal{P}' = 1$. These numerical normalizations also follow directly from the specified basic generator: Karmazyn's graded choice completes to the basic generator at the central target point [21, Remark 3.5]; Van den Bergh's complete-local classification gives determinant degree one for its nonfree positive summand [37, Proposition 3.5.4 and

Theorem 3.5.5], and its rank is ℓ [21, Remark 3.11]. The generator $\mathcal{P}$ belongs to the (-1)-perverse heart, while $\mathcal{T}$ belongs to the zero-perverse heart. Duality preserves the distinguished free projector and replaces the endomorphism algebra by its opposite.

The sequence (2.3) can be obtained on the universal family itself. Put $E = \mathcal{P}'$, and let $\mathcal{J}_C$ be the ideal of the reduced central curve. The structure-sheaf vanishing gives $H^1(\mathcal{J}_C) = 0$; global generation of E, together with cohomological dimension one, then gives $H^1(\mathcal{J}_C E) = 0$. Hence global sections of E surject onto its sections on C. The globally generated restriction of rank ℓ and degree one is $\mathcal{O}_C(1) \oplus \mathcal{O}_C^{\ell-1}$: general $\ell - 1$ sections give a line quotient of degree one, and $H^1(\mathcal{O}_C(-1)) = 0$ splits that extension. The positive line is invariant under the grading action. Since $\mathbb{G}_m$ is linearly reductive, choose an equivariant splitting of the resulting sequence of global sections on C. A weight basis of the complementary trivial bundle then lifts to homogeneous global sections. Their degeneracy locus is closed and invariant. Its image under the projective contraction, if nonempty, would contain the vertex of the positively graded affine target, contradicting their independence on C. Their cokernel is therefore the line bundle $\mathcal{L} = \det E$ everywhere. Finally, $H^1(E^\vee) = 0$ shows that the extension classes generate $H^1(\mathcal{L}^{-1})$; after completion they are minimal because the nonfree summand is indecomposable [37, Lemma 3.5.2].

Lemma 2.22 (Tor-independence for arbitrary classifying pullback)**.** *Let B be a Noetherian $\mathbb{H}_\ell$-algebra, and form*

$$\begin{array}{ccc} \mathcal{Y}_B & \xrightarrow{g} & \mathcal{Y}_\ell \\ \pi_B \downarrow & & \downarrow \pi \\ \mathcal{X}_B & \xrightarrow{h} & \mathcal{X}_\ell. \end{array}$$

This square is Tor-independent. Consequently

$$R\pi_{B*} L g^* E \simeq L h^* R\pi_* E \tag{2.4}$$

for every perfect E on $\mathcal{Y}_\ell$, and $R\pi_{B*}\mathcal{O}_{\mathcal{Y}_B} = \mathcal{O}_{\mathcal{X}_B}$.

Proof. Write the local rings as $C \to A \to Q$, where A and Q are flat over $C = \mathcal{O}_H$. Then

$$Q \otimes^L_A (A \otimes_C B) \simeq Q \otimes^L_C B \simeq Q \otimes_C B.$$

Thus the displayed Cartesian square is Tor-independent. The morphism π is projective, hence quasi-compact and quasi-separated. Independently, a perfect complex E has quasi-coherent cohomology. The derived base-change isomorphism of [35, Lemma 36.22.5, Tag 08IB] therefore applies: its canonical map

$$L h^* R\pi_* E \longrightarrow R\pi_{B*} L g^* E$$

is an isomorphism, giving (2.4). Taking $E = \mathcal{O}$ and using $R\pi_*\mathcal{O} = \mathcal{O}$ gives the final assertion. □

Lemma 2.23 (Base change of the perverse generator)**.** *The pullbacks $\mathcal{P}_B = g^*\mathcal{P}$ and $\mathcal{T}_B = g^*\mathcal{T}$ are projective generators of the (-1)- and zero-perverse hearts of π_B respectively, and they induce the corresponding equivalences on bounded coherent derived categories.*

Proof. We verify the hypotheses of Van den Bergh's projective-generator criterion after pullback. Write $\mathcal{X}_\ell = \operatorname{Spec} \mathcal{R}_\ell$. Since $\mathcal{R}_\ell$ is of finite type over $\mathbb{H}_\ell$, $\mathcal{X}_B = \operatorname{Spec}(\mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B)$ is Noetherian.

The morphism π_B is projective, its fibres have dimension at most one, and the preceding lemma gives $R\pi_{B*}\mathcal{O} = \mathcal{O}$. These are the geometric hypotheses of the criterion; no smoothness or reducedness of B is required. His category $\mathfrak{V}$ consists of globally generated bundles M satisfying $R^1\pi_*(M^\vee) = 0$. A finite evaluation surjection for $\mathcal{P}$ remains surjective after pullback. Since $\mathcal{P} \in \mathfrak{V}$, $R\pi_*\mathcal{T} = (\pi_*\mathcal{T})[0]$. By Lemma 2.22,

$$R\pi_{B*}\mathcal{T}_B \simeq Lh^*(\pi_*\mathcal{T}).$$

The left side lies in $D^{\geq 0}$, while the right side lies in $D^{\leq 0}$. Both are therefore concentrated in degree zero, giving $R^1\pi_{B*}\mathcal{T}_B = 0$. Relative ampleness of $\det \mathcal{P}$, global generation, and the direct $\mathcal{O}$-summand also survive. Van den Bergh's intrinsic projective-generator criterion now applies [37, Propositions 3.2.5–3.2.7], and the derived equivalences follow from [37, Corollary 3.2.8]. □

Proposition 2.24 (Ordinary endomorphism base change)**.** *There is a canonical isomorphism of marked algebras*

$$\mathrm{End}_{\mathcal{Y}_B}(\mathcal{T}_B)^{\mathrm{op}} \cong \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B. \tag{2.5}$$

Moreover, all higher self-Ext groups vanish. The isomorphism preserves e_0, and the bundle pullbacks preserve ranks and determinants. The algebra identification remains valid after central localization and Noetherian completion.

Proof. Apply derived base change to the vector bundle $\mathcal{E}nd(\mathcal{T})$. The tilting property gives

$$R\pi_*\mathcal{E}nd(\mathcal{T}) = \pi_*\mathcal{E}nd(\mathcal{T})[0].$$

Put $M = \Gamma(\mathcal{Y}_\ell, \mathcal{E}nd(\mathcal{T})) = \mathcal{A}_\ell^{\mathrm{op}}$, and write $\mathcal{X}_\ell = \operatorname{Spec} \mathcal{R}_\ell$. The flatness of both $\mathcal{R}_\ell$ and M over $\mathbb{H}_\ell$, asserted for the universal family in [21, Section 3.3, the defining paragraph for $\mathcal{A}_\ell$], gives

$$M \otimes^L_{\mathcal{R}_\ell} (\mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B) \simeq M \otimes^L_{\mathbb{H}_\ell} B \simeq M \otimes_{\mathbb{H}_\ell} B.$$

Consequently the pulled-back endomorphism bundle has derived pushforward concentrated in degree zero. Taking sections over the affine target proves (2.5) and the vanishing of all higher self-Ext groups. The degree-zero map is the pullback of endomorphisms, so it respects composition and the identity. Pullback also preserves the inclusion and retraction of the distinguished free summand, hence its projector e_0; rank and determinant commute with bundle pullback.

The endomorphism algebra is finite over the Noetherian target ring by properness. A central localization and the completion of a Noetherian localization at an ideal are flat. Applying flat base change to the endomorphism bundle therefore identifies its endomorphism algebra with the corresponding ordinary scalar extension. Finiteness identifies this scalar extension with the completed algebra when taking a completion. The same argument preserves the higher self-Ext vanishing and e_0. □

The affine target ring is also the centre of this endomorphism algebra. Indeed, a central endomorphism of $\mathcal{P}_B = \mathcal{O} \oplus \mathcal{P}'_B$ commutes with the free projector and is therefore diagonal, with a scalar r on $\mathcal{O}$. Commuting with every map $\mathcal{O} \to \mathcal{P}'_B$ forces the other diagonal block to act by r on all global sections. Global generation then makes that block r times the identity. Hence the centre is $\Gamma(\mathcal{Y}_B, \mathcal{O}) = \Gamma(\mathcal{X}_B, \mathcal{O})$. This argument also applies after the stated localizations and completions, and duality gives the same centre for the zero-perverse convention.

Write $\Lambda_B = \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B$. The marked endomorphism comparison precedes formation of the relative contraction quotient $\Lambda_B/\Lambda_B e_0 \Lambda_B$.

Proposition 2.25 (The special fibre is the basic NCCR)**.** *Let $B = \mathbb{C}[[s, w]]$ carry a classifying map whose restriction modulo s is the central arc. Localize Λ_B at the central contraction point of its centre and complete there. The reduction $\widehat{\Lambda_B}/(s)$ is isomorphic to the basic NCCR of the original flop, preserving the free idempotent e_0.*

Proof. The algebra is finite over its Noetherian centre, so completion commutes with reduction modulo s. Lemma 2.3 then identifies the completed geometry with the original flop.

It remains to identify the transported generator. On the completed one-curve flop, Van den Bergh's classification gives exactly two indecomposable positive perverse projectives: $\mathcal{O}$ and the maximal extension $\mathcal{M}$, with $\operatorname{rk}\mathcal{M} = \ell$ and $\deg_C \det \mathcal{M} = 1$ [37, Proposition 3.5.4 and Theorem 3.5.5]. Write the reduced nonfree universal summand as $\mathcal{O}^{\oplus a} \oplus \mathcal{M}^{\oplus b}$. Determinant degree gives $b = 1$, while rank gives $\ell = a + \ell$, hence $a = 0$. The reduced positive generator is therefore $\mathcal{O} \oplus \mathcal{M}$. Dualizing gives the selected zero-perverse basic generator and fixes e_0. □

2.2.2 The generator at a transverse root

After the finite extension of Proposition 2.20, let λ be a geometric generic coefficient-j point, let $\widehat{Y}_\lambda \to \operatorname{Spec}\widehat{R}_\lambda$ be its completed small resolution, and let C_λ be its exceptional curve. The local model is split over K'.

Lemma 2.26 (The split ODP projectives over the splitting field)**.** *Let F be any characteristic-zero field and put*

$$R_F = F[[a_1, a_2, b_1, b_2]]/(a_1a_2 - b_1b_2), \qquad I = (a_1, b_1).$$

On the small resolution $\pi_Z \colon Z = \operatorname{Proj}_{R_F} \bigoplus_{n\geq 0} I^n \to \operatorname{Spec} R_F$, let $\mathcal{Q} = \mathcal{O}_Z(1)$. Then $\mathcal{O}_Z \oplus \mathcal{Q}$ is the basic positive perverse projective generator. Every positive perverse projective vector bundle is a direct sum of $\mathcal{O}_Z$ and $\mathcal{Q}$, and

$$\operatorname{rk}\mathcal{Q} = 1, \qquad \deg_C \mathcal{Q} = 1, \qquad \pi_{Z*}\mathcal{Q} = I.$$

These assertions remain true after completion along the exceptional curve.

Proof. Write $\theta = b_1/a_1$ on the a_1-chart and $\theta^{-1} = a_1/b_1$ on the b_1-chart. After removing the Rees torsion and completing along the exceptional curve, the charts and their gluing are

$$\widehat{U}_{a_1} = \operatorname{Spf} F[\theta][[a_1, b_2]], \quad b_1 = a_1\theta, \quad a_2 = \theta b_2,$$

$$\widehat{U}_{b_1} = \operatorname{Spf} F[\theta^{-1}][[b_1, a_2]], \quad a_1 = \theta^{-1}b_1, \quad b_2 = \theta^{-1}a_2.$$

Thus Z is the split Atiyah small resolution. Its exceptional curve is given by $a_1 = b_2 = 0$ on $\widehat{U}_{a_1}$ and $b_1 = a_2 = 0$ on $\widehat{U}_{b_1}$, so $C \cong \mathbb{P}^1_F$. The transition of the tautological quotient is that of $\mathcal{O}_{\mathbb{P}^1}(1)$, hence $\mathcal{Q}|_C = \mathcal{O}_C(1)$. The quotient $R_F/I \cong F[[a_2, b_2]]$ and the depth lemma show that I is maximal Cohen–Macaulay, hence reflexive. The pushforward of $\mathcal{Q}$ is reflexive by Hartogs extension across C in the smooth small resolution. It agrees with I away from the vertex, so $\pi_{Z*}\mathcal{Q} = I$.

To verify the perverse-generator conditions, use $N_{C/Z} \cong \mathcal{O}_C(-1)^{\oplus 2}$. On the n-th infinitesimal graded piece of the formal neighbourhood of C, the restrictions of $\mathcal{O}_Z$ and $\mathcal{Q}^{-1}$ are respectively direct sums of

$$\operatorname{Sym}^n N_{C/Z}^\vee \cong \mathcal{O}_C(n)^{\oplus(n+1)} \quad \text{and} \quad \mathcal{O}_C(n-1)^{\oplus(n+1)}.$$

Their first cohomology vanishes for every $n \geq 0$, as follows from the two-affine-chart computation of $H^1(\mathbb{P}^1_F, \mathcal{O}(k)) = 0$ for $k \geq -1$. The chart equations identify the ideal of C with $\mathfrak{m}_{R_F}\mathcal{O}_Z$. Thus its infinitesimal thickenings are exactly the thickenings used by the theorem on formal functions. Induction over these thickenings, followed by [35, Theorem 30.20.5, Tag 02OC], gives

$$R^1\pi_{Z*}\mathcal{O}_Z = R^1\pi_{Z*}\mathcal{Q}^{-1} = 0.$$

The algebraic small resolution of the affine quadratic cone is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$; the charts above describe its completion along the zero section. The bundle $\mathcal{Q}$ is relatively ample and globally generated. Together with the cohomology vanishings and the structure-sheaf summand, these properties verify Van den Bergh's projective-generator criterion [37, Propositions 3.2.5–3.2.7], both over the complete local target and on the formal completion.

For the classification of projectives, the endomorphism algebra of $\mathcal{O}_Z \oplus \mathcal{Q}$ is finite over the complete local ring R_F and is semiperfect. The two summands have local endomorphism rings $\mathrm{End}(\mathcal{O}_Z) = R_F = \mathrm{End}(\mathcal{Q})$, and their degrees on C distinguish them. Their projectors form a complete set of pairwise inequivalent primitive idempotents. Projectives over a semiperfect ring are direct sums of the corresponding indecomposable projectives. The tilting equivalence therefore gives the decomposition of every projective in the heart. Both summands and their idempotents are defined over F. Since Z is projective over the complete Noetherian ring R_F, Grothendieck existence identifies coherent sheaves on Z with coherent sheaves on its completion along C [35, Lemma 30.24.3, Tag 0885]. Formal functions preserves the cohomological conditions defining the perverse hearts, and full faithfulness preserves their Hom conditions. Hence this equivalence preserves projectives and the displayed decomposition on the formal completion. □

Proposition 2.27 (Decomposition of the local projective generator)**.** *Let $\mathcal{Q}$ be the unique nonfree indecomposable positive perverse projective on the completed ODP small resolution, normalized by* $\mathrm{rk}\,\mathcal{Q} = 1$ *and* $\deg \det \mathcal{Q}|_{C_\lambda} = 1$*. Then*

$$\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus(\ell-j)} \oplus \mathcal{Q}^{\oplus j}. \tag{2.6}$$

After pushforward to the ODP centre,

$$N_\lambda \cong \widehat{R}_\lambda^{\oplus(\ell-j)} \oplus I_\lambda^{\oplus j},$$

where I_λ is the rank-one Atiyah MCM module.

Proof. Proposition 2.24 and Lemma 2.23 transport the universal positive generator to the completed local model after field specialization and completion. The roots have been split over K', so the local model is the split ODP of Lemma 2.26 with $F = K'$. That lemma and Krull–Schmidt give $\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus a} \oplus \mathcal{Q}^{\oplus b}$. Rank gives $a + b = \ell$. Determinant commutes with pullback, and Lemma 2.10 gives

$$\deg_{C_\lambda} \det \widehat{\mathcal{P}}'_\lambda = j \deg_C \det \mathcal{P}' = j.$$

The degree of the decomposition is b. Thus $b = j$ and $a = \ell - j$. Lemma 2.26 identifies the pushforward of $\mathcal{Q}$ with the Atiyah module I_λ. □

2.2.3 The local contraction quotient

The preceding decomposition identifies the local contraction quotient with stable endomorphisms of the Atiyah module. For a module over a local ring R, write $\underline{\mathrm{End}}_R$ for endomorphisms modulo maps factoring through finite free R-modules.

Lemma 2.28 (Stable endomorphisms of the Atiyah module)**.** *Let*

$$R_{\mathrm{odp}} = F[[a_1, a_2, b_1, b_2]]/(a_1a_2 - b_1b_2), \qquad I = (a_1, b_1),$$

where F has characteristic zero. Then

$$\underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I) \cong F, \qquad \underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I^{\oplus j}) \cong M_j(F).$$

Proof. Write $R = R_{\mathrm{odp}}$. Since I is rank-one reflexive over the normal domain R, its endomorphisms are multiplications by fractions. At each height-one prime, I is free of rank one; hence $(I : I) \subseteq \bigcap_{\mathrm{ht}\, p=1} R_p = R$. Thus $\mathrm{End}_R(I) = R$.

To identify the maps that factor through a free module, we claim that

$$I^{-1} = R + R(b_2/a_1).$$

The displayed right-hand side lies in I^{-1}, since $(b_2/a_1)a_1 = b_2$ and $(b_2/a_1)b_1 = a_2$. Conversely, if $qI \subset R$, put $r = qa_1 \in R$. Then $b_1 r \in a_1 R$. In $R/(a_1) = F[[a_2, b_1, b_2]]/(b_1b_2)$, the annihilator of b_1 is (b_2). Thus $r = a_1r_0 + b_2r_1$, and $q = r_0 + (b_2/a_1)r_1$, proving the claim. Endomorphisms factoring through a free module form the ideal

$$II^{-1} = (a_1, b_1) + (b_2, a_2) = (a_1, a_2, b_1, b_2) = \mathfrak{m}.$$

Therefore $\underline{\mathrm{End}}_R(I) = R/\mathfrak{m} = F$; taking matrices gives the second assertion. □

Proposition 2.29 (Comparison with the MCM generator)**.** *For the completed ODP small resolution, pushforward gives an isomorphism*

$$\mathrm{End}_{\widehat{Y}_\lambda}(\widehat{\mathcal{P}}_\lambda) \cong \mathrm{End}_{\widehat{R}_\lambda}\big(\widehat{R}_\lambda^{\oplus(\ell-j+1)} \oplus I_\lambda^{\oplus j}\big) \tag{2.7}$$

which preserves the projector onto the distinguished free summand.

Proof. The completed ODP is a normal complete local Gorenstein threefold, and its small resolution is regular, projective, birational, and crepant, with fibres of dimension at most one and $R\widehat{\pi}_*\mathcal{O} = \widehat{R}$. The positive bundle is the perverse projective generator by Lemma 2.23; its two required $R^1\widehat{\pi}_*$-vanishings follow from the defining $\mathfrak{V}$-condition and global generation. Van den Bergh's comparison theorem identifies the resolved endomorphism algebra with that of the pushed-forward MCM generator [37, Lemma 3.2.9 and Proposition 3.2.10]. Naturality sends the direct-summand projector to the distinguished free projector. □

Theorem 2.30 (The completed coefficient-j factor)**.** *At every transverse coefficient-j point produced by the formal marked deformation of the classifying arc, the completion of the relative contraction quotient is the matrix algebra $M_j(K')$. The maximal ideal of the completed centre acts trivially on this factor.*

Proof. After the arbitrary classifying pullback, central localization and Noetherian completion are flat. For a finite algebra Λ with idempotent e_0,

$$\widehat{\Lambda/\Lambda e_0\Lambda} \cong \widehat{\Lambda}/\widehat{\Lambda}e_0\widehat{\Lambda}.$$

By Proposition 2.29, the completed quotient is the endomorphism algebra of $\widehat{R}^{\ell-j+1} \oplus I^j$ modulo the ideal generated by one distinguished free projector. Every other free projector factors through that copy, so the quotient is the stable endomorphism algebra of I^j. Lemma 2.28 gives $M_j(K')$. The centre acts through the residue field, proving the last assertion. □

2.3 Assembling a finite free deformation

The preceding two subsections provide the geometric points and their local matrix factors. The root multiplicities come from the Bryan–Katz–Leung divisor count and the GV interpretation used above [6, Lemma 2.8]; Toda's formula supplies the final dimension comparison [36, Theorem 1.1]. We assemble the local factors inside one finite relative quotient. Put $B = \mathbb{C}[[s, w]]$, and let $\nu : \operatorname{Spec} B \to H$ be the completion of the constant-translation family $F(s, w) = f_0(w)+sv$ from Lemmas 2.15 and 2.16. Set

$$R_B = \mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B.$$

Let $\mathfrak{n} \subset R_B$ be the maximal ideal of the central contraction point (it contains s), and define the central completion

$$S_\nu = \widehat{(R_B)_{\mathfrak{n}}}.$$

The universal source and target are flat over the marked base. After the pullback to B, localization and completion of the affine target are flat, as are their base changes on the source. Thus this completed geometric family is flat over B, and in particular over $\mathbb{C}[[s]]$, with the original completed flop as its special fibre. Let $\mathcal{T}_\nu$ be the completed pullback of the tilting bundle. The endomorphism comparison identifies the corresponding algebra before forming its free-vertex quotient:

$$\Lambda_\nu = \mathcal{A}_\ell \otimes_{\mathcal{R}_\ell} S_\nu \cong \operatorname{End}(\mathcal{T}_\nu)^{\mathrm{op}}, \qquad \mathscr{A} = \Lambda_\nu/\Lambda_\nu e_0 \Lambda_\nu. \tag{2.8}$$

Proposition 2.31 (Identification of the special fibre)**.** *There is a marked algebra isomorphism*

$$\mathscr{A}/(s) \cong A_{\mathrm{con}}.$$

Proof. The generator comparison is made before the idempotent quotient. Proposition 2.25 gives $\Lambda_\nu/(s) \cong \Lambda$, the selected basic NCCR of the original flop, preserving e_0. Right exactness then gives

$$\mathscr{A}/(s) \cong \frac{\Lambda}{\Lambda e_0 \Lambda} = A_{\mathrm{con}}.$$

□

Lemma 2.32 (Topological finiteness)**.** *Let $R_0 = \mathbb{C}[[s]]$, and define S as follows:*

$$S = R_0[[x_1, \dots, x_m]]/J.$$

Then S is complete Noetherian. Let M be finite over S. If M/sM is finite-dimensional over $\mathbb{C}$, then M is finite over R_0.

Proof. Choose lifts $m_1, \dots, m_d$ of a $\mathbb{C}$-basis of M/sM. For $m \in M$, successive reduction modulo sM gives

$$m = \sum_{i=1}^{d} \left(\sum_{a=0}^{q-1} c_{ia} s^a \right) m_i + s^q m^{(q)}, \qquad c_{ia} \in \mathbb{C}.$$

The series $c_i(s) = \sum_{a \geq 0} c_{ia} s^a$ belong to R_0. For each q, their tails are divisible by s^q, so

$$m - \sum_i c_i(s) m_i \in s^q M.$$

The ring S is Noetherian local, s lies in its maximal ideal, and M is finite over S. Krull intersection gives $\bigcap_q s^q M = 0$. Hence the m_i generate M over R_0. □

Proposition 2.33 (Finiteness of the relative contraction algebra)**.** *The algebra $\mathscr{A}$ is a finite $\mathbb{C}[[s]]$-module.*

Proof. The universal target is of finite type over H, so its pullback is of finite type over $B = \mathbb{C}[[s, w]]$. Completion at the central contraction point gives a ring of the form

$$S_\nu \cong \mathbb{C}[[s, x_1, \dots, x_m]]/J$$

for finitely many generators. Proposition 2.4(i) states that $\mathcal{A}_\ell$ is finite over $\mathcal{R}_\ell$. Base change and completion make Λ_ν finite over S_ν, and its idempotent quotient $\mathscr{A}$ remains finite over S_ν.

By Proposition 2.31, $\mathscr{A}/(s) \cong A_{\mathrm{con}}$ is finite-dimensional. The topological finiteness lemma now gives finiteness over $\mathbb{C}[[s]]$. □

Lemma 2.34 (Fullness away from the root support)**.** *Over the geometric generic marked slice, the relative contraction quotient vanishes away from the finitely many transverse curve-root points. Consequently it is a finite algebra supported on exactly those closed points.*

Proof. The root-divisor description identifies D^{curv} with the full locus where a complete curve occurs [29, Theorem 2.6]. Away from D^{curv}, a positive-dimensional fibre of the projective morphism would contain a complete curve. The morphism is therefore quasi-finite and hence finite. Since $R\pi_{\ell *}\mathcal{O}_{\mathcal{Y}_\ell} = \mathcal{O}_{\mathcal{X}_\ell}$, a finite base-changed morphism is an isomorphism. Proposition 2.20 meets D^{curv} only at the good transverse roots and supplies exactly the curves already counted. Away from those points, every summand of the pushed-forward vector bundle is locally free and, after shrinking, finite free. Every matrix unit in the endomorphism algebra of the pushed-forward vector bundle then factors through the distinguished rank-one free summand. Thus e_0 is full and the idempotent quotient vanishes.

We must also check that completion at the central target point has retained every moving root. Let R_H be the completed marked base and $\mathcal{C}_H$ the completed universal contraction quotient. The actual classifying-pullback comparison and the universal weighted presentation, proved independently in Propositions 4.6 and 4.5, give

$$\mathscr{A} \cong \mathcal{C}_H \otimes_{R_H} B, \qquad \mathrm{Fitt}_0^B(\mathscr{A}) = \left(\prod_j G_j(s, w)^{j^2} \right).$$

Their proofs use universal endomorphism base change and the Cartan algebra presentation, and do not use the semisimple-existence theorem being proved here. The flat source and target hypotheses for the classifying comparison were checked above.

After the finite splitting extension, each root is a rational point of $B[1/s] \otimes_{\mathbb{C}((s))} K'$. At a coefficient-$j$ root its local parameter is $w - w_{j,a}$, and the displayed Fitting ideal has order $j^2 > 0$. The localization of the actual quotient is therefore nonzero there. Since the free projector is full at every isomorphism point of the contraction, this nonzero part lies over an exceptional curve. The good single-root local model has just one such curve. Thus every listed root survives in the central completion, and no other support point is possible.

At each of these actual points, Theorem 2.30 gives the full nonzero matrix algebra. Proposition 2.33 makes the generic quotient finite-dimensional over its coefficient field and finite over its Artinian central image. Hence its support is precisely the asserted finite set of closed points. □

Lemma 2.35 (Assembly of the local factors)**.** *Let S be a Noetherian K'-algebra and A a finite S-algebra supported at K'-rational maximal ideals $p_1, \dots, p_r$, so $S/p_i = K'$. Suppose*

$$\widehat{A_{p_i}} \cong M_{d_i}(K')$$

and the maximal ideal of $\widehat{S_{p_i}}$ acts trivially. Then

$$A \cong \prod_{i=1}^{r} M_{d_i}(K').$$

In particular its scheme-theoretic support is reduced.

Proof. Completion $S_{p_i} \to \widehat{S_{p_i}}$ is faithfully flat. Since $p_i \widehat{A_{p_i}} = 0$, faithful flatness gives $p_i A_{p_i} = 0$. Hence A_{p_i} already equals its completion and is $M_{d_i}(K')$. Put $J = \bigcap_i p_i$. The finite module JA localizes to zero away from the support and also at every p_i, where all other maximal ideals become units. Thus $JA = 0$. The Chinese remainder theorem gives $S/J \cong (K')^r$; its central idempotents split A, and the local factors are the stated matrix algebras. □

Proposition 2.36 (The geometric generic fibre)**.** *After the finite splitting extension $K'/\mathbb{C}((s))$ of Proposition 2.20,*

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \tag{2.9}$$

Proof. Propositions 2.19 and 2.20 give exactly n_j coefficient-j points. In the normalized ramified model of that proposition, $K' = \mathbb{C}((u))$, $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. The split points are therefore K'-rational points of the formal w-disk, all specializing to its central maximal ideal. Lemma 2.34 gives the full support, and Theorem 2.30 identifies every completed factor with $M_j(K')$. Apply Lemma 2.35 simultaneously to all support points. □

Lemma 2.37 (Freeness from the fibre dimensions)**.** *Let M be a finite module over the DVR $\mathbb{C}[[s]]$. If*

$$\dim_{\mathbb{C}} M/sM = \dim_{\mathbb{C}((s))} M[1/s] = r,$$

then M is free of rank r.

Proof. The structure theorem gives

$$M \cong \mathbb{C}[[s]]^{\oplus r} \oplus \bigoplus_{h=1}^{m} \mathbb{C}[[s]]/(s^{e_h}).$$

The generic dimension is the free rank, while reduction modulo s has dimension $r+m$. Equality forces $m=0$. □

Proof of Theorem 2.1. Take the relative contraction quotient defined in (2.8). Proposition 2.31 gives its special fibre, and Proposition 2.36 gives its geometric generic product. In particular

$$\dim_{\mathbb{C}((s))} \mathscr{A}[1/s] = \sum_{j=1}^{\ell} j^2 n_j.$$

Proposition 2.33 gives finiteness over $\mathbb{C}[[s]]$. On the other hand, Proposition 2.4(v) and the special-fibre identification give

$$\dim_{\mathbb{C}} \mathscr{A}/(s) = \dim_{\mathbb{C}} A_{\mathrm{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Lemma 2.37 now removes the possible torsion and proves that $\mathscr{A}$ is finite free of this rank. □

3 The cocenter and uniqueness of the semisimple generic fibre

Section 2 constructs a deformation whose geometric generic fibre is a specified product of matrix algebras and puts the contraction algebra in the closure of the corresponding change-of-basis orbit. Here we prove that this orbit closure is the unique irreducible component of the algebra-law scheme through the contraction algebra. Comparing the orbit and tangent dimensions reduces the claim, by symmetry and periodicity, to a cocenter count on the marked ADE base.

3.1 Statement and proof strategy

For the completed simple one-curve flop $f\colon X \to \operatorname{Spec} R$, let $A = A_{\mathrm{con}}$ be its contraction algebra with $d = \dim_{\mathbb{C}} A$, and let n_j be its genus-zero GV invariants. Put

$$S_{\mathrm{GV}} := \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}.$$

Theorem 3.1 (Uniqueness of the semisimple generic fibre for one curve)**.** *For every completed simple one-curve flop of a smooth complex threefold,*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j. \tag{3.1}$$

Consequently A is a smooth point of the fixed-unit associative-law scheme X_d, the orbit closure $\overline{G \cdot S_{\mathrm{GV}}}$ is the unique irreducible component of X_d through A, and every complex semisimple algebra to which A deforms is isomorphic to S_{GV}. More generally, if $\mathscr{D}$ is a finite free unital $\mathbb{C}[[u]]$-algebra with $\mathscr{D}/u\mathscr{D} \cong A$ and geometrically semisimple generic fibre, then $\mathscr{D} \otimes_{\mathbb{C}[[u]]} \overline{\mathbb{C}((u))} \cong S_{\mathrm{GV}} \otimes_{\mathbb{C}} \overline{\mathbb{C}((u))}$.

The theorem identifies the split generic fibre. Section 2 separately constructs a finite free integral $\mathbb{C}[[u]]$-deformation that realizes it.

The orbit criterion requires equality of the tangent and orbit dimensions. The author first "exhausts" all the author can do on different aspects of non-commutative algebraic geometry to get symmetry, periodicity of contraction algebra, and centre–cocenter duality, and we use those to reduce that equality we needed for applying theorem 3.6 to contraction algebras to figuring out the actual dimension of $HH_0(A)$, which we expects to be equal to $\sum_{j=1}^{\ell} n_j$. And in next few subsections, we compute this cocenter from a universal presentation on the Cartan cover, descend the presentation to the marked base, and restrict it to the original classifying arc. Each reduced curve divisor occurs once in its determinant, so the arc length counts the simple factors in the resulting product of matrix algebras. Then finally getting the numerical equality we want.

3.2 The orbit criterion

Jelisiejew–Smoktunowicz study the unframed scheme of algebra laws; their orbit analysis gives local closedness and openness for semisimple laws [20, Section 1 and Theorem 2.14]. We need the corresponding statements for a fixed identity. Let k be algebraically closed of characteristic zero, let $V = k \cdot 1 \oplus W$ have dimension d, and let X_d be the affine scheme of associative multiplications on V for which the distinguished vector 1 is the identity. The scheme X_d is defined by the associativity equations. The group preserving the identity is

$$G = \mathrm{Stab}_{\mathrm{GL}(V)}(1).$$

For a unital algebra Λ, the normalized Hochschild cochains are

$$C^n_{\mathrm{norm}}(\Lambda, \Lambda) = \mathrm{Hom}_k((\Lambda/k \cdot 1)^{\otimes n}, \Lambda).$$

We write Z^n_{norm} and B^n_{norm} for the cocycles and coboundaries of this complex.

Lemma 3.2 (Fixed-unit tangent spaces)**.** *For $\Lambda \in X_d(k)$,*

$$T_\Lambda X_d = Z^2_{\mathrm{norm}}(\Lambda, \Lambda), \qquad T_\Lambda(G \cdot \Lambda) = B^2_{\mathrm{norm}}(\Lambda, \Lambda),$$

and

$$\mathrm{Lie}\,\mathrm{Stab}_G(\Lambda) = \mathrm{Der}(\Lambda).$$

Moreover, G is smooth, connected, and irreducible of dimension $d(d-1)$, and its orbits are precisely the unital isomorphism classes.

Proof. A dual-number multiplication $m + \varepsilon\phi$ has the prescribed unit precisely when ϕ is normalized. Linearized associativity is then the normalized Hochschild cocycle equation. Differentiating transport by $1 + \varepsilon f$, with $f(1) = 0$, gives the normalized coboundaries; the kernel of the resulting linear map is the derivation space.

Relative to $k \cdot 1 \oplus W$, every element of G has the unique form

$$\begin{pmatrix} 1 & \lambda \\ 0 & h \end{pmatrix}, \qquad \lambda \in W^*, \quad h \in \mathrm{GL}(W).$$

Thus $G \cong W^* \rtimes \mathrm{GL}(W)$, which proves the assertions about the group.

Let $O = G \cdot \Lambda$ have its reduced locally closed orbit structure, and write $q : G \to O$ for the orbit map. The irreducibility of G implies that of O. Its smooth locus is nonempty over the perfect field k, and is invariant under the transitive $G(k)$-action; hence O is smooth. In characteristic zero the dominant morphism q is generically separable. Thus its differential has rank $\dim O$ at some k-point $g \in G$. The identity $q \circ L_g = a_g \circ q$, with a_g the action of g on O, shows that its rank at g equals its rank at the identity, since both translations have invertible differentials. Consequently $d_e q$ surjects onto $T_\Lambda O$. Its image is the normalized coboundary space computed above, proving the claimed orbit tangent identity even when the ambient law scheme is nonreduced.

A unital algebra isomorphism fixes the identity, so its change of basis lies in G. □

Lemma 3.3 (A family gives fixed-unit orbit closure)**.** *Let $\mathfrak{o}$ be a complete equicharacteristic DVR with algebraically closed residue field k, coefficient field $k \subset \mathfrak{o}$ and fraction field K. Let $\mathscr{C}$ be a finite free unital $\mathfrak{o}$-algebra with special fibre A. If S is a d-dimensional k-algebra and*

$$\mathscr{C}_K \otimes_K \overline{K} \cong S \otimes_k \overline{K},$$

then

$$A \in \overline{G \cdot S} \subset X_d.$$

Proof. The unit of $\mathscr{C}$ is unimodular because it reduces to the nonzero unit of A. Extend it to an $\mathfrak{o}$-basis lifting the prescribed basis of A. The structure constants define a morphism $\operatorname{Spec} \mathfrak{o} \to X_d$ whose closed point is A.

Let $Y = \overline{G \cdot S}$ with its reduced closed structure. The assumed geometric generic-fibre isomorphism preserves the identity, so the geometric generic point of the morphism defined by structure constants above is a $G(\overline{K})$-translate of S. Every equation of Y therefore maps to zero in $\overline{K}$. Since $\mathfrak{o} \hookrightarrow \overline{K}$ is injective, it already maps to zero in $\mathfrak{o}$. The entire morphism factors through Y, and its closed point A belongs to Y. □

The next lemma gives the boundary form of the orbit argument. The algebraic-group action makes the orbit smooth, and a local dimension comparison controls a point in its closure.

Lemma 3.4 (Smoothness at an open-orbit boundary)**.** *Let a smooth irreducible algebraic group H act on a finite-type scheme X over an algebraically closed field k of characteristic zero. Let $y \in X(k)$, and write $O = H \cdot y$.*

(i) *If $\dim T_y X = \dim O$, then y is smooth and O is open in X.*

(ii) *Suppose O is open. If a closed point $x \in \overline{O}$ satisfies $\dim T_x X = \dim O$, then x is smooth and $\overline{O}$ is its unique irreducible component. Every open orbit whose closure contains x equals O.*

Proof. Work first in the underlying reduced scheme X_{red}. The orbit map has constructible image, so that image contains a nonempty open subset of its irreducible reduced closure. Translates of this subset show that O is locally closed. Since k is perfect, this reduced orbit has a nonempty smooth locus; transitivity makes the whole orbit smooth. This is the orbit argument of [4, Proposition 1.11], applied over the stated algebraically closed characteristic-zero field. It does not require X to be reduced; the local rings in the dimension comparison below

are those of the original scheme X. For (i), let $R = \mathcal{O}_{X,y}$ and write the orbit local ring as R/I. With $\delta = \dim O$,

$$\delta = \dim(R/I) \leq \dim R \leq \operatorname{edim} R = \delta.$$

Thus R is regular, hence a domain. The prime ideal I has height zero, so it is zero. The locally closed orbit therefore contains an open neighbourhood of y; its translates show that O is open in the original scheme X. For (ii), irreducibility of H makes $\overline{O}$ an irreducible component of dimension δ. At the closed point x, the same dimension inequalities give regularity and hence smoothness. A regular local ring has one minimal prime, so this is the unique component through x. The closure of any other open orbit through x is the same component. The two nonempty open orbits must intersect, and hence coincide. □

Proposition 3.5 (Openness of semisimple orbits)**.** *If $S \in X_d(k)$ is split semisimple, then S is a smooth point of the scheme X_d and $G \cdot S$ is open in X_d.*

Proof. The algebra S is separable, hence projective as an S^e-module, so $HH^i(S,S) = 0$ for $i > 0$. In particular its normalized cocycles and coboundaries in degree two agree. By Lemma 3.2, the tangent dimension at S is the orbit dimension. Lemma 3.4(i) proves smoothness and openness. □

Theorem 3.6 (Orbit-theoretic smoothness and uniqueness)**.** *Let $A, S \in X_d(k)$, with S split semisimple. Assume*

$$A \in \overline{G \cdot S}, \qquad \dim T_A X_d = \dim(G \cdot S). \tag{3.2}$$

Then:

(i) *A is a smooth point of X_d;*

(ii) *$\overline{G \cdot S}$ is the unique irreducible component of X_d through A;*

(iii) *if S' is split semisimple and $A \in \overline{G \cdot S'}$, then $S' \cong S$.*

Proof. Proposition 3.5 makes both semisimple orbits open, and G is smooth and irreducible by Lemma 3.2. Apply Lemma 3.4(ii). The equality of the two orbits yields a unital algebra isomorphism, again by Lemma 3.2. □

Corollary 3.7 (Hochschild numerical criterion)**.** *Let $A, S \in X_d(k)$, with S semisimple and $A \in \overline{G \cdot S}$. The dimension equality in Theorem 3.6 is equivalent to*

$$\dim \operatorname{Der}(A) - \dim HH^2(A) = d - \dim Z(S). \tag{3.3}$$

In particular, it holds if

$$\dim Z(A) = \dim Z(S), \qquad \dim HH^1(A) = \dim HH^2(A).$$

Proof. Lemma 3.2 gives

$$\dim T_A X_d = d(d-1) - \dim \operatorname{Der}(A) + \dim HH^2(A).$$

For split semisimple S, all derivations are inner and $\dim \operatorname{Der}(S) = d - \dim Z(S)$. Therefore

$$\dim(G \cdot S) = d(d-1) - (d - \dim Z(S)),$$

where characteristic zero ensures that the stabilizer group scheme is smooth, so its dimension equals the dimension of its Lie algebra. This proves the equivalence. Finally, $\dim \operatorname{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$. □

3.3 Symmetry, periodicity, and Tate duality

The deformation of Section 2 supplies the existence and orbit-closure hypothesis. In 3.3, the auther uses symmetry and four-periodicity of contraction algebra to give $\dim HH^1(A) = \dim HH^2(A)$, and a symmetrizing trace identifies the centre with the dual cocenter, and then the orbit criterion is therefore reduced to a cocenter-dimension calculation.

Proposition 3.8 (Symmetry)**.** *The contraction algebra is symmetric:* $A_{\mathrm{con}} \cong DA_{\mathrm{con}}$ *as* A_{con}*-bimodules.*

Proof. Let N be the nonfree Van den Bergh summand. The completed base R is an isolated cDV singularity and $A = \underline{\mathrm{End}}_R(N)^{\mathrm{op}}$. The stable endomorphism algebra is symmetric by [2, Theorem 2.6], and the opposite of a symmetric algebra is symmetric. □

Proposition 3.9 (Bimodule four-periodicity)**.** *In the stable category of* A*-bimodules,*

$$\Omega^4_{A^e}(A) \simeq A.$$

Proof. Put $\Lambda = \underline{\mathrm{End}}_R(N)$, so that $A = \Lambda^{\mathrm{op}}$ in our convention. Jasso–Keller–Muro's Theorem 3.2.5 applies because R is an isolated cDV singularity with a crepant resolution. Their equation (3.2.4) identifies the coefficient bimodule with the diagonal Λ-bimodule, and their theorem gives an exact sequence with projective middle terms and Λ at both ends. Exchanging the left and right actions therefore gives the exact sequence of A-bimodules

$$0 \longrightarrow A \longrightarrow P_3 \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

with projective middle terms [19, Theorem 3.2.5]. The displayed sequence is the image, under passage to opposite bimodules, of the representative of the restricted universal Massey product supplied there. It therefore yields the stated stable bimodule isomorphism.

□

Lemma 3.10 (Functorial hypersurface periodicity)**.** *Let* Q *be a Noetherian regular local ring of finite dimension, let* $f \in Q$ *be a nonunit nonzerodivisor, and put* $R = Q/(f)$*. On the ungraded stable category of finitely generated maximal Cohen–Macaulay* R*-modules there is a natural isomorphism*

$$\Omega^2_R \simeq \mathrm{Id}.$$

Proof. For a nonzero maximal Cohen–Macaulay R-module M, $\operatorname{depth}_Q M = \dim Q - 1$. The regular-local resolution bound [35, Proposition 10.110.1, Tag 00O7] therefore gives

$$0 \longrightarrow F_{1,M} \xrightarrow{\varphi_M} F_{0,M} \longrightarrow M \longrightarrow 0$$

with finite free Q-modules and injective φ_M. Since $fM = 0$, multiplication by f factors uniquely through φ_M, giving $\psi_M : F_{0,M} \to F_{1,M}$ with

$$\varphi_M\psi_M = fI, \qquad \psi_M\varphi_M = fI.$$

The second identity follows by applying the injective φ_M. The zero module is handled by zero free modules.

Write a bar for reduction modulo f. These maps give an exact periodic resolution over R. For example, if $\varphi_M(x) = fy = \varphi_M\psi_M(y)$, injectivity gives $x = \psi_M(y)$; hence $\ker \bar{\varphi}_M = \operatorname{im} \bar{\psi}_M$.

The other kernel identity follows in the same way using injectivity of ψ_M, which follows from $\varphi_M\psi_M = fI$. Consequently

$$\vartheta_M : M \xrightarrow{\sim} K_2(M) := \ker \bar{\varphi}_M, \qquad [x] \longmapsto \bar{\psi}_M(x),$$

identifies M with a second syzygy.

For an arbitrary homomorphism $a : M \to N$, lift it to $a_0 : F_{0,M} \to F_{0,N}$ and then choose a_1 with $\varphi_N a_1 = a_0\varphi_M$. Injectivity of φ_N and the common equation f give

$$a_1\psi_M = \psi_N a_0.$$

Thus $K_2(a)\vartheta_M = \vartheta_N a$, where $K_2(a)$ is the restriction of $\bar{a}_1$ to the second kernel. Changing the first lift by $\varphi_N b$ changes the second by $b\varphi_M$, whose reduction is zero on $K_2(M)$. This proves independence of the lift and compatibility with addition, identity maps and composition. Maps factoring through a free module still factor through a free module, since $K_2(P) \cong P$ for free P, so these maps and isomorphisms descend to the stable category.

Finally this functor agrees naturally there with the second syzygy formed using any other free presentations. Write $d_{1,M}$ and $d_{1,N}$ for their degree-one differentials. Lift the identity in both directions between the two presentations. If two chain lifts induce the same module map, their differences satisfy $\delta_0 = d_{1,N}h_0$; then $\delta_1 - h_0 d_{1,M}$ takes values in the target second kernel. Its restriction to the source second kernel factors through the source degree-one free module. Thus the induced differences vanish stably. The comparison maps are consequently inverse stable isomorphisms and are natural in every module homomorphism. Composing them with ϑ_M proves the asserted natural isomorphism, including its restriction to any stable additive subcategory. □

Alternative cluster-tilting argument. Dugas's theorem provides a second route to four-periodicity from a cluster-tilting object in a Frobenius category. Let $\mathcal{B} = \mathrm{CM}(R)$, whose projective-injective objects are $\operatorname{add} R$. Write the Van den Bergh tilting bundle on the smooth source X as $\mathcal{O}_X \oplus \mathcal{N}$, and put $N = f_*\mathcal{N}$. The nonfree summand in [19, Theorem 1.3.1] is a $2\mathbb{Z}$-cluster-tilting object of $\underline{\mathrm{CM}}(R)$. This property also holds for its R-dual: duality is an exact contravariant equivalence on maximal Cohen–Macaulay modules, interchanges the two Ext^1-orthogonality conditions, and interchanges left and right approximations. Thus it applies to our perverse convention, with $\underline{\mathrm{End}}_R(N) = A^{\mathrm{op}}$. Adding the projective-injective summand R gives the 2-cluster-tilting object $R \oplus N$ in the Frobenius category $\mathcal{B}$. Write the completed cDV hypersurface as $R = \mathbb{C}[[x_1, x_2, x_3, x_4]]/(f)$. Lemma 3.10 gives $\Omega_R^2 \simeq \mathrm{Id}$ naturally on its stable category, and hence on $\operatorname{add}(R \oplus N)$ modulo projectives. This supplies the parameters $d_{\mathrm{D}} = 2$ and $r_{\mathrm{D}} = 1$ in [9, Theorem 1.1(3)]. If $A \neq \mathbb{C}$, the one-curve contraction algebra is local and nonsemisimple, so it has no semisimple block, and Dugas's theorem gives a periodic projective bimodule resolution of A^{op}, hence of A, with period dividing $(d_{\mathrm{D}} + 2)r_{\mathrm{D}} = 4$. If $A = \mathbb{C}$, its stable bimodule class is zero and its positive Hochschild groups vanish.

Lemma 3.11 (Tate–Hochschild duality)**.** *For every symmetric finite-dimensional algebra A over $\mathbb{C}$,*

$$\big(\widehat{HH}^{-n}(A)\big)^\vee \cong \widehat{HH}^{n-1}(A) \qquad (n \in \mathbb{Z}).$$

If A is four-periodic in the stable bimodule category, then

$$DHH^1(A) \cong HH^2(A), \qquad \dim HH^1(A) = \dim HH^2(A). \tag{3.4}$$

Proof. For a symmetric algebra A, the enveloping algebra A^e is symmetric, and $\widehat{HH}^*(A) = \widehat{\mathrm{Ext}}^*_{A^e}(A, A)$. The displayed duality is the Tate–Hochschild duality of [28, equation (2.11)]. Taking $n = 3$ and using the period-four bimodule isomorphism gives

$$DHH^1(A) \cong D\widehat{HH}^{-3}(A) \cong \widehat{HH}^2(A) = HH^2(A),$$

because positive Tate degrees agree with ordinary Hochschild cohomology. □

Etingof–Latour–Rains use the same trace-space duality for their central extensions [11, Corollary 2.2]. The following symmetric-algebra form is the one needed here and later for relative traces.

Lemma 3.12 (Centre–cocenter duality)**.** *If A is finite-dimensional symmetric with symmetrizing trace $\tau\colon A \to \mathbb{C}$, then*

$$Z(A) \xrightarrow{\sim} HH_0(A)^\vee, \qquad z \mapsto \big(\bar{a} \mapsto \tau(za)\big).$$

Consequently,

$$\dim Z(A) = \dim HH_0(A). \tag{3.5}$$

Proof. For a central element z,

$$\tau(z[a, b]) = \tau(zab - zba) = \tau(zab - azb) = 0,$$

so the functional factors through $A/[A, A]$. Conversely, identify $A \cong DA$ by the symmetrizing form. If the functional associated to $c \in A$ annihilates every commutator, then for all a, b,

$$0 = \tau(c[a, b]) = \tau((ca - ac)b).$$

Nondegeneracy forces $ca = ac$ for all a, so $c \in Z(A)$. The annihilator of the commutator subspace is therefore exactly the centre, which proves perfection. □

3.4 Reduction to the cocenter dimension

For $G = \mathrm{Stab}_{\mathrm{GL}_d}(1)$, Lemma 3.2 and Corollary 3.7 give the tangent and orbit formulas below. The second equality uses $\dim \mathrm{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$ and Lemma 3.11; the last uses $\dim Z(S_{\mathrm{GV}}) = \sum_j n_j$ and the fact that all derivations of a semisimple complex algebra are inner:

$$\dim T_A X_d = d(d-1) - \dim \mathrm{Der}(A) + \dim HH^2(A) \tag{3.6}$$

$$= d(d-1) - d + \dim Z(A), \tag{3.7}$$

$$\dim(G \cdot S_{\mathrm{GV}}) = d(d-1) - d + \sum_j n_j. \tag{3.8}$$

Subtracting and applying Lemma 3.12 gives

$$\dim T_A X_d - \dim(G \cdot S_{\mathrm{GV}}) = \dim HH_0(A) - \sum_j n_j. \tag{3.9}$$

Theorem 2.1 and Lemma 3.3 already give $A \in \overline{G \cdot S_{\mathrm{GV}}}$. Thus the only remaining hypothesis of Theorem 3.6 is that the right-hand side of (3.9) vanish. This reduces Theorem 3.1 to the cocenter count $\dim HH_0(A) = \sum_j n_j$. The following subsections 3.5 to 3.9 deal with getting this numerical equality in several steps.

3.5 Universal marked cocenter on the Cartan cover

The Cartan cover separates the individual root factors; descent to the marked base will group them by curve coefficient. Fix one of the six primitive marked ADE pairs (Φ, p). Here p is the retained vertex α_* of Section 2, so $m_p(\alpha) = c_*(\alpha)$. Let $\mathfrak{h}$ be the Cartan space of rank r, and put

$$W_0 = \langle s_i : i \neq p \rangle, \qquad H = \mathfrak{h}/W_0.$$

Write

$$\widetilde{R} = \widehat{\mathcal{O}}_{\mathfrak{h},0} \cong \mathbb{C}[[\lambda_1, \dots, \lambda_r]].$$

For a positive root $\alpha = \sum_i m_i(\alpha)\alpha_i$, put

$$N_p := \#\{\alpha \in \Phi^+ : m_p(\alpha) > 0\}. \tag{3.10}$$

Choose an orientation Q of the finite Dynkin diagram, and let $\overline{Q}$ be its double. Over $\widetilde{R}$, set the path algebra

$$\Pi_\lambda = \widetilde{R}\,\overline{Q} / \left(\sum_{a \in Q_1} [a, a^*] - \sum_i \lambda_i e_i \right).$$

We use the convention of [21, Proposition 4.15] and write $(\lambda, \alpha) = \sum_i m_i(\alpha)\lambda_i$. Let

$$\mathcal{D}_p = e_p \Pi_\lambda e_p.$$

Theorem 3.13 (The marked cocenter on the Cartan cover)**.** *For every primitive marked ADE pair (Φ, p), there is an exact sequence*

$$0 \longrightarrow \widetilde{R}^{N_p} \xrightarrow{\Psi_p} \widetilde{R}^{N_p} \longrightarrow HH_0(\mathcal{D}_p/\widetilde{R}) \longrightarrow 0 \tag{3.11}$$

such that

$$\det \Psi_p = \eta \Delta_p, \qquad \Delta_p := \prod_{\substack{\alpha \in \Phi^+ \\ m_p(\alpha) > 0}} (\lambda, \alpha), \qquad \eta \in \widetilde{R}^\times. \tag{3.12}$$

To prove Theorem 3.13, we realize its presentation as multiplication by the central element on the cocenter of a finite central extension. We first compute the generic rank and the closed-fibre dimension of that cocenter. The coefficient-five E_8 case is completed in the next subsection; the proof of the presentation then uses only the resulting equality.

The central extension. In the notation of Etingof–Rains [10], take $\mu = \rho$, normalized by $\rho_i = (\rho, \alpha_i) = 1$. The universal parameters x_i satisfy $\sum_a [a, a^*] = \sum_i x_i e_i$, and z is adjoined centrally. The coefficient ring for the regular weight ρ is

$$K_\rho = \mathbb{C}[x_1 - \rho_1 z, \dots, x_r - \rho_r z].$$

The fibre at a parameter λ is the algebra Π^ρ_λ with relations

$$\sum_a [a, a^*] = \sum_i (\rho_i z + \lambda_i) e_i.$$

Via the coordinates $\lambda_i := x_i - \rho_i z$, completing K_ρ at its augmentation gives the formal power-series ring $\mathbb{C}[[\lambda_1, \dots, \lambda_r]]$, namely the completed Cartan parameter ring $\widetilde{R}$. The λ_i are base coordinates, whereas z remains a central element of the algebra. The finite-free statement over K_ρ is exactly [10, Theorem 3.2(i)]; passing to the completed base and the marked corner is part of the present construction.

Define the marked corner of the completed universal central extension by

$$E_p := e_p(\Pi[z] \widehat{\otimes}_{K_\rho} \widetilde{R}) e_p, \qquad \mathsf{M}_p := HH_0(E_p / \widetilde{R}).$$

Since the Etingof–Rains central extension is finite free over K_ρ, the corner E_p is finite over $\widetilde{R}$. Its cocenter M_p, being a quotient of E_p, is finite as well.

The closed fibre of E_p is $e_p \Pi^\rho_0 e_p$. Since HH_0 is the cokernel of the commutator map, it commutes with arbitrary base change. In particular,

$$\mathsf{M}_p \otimes_{\widetilde{R}} \mathbb{C} \cong HH_0(e_p \Pi^\rho_0 e_p).$$

Generic rank. Etingof–Rains prove generic semisimplicity in [10, Theorem 3.2(v) and the display immediately following it]. Their positive-root-indexed decomposition has ordinary deformed preprojective factors. The Crawley–Boevey criterion supplies a simple module of dimension vector α [40, Theorem 1.2]. Combined with the Etingof–Rains dimension calculation, this criterion identifies these factors as matrix algebras. After passing to a splitting field $L / \operatorname{Frac}(\widetilde{R})$,

$$(E_p)_L \cong \prod_{m_p(\alpha) > 0} M_{m_p(\alpha)}(L). \tag{3.13}$$

Each marked matrix factor contributes one dimension to its cocenter, so

$$\dim_L (\mathsf{M}_p)_L = N_p. \tag{3.14}$$

Lemma 3.14 (The algebra at a single-root parameter)**.** *Let k be algebraically closed of characteristic zero and let $\eta \in k^r$ annihilate exactly one positive root α. Then the ordinary deformed preprojective algebra Π_η of the finite Dynkin diagram is $M_{\mathrm{ht}(\alpha)}(k)$, and its vertex projector e_i has rank $m_i(\alpha)$.*

Proof. The path-length filtration makes $\operatorname{gr}\Pi_\eta$ a quotient of the finite-dimensional ordinary preprojective algebra, so Π_η is finite-dimensional [10, Section 2.1]. The simple-module criterion [40, Theorem 1.2] shows that every simple module has dimension vector α, and that such a simple exists: there is no decomposition of α into two or more positive roots annihilated by η.

For two representations U, V, splitting an extension on the vertex spaces gives the complex

$$\bigoplus_i \operatorname{Hom}_k(U_i, V_i) \xrightarrow{d^0} \bigoplus_{a \in \overline{Q}_1} \operatorname{Hom}_k(U_{\mathrm{tail}(a)}, V_{\mathrm{head}(a)}) \xrightarrow{d^1} \bigoplus_i \operatorname{Hom}_k(U_i, V_i).$$

Here d^0 changes the splitting and d^1 is the linearized preprojective relation; its middle cohomology is $\operatorname{Ext}^1_{\Pi_\eta}(U, V)$. The trace pairings identify $d^1(U, V)$ with the transpose, up to sign, of $d^0(V, U)$. Taking dimensions therefore gives

$$\dim \operatorname{Ext}^1(U, V) = \dim \operatorname{Hom}(U, V) + \dim \operatorname{Hom}(V, U) - (\dim U, \dim V),$$

where the last term is the symmetric Cartan pairing. Since $(\alpha, \alpha) = 2$, two nonisomorphic simples of this vector would give a negative extension dimension. Thus the simple S is unique, and the same formula gives $\operatorname{Ext}^1(S, S) = 0$. Its indecomposable projective cover cannot have nonzero radical: otherwise that radical has a quotient S, giving a nonzero self-extension of S. Hence every finite-length module is semisimple. It follows that $\Pi_\eta = \operatorname{End}_k(S)$, with the stated matrix size and projector ranks. □

The closed-fibre dimension. To prove freeness, we also need the special-fibre equality:

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) = N_p. \tag{3.15}$$

At $\lambda = 0$, our normalization of ρ gives the central-extension relation used by Etingof–Latour–Rains (ELR) [11]. Their case analysis at $\mu = \rho$ computes the nodal corners [11, Section 3.2]. ELR state the calculation for nodal vertices, so the comparison with the six primitive flop markings and their deleted Levi diagrams uses Karmazyn's presentations below.

ELR call the branching vertex of a Dynkin diagram of type D or E the *nodal vertex*. Karmazyn's six primitive flop presentations determine the retained vertex and the Levi Weyl subgroup W_0. The retained vertices and Levi types follow from [21, Examples 4.22–4.27]. With that numbering, they are:

length	Dynkin type	retained vertex	W_0 in Karmazyn	deleted diagram	ELR status
1	A_1	unique	trivial	empty	immediate
2	D_4	$p = 4$	S_2^3	A_1^3	nodal D
3	E_6	$p = 6$	$S_2 \times S_3^2$	$A_1 \sqcup A_2^2$	nodal E_6
4	E_7	$p = 7$	$S_3 \times S_2 \times S_4$	$A_2 \sqcup A_1 \sqcup A_3$	nodal E_7
5	E_8	$p = 4$	$S_4 \times S_5$	$A_3 \sqcup A_4$	non-nodal
6	E_8	$p = 8$	$S_5 \times S_2 \times S_3$	$A_4 \sqcup A_1 \sqcup A_2$	nodal E_8

For a positive root α, ELR's phrase "the root contains the nodal simple root" is equivalent to $m_p(\alpha) > 0$. Hence their root count is N_p.

For type D, ELR [11, Section 3, Lemma 3.4] give degreewise upper bounds for the cocenter of the nodal corner. Summing them gives

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) \leq N_p.$$

The generic decomposition (3.14) gives a finite $\widetilde{R}$-module M_p of generic rank N_p. For a finite module over the local domain $\widetilde{R}$, the closed-fibre dimension is at least the generic rank. This supplies the reverse inequality and hence equality.

For the nodal E_6, E_7, E_8 corners, ELR [11, Section 3, Case 2b] compute the same graded cocenter dimensions by exact Magma calculations and obtain equality with the positive-root count. These are the primitive flop lengths $3, 4, 6$ in the table.

For A_1, the closed corner $e_p \Pi_0^\rho e_p = \mathbb{C}[z]/(z)$ has one-dimensional cocenter. The remaining coefficient-five E_8 corner is not a nodal ELR case. Subsection 3.6 and Appendix B establish its special-fibre cocenter dimension 104 by a direct relation calculation and exact certificate. The following subsection supplies this remaining case before we construct the universal cocenter presentation.

3.6 The coefficient-five E_8 computation

The coefficient-five corner supplies the final closed-fibre dimension needed to prove the single-curve cocenter formula. Unlike the nodal cases, it is computed here from explicit homogeneous relations. The exact sequence for the cocenters of a retained corner and of the quotient after deleting vertices, proved in Subsection 4.3.3, later gives the same value in the multi-curve setting. The direct calculation here establishes the special-fibre equality used in the current single-curve proof.

Deleting the coefficient-five vertex of E_8 leaves $A_3 \times A_4$, so

$$N_p = |\Phi^+(E_8)| - |\Phi^+(A_3)| - |\Phi^+(A_4)| = 120 - 6 - 10 = 104. \tag{3.16}$$

Let $B_5 = e_p \Pi_0^\rho e_p$ be the closed marked corner. Appendix B.1 derives its relations directly from the preprojective relations on the ρ-line. It constructs elements $\beta, \gamma \in B_5$ that, together with z, generate B_5 (Lemma B.2). The three relations

$$\begin{aligned} \mathsf{P} &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z), \\ \mathsf{R}_4 &= (\gamma + \beta^2)^2 + \beta\gamma\beta - 5z^2(\gamma + \beta^2) + 4z^4, \\ \mathsf{R}_5 &= \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 - 5z^2\gamma\beta \end{aligned}$$

vanish in B_5 (Lemma B.3). Give $\mathbb{C}[z]\langle\beta, \gamma\rangle$ and B_5 the half-path grading

$$\deg z = \deg \beta = 1, \qquad \deg \gamma = 2.$$

The three relations are homogeneous, so there is a surjection of graded algebras

$$Q_\rho := \frac{\mathbb{C}[z]\langle\beta, \gamma\rangle}{(\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5)} \twoheadrightarrow B_5. \tag{3.17}$$

The induced surjection $HH_0(Q_\rho) \twoheadrightarrow HH_0(B_5)$ transfers cocenter upper bounds from Q_ρ to B_5. Thus it is enough first to bound the quotient defined by the three displayed relations.

Remark B.4 explains why the signs of the parameter terms differ from a direct substitution into the displayed universal relations in the arXiv version of [21] specified in that remark.

Proposition 3.15 (Exact modular calculation)**.** *Let* $h_m = \dim_{\mathbb{C}} HH_0(B_5)_m$*. In degrees* $0, \dots, 23$*, the following sequence gives upper bounds for* h_m*:*

$$(1, 2, 4, 6, 7, 7, 7, 6, 6, 6, 6, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 2, 2, 1).$$

The sum of these bounds is 99*.*

Proof. We bound each cocenter piece using exact modular row ranks. In degree m, use cyclic-word columns $z^a[w]_{\mathrm{cyc}}$ with $a + \mathrm{wt}(w) = m$. For every $\mathsf{R} \in \{\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5\}$, $a \geq 0$, and word w in β, γ satisfying $a + \deg \mathsf{R} + \mathrm{wt}(w) = m$, include the row $z^a[\mathsf{R}w]_{\mathrm{cyc}}$. The empty word, of weight zero, is included in both the column and row enumerations. These span the relations in the cocenter: a two-sided multiple $u\mathsf{R}v$ has the same cyclic class as $\mathsf{R}vu$, and central powers of z can be moved to the front. The cokernel in degree m is therefore $HH_0(Q_\rho)_m$.

The relation matrices have integer entries. Exact Gaussian elimination over $\mathbb{F}_{1000003}$ gives the ranks recorded in Appendix B.2; running the same computation over $\mathbb{F}_{1000033}$ gives the same ranks as an additional modular check. The conclusion over characteristic zero uses each nonzero modular minor: that minor is also nonzero over $\mathbb{Q}$, so the modular rank is a lower bound for the characteristic-zero rank. Subtracting from the column count gives an upper bound for $\dim HH_0(Q_\rho)_m$, and the surjection (3.17) gives the stated upper bound for h_m. □

3.6.1 Tail bound and exact grading comparison

The finite calculation controls degrees through 23. To bound the tail, write h for the Coxeter number. In the Etingof–Rains grading, arrows have degree 1 and z has degree 2. Their Hilbert-series formulas [10, Section 2.1, equation (1), and Theorem 3.2(iii)–(iv)] show that the ordinary preprojective algebra has top arrow degree $h - 2$, while the ρ-special central extension has top degree $2(h - 2)$. On the marked corner (z, β, γ) have ER degrees $(2, 2, 4)$. The finite computation uses exactly half this grading. For E_8, $h = 30$, so the top ER degree is $2(h - 2) = 56$, which is computational half-degree 28. Setting $z = 0$ leaves the ordinary preprojective corner. Because the E_8 graph is bipartite, its closed paths have even arrow length; the top arrow-degree bound $h - 2 = 28$ therefore forces the corner to vanish in half-degrees above 14.

Since z is central,

$$HH_0(B_5)/(z) \cong HH_0(B_5/(z)).$$

Consequently multiplication by z is surjective

$$HH_0(B_5)_{m-1} \twoheadrightarrow HH_0(B_5)_m \qquad (m > 14).$$

The finite calculation gives $\dim HH_0(B_5)_{23} \leq 1$, so every degree $24, \dots, 28$ has dimension at most one. Together with the degreewise bounds of Proposition 3.15, this gives

$$\dim HH_0(B_5) \leq 99 + 5 = 104.$$

The finite module M_p has generic rank $N_p = 104$ by (3.14), and its closed fibre is $HH_0(B_5)$. The closed-fibre dimension is at least the generic rank, which supplies the opposite inequality:

$$\dim HH_0(B_5) \geq 104.$$

The two inequalities therefore give

$$\dim HH_0(B_5) = 104 = N_p. \tag{3.18}$$

This proves (3.15) in the remaining case. Only after the tail bound and generic-rank comparison do the degreewise upper bounds become equalities. Since their sum is the exact dimension, every bound of Proposition 3.15 is attained, each degree $24, \dots, 28$ contributes exactly one, and $HH_0(Q_\rho)_m \to HH_0(B_5)_m$ is an isomorphism for $m \le 23$.

3.7 Constructing the cocenter presentation

Proof of Theorem 3.13. The preceding case analysis, completed by (3.18), gives (3.15) for all six primitive markings. There is also an independent structural proof of this equality: Theorem 4.15, applied to the closed ρ-central extension with retained set $\{p\}$, gives $\dim HH_0(e_p\Pi_0^\rho e_p) = N_p$, since the roots of the deleted subdiagram are exactly those with marked coefficient zero. That closed-algebra proof does not use the universal presentation constructed here. In particular it includes the non-nodal coefficient-five marking.

Freeness. Choose lifts of a basis of the special fibre in (3.15). By (3.15), the special fibre of M_p has dimension N_p. Nakayama gives a surjection

$$\widetilde{R}^{N_p} \twoheadrightarrow \mathsf{M}_p.$$

After tensoring with $\mathrm{Frac}(\widetilde{R})$, (3.14) shows that this map is an isomorphism. Its kernel is a submodule of a free module over the domain $\widetilde{R}$, hence torsion-free; since its generic fibre is zero, the kernel is zero. Therefore

$$\mathsf{M}_p \cong \widetilde{R}^{N_p}. \tag{3.19}$$

Multiplication by the central element. Using the free $\widetilde{R}$-module in (3.19), multiplication by z induces

$$\mu_z \colon \mathsf{M}_p \to \mathsf{M}_p.$$

Setting $z = 0$ recovers the ordinary deformed preprojective corner. Since z is central,

$$HH_0(\mathcal{D}_p/\widetilde{R}) \cong \mathsf{M}_p/z\mathsf{M}_p = \operatorname{coker} \mu_z. \tag{3.20}$$

On the root factor α, the decomposition displayed immediately after [10, Theorem 3.2(v)] gives

$$z|_{V_\alpha} = -\frac{(\lambda, \alpha)}{(\rho, \alpha)}.$$

Therefore μ_z has the same eigenvalue on the one-dimensional cocenter of that factor, and over the fraction field

$$\det \mu_z = \prod_{m_p(\alpha)>0} \left(-\frac{(\lambda, \alpha)}{(\rho, \alpha)} \right).$$

The denominators are nonzero complex constants, so

$$\det \mu_z = \eta \Delta_p, \qquad \eta \in \widetilde{R}^\times.$$

This determinant is nonzero, so μ_z is injective over the domain $\widetilde{R}$. Together with (3.19)–(3.20), this gives the required presentation and determinant. □

3.8 Descent to the marked base

The Cartan-cover calculation distinguishes individual roots. Descending through the Levi Weyl group records roots with the same marked coefficient by one divisor on the marked base.

Write

$$\mathbb{H}_\Gamma = \mathcal{O}(\mathfrak{h}), \qquad \mathbb{H}_\ell = \mathbb{H}_\Gamma^{W_0}, \qquad R_H = \widehat{\mathbb{H}}_\ell, \qquad \widetilde{R} = \widehat{\mathbb{H}}_\Gamma,$$

with completions at the graded origins. The extension $R_H \to \widetilde{R}$ is finite free and faithfully flat. Here this can be seen directly from the type-A factors of the deleted Levi diagram in the table above. The Cartan representation of W_0 is a fixed line together with sum-zero permutation representations of S_q, one for each component A_{q-1}. On such a block the invariant ring is $\mathbb{C}[e_2, \dots, e_q]$, where e_i are the elementary symmetric polynomials in $y_1, \dots, y_q$ with $\sum_i y_i = 0$. It has the free module basis

$$\{y_1^{a_1} \cdots y_q^{a_q} : 0 \le a_i < q - i + 1\}.$$

Indeed, start with a generic monic polynomial of degree q, adjoin its first root, divide by the resulting linear factor, and adjoin the next root. The successive monic quotients have degrees $q, q-1, \dots, 1$, giving this basis over the coefficient ring. The resulting splitting algebra is the polynomial ring in the ordered roots; setting their sum equal to zero preserves its free basis. The elementary symmetric polynomials generate the invariants by successive cancellation of the leading monomial of a symmetric polynomial. Tensoring the block bases and adjoining the fixed coordinate proves freeness of $\mathbb{H}_\Gamma$ over $\mathbb{H}_\ell$ with rank $|W_0| > 0$. Over the invariant origin each root coordinate is nilpotent, so the invariant-origin and Cartan-origin adic topologies are equivalent. Completion preserves the displayed finite free basis, which proves the assertion for $R_H \to \widetilde{R}$.

3.8.1 The marked algebra on the Cartan cover

Let $\mathcal{A}_\Gamma$ be Karmazyn's universal extended-Dynkin algebra and put $e_{0p} = e_0 + e_p$. Write $\mathcal{T}_{\Gamma,0p}$ for the two-summand generator on the Cartan partial resolution associated to this corner [21, Proposition 4.21], and $\mathcal{T}_\ell$ for the generator of Subsection 2.2. Completions below are taken at the central graded origins.

Proposition 3.16 (Comparison of the marked generators)**.** *On the completed Cartan partial resolution there is an isomorphism*

$$\theta : \widehat{\mathcal{T}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{\mathcal{T}}_{\Gamma,0p} \tag{3.21}$$

preserving the structure-sheaf and retained-vertex summands. It induces an isomorphism

$$\widehat{\mathcal{A}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{e_{0p} \mathcal{A}_\Gamma e_{0p}} \tag{3.22}$$

sending e_0 to e_0 and e_p to e_p.

Proof. The geometric comparison in [21, Section 2.3 and the proof of Proposition 4.21] places both generators on the same Cartan partial-resolution family, obtained by pulling back the marked family and contracting the unretained curves of the simultaneous resolution. We use the forward coefficient map $\mathbb{H}_\ell \to \mathbb{H}_\Gamma$. Write U for the affine target ring completed at its full

central point and put $S = \widetilde{R}$, $U_S = U \otimes_{R_H} S$. The finite free parameter extension makes U_S finite over U, and

$$U_S/\mathfrak{m}_U U_S \cong S/\mathfrak{m}_H S$$

is local Artinian. Hence U_S is local, with maximal ideal $\mathfrak{n}_S$, and for some $c \geq 1$

$$\mathfrak{n}_S^c \subseteq \mathfrak{m}_U U_S \subseteq \mathfrak{n}_S.$$

It is therefore complete at the full new target point. As a complete Noetherian local ring it is also complete for the parameter ideal $\mathfrak{m}_S U_S$: compatible representatives form a $\mathfrak{n}_S$-adic Cauchy sequence, and each fixed power of the parameter ideal is closed. These two ideal topologies need not coincide. The source is projective over U_S, and source and target remain flat over S.

Fix the full-to-partial morphism

$$q : \mathcal{Z}_S \longrightarrow \mathcal{Y}_{\ell,S}$$

for the same chosen simultaneous resolution and the same side of the partial resolution [29, Section 2.4.2 and the proof of Theorem 2.6]. The normalized $\mathcal{T}_{\Gamma,0p}$ is a bundle on this *partial* resolution, obtained by lifting its labelled central basic generator; it is not defined merely as a direct sum upstairs.

On the completed central surface the retained-corner construction of [18, Theorem 4.6 and its proof], dualized to the zero-perverse convention, gives the labelled basic generator

$$\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_{0,p}.$$

The restriction of the fixed $\mathcal{T}_\ell$ is this same basic generator. Indeed, its completed summand endomorphism rings are local by the basic normalization in [21, Remark 3.5], and Proposition 2.24 identifies the specialized endomorphism rings with their quotients by the parameter ideal. The summands therefore remain indecomposable. The positive nonfree summand $(\mathcal{T}'_\ell)^\vee$ has determinant degree 1 on the retained curve, so $\mathcal{T}'_\ell$ has determinant degree -1; this nonzero degree persists on the central curve and distinguishes it from $\mathcal{O}$. The specialized bundle is a projective generator of the same zero-perverse heart, so its two summands are the indicated labelled basic projectives.

The same central construction gives the actual labelled pullback

$$q_0^* \mathcal{T}_0 \cong \mathcal{T}_{\mathrm{full},0,0} \oplus \mathcal{T}_{\mathrm{full},0,p}$$

on the full central resolution. It also gives $Rq_{0*}\mathcal{O} = \mathcal{O}$, since the contracted central surface singularities are rational. This identity persists over successive Artin thickenings of S: for a small extension, parameter flatness identifies the kernels of structure-sheaf reduction on source and target with their central structure sheaves tensored with the extension kernel. Pushing forward these exact sequences proves the identity and higher-direct-image vanishing on the next thickening. Induction and formal functions on the completed target model [35, Tag 02OC] give $Rq_*\mathcal{O} = \mathcal{O}$.

Each summand of $\mathcal{T}_0$ has vanishing higher self-Ext. Choose an isomorphism for each pair of corresponding central summands. Across the parameter thickening from order n to order $n+1$, the obstruction to lifting such an isomorphism lies in Ext^1 of that central summand with itself, tensored with $\mathfrak{m}_{\widetilde{R}}^n/\mathfrak{m}_{\widetilde{R}}^{n+1}$; it vanishes. A lift remains invertible modulo the nilpotent thickening. Induction therefore gives compatible summand-wise isomorphisms on all parameter

thickenings. The completed affine target is also complete for the parameter ideal. Since the source is projective over that target, coherent formal existence and full faithfulness algebraize these compatible isomorphisms and their inverses [22, Lemma 3.2], [35, Tag 088F]. Their direct sum is θ, and it preserves both projectors. The choices differ by automorphisms of the two summands.

On the full family, apply the same compatible lifting argument to $q^*\mathcal{T}_{\ell,S}$ and the retained direct sum of the full generator. Their common central bundle is a direct summand of a tilting bundle, so its higher self-Ext vanishes. This gives a labelled bundle isomorphism upstairs. Adjunction and the projection formula, using $Rq_*\mathcal{O} = \mathcal{O}$, now give

$$\begin{aligned}\widehat{\mathcal{A}}_\ell \otimes_{R_H} S &\cong \operatorname{End}_{\mathcal{Y}_{\ell,S}}(\mathcal{T}_{\ell,S})^{\mathrm{op}} \\ &\cong \operatorname{End}_{\mathcal{Z}_S}(q^*\mathcal{T}_{\ell,S})^{\mathrm{op}} \\ &\cong e_{0p}\widehat{\mathcal{A}}_\Gamma e_{0p}.\end{aligned}$$

The first arrow is the actual endomorphism base change in Proposition 2.24; the last is [21, Proposition 4.15] applied to the full labelled generator. The same comparison applies to $\mathcal{T}_{\Gamma,0p}$, while θ was constructed on the partial family itself. Every arrow is S-linear and preserves both projectors. The cofinal target ideals checked above identify the ordinary and completed tensors, proving (3.22) with the stated completion convention. □

For an algebra Λ with idempotent e_0, scalar extension commutes with the quotient $\Lambda/\Lambda e_0\Lambda$ by right exactness. The corner identity

$$\frac{e_{0p}\mathcal{A}_\Gamma e_{0p}}{(e_{0p}\mathcal{A}_\Gamma e_{0p})e_0(e_{0p}\mathcal{A}_\Gamma e_{0p})} \cong e_p(\mathcal{A}_\Gamma/\mathcal{A}_\Gamma e_0\mathcal{A}_\Gamma)e_p \tag{3.23}$$

then identifies the marked quotient upstairs. By [21, Proposition 4.15], $\mathcal{A}_\Gamma$ is the universal extended-Dynkin deformed preprojective algebra. Scalar extension and the idempotent quotient therefore identify the completed marked algebra as follows. Write $(1, d_1, \dots, d_r)$ for the minimal positive imaginary root of the extended diagram. The affine parameter relation is $t_0 + \sum_{i=1}^r d_i t_i = 0$. Killing the affine vertex of $\mathcal{A}_\Gamma$ leaves the finite-Dynkin algebra Π_λ, with the fixed coordinates

$$\lambda_i = t_i \quad (1 \le i \le r), \qquad \lambda_0 = -\sum_{i=1}^r d_i\lambda_i.$$

In particular this comparison introduces no parameter automorphism. Thus

$$\left(\widehat{\mathcal{A}}_\ell/\widehat{\mathcal{A}}_\ell e_0\widehat{\mathcal{A}}_\ell\right)\widehat{\otimes}_{R_H}\widetilde{R} \cong e_p\Pi_\lambda e_p = \mathcal{D}_p. \tag{3.24}$$

3.8.2 The cocenter on the marked base

Put

$$\mathcal{C}_H = \widehat{\mathcal{A}}_\ell/\widehat{\mathcal{A}}_\ell e_0\widehat{\mathcal{A}}_\ell, \qquad \mathsf{M}_H = HH_0(\mathcal{C}_H/R_H).$$

The completion is at the graded origin. Since $\widetilde{R}$ is finite free over R_H, tensoring a complete module with $\widetilde{R}$ gives a finite direct sum of copies of that module and is already complete. Thus (3.24) gives

$$\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p. \tag{3.25}$$

Theorem 3.17 (The universal marked cocenter)**.** *The algebra $\mathcal{C}_H$ is finite over R_H. Its cocenter has a free resolution*

$$0 \longrightarrow R_H^q \xrightarrow{\Theta} R_H^q \longrightarrow \mathsf{M}_H \longrightarrow 0 \tag{3.26}$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H) = (\det \Theta) = (P_1 \cdots P_\ell),$$

where $P_j = 0$ defines the reduced coefficient-j curve divisor.

Proof. We descend the Cartan-cover presentation in two stages: first its square resolution, then its determinant. The algebra $\mathcal{D}_p$ is finite over $\widetilde{R}$, so finiteness descends along the faithfully flat Cartan cover to $\mathcal{C}_H$ and M_H over R_H. The commutator-cokernel description gives

$$\mathsf{M}_H \otimes_{R_H} \widetilde{R} \cong HH_0(\mathcal{D}_p/\widetilde{R}). \tag{3.27}$$

Take a finite free surjection onto M_H, with kernel K. After base change, Theorem 3.13 makes $K \otimes_{R_H} \widetilde{R}$ projective, by Schanuel's lemma applied to the two free presentations. Finite projectivity descends along the faithfully flat cover [35, Tag 058S], and a finite projective module over the local ring R_H is free. The Cartan pullback has a square presentation with nonzero determinant, so M_H has generic rank zero. The two free ranks therefore agree, giving (3.26).

For the determinant, Fitting ideals commute with base change, so

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H)\widetilde{R} = (\Delta_p). \tag{3.28}$$

For $i \neq p$, the simple reflection s_i preserves m_p and sends every positive root other than α_i to a positive root. Thus W_0 permutes the factors of Δ_p without signs, and $\Delta_p \in \widetilde{R}^{W_0} = R_H$. Contracting the ideal in (3.28) along the faithfully flat extension gives

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H) = (\Delta_p). \tag{3.29}$$

Grouping the roots by marked coefficient and using Lemmas 2.5 and 2.6 gives

$$\Delta_p = \eta P_1 \cdots P_\ell, \qquad \eta \in R_H^\times. \tag{3.30}$$

□

The square presentation expresses the cocenter as a torsion module whose determinant records every reduced curve divisor once. Restriction to the classifying arc therefore converts its determinant into the required number of simple factors.

3.9 Restriction to the central arc

Let $R_1 = \mathbb{C}[[w]]$, and let $f_0 : \operatorname{Spec} R_1 \to H$ be the classifying arc of the original flop. Pulling the descended presentation back to this arc will compute the cocenter length of the original contraction algebra as the sum of the orders of intersection with the curve divisors. Set

$$\mathcal{C}_0 = \mathcal{C}_H \otimes_{R_H, f_0} R_1, \qquad \mathsf{M}_0 = HH_0(\mathcal{C}_0/R_1).$$

By Lemma 2.12 and Proposition 2.19, the arc meets the curve discriminant only at the origin, and $\operatorname{ord}_w P_j(f_0(w)) = n_j$.

3.9.1 The contraction algebra on the arc

Let $\widehat{\mathcal{R}}_\ell$ be the completion, at the graded origin, of the affine target ring $\mathcal{R}_\ell$ of the universal family, and put

$$Z_0 = \widehat{\mathcal{R}}_\ell \otimes_{R_H} R_1, \qquad \Lambda_0 = \widehat{\mathcal{A}}_\ell \otimes_{R_H} R_1.$$

Then $\mathcal{C}_0 = \Lambda_0/\Lambda_0 e_0 \Lambda_0$. Let $\mathfrak{m}$ denote the contraction maximal ideal of Z_0.

Lemma 3.18 (The central-arc quotient)**.** *The natural completion map $\mathcal{C}_0 \to \widehat{\mathcal{C}_{0\mathfrak{m}}}$ is an isomorphism, and*

$$\mathcal{C}_0 \cong \widehat{\mathcal{C}_{0\mathfrak{m}}} \cong A. \tag{3.31}$$

In particular,

$$HH_0(\mathcal{C}_0) \cong HH_0(A). \tag{3.32}$$

Proof. The classifying arc is nonconstant: the power series $g(w) = P_1(f_0(w))$ has positive finite order. Weierstrass division makes $\mathbb{C}[[w]]$ finite over $\mathbb{C}[[g(w)]]$. The latter subring lies in the image of R_H, so R_1 is finite over R_H.

Consequently Z_0 is finite over the complete Noetherian local ring $\widehat{\mathcal{R}}_\ell$. Modulo the maximal ideal of $\widehat{\mathcal{R}}_\ell$, the fibre is $R_1/\mathfrak{m}_H R_1$, where $\mathfrak{m}_H$ is the maximal ideal of R_H. This fibre is local Artinian with residue field $\mathbb{C}$. Thus Z_0 is local and complete at $\mathfrak{m}$. The finite Z_0-algebra Λ_0 and its quotient $\mathcal{C}_0$ are also complete.

The ordinary tensor products above therefore already compute the completed pullback along the central arc. Proposition 2.25 identifies Λ_0 with the original basic NCCR, preserving e_0. Taking the idempotent quotient gives $\mathcal{C}_0 \cong A$, and (3.32) follows on passing to cocenters. □

3.9.2 The length calculation

Base change gives $\mathsf{M}_0 = \mathsf{M}_H \otimes_{R_H} R_1$. The square presentation therefore becomes a finite-length presentation over the DVR R_1. Write $\Theta_0 = f_0^*\Theta$. Its determinant is

$$\det \Theta_0 = \eta(w) \prod_{j=1}^{\ell} P_j(f_0(w)) \neq 0. \tag{3.33}$$

Thus the base change of (3.26) is exact:

$$0 \longrightarrow R_1^q \xrightarrow{\Theta_0} R_1^q \longrightarrow \mathsf{M}_0 \longrightarrow 0. \tag{3.34}$$

Smith normal form over R_1 yields

$$\dim_{\mathbb{C}} \mathsf{M}_0 = \operatorname{length}_{R_1} \mathsf{M}_0 = \operatorname{ord}_w \det \Theta_0. \tag{3.35}$$

The relative and absolute cocenters have the same underlying $\mathbb{C}$-vector space, since $r[a,b] = [ra,b]$ for central r. Combining Lemma 3.18, (3.33), and Proposition 2.19 gives

$$\dim_{\mathbb{C}} HH_0(A) = \sum_j \operatorname{ord}_w P_j(f_0(w)) = \sum_j n_j. \tag{3.36}$$

Centre–cocenter duality then gives

$$\dim_{\mathbb{C}} Z(A) = \sum_j n_j. \tag{3.37}$$

3.10 Completion of the uniqueness proof

The cocenter count gives the remaining tangent-dimension equality. The orbit-closure relation can be supplied by Theorem 2.1; the proof below instead constructs it directly from the universal algebra, using the dependency discussion below. Together these inputs allow the orbit criterion to identify the semisimple algebra obtained by splitting the generic fibre.

Proof of Theorem 3.1. Equations (3.9) and (3.36) give

$$\dim T_A X_d = \dim(G \cdot S_{\mathrm{GV}}). \tag{3.38}$$

We construct the orbit-closure input directly from the universal algebra. Choose a constant translation $F(s,w) = f_0(w) + sv$ as in Lemmas 2.16 and 2.17. Its distinguished root polynomials $W_j(s,w)$ have degrees n_j, are separable and pairwise coprime over $\mathbb{C}((s))$, and their roots avoid Σ^{bad}. Put

$$\mathscr{A} = \mathcal{C}_H \otimes_{R_H,F} \mathbb{C}[[s,w]].$$

The square algebra presentation in Proposition 4.5, together with Lemma 4.2, makes $\mathscr{A}$ finite free over $\mathbb{C}[[s]]$; its special fibre is the actual algebra A by Lemma 3.18. The reduced annihilator in Proposition 4.14, specialized to $J = \{p\}$, says that $\delta = \prod_j P_j$ annihilates $\mathcal{C}_H$. Thus $\mathscr{A}$ is an algebra over the finite root algebra

$$\mathbb{C}[[s,w]]/(\delta(F)) \cong \mathbb{C}[[s]][w]/\left(\prod_j W_j(s,w)\right).$$

This finite presentation is used before inverting s.

Over $L = \overline{\mathbb{C}((s))}$, that central algebra splits into one copy of L for each root of the W_j. Each resulting factor of $\mathscr{A}_L$ is the fibre of $\mathcal{C}_H$ at its parameter point. The finite faithfully flat Cartan cover supplies a lift of that point to $\widetilde{R}$ with residue field L. A root of W_j lies on a non-Levi root hyperplane of coefficient j; avoidance of Σ^{bad} excludes every other root hyperplane. The marked comparison (3.25) and Lemma 3.14 identify the entire factor with $M_j(L)$. In particular, there is no additional nilpotent contribution at that point. Consequently

$$\mathscr{A}_L \cong \prod_{j=1}^{\ell} M_j(L)^{\times n_j}.$$

Lemma 3.3 now gives

$$A \in \overline{G \cdot S_{\mathrm{GV}}}. \tag{3.39}$$

Theorem 3.6 now shows that A is smooth, that $\overline{G \cdot S_{\mathrm{GV}}}$ is its unique component, and that any split semisimple algebra whose orbit closure contains A is isomorphic to S_{GV}.

Let $\mathscr{D}$ be any finite free formal deformation of A with geometrically semisimple generic fibre, and let S' be a product of complex matrix algebras such that $\mathscr{D} \otimes_{\mathbb{C}[[u]]} \overline{\mathbb{C}((u))} \cong S' \otimes_{\mathbb{C}} \overline{\mathbb{C}((u))}$. Thus S' records the matrix sizes and multiplicities of the split generic fibre. Lemma 3.3 gives $A \in \overline{G \cdot S'}$, and the orbit criterion yields

$$S' \cong S_{\mathrm{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \tag{3.40}$$

Tensoring this isomorphism with $\overline{\mathbb{C}((u))}$ gives the asserted description of the geometric generic fibre. □

Corollary 3.19. *Let $\mathscr{B}$ be a finite free $\mathbb{C}[[s]]$-algebra with special fibre A and geometrically semisimple generic fibre. Then $HH_0(\mathscr{B}/\mathbb{C}[[s]])$ is free of rank $\sum_j n_j$.*

Proof. The relative cocenter is finite and commutes with arbitrary base change. Theorem 3.1 gives dimension $\sum_j n_j$ on the closed fibre and, after a splitting extension, on the generic fibre. Lemma 2.37 proves freeness. □

The divisor presentation computes the single-curve cocenter, and orbit geometry turns its dimension into the generic-fibre statement. For several retained curves, Section 4 uses the exact cocenter sequence associated with deleting vertices. Lemma 4.2 controls pullback of the square presentations along classifying maps $R \to B[[w]]$, yielding the numerical statements used in Sections 5 and 6.

4 Universal presentations and flat families

Let a smooth threefold flop vary over a complete Noetherian local complex base. Using a lift of its fixed basic generator, we form the relative contraction quotient. We prove that this quotient, its cocenter, and its centre are finite free over the base, with ranks determined by the central flop. This is the content of the three flatness assertions in [17, Conjecture 4.3], in the completed smooth-threefold setting. Kawamata established algebra flatness over a disk for configurations of exceptional curves [23, Theorem 3.1(2)].

Under the hypotheses verified below, the simultaneous partial resolutions recalled in [21, Section 2.3] and the tilting-bundle theorem of [22, Theorem 3.4] identify the geometric family and its endomorphism algebra. The quotient by the free projector and its cocenter have square presentations derived from the Etingof–Rains central extension [10, Theorem 3.2]. The determinants of these two square presentations generate the zeroth Fitting ideals, with factors given by the curve-divisor equations.

Pullback gives matrices over $B[[w]]$, where B is the external base and w is the surface parameter. Lemma 4.2 shows that finiteness of the cokernel on the closed arc forces freeness over B, even when B has nilpotents. The rank of that cokernel is the order of the closed-arc determinant. We apply the lemma separately to the algebra and its cocenter, then use a symmetrizing trace to obtain the centre.

The one-curve cocenter presentation is available from Section 3. The pullback and trace arguments below are independent of the number of retained curves. For several curves, the corresponding presentations are constructed in Subsection 4.3; the Peirce and cocenter ranks of the multi-curve contraction quotient enter Section 5.

The two existence arguments and their dependencies. The geometric proof of Theorem 2.1 identifies its local matrix factors in Theorem 2.30 and assembles them in Proposition 2.36. It also uses Proposition 4.5 in Lemma 2.34 to retain every moving support point. The proof of Theorem 3.1 gives another construction of the orbit-closure input directly from $\mathcal{C}_H$: Proposition 4.5 and Lemma 4.2 give a finite free family; Proposition 4.14, with $J = \{p\}$, and Lemma 3.14 identify its split factors. This second construction uses neither the geometric local-factor calculation nor the cocenter count. Both constructions use the marked Cartan comparison, the actual central-arc quotient, and the translated-root geometry and GV identification from Proposition 2.19. The algebra presentation and reduced-annihilator proofs below use the

Etingof–Rains central extension independently of semisimple existence and uniqueness. The separate cocenter presentation supplies the additional input needed for the tangent-dimension calculation and for centre and cocenter flatness.

Remark 4.1 (Relative-algebra convention). A geometric deformation over $(B, \mathfrak{m}, \mathbb{C})$ means a projective contraction with affine Noetherian target, source and target flat over B, and the prescribed central contraction as closed fibre. We complete along the central contraction point and its inverse image. The relative algebra is the idempotent quotient of a lift of the fixed normalized basic Van den Bergh generator, with its structure-sheaf summand and all curve labels retained.

The finite relative algebra carries all support points that move away from the central point. Base change is performed on this algebra. For example, $B[[w]]/(w^2 - t)$, with $B = \mathbb{C}[[t]]$, is free of rank two over B, whereas $\mathbb{C}((t))[[w]]/(w^2 - t) = 0$. Thus the generic fibre of the finite relative algebra is obtained by localizing that algebra; completing again at $w = 0$ would remove its moving points. Fixing the generator is equally significant: its original summand multiplicities give the matrix blocks in the split fibres. A block M_j contributes j^2 to the algebra dimension, while its basic Morita representative $\mathbb{C}$ has dimension one.

4.1 Freeness and traces over local bases

Two algebraic facts turn the universal presentations into flat families. A square matrix over $B[[w]]$ with finite-dimensional closed-arc cokernel gives a finite free B-module. Once both an algebra and its cocenter are finite free, a symmetrizing trace on the closed algebra lifts and identifies the centre with the dual cocenter. We prove these facts before applying them to contraction quotients.

4.1.1 Freeness over a possibly nonreduced base

On the closed classifying arc, Smith normal form computes the cokernel length from the determinant. Freeness over B requires control through every power of its maximal ideal, including its nilpotent directions.

Lemma 4.2 (Freeness of the cokernel over the coefficient ring). *Let $(B, \mathfrak{m}, k)$ be a complete Noetherian commutative local ring, put $D_B = B[[w]]$, and let $\Phi : D_B^a \to D_B^a$ be a square matrix. If $\det \overline{\Phi} \neq 0$ in $k[[w]]$, then Φ is injective and its cokernel is finite free over B, of rank $\operatorname{ord}_w \det \overline{\Phi}$.*

Proof. Put $M = \operatorname{coker} \Phi$. Smith normal form over $k[[w]]$ makes its closed fibre finite-dimensional. Lift a basis to $m_1, \dots, m_b \in M$. Successive reduction modulo $\mathfrak{m}M$ writes every $m \in M$ as

$$m = \sum_i b_i^{(n)} m_i + r_n, \qquad r_n \in \mathfrak{m}^n M, \quad b_i^{(n+1)} - b_i^{(n)} \in \mathfrak{m}^n.$$

Completeness of B gives limits for the coefficients. Since M is finite over the Noetherian local ring D_B, Krull intersection [35, Tag 00IP] gives $\bigcap_n \mathfrak{m}^n M = 0$. The limiting coefficients express m as a B-linear combination of the m_i. Thus M is finite over B.

If $\Phi v = 0$, injectivity of $\overline{\Phi}$ gives $v \in \mathfrak{m} D_B^a$. On each associated graded piece the map is

$$1 \otimes \overline{\Phi} : (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_k k[[w]]^a \longrightarrow (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_k k[[w]]^a,$$

which is injective. The identification of each graded piece uses the flatness of $B[[w]]$ over the Noetherian ring B: it is the (w)-adic completion of the polynomial ring $B[w]$, and Noetherian completion is flat [35, Tag 00MB]. Induction places v in every power of $\mathfrak{m}$, and Krull intersection gives $v = 0$.

The exact sequence just obtained is a flat resolution over B, so

$$\operatorname{Tor}_1^B(M, k) = \ker \overline{\Phi} = 0.$$

Take a minimal free surjection $B^b \to M$, with finite kernel L. The Tor vanishing makes the reduction of L inject into the kernel of the closed-fibre isomorphism. Thus $L/\mathfrak{m}L = 0$, and Nakayama gives $L = 0$. The rank is the closed-fibre dimension; Smith normal form gives $b = \dim_k(M/\mathfrak{m}M) = \operatorname{ord}_w \det \overline{\Phi}$. □

The finite-slice argument applied to a free resolution on the marked base also identifies the derived pullback with the ordinary one. In the following corollary, $(B, \mathfrak{m}, k)$ remains a complete Noetherian local ring as in Lemma 4.2.

Corollary 4.3 (Derived pullback along a classifying map)**.** *Suppose a finite module C over a Noetherian local ring R has an exact sequence $0 \to R^a \xrightarrow{\Theta} R^a \to C \to 0$. For a local map $F : R \to B[[w]]$, if $C \otimes_R k[[w]]$ is finite-dimensional, then*

$$C \otimes_R^L B[[w]] \simeq (C \otimes_R B[[w]])[0].$$

The ordinary pullback is finite free over B, with rank equal to the dimension of that closed fibre.

Proof. Compute the derived pullback with the displayed free resolution. Its closed-fibre matrix $\overline{\Theta(F)}$ has finite-dimensional cokernel, hence is invertible over $k((w))$. Lemma 4.2 makes $\Theta(F)$ injective and its cokernel finite free over B. The pulled-back resolution therefore has cohomology only in degree zero, as asserted. □

4.1.2 Symmetrizing traces and base change

The trace pairing identifies the centre with the dual cocenter of a symmetric algebra; see [11, Corollary 2.2] for the central extension. The relative assertion requires lifting a given closed-fibre trace and preserving its perfect pairing under scalar extension. Here the base need only be local.

Lemma 4.4 (Lifting a symmetrizing trace)**.** *Let $(B, \mathfrak{m}, k)$ be a commutative local ring, and let $\mathscr{A}$ be a finite free B-algebra. Suppose $M = HH_0(\mathscr{A}/B)$ is finite free and the closed fibre $A_0 = \mathscr{A} \otimes_B k$ of $\mathscr{A}$ is symmetric. Every symmetrizing trace of A_0 lifts to a perfect B-linear symmetrizing trace τ on $\mathscr{A}$. It induces an isomorphism*

$$Z(\mathscr{A}) \xrightarrow{\sim} M^\vee, \qquad z \longmapsto (\bar{a} \longmapsto \tau(za)).$$

The centre and cocenter commute with every commutative scalar extension $B \to B'$. The lifted trace determines the displayed duality, while the base-change maps are canonical.

Proof. As a commutator cokernel, M commutes with arbitrary scalar extension. The given closed trace is a functional on $M \otimes_B k$. Freeness of M lifts it to a functional $M \to B$, hence to

a trace $\tau : \mathscr{A} \to B$. The determinant of the pairing $(a, b) \mapsto \tau(ab)$ has nonzero reduction, so it is a unit in the local ring B. The lifted trace is therefore symmetrizing.

To identify the centre, take the annihilator of the commutator submodule under this perfect pairing. We have

$$\tau(z[a, b]) = \tau((za - az)b).$$

This identity shows that a central element annihilates every commutator under the pairing. Conversely, if $\tau(z[a, b]) = 0$ for all a, b, perfection forces $za - az = 0$ for every a. The annihilator is thus the centre, giving the stated duality and centre freeness.

After any scalar extension to B', the pairing remains perfect. Base change for the cocenter and for the dual of a finite free module identifies the trace formula over B' with the natural centre comparison. The natural centre comparison is therefore an isomorphism for every such extension, including a nonflat one. □

4.2 One curve: presentations and pullback along classifying maps

At a transverse point of D_j, the matrix algebra M_j contributes j^2 to the algebra dimension and 1 to the cocenter dimension. These are the exponents in the two determinant formulas.

4.2.1 A square presentation of the universal quotient

Recall the notation of Subsection 3.8. The ring R_H is the completed marked base, $\widetilde{R}$ is its finite faithfully flat Cartan cover, and $\mathcal{C}_H$ is the universal contraction quotient. The equation $P_j = 0$ defines the reduced coefficient-j curve divisor. The algebra presentation is derived below from the same central extension that gives the cocenter presentation.

Proposition 4.5 (A square presentation of the universal algebra)**.** *The finite R_H-module $\mathcal{C}_H$ has a free resolution of the form*

$$0 \longrightarrow R_H^a \xrightarrow{\Theta_A} R_H^a \longrightarrow \mathcal{C}_H \longrightarrow 0 \tag{4.1}$$

with

$$\operatorname{Fitt}_0^{R_H}(\mathcal{C}_H) = (\det \Theta_A) = \left(\prod_{j=1}^{\ell} P_j^{j^2} \right). \tag{4.2}$$

In particular, $\mathcal{C}_H$ is a Cohen–Macaulay module of codimension one over R_H.

Proof. Let E_p be the central extension of Subsection 3.5. Theorem 3.2(i) of [10] makes $\Pi[z]$ finite free over $K_\rho = \mathbb{C}[x_i - \rho_i z]$. After flat completion, its retained corner is a direct summand of a finite free $\widetilde{R}$-module. It is therefore free, since $\widetilde{R}$ is local. Before the change of coefficient ring, z is a polynomial variable, so multiplication by z is injective. Flat completion and the corner projection preserve this injection. Its cokernel is obtained by setting $z = 0$:

$$0 \longrightarrow E_p \xrightarrow{z} E_p \longrightarrow \mathcal{D}_p \longrightarrow 0. \tag{4.3}$$

To compute the determinant, use the generic decomposition (3.13). The factor of $(E_p)_L$ indexed by a positive root with $m_p(\alpha) > 0$ is $M_{m_p(\alpha)}(L)$, on which z acts as $-(\lambda, \alpha)/(\rho, \alpha)$ [10,

Theorem 3.2(v) and the following display]. This scalar occurs with multiplicity $m_p(\alpha)^2$, the dimension of the matrix algebra. Hence, in the domain $\widetilde{R}$,

$$\det(z \mid E_p) = \eta \prod_{\substack{\alpha\in\Phi^+\\ m_p(\alpha)>0}} (\lambda,\alpha)^{m_p(\alpha)^2}, \qquad \eta \in \mathbb{C}^\times.$$

The determinant is nonzero, so $\mathcal{D}_p$ is torsion and has projective dimension one over $\widetilde{R}$.

By (3.25), $\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p$. Choose a finite free surjection $R_H^a \twoheadrightarrow \mathcal{C}_H$ with kernel K. The resolution (4.3) makes $K \otimes_{R_H} \widetilde{R}$ projective. Finite projectivity descends along the faithfully flat map $R_H \to \widetilde{R}$, and projective modules over the local ring R_H are free. Thus K is free. Since $\mathcal{C}_H$ is torsion, $\operatorname{rk} K = a$, proving (4.1).

Finally, Fitting ideals commute with scalar extension. The root-orbit and reduced-divisor identifications in Lemmas 2.5 and 2.6 identify the displayed determinant ideal with the extension of $(\prod_j P_j^{j^2})$. Faithful flatness then gives (4.2) downstairs. Since R_H is regular local, Auslander–Buchsbaum [35, Proposition 10.111.1, Tag 090V] shows that the nonzero torsion module $\mathcal{C}_H$, of projective dimension one, has depth and dimension $\dim R_H - 1$. □

The two universal presentations are supported on the same root divisors. Up to units, their determinants are $\prod_j P_j^{j^2}$ for the algebra and $\prod_j P_j$ for the cocenter, the latter by Theorem 3.17. Their exponents measure the dimension of a matrix block and the number of blocks, respectively.

4.2.2 Classifying an arbitrary geometric deformation

Fix a completed simple flop $f_0 : X_0 \to Y_0$ of a smooth complex threefold. Let $\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_0'$ be its selected basic tilting generator, as in Section 2, and A_0 its contraction algebra.

The geometric families and their relative quotients are those of Remark 4.1.

Proposition 4.6 (Families of classifying arcs)**.** *Let B be a complete Noetherian local $\mathbb{C}$-algebra with residue field $\mathbb{C}$, with the residue identification compatible with its $\mathbb{C}$-algebra structure, and let f_B be such a completed deformation. A lift of a general surface-section parameter supplies a continuous local map*

$$F : R_H \longrightarrow D_B = B[[w]]$$

and identifies f_B with the completed pullback of the universal marked partial resolution. There is a relative tilting generator $\mathcal{T}_B$ lifting $\mathcal{T}_0$. For any such lift with the fixed summands, its relative contraction quotient satisfies

$$\mathscr{A}_B := \frac{\operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}}{\operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}} e_0 \operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}} \cong \mathcal{C}_H \otimes_{R_H,F} D_B. \tag{4.4}$$

Its reduction modulo $\mathfrak{m}$ is A_0.

Proof. Write the completed target as $\operatorname{Spec} R$, with maximal ideal $\mathfrak{n}$, and retain the projective R-scheme whose formal completion is the given source. Choose a general central section parameter w_0 and a lift $w \in \mathfrak{n}$. Evaluation of power series defines $D_B = B[[w]] \to R$. The element w_0 is regular on the central target and source; this remains true after their Noetherian completions. At a source point over the closed target point, let Q be its local ring. The maps $B \to D_B \to Q$ are local, Q is B-flat, and $Q/\mathfrak{m}Q$ is torsion-free, hence flat, over $\mathbb{C}[[w]]$. The

fibrewise criterion with module Q, finite over itself, gives D_B-flatness [35, Tag 00MP]. Every source point specializes to a point over the closed target point by properness over R; localization therefore proves flatness at all points. The same criterion applies to R.

Let $p : S \to V = \operatorname{Spec} A$ be the algebraic marked partial ADE resolution modeling the closed surface, and let $\widehat{S}$ be its completion over the vertex ideal $\mathfrak{q} \subset A$. Reid's section lemma gives this partial resolution [27, Section 1, preprint p. 5]. Its strict transform is the scheme-theoretic inverse section: smallness excludes an exceptional divisorial component, while the Cartier divisor on the smooth central source has no embedded component. Fix the central marked identification with $\widehat{S}$. The standard simultaneous partial-resolution source family over $H = \mathfrak{h}/W_0$ is semiuniversal for ordinary deformations of S, its Kodaira–Spencer map is an isomorphism, and H is smooth [27, Section 6, preprint p. 38] [21, Section 2.3, preprint pp. 9–10]. The equivariant terminology in these references includes ordinary deformations; no action on the given family is assumed.

We first check the completion interface. Since S has only ADE singularities, $L_{S/\mathbb{C}}$ is perfect of amplitude $[-1, 0]$. Put $K = R\mathcal{H}om(L_{S/\mathbb{C}}, \mathcal{O}_S)$. The finite A-module $H^1(S, K)$ is supported at the vertex: away from it, p is an isomorphism and K is a tangent bundle in degree zero. Thus it is unchanged by completion. The continuous cotangent complex of $\widehat{S}$ is the completion of $L_{S/\mathbb{C}}$ [41, Proposition 3.9, preprint p. 16]. Dualization commutes with this completion because $L_{S/\mathbb{C}}$ is perfect. Proper formal functions therefore gives

$$T^1_{\mathrm{cont}}(\widehat{S}) = H^1(\widehat{S}, \widehat{K}) = H^1(S, K)^{\wedge}_{\mathfrak{q}} = T^1(S)$$

[35, Tags 0A0G and 0A0H]. The comparison respects Kodaira–Spencer maps.

Put $\mathfrak{a} = (\mathfrak{m}, w)$ and $D_n = D_B/\mathfrak{a}^n$. Inductively choose both $F_n : R_H \to D_n$ and an isomorphism ϕ_n of the completed source pullback with the given completed source over D_n. Suppose the pair is chosen at stage n. The power-series ring R_H allows an arbitrary lift of F_n to D_{n+1}. Compare its source with the given source, using ϕ_n to fix the identification over D_n. For the square-zero extension with kernel $J_n = \mathfrak{a}^n/\mathfrak{a}^{n+1}$, annihilated by the maximal ideal of D_{n+1}, the classes of such liftings with this reduction identification fixed form a torsor under $T^1_{\mathrm{cont}}(\widehat{S}) \otimes_{\mathbb{C}} J_n$ [41, Propositions 4.2–4.3 and Corollary 4.4, preprint pp. 19–20]. Here the formal sources are locally Noetherian, and their reductions with respect to the full target ideals of definition are of finite type over $\mathbb{C}$. Their maps to the Artin bases are therefore pseudo-finite-type in the sense of [41, Section 1.1.1]. No properness or adicness over D_n is required. In Corollary 4.4 of that paper, prescribe the extension kernel $\mathcal{O}_{\widehat{S}} \otimes_{\mathbb{C}} J_n$ and its canonical coefficient map. For a flat lifting this map is an isomorphism; conversely, an extension with this prescribed map is flat by [35, Lemma 91.11.2, Tag 08MQ]. Thus the classification is of flat liftings with the specified reduction isomorphism, rather than arbitrary coherent extensions.

Write $\mathfrak{X}_n$ for the identified formal source over D_n and $\widehat{L}_{\mathfrak{X}_n/D_n}$ for its completed cotangent complex. Ordinary flat cotangent base change on the already identified algebraic standard pullback, followed by completion, gives

$$\operatorname{Ext}^1_{\mathfrak{X}_n}(\widehat{L}_{\mathfrak{X}_n/D_n}, \mathcal{O}_{\widehat{S}} \otimes_{\mathbb{C}} J_n) \cong T^1_{\mathrm{cont}}(\widehat{S}) \otimes_{\mathbb{C}} J_n$$

[35, Tag 08QQ]. This uses the standard pullback at stage n, not an algebraization of the unknown lifting. The extension classification fixes the specified reduction ϕ_n, so equality of its classes supplies an isomorphism extending that same ϕ_n. Changing the lifted base coordinates changes the torsor class by the Kodaira–Spencer map. Its surjectivity cancels the difference,

yielding F_{n+1} and an isomorphism ϕ_{n+1} extending the specified ϕ_n. Hence $F = \varprojlim F_n$ is a continuous local map $R_H \to D_B$, with compatible formal source isomorphisms.

We recover the targets while distinguishing parameter completion from completion at the full target maximal ideal. Set $I = \mathfrak{a}R$, $R_n = R/I^n$, and let X_n be the actual projective R_n-scheme. The closed D_B-fibre is the partial resolution of a rational double point, so $H^0(X_1, \mathcal{O}) = R_1$ and $H^{>0}(X_1, \mathcal{O}) = 0$. Flatness identifies these square-zero extension kernels with $\mathcal{O}_{X_1} \otimes_{\mathbb{C}} J_n$ and $R_1 \otimes_{\mathbb{C}} J_n$, respectively. Induction on these exact sequences gives $H^0(X_n, \mathcal{O}) = R_n$ and $H^{>0}(X_n, \mathcal{O}) = 0$, also for the standard pullback. Since R_n is complete for its full maximal ideal, formal functions identifies the global functions on the completed source at this stage with R_n [35, Tag 02OC]. Thus ϕ_n induces an isomorphism of the complete affine targets, and respects their canonical contraction maps. After identifying those targets, Grothendieck's full faithfulness for morphisms, applied over R_n with its full maximal ideal, algebraizes ϕ_n and its inverse to scheme isomorphisms [35, Tag 0A4Z]. Uniqueness makes these algebraizations compatible in n.

A complete Noetherian local ring $(R, \mathfrak{n})$ is also complete for $I \subset \mathfrak{n}$: compatible representatives are $\mathfrak{n}$-adically Cauchy, and each I^r is $\mathfrak{n}$-adically closed. Separatedness follows from Krull intersection. This does not assert cofinality of the two topologies. The target isomorphisms therefore pass to the actual complete target rings. Applying the same full-faithfulness theorem again, now with ideal I, identifies the projective sources and their contraction morphisms, hence their formal completions. Properness here is over the affine target throughout. The chosen central identification preserves the curve labels. If the marking is expressed by degree line bundles, their lifted classes are unique since $H^1(\widehat{S}, \mathcal{O}) = H^2(\widehat{S}, \mathcal{O}) = 0$; the square-zero units sequence also makes their isomorphisms compatible [35, Tag 0C6R]. Indeed, for the square-zero coefficient ideal $\mathcal{J}$, the units sequence has kernel $1 + \mathcal{J} \cong \mathcal{J}$. It is exact on the same underlying formal topological space, just as for schemes. The vanishing of $H^2(\mathcal{J})$ lifts line bundles, and that of $H^1(\mathcal{J})$ gives uniqueness and surjectivity on the units needed to lift their isomorphisms. This retains the fixed marked source family and its positive class, without choosing a lift to the full Weyl cover.

The universal generator pulls back by Lemmas 2.22 and 2.23. Its central restriction is $\mathcal{T}_0$, with its specified summands, by Proposition 2.25 and the same rank-and-determinant argument on the central arc. Vanishing of the first and second self-Ext groups of $\mathcal{T}_0$ lifts the summands and the isomorphisms between their lifts through each square-zero thickening. As in Proposition 3.16, the compatible isomorphisms pass to the formal limit and preserve e_0. Thus any other lift of the fixed generator gives an isomorphic marked quotient. Ordinary endomorphism base change (Proposition 2.24) and right exactness of the idempotent quotient now give (4.4).

The reduction F_0 modulo $\mathfrak{m}$ classifies the original flop, and Lemma 3.18 identifies the corresponding central-arc quotient with A_0. This closed fibre is finite-dimensional, so Proposition 4.5 and Corollary 4.3 make the right side of (4.4) finite over B. On this finite module, w acts nilpotently modulo $\mathfrak{m}$. Consequently the $(\mathfrak{m}, w)$-adic and $\mathfrak{m}$-adic topologies agree. The ordinary and completed quotient expressions therefore coincide, with all moving support points retained. □

4.2.3 The one-curve application

Proposition 4.7 (The one-curve application)**.** *Let f_0 and f_B be as in Proposition 4.6. Put*

$$d = \dim_{\mathbb{C}} A_0 = \sum_j j^2 n_j, \qquad N = \sum_j n_j.$$

Then $\mathscr{A}_B$ is finite free over B of rank d, and $HH_0(\mathscr{A}_B/B)$ and $Z(\mathscr{A}_B)$ are finite free of rank N. For every commutative B-algebra B', the canonical maps

$$HH_0(\mathscr{A}_B/B) \otimes_B B' \xrightarrow{\sim} HH_0(\mathscr{A}_B \otimes_B B'/B'),$$
$$Z(\mathscr{A}_B) \otimes_B B' \xrightarrow{\sim} Z(\mathscr{A}_B \otimes_B B')$$

are isomorphisms. Moreover

$$\mathcal{C}_H \otimes^L_{R_H,F} B[[w]] \simeq \mathscr{A}_B[0].$$

The conclusion holds for every complete Noetherian local base B and every geometric family specified in the hypotheses.

Proof. For the algebra, pull back (4.1) along F. The closed-fibre cokernel is A_0, so Corollary 4.3 gives algebra freeness, rank d, and the derived statement. In terms of the divisor equations,

$$d = \operatorname{ord}_w \det \Theta_A(F_0) = \sum_j j^2 \operatorname{ord}_w P_j(F_0) = \sum_j j^2 n_j,$$

where Corollary 4.25, applied with $J = \{p\}$, identifies the orders along the possibly nonconvergent central classifying arc F_0 with the same GV multiplicities. Its comparison through the intrinsic local Hilbert length does not require F_0 to equal the analytic classifying arc.

For the cocenter, use (3.26) together with the commutator-cokernel description of the relative cocenter. Together, these give

$$HH_0(\mathscr{A}_B/B) \cong \operatorname{coker}\big(\Theta(F) : D_B^q \to D_B^q\big).$$

Here the relative cocenter can be computed over either B or D_B: the B-span of commutators is already a D_B-module, since $r[a,b] = [ra,b]$ for a central scalar r. Lemma 4.2 and $\operatorname{ord}_w \prod_j P_j(F_0) = N$ make the cocenter free. To obtain the centre, recall that A_0 is symmetric by Proposition 3.8. Lemma 4.4 then supplies the centre, the lifted trace, and arbitrary centre base change. Cocenter base change already follows from its cokernel presentation. □

The scalar presentations of the individual root divisors give a further consequence: each intersection multiplicity is conserved over the whole base.

Corollary 4.8 (Constant degree of intersections with each root divisor)**.** *For each j, the scheme*

$$\operatorname{Spec}(B[[w]]/(P_j(F))) \longrightarrow \operatorname{Spec} B$$

is finite flat of degree n_j.

Proof. Apply Lemma 4.2 to the one-by-one matrix $P_j(F)$, whose closed specialization has order n_j. □

For several curves, the remaining task is to construct the universal presentations with every retained projector kept in the comparison.

4.3 Idempotent subalgebras associated to several curves

Let $f_0 : X_0 \to Y_0$ be the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold, with any finite exceptional configuration. We use the relative-algebra convention of Remark 4.1. For one curve, a root contributes one matrix size. For several curves, it contributes a vector of ranks for the retained projectors. The universal algebra and its Peirce entries record these dimensions; the cocenter records the number of simple factors.

4.3.1 Restricted roots and the marked universal quotient

Let Δ be the ADE diagram of a general surface section, and let $J \subseteq \Delta$ be the nonempty set of retained vertices. We use the Katz–Morrison discriminant description in the restricted-root form of Nabijou–Wemyss [29, Theorem 2.6]. Here a retained vertex labels a curve that remains on the partial resolution. The complementary set I labels the curves contracted from the minimal resolution. Put

$$I = \Delta \setminus J, \qquad W_I = \langle s_i : i \in I \rangle, \qquad S = \widehat{\mathcal{O}}_{\mathfrak{h},0}, \qquad R_J = S^{W_I} = \widehat{\mathcal{O}}_{\mathfrak{h}/W_I,0}.$$

Write $S = \mathbb{C}[[\lambda_i : i \in \Delta]]$, and let $q : \mathfrak{h} \to \mathfrak{h}/W_I$ denote the quotient map. In the notation of [29], the contracted subset is I and the retained subset is J. Each generator s_i, $i \in I$, is an order-two reflection on the full Cartan space $\mathfrak{h}$, and W_I is finite. Chevalley's theorem therefore gives a polynomial ring $H = \mathbb{C}[\mathfrak{h}]^{W_I}$, with $\mathbb{C}[\mathfrak{h}]$ finite free of rank $|W_I|$ over H [7, Theorem (A), Lemma 3 and the proof of Theorem (B)]. This applies to disconnected complements, their fixed subspaces and components of types D and E as well.

Let $\mathfrak{m} = H_{>0}$ and $\mathfrak{n} = \mathbb{C}[\mathfrak{h}]_{>0}$. The quotient fibre over $\mathfrak{m}$ is supported only at the origin: for a nonzero v, choose a linear form nonzero at all points of its finite W_I-orbit and take the product of its translates. This is a positive-degree invariant nonzero at v. Thus $\mathfrak{n}^c \subseteq \mathfrak{m}\mathbb{C}[\mathfrak{h}] \subseteq \mathfrak{n}$ for some $c \geq 1$. Completing the finite free module therefore gives $S = \mathbb{C}[\mathfrak{h}] \otimes_H \widehat{H}_{\mathfrak{m}}$. The Reynolds projector $|W_I|^{-1} \sum_{g \in W_I} g$ commutes with this scalar extension, so $S^{W_I} = \widehat{H}_{\mathfrak{m}} = R_J$. Consequently both completed rings are regular and $R_J \to S$ is finite free of rank $|W_I|$, hence faithfully flat. For $\alpha = \sum_i m_i(\alpha)\alpha_i \in \Phi^+(\Delta)$, set

$$\pi_I(\alpha) = (m_j(\alpha))_{j \in J}, \qquad \mathcal{B}_J = \pi_I(\Phi^+(\Delta)) \setminus \{0\}, \qquad s_\beta = \sum_{j \in J} \beta_j.$$

Thus $\pi_I(\alpha)$ retains exactly the coefficients relevant to the partial resolution, while s_β will be the size of the corresponding generic matrix block. Proportional but distinct restricted roots remain distinct elements of $\mathcal{B}_J$.

Lemma 4.9 (Equations of the restricted-root divisors)**.** *For $\beta \in \mathcal{B}_J$, the product*

$$p_\beta = \prod_{\pi_I(\alpha)=\beta} \ell_\alpha, \qquad \ell_\alpha = \sum_i m_i(\alpha)\lambda_i,$$

is invariant under W_I. It descends to a reduced irreducible element $P_\beta \in R_J$, whose zero locus is the coefficient-β curve divisor of the marked partial-resolution family. Distinct β's give distinct divisors, even when they are proportional.

Proof. The fibres of π_I on non-Levi positive roots are single W_I-orbits [29, Lemma 2.3]. For $i \in I$, s_i preserves the retained coefficients and sends every positive root other than α_i to a positive root. It therefore permutes the factors of p_β without signs.

Those factors are pairwise nonassociated linear primes in the factorial ring S. A proper factorization in R_J would select a proper nonempty invariant subset of them, contrary to transitivity. The same argument in the formal power-series ring proves irreducibility after completion and reducedness. The geometric interpretation of the descended divisor, including the distinction between proportional restricted roots, is [29, Theorem 2.6 and Remark 2.7]. Finally, at a general point of a non-Levi root hyperplane the W_I-stabilizer is trivial. The quotient is therefore generically étale there, so the descended equation has multiplicity one. □

To identify the algebra supported on these divisors, let $\mathcal{X}_J \to \mathcal{Y}_J$ be the completed universal family over R_J. Write the normalized generator and contraction quotient of this completed universal family as

$$T_J = \mathcal{O} \oplus \bigoplus_{j\in J} T_j, \qquad \Lambda_J = \mathrm{End}(T_J)^{\mathrm{op}}, \qquad C_J = \Lambda_J/\Lambda_J e_0 \Lambda_J.$$

All projectors remain fixed throughout the comparison. Put $e_J = \sum_{j\in J} e_j$ and $e_{0J} = e_0 + e_J$.

On the closed surface, Iyama–Kalck–Wemyss–Yang identify the tilting bundle on a partial resolution with the retained summands on the full resolution [18, Theorem 4.6]. Karmazyn's uniqueness theorem for lifts of the bundles carries this comparison through the formal family.

For ordinary classifying pullbacks, also retain the algebraic model before completion. Put $H_J^{\mathrm{alg}} = \mathbb{C}[\mathfrak{h}]^{W_I}$, and write

$$\pi_J^{\mathrm{alg}} : \mathcal{X}_J^{\mathrm{alg}} \longrightarrow \operatorname{Spec} \mathcal{R}_J^{\mathrm{alg}}$$

for the standard simultaneous partial resolution over this polynomial marked base. Its labelled algebraic generator, constructed in the proof, is denoted T_J^{alg}, with $\Lambda_J^{\mathrm{alg}} = \mathrm{End}(T_J^{\mathrm{alg}})^{\mathrm{op}}$. The parameter-completed, target-uncompleted model has

$$\mathcal{R}_J^{\mathrm{par}} = \mathcal{R}_J^{\mathrm{alg}} \otimes_{H_J^{\mathrm{alg}}} R_J, \qquad \Lambda_J^{\mathrm{par}} = \Lambda_J^{\mathrm{alg}} \otimes_{H_J^{\mathrm{alg}}} R_J.$$

Completion at its full central target point gives the previously defined $\mathcal{X}_J, T_J, \Lambda_J$.

Proposition 4.10 (The marked Cartan comparison)**.** *There is an isomorphism preserving every labelled projector*

$$\Lambda_J \widehat{\otimes}_{R_J} S \cong e_{0J} \widehat{\Pi}_{\widetilde{\Delta},\lambda} e_{0J}.$$

Killing the free projector gives

$$C_J \otimes_{R_J} S \cong D_J := e_J \Pi_{\Delta,\lambda} e_J. \tag{4.5}$$

For every commutative Noetherian R_J-algebra B, form the ordinary scheme pullback of the parameter-completed, target-uncompleted model, and write $T^{\mathrm{ord}}_{J,B}$ for its pulled-back labelled generator. It remains a tilting generator, and the canonical marked algebra map gives

$$\mathrm{End}(T^{\mathrm{ord}}_{J,B})^{\mathrm{op}} \cong \Lambda^{\mathrm{par}}_J \otimes_{R_J} B.$$

These assertions persist under central localization and ideal-adic completion of the resulting Noetherian affine target. On a completed formal classifying pullback the full endomorphism algebra is the completed parameter tensor; the parameter-finite contraction quotient retains its ordinary tensor identification.

Proof. The algebraic model in [21, Section 2.3] is defined for every coloured ADE diagram. Its affine target is finite type over H^{alg}_J; source and target are flat over that base, the morphism is projective, its fibres have dimension at most one, and $R\pi_*\mathcal{O} = \mathcal{O}$, as specified at the start of [21, Section 3.1]. The weighted ADE construction gives good $\mathbb{C}^*$-actions on the affine target and parameter base: their coordinate rings have positive weights and degree-zero part $\mathbb{C}$. The source is normal, since it is flat over the regular parameter base with geometrically normal partial-resolution surface fibres. Locally this follows from Serre's criterion: base parameters provide the required depth, while a height-one point lies either in a normal generic fibre or over a height-one base point with a zero-dimensional normal fibre.

Since $\mathcal{X}^{\mathrm{alg}}_J$ is projective over an affine variety, it is quasi-projective. Choose an auxiliary ample line bundle $\mathcal{L}$ on it. By normality, a positive tensor power of $\mathcal{L}$ admits a $\mathbb{C}^*$-linearization [22, Lemma 4.10 and Remark 4.11]. Replace $\mathcal{L}$ by this power. Its pullback to the completion at the central target point is relatively ample and linearized, as required by [30, Appendix, Propositions (A.5)–(A.6)].

The normalized central bundles have equivariant structures. In the positive convention they are the torsion-free pullbacks of the retained graded special McKay modules, and the zero-perverse bundles are their duals; see [22, Theorem 6.2 and Corollary 6.6] and [18, Theorem 4.6 and its proof]. Apply [22, Lemma 5.3 and Theorem 5.4] to the algebraic equivariant family. Here "graded complete" has the graded meaning of [22, Definitions 4.1 and Remark 4.2], which includes the positively graded polynomial ring H^{alg}_J. The good actions, normality, projectivity and flat affine target required by that theorem have just been checked. It gives an actual algebraic tilting bundle and an H^{alg}_J-flat endomorphism algebra. The summands and their inclusion and retraction maps may be lifted and algebraized equivariantly, preserving all labels: positive self-Ext vanishes for each summand, taking $\mathbb{C}^*$-invariants is exact, and equivariant coherent algebraization preserves the specified maps [30, Appendix, Lemma (A.4), Propositions (A.5)–(A.6)]. Thus the resulting bundle is T^{alg}_J, and its completion is the fixed T_J by compatible summandwise lift uniqueness. The structure-sheaf summand is kept throughout. Properness makes Λ^{alg}_J finite over $\mathcal{R}^{\mathrm{alg}}_J$.

We establish positivity on this actual algebraic lift before taking any classifying pullback. In this paragraph abbreviate

$$H = H^{\mathrm{alg}}_J, \quad U = \mathcal{R}^{\mathrm{alg}}_J, \quad X = \mathcal{X}^{\mathrm{alg}}_J, \quad T = T^{\mathrm{alg}}_J, \quad P = T^{\vee}, \quad E = \mathrm{End}_X(T).$$

Let $\mathfrak{m} = H_{>0}$, set $U_c = U/\mathfrak{m}U$ and $X_c = X \times_H \mathbb{C}$, and let $o \in \mathrm{Spec}\, U$ be the homogeneous origin. Write $T_c = T|_{X_c}$ and $P_c = T_c^{\vee}$. The specialization in [22, Theorem 5.4 and its proof] is induced by restriction of endomorphisms. It therefore preserves the fixed projector e_0. With

composition multiplication on E, its direct summand

$$M = \mathrm{Hom}_X(T, \mathcal{O}_X) = \Gamma(X, P) \cong e_0 E$$

satisfies

$$M \otimes_H \mathbb{C} \xrightarrow{\sim} \Gamma(X_c, P_c)$$

by the actual restriction map. Properness makes M a finite U-module. Consider the equivariant evaluation and its coherent cokernel:

$$\mathcal{O}_X \otimes_U M \longrightarrow P \longrightarrow \mathcal{Q} \longrightarrow 0.$$

Its restriction to X_c is precisely the evaluation of P_c.

Let $\mathfrak{n}_c$ be the homogeneous maximal ideal of U_c, and let $\widehat{U}_c = \widehat{(U_c)_{\mathfrak{n}_c}}$. Proper flat base change identifies

$$\Gamma(X_c, P_c) \otimes_{U_c} \widehat{U}_c \cong \Gamma(X_c \times_{U_c} \widehat{U}_c, \widehat{P}_c),$$

compatibly with evaluation. The specified labelled normalization on this completed *central surface* identifies $\widehat{P}_c$ with its positive generator, which is globally generated. Consequently $\mathcal{Q}|_F = 0$ for the scheme-theoretic fibre $F = \pi^{-1}(o)$; Nakayama then makes $\operatorname{Supp} \mathcal{Q}$ disjoint from F. The image $\pi(\operatorname{Supp} \mathcal{Q})$ is a closed invariant subset of the affine target, because π is proper. Every nonempty such subset contains o: the strictly positive grading extends the orbit of any closed point to $\mathbb{A}^1$, with value o at zero. Thus this image is empty and $\mathcal{Q} = 0$. A finite set of U-generators of M gives a surjection $\mathcal{O}_X^{\oplus N} \twoheadrightarrow P$.

For the total determinant $L = \det P$, the central normalization gives

$$\deg_{C_i} L = \sum_{j \in J} \deg_{C_i} \det(T_j^\vee) = \sum_{j \in J} \delta_{ij} = 1.$$

The disjoint union of the projective-line normalizations of the retained curves maps finitely and surjectively to F, through its reduction. The pullback of $L|_F$ is ample, so $L|_F$ is ample even if F is nonreduced [35, Tag 0B5V]. For the proper morphism π, fibrewise ampleness extends to relative ampleness over a target neighbourhood [35, Tag 0D2S]. Thus the locus $\{y \in \operatorname{Spec} U : L|_{X_y} \text{ is ample}\}$ is open. It is invariant and contains o, so its closed invariant complement is empty. Consequently L is π-ample. This concerns the total determinant; the individual summand determinants can have degree zero on other retained curves. Finally,

$$H^1(X, T) = \mathrm{Ext}^1_X(\mathcal{O}_X, T) = 0,$$

as a direct summand of the positive self-Ext of T. The structure-sheaf summand is fixed, so [37, Propositions 3.2.6–3.2.7 and Corollary 3.2.8] applies: P and T are the negative-one and zero perverse projective generators, respectively, with their bounded coherent derived equivalences.

We next carry out the Cartan comparison on the chosen completed models. Let $(U, \mathfrak{n})$ be the full central affine target completion. Since S is finite free over R_J, the ring $U_S = U \otimes_{R_J} S$ is finite over U, with closed fibre $S/\mathfrak{m}_{R_J} S$. This fibre is local Artinian, so U_S has one maximal ideal $\mathfrak{n}_S$, and for some $c \geq 1$,

$$\mathfrak{n}_S^c \subseteq \mathfrak{n} U_S \subseteq \mathfrak{n}_S.$$

It is therefore complete for its full target maximal ideal. The finite endomorphism modules are complete as well. This also identifies the ordinary Cartan tensor with the completed one in the first display.

For parameter thickenings we use the ideal $\mathfrak{m}_S U_S$ on this already completed target. A complete Noetherian local ring is complete for every ideal contained in its maximal ideal: a compatible sequence of coset representatives is maximal-ideal Cauchy, and each ideal power is closed because its finite quotient is complete [35, Tag 00MA]. The same holds for finite modules. This justifies parameter-adic formal existence here without identifying the parameter and target topologies.

On the central ADE surface, let $q_0 : Z_0 \to X_{J,0}$ be the map from the minimal to the partial resolution. The tilting-bundle construction in [18, Theorem 4.6 and its proof] gives

$$q_0^* T_{J,0} \cong T_{\mathrm{full},0,0} \oplus \bigoplus_{j \in J} T_{\mathrm{full},0,j}.$$

In the positive-dual convention, one pulls back the determinant-degree-one line bundles and their maximal extensions. The vanishing of $\mathrm{Ext}^1(-, \mathcal{O})$ makes the pulled-back extensions maximal. Dualizing gives the displayed zero-perverse convention. In particular its nonfree determinant degrees are $-\delta_{ij}$.

Fix the full simultaneous resolution compatible with the selected standard partial resolution, and write q for the resulting morphism after Cartan base change. The rational central surface gives $Rq_{0*}\mathcal{O} = \mathcal{O}$. On a small parameter thickening, flatness identifies the structure-sheaf kernel with the central structure sheaf tensored by the square-zero kernel of the base. Proper pushforward and the central higher-image vanishing therefore extend this identity through every Artin thickening. Formal functions [35, Tag 02OC] give $Rq_*\mathcal{O} = \mathcal{O}$ on the completed family.

The pullback of T_J and the retained full-resolution summands are now bundles on the same S-flat model with identified central restrictions. Vanishing of the first self-Ext extends each specified labelled isomorphism compatibly to the next thickening; its inverse lifts with it. Grothendieck existence and full faithfulness [35, Tag 088F] produce the summand-preserving isomorphism on the completed model. These are chosen isomorphisms, determined up to automorphisms of the labelled summands. Projection formula and $Rq_*\mathcal{O} = \mathcal{O}$ then identify their actual endomorphism algebras. The full universal McKay algebra is the extended-Dynkin deformed preprojective algebra [21, Proposition 4.15]. Use its non-affine parameters $\lambda_i = t_i$ and $\lambda_0 = -\sum_i d_i \lambda_i$, where $(1, d_i)$ is the minimal positive imaginary root. Retaining the fixed summands gives the first displayed corner isomorphism with these coordinates.

For the unrestricted ordinary pullback, return to the finite-type model. The ring $\mathcal{R}_J^{\mathrm{par}}$ is finite type over R_J, so $\mathcal{R}_J^{\mathrm{par}} \otimes_{R_J} B$ is Noetherian for *every* Noetherian B in the statement. Its projective source is Noetherian too. For affine source and target rings $R_J \to A \to Q$ of this model, flatness over R_J gives

$$Q \otimes_A^L (A \otimes_{R_J} B) \simeq Q \otimes_{R_J}^L B \simeq Q \otimes_{R_J} B.$$

Derived base change [35, Tag 08IB], applied to the endomorphism vector bundle, together with tilting and R_J-flatness of Λ_J^{par}, concentrates the comparison in degree zero. The degree-zero map is the ordinary endomorphism pullback, and preserves multiplication and every projector. The same comparison for $\mathcal{O}$ gives $R\pi_{B*}\mathcal{O} = \mathcal{O}$. The finite evaluation surjection for $(T_J^{\mathrm{alg}})^\vee$, proved on the algebraic model above, pulls back to a surjection onto $(T_{J,B}^{\mathrm{ord}})^\vee$. Its total determinant remains relatively ample, and its fixed structure-sheaf summand remains present. Moreover $H^1(T_{J,B}^{\mathrm{ord}}) = 0$ is the structure-sheaf-to-generator direct summand of the already vanishing first self-Ext group. Projectivity, the fibre-dimension bound and the structure-sheaf pushforward

identity have all been checked on this new Noetherian target. Thus [37, Propositions 3.2.6–3.2.7 and Corollary 3.2.8] applies there anew and gives both specified perverse projective generators and their bounded coherent derived equivalences. No flatness, locality, reducedness or equivariance hypothesis on B is used.

Central localization and ideal-adic completion of the new Noetherian affine target are flat ring maps [35, Tag 00MB]. Proper flat base change computes their endomorphism algebras by ordinary scalar extension over that target ring. For a completed formal classifying pullback, the same calculation on finite Artin parameter quotients and formal functions gives the completed parameter tensor of the full endomorphism algebra.

Finally, scalar extension commutes with the idempotent quotient, and

$$\frac{e_{0J}\Lambda e_{0J}}{e_{0J}\Lambda e_0\Lambda e_{0J}} \cong e_J(\Lambda/\Lambda e_0\Lambda)e_J.$$

Removing the affine vertex leaves the finite-Dynkin corner. That quotient is finite over the Cartan parameter ring by the central extension freeness in [10, Theorem 3.2(i)], followed by setting its central variable to zero. Its closed parameter fibre has nilpotent positive-degree ideal, so its parameter and full-target module topologies are cofinal. Exactness of Noetherian completion therefore leaves this already complete finite module unchanged, proving the ordinary tensor formula (4.5). Faithful flatness gives parameter-finiteness downstairs. On the algebraic model this also follows by graded Nakayama: a homogeneous basis of the finite central quotient lifts to generators over H_J^{alg}, since successive positive-parameter corrections strictly lower weight. Consequently completing its quotient in the parameters gives precisely C_J, whose further finite-support pullbacks retain ordinary tensors. □

4.3.2 The presentation of the algebra and its Fitting ideal

Adjoin a polynomial central variable z to the universal finite-Dynkin preprojective algebra Π_x. This makes the divisor multiplicities visible on the retained corner. Set $\rho_i = 1$, $\lambda_i = x_i - \rho_i z$, and

$$\mathscr{E} = \Pi_x[z] \otimes_{\mathbb{C}[\lambda_i]} S, \qquad \mathscr{E}_J = e_J \mathscr{E} e_J.$$

Etingof–Rains prove freeness over the shifted parameter ring and semisimplicity over its generic point [10, Theorem 3.2(i),(v)]. The simple factors of the generic central extension are indexed by positive roots, and the scalar action of z locates the corresponding root hyperplanes. The retained corner records the part of each factor seen by the projectors in J.

Lemma 4.11 (The central extension and its generic simple factors)**.** *The module $\mathscr{E}_J$ is finite free over S, and*

$$0 \longrightarrow \mathscr{E}_J \xrightarrow{z} \mathscr{E}_J \longrightarrow D_J \longrightarrow 0 \tag{4.6}$$

is exact. If $K = \operatorname{Frac} S$, then

$$(\mathscr{E}_J)_{\overline{K}} \cong \prod_{\substack{\alpha\in\Phi^+ \\ \pi_I(\alpha)\neq 0}} M_{s_{\pi_I(\alpha)}}(\overline{K}). \tag{4.7}$$

On the factor indexed by α, the central element z acts as $-\ell_\alpha/\langle\rho,\alpha\rangle$.

Proof. The whole extension is finite free over the shifted parameter ring [10, Theorem 3.2(i)]. The corner projection $a \mapsto e_J a e_J$ is S-linear and idempotent, so its image is projective and hence free over the local ring S. Multiplication by the polynomial variable z is injective before completion and remains injective after flat completion and on the corner. Its cokernel is obtained by setting $z = 0$, which proves (4.6).

The generic simple factors of the whole extension are $\operatorname{End}(V_\alpha)$, with $\dim e_i V_\alpha = m_i(\alpha)$ [10, Theorem 3.2(v) and its proof]. Orthogonality of the projectors gives

$$e_J V_\alpha = \bigoplus_{j\in J} e_j V_\alpha, \qquad \dim e_J V_\alpha = s_{\pi_I(\alpha)}.$$

Taking the corner of $\operatorname{End}(V_\alpha)$ gives $\operatorname{End}(e_J V_\alpha)$. Taking traces in the preprojective relation gives $0 = \ell_\alpha + z\langle\rho,\alpha\rangle$, determining the scalar action of z. □

Lemma 4.12 (The labelled algebra at a single-root point)**.** *Let $S \to \Omega$ be a geometric Cartan point, with Ω an algebraically closed extension of $\mathbb{C}$. Suppose exactly one positive root α has $\ell_\alpha = 0$ at this point, and put $\beta = \pi_I(\alpha) \neq 0$. Then*

$$D_J \otimes_S \Omega \cong \operatorname{End}_\Omega\left(\bigoplus_{j\in J} \Omega^{\beta_j}\right),$$

with each labelled projector e_j of rank β_j. The same is the geometric fibre of the actual universal quotient C_J at the corresponding marked-base point.

Proof. We first work with the full central extension before taking the retained corner. Write $h_\gamma = \langle\rho,\gamma\rangle$ and $c_\gamma = -\ell_\gamma/h_\gamma$ for each positive root γ. Freeness and the generic decomposition in Lemma 4.11, applied to the full vertex set, show that

$$\prod_{\gamma\in\Phi^+} (z - c_\gamma) = 0$$

in the central extension: the identity holds over its fraction field and hence over its coefficient domain.

Restrict to $H_\alpha = (\ell_\alpha = 0)$ and invert every other root equation. On this integral open set put

$$q_\alpha(Z) = \prod_{\gamma\neq\alpha} (Z - c_\gamma), \qquad \varepsilon_\alpha = \frac{q_\alpha(z)}{q_\alpha(0)}.$$

The denominator is a unit. Since $zq_\alpha(z) = 0$ and $q_\alpha(Z) - q_\alpha(0)$ is divisible by Z, ε_α is a central idempotent and $z\varepsilon_\alpha = 0$. Bézout for Z and $q_\alpha(Z)$ shows that z is invertible on the complementary summand. Thus setting $z = 0$ leaves precisely the ε_α-summand, a finite projective module on this integral open set.

Its rank is h_α^2. To compute it at the generic point of H_α, write $\lambda = v + t\rho$, with $v \in H_\alpha$ and $t = \ell_\alpha(\lambda)/h_\alpha$. The change $z_\alpha = z + t$ identifies the full central extension with the extension restricted to H_α, followed by adjoining this parameter. In its generic decomposition, z_α acts on the γ-block by $-\ell_\gamma(v)/h_\gamma$. These functions are distinct on H_α: if $\gamma/h_\gamma - \eta/h_\eta$ were a multiple of α, pairing with ρ would make that multiple zero, forcing $\gamma = \eta$. Only the α-block has scalar zero, and its dimension is h_α^2. The rank of the projective summand is constant on the integral open set, so every single-root fibre of the finite-Dynkin algebra Π_λ has this dimension.

Crawley–Boevey's simple-module criterion [40, Theorem 1.2] supplies a simple Π_λ-module of dimension vector α. Indeed α is positive, $\lambda \cdot \alpha = 0$, and there is no decomposition of α into two or more positive roots all annihilated by λ: each such root would have to equal α. This verifies the criterion's strict inequality condition vacuously; in Dynkin type its function $p(\gamma) = 1 - (\gamma, \gamma)/2$ is zero on every root. The action on this simple module maps Π_λ onto its full endomorphism algebra. Both algebras have dimension h_α^2, so the map is an isomorphism. Its vertex projectors have ranks $m_i(\alpha)$ by the dimension-vector statement. Taking the retained corner gives the displayed marked matrix algebra. Finally, Proposition 4.10 identifies this corner with the fibre of the actual C_J, preserving every projector. □

Theorem 4.13 (Presentations and Fitting ideals for several retained vertices)**.** *There is a free resolution of the form*

$$0 \longrightarrow R_J^a \xrightarrow{\Theta_J} R_J^a \longrightarrow C_J \longrightarrow 0 \tag{4.8}$$

with

$$\operatorname{Fitt}_0^{R_J}(C_J) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta^{s_\beta^2} \right). \tag{4.9}$$

For $i, j \in J$, the module $e_i C_J e_j$ also has a free resolution with two free modules of equal rank, and

$$\operatorname{Fitt}_0^{R_J}(e_i C_J e_j) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta^{\beta_i \beta_j} \right).$$

For the zero module, the zeroth Fitting ideal is the unit ideal. The nonzero module C_J is Cohen–Macaulay of codimension one.

Proof. Over the algebraic closure of the fraction field, the factor belonging to α in (4.7) has algebra dimension $s_{\pi_I(\alpha)}^2$. Therefore

$$\det(z \mid \mathscr{E}_J) = u \prod_{\pi_I(\alpha) \neq 0} \ell_\alpha^{s_{\pi_I(\alpha)}^2} = u \prod_\beta (q^* P_\beta)^{s_\beta^2}, \qquad u \in \mathbb{C}^\times.$$

The determinant is nonzero, so D_J is a finite torsion S-module with a free resolution consisting of two free modules of equal rank. Apply the faithfully flat syzygy descent from the proof of Proposition 4.5 to $R_J \to S$ and (4.5). After base change to S, the first syzygy is projective. Faithfully flat descent [35, Tag 058S] makes it projective over R_J, hence free because R_J is local;

torsion forces the two ranks to agree. This proves (4.8). Base change and faithful contraction of Fitting ideals give (4.9).

For the Peirce entries, restrict multiplication by z to the free direct summand $e_i\mathscr{E}e_j$. The cokernel of this restricted map is $e_iD_Je_j$, whose square presentation descends by the same syzygy construction. The α-summand of the generic fibre of $e_i\mathscr{E}e_j$ is $\mathrm{Hom}(e_jV_\alpha, e_iV_\alpha)$, of dimension $m_i(\alpha)m_j(\alpha) = \beta_i\beta_j$, giving the entrywise formula. Finally, regularity of R_J and Auslander–Buchsbaum [35, Proposition 10.111.1, Tag 090V] give $\operatorname{depth}_{R_J} C_J = \dim R_J - 1 = \dim C_J$. □

Proposition 4.14 (The reduced annihilator)**.** *Put $\delta_J = \prod_{\beta\in\mathcal{B}_J} P_\beta$. Then*

$$\mathrm{Ann}_{R_J}(C_J) = (\delta_J).$$

There is a matrix Ξ_J such that $\Theta_J\Xi_J = \Xi_J\Theta_J = \delta_J I_a$.

Proof. Consider the polynomial

$$Q(Z) = \prod_{\pi_I(\alpha)\neq 0} \left(\langle\rho,\alpha\rangle Z + \ell_\alpha\right)$$

which annihilates $\mathscr{E}_J$ over the fraction field, by Lemma 4.11. Freeness over the domain S therefore implies $Q(z)\mathscr{E}_J = 0$. Setting $z = 0$ gives $(q^*\delta_J)D_J = 0$. The Fitting ideal shows that the support is exactly the union of the indicated hyperplanes, whose reduced equation is $q^*\delta_J$. Hence

$$(q^*\delta_J) \subseteq \mathrm{Ann}_S(D_J) \subseteq \sqrt{\mathrm{Ann}_S(D_J)} = (q^*\delta_J).$$

Faithful flatness gives the annihilator equality downstairs. Since δ_J annihilates the cokernel of Θ_J, lift $\delta_J I_a$ through Θ_J to obtain Ξ_J. Injectivity of Θ_J gives the second matrix identity. □

4.3.3 An exact sequence for cocenters after deleting vertices

Each matrix factor contributes one dimension to the cocenter. To obtain the cocenter presentation over the Cartan base, we compute the closed-fibre dimension of the retained corner. Write $P_\Delta = \Pi_0^\rho$ for the ordinary central extension specialized at $\lambda = 0$, with $\rho_i = 1$; the element z remains part of this algebra. Put

$$Q_I = P_\Delta/P_\Delta e_J P_\Delta.$$

The quotient deletes the retained vertices, leaving the components of I. Etingof–Latour–Rains compute the full-algebra trace count for each of these diagrams [11, Theorem 3.1]. The following exact sequence identifies the cocenter of the retained corner with the kernel of the quotient map.

Theorem 4.15 (An exact sequence for cocenters of ADE central extensions)**.** *For every nonempty $J \subseteq \Delta$, there is an exact sequence*

$$0 \longrightarrow HH_0(e_J P_\Delta e_J) \longrightarrow HH_0(P_\Delta) \longrightarrow HH_0(Q_I) \longrightarrow 0. \tag{4.10}$$

In particular, if $N_J = |\Phi^+(\Delta)| - |\Phi^+(I)|$, then

$$\dim_{\mathbb{C}} HH_0(e_J P_\Delta e_J) = N_J. \tag{4.11}$$

With arrows of internal degree one and z of degree two, the odd pieces of $HH_0(e_J P_\Delta e_J)$ vanish and

$$\dim HH_0(e_J P_\Delta e_J)_{2p} = \#\{\alpha \in \Phi^+(\Delta) : \pi_I(\alpha) \neq 0, \mathrm{ht}(\alpha) = p+1\}.$$

The main point is injectivity of the corner map. In the exact sequence of Lemma 4.17, the kernel of that corner map is the image of the boundary from $HH_1(Q_I)$. This boundary vanishes once $HH_1(P_\Delta) \to HH_1(Q_I)$ is surjective. The central element z provides the cycles used to prove that surjectivity.

Keller's localization theorem gives a sequence for dg Hochschild homology. The following comparison relates the low-degree dg Hochschild groups in that sequence to those of the ordinary Jacobi algebras. We use polynomial Ginzburg dg algebras before completion over the Cartan base, with cohomological grading for dg algebras and homological grading for Hochschild homology.

Lemma 4.16 (A low-degree comparison for self-injective Jacobi algebras)**.** *Let Γ be a nonpositive, homologically smooth, bimodule 3-Calabi–Yau dg algebra over $\mathbb{C}$, with $A = H^0(\Gamma)$ finite-dimensional, basic, and self-injective. Then $H^{-1}(\Gamma) = 0$, and the canonical projection gives natural isomorphisms*

$$HH_i(\Gamma) \cong HH_i(A), \qquad i = 0, 1.$$

For any nonpositive dg algebra, the degree-zero comparison holds and its negative Hochschild homology vanishes.

Proof. Amiot's theorem makes $\mathcal{C} = \operatorname{per}\Gamma/\mathcal{D}_{\mathrm{fd}}\Gamma$ a Hom-finite 2-Calabi–Yau category. The image T of Γ is cluster tilting, with endomorphism algebra A, [1, Theorem 2.1]. The projection to $\mathcal{C}$ is fully faithful on the fundamental domain

$$\mathcal{F} = \operatorname{per}\Gamma \cap \mathcal{D}^{\leq 0} \cap {}^{\perp}\mathcal{D}^{\leq -2} \qquad \text{[1, Proposition 2.9]}.$$

Both Γ and $\Gamma[1]$ lie in this domain. Indeed, they are perfect and have cohomology in nonpositive degrees. For $M \in \mathcal{D}^{\leq -2}$, the two groups $\mathrm{Hom}_{\mathcal{D}(\Gamma)}(\Gamma, M) = H^0(M)$ and $\mathrm{Hom}_{\mathcal{D}(\Gamma)}(\Gamma[1], M) = H^{-1}(M)$ vanish, giving the required left orthogonality. Since A is basic and self-injective, the criterion of Iyama–Oppermann [39, Proposition 3.6], in the formulation [16, Proposition 4.4], gives $T \cong T[2]$. Hence

$$H^{-1}(\Gamma) = \mathrm{Hom}_{\operatorname{per}\Gamma}(\Gamma[1], \Gamma) \cong \mathrm{Hom}_{\mathcal{C}}(T[1], T) \cong \mathrm{Hom}_{\mathcal{C}}(T, T[1]) = 0.$$

The last equality holds because T is cluster tilting.

For the Hochschild comparison, use the reversed internal grading $H_i\Gamma = H^{-i}(\Gamma)$ and filter the Hochschild bicomplex by chain length. Over $\mathbb{C}$, this produces the first-quadrant spectral sequence

$$E^2_{p,q} = HH_p(H_*\Gamma)_q \Longrightarrow HH_{p+q}(\Gamma).$$

Under this reversed convention, the vanishing just proved is $H_1\Gamma = 0$, so the row of internal degree one is zero. In total degrees zero and one, the only possible terms are $HH_0(H_0\Gamma)$ and $HH_1(H_0\Gamma)$, at $(0,0)$ and $(1,0)$, and no higher differential enters or leaves them. The edge maps are induced by the canonical projection, which proves naturality and both comparisons. The degree-zero statement does not use the H_1-vanishing. Finally, there are no chains of negative total degree, so negative Hochschild homology vanishes. □

Lemma 4.17 (The vertex-deletion trace sequence)**.** *Let $P = \mathrm{Jac}(Q, W)$ be the ordinary Jacobi algebra of a finite quiver with polynomial potential. Delete a set of vertices, with sum of projectors e, and denote the resulting Jacobi algebra by $Q' = P/PeP$. Suppose P and Q' are finite-dimensional, basic, and self-injective. There is a natural exact sequence*

$$HH_1(P) \longrightarrow HH_1(Q') \longrightarrow HH_0(ePe) \longrightarrow HH_0(P) \longrightarrow HH_0(Q') \longrightarrow 0. \tag{4.12}$$

The map from the corner sends $[a]$ to $[a]$.

Proof. Let Γ and Γ' denote the polynomial Ginzburg dg algebras before and after the deletion. Both are nonpositive, homologically smooth, and bimodule 3-Calabi–Yau [24, Theorem 6.3]. The finite path dg categories are finitely cellular, and the potential supplies the negative cyclic lift of the deformation class required by the corrected theorem [25, Theorem 2.1]. Vertex deletion is a dg localization [24, Corollary 7.3]. Its derived kernel is the localizing subcategory $\mathrm{Loc}(e\Gamma)$, by [24, Theorem 5.8(b)]. For several deleted vertices, compose the single-vertex localizations: each next deleted representable is the image of its original representable, so the composite kernel is generated by their union.

Put $E = e\Gamma e$. The bimodule $e\Gamma$ gives an adjunction

$$- \otimes^L_E e\Gamma : \mathcal{D}(E) \rightleftarrows \mathcal{D}(\Gamma) : R\,\mathrm{Hom}_\Gamma(e\Gamma, -).$$

The right adjoint is $(-)e$ and preserves coproducts, since $e\Gamma$ is a summand of the free right Γ-module. The adjunction unit is an isomorphism on the generator E, hence on all of $\mathcal{D}(E)$. Thus the left adjoint is fully faithful with image $\mathrm{Loc}(e\Gamma)$. Consequently $E \to \Gamma \to \Gamma'$ is an exact sequence in the Morita category by [26, Theorem 4.11(iii)].

Morita invariance and localization of Hochschild mixed complexes [26, Theorem 5.2] give a triangle

$$C_*(e\Gamma e) \longrightarrow C_*(\Gamma) \longrightarrow C_*(\Gamma') \longrightarrow C_*(e\Gamma e)[1].$$

Lemma 4.16 identifies the two degree-one terms with $HH_1(P)$ and $HH_1(Q')$, and the degree-zero terms with the respective ordinary cocenters. For the corner, $H^0(e\Gamma e) = ePe$ and $HH_{-1}(e\Gamma e) = 0$. Thus the corresponding segment of the long exact sequence is (4.12). Under Morita invariance, the trace of an endomorphism of the projective $e\Gamma$ is its cyclic class in Γ. The corner map is therefore the stated inclusion on cyclic classes, and this construction remains meaningful for the generally nonunital corner inclusion. □

Lemma 4.18 (Central cycles generate first Hochschild homology)**.** *Let Q be a finite-dimensional symmetric complex algebra and $c \in Z(Q)$. If every derivation of Q annihilating c is inner, then the map*

$$Q \longrightarrow HH_1(Q), \qquad q \longmapsto [q \otimes c]$$

is surjective. If $P \twoheadrightarrow Q$ lifts c to a central element $z \in P$, then $HH_1(P) \to HH_1(Q)$ is surjective.

Proof. The cycles in the statement are available because the Hochschild boundary of $q \otimes c$ is $qc - cq = 0$. A symmetrizing trace τ identifies the dual chain complex with the Hochschild cochain complex and gives a perfect pairing

$$HH_1(Q) \times HH^1(Q) \longrightarrow \mathbb{C}, \qquad \langle [q \otimes c], [D] \rangle = \tau(qD(c)).$$

Inner derivations kill c, so this formula depends only on the cohomology class of D. If a class pairs trivially with every displayed cycle, then $\tau(qD(c)) = 0$ for every $q \in Q$. Nondegeneracy of the symmetrizing form gives $D(c) = 0$. The hypothesis then makes D inner, so its cohomology class is zero. Finite-dimensional duality proves the asserted surjectivity. For the second assertion, a lift a of q gives the cycle $a \otimes z$, which maps to $q \otimes c$. □

Proof of Theorem 4.15. For the vertex-deletion sequence, use the following polynomial potential. Fix an orientation of Δ, use left-to-right path composition, and put

$$r_i = e_i \left(\sum_a [a, a^*] \right) e_i, \qquad W_\Delta = \frac{1}{2} \sum_i r_i^2.$$

For an arrow $a : i \to j$, its two cyclic derivatives are

$$\partial_{a^*} W_\Delta = r_i a - a r_j, \qquad \partial_a W_\Delta = a^* r_i - r_j a^*.$$

It follows that in the Jacobi algebra $z = \sum_i r_i$ commutes with every arrow and that $r_i = z e_i$. Conversely, these central-extension relations recover both displayed cyclic derivatives. The resulting polynomial algebra maps are inverse, and hence

$$\mathrm{Jac}(\overline{Q}, W_\Delta) \cong P_\Delta.$$

Deleting J removes the incident arrows. The potential restricts to the sum of the analogous potentials on the connected components I_ν of I, and consequently

$$Q_I \cong \prod_{I_\nu} P_{I_\nu},$$

compatibly with the common central element z.

For every connected ADE diagram, Etingof–Latour–Rains prove symmetry [11, Proposition 2.1], the full-algebra trace formula [11, Theorem 3.1], and that derivations annihilating z are inner [11, Corollary 3.5 and the following remark]. The normalization $\rho_i = 1$ induces the same normalization on every component. The one-vertex algebra is $\mathbb{C}$, where the corresponding claims are verified directly. Hence P_Δ and Q_I are finite, basic, symmetric Jacobi algebras. A derivation of a finite product preserves its central idempotents, so the condition on derivations holds componentwise for Q_I. Apply Lemma 4.18 to $P_\Delta \twoheadrightarrow Q_I$, with c the image of z. It

gives $HH_1(P_\Delta) \twoheadrightarrow HH_1(Q_I)$. In (4.12), this makes the boundary from $HH_1(Q_I)$ vanish. The remaining sequence is (4.10). If I is empty, it is the identity sequence on the full algebra.

The full trace dimensions in [11, Theorem 3.1] are the numbers of positive roots, and degree $2p$ counts the roots of height $p+1$. Subtract the formula for the components of I from the formula for Δ. The roots removed in this subtraction are precisely those with zero retained coefficients, equivalently those supported on I. This gives both asserted dimensions. Since the ordinary corner and quotient maps preserve the internal grading, the same subtraction gives the degreewise statement. □

The dimension of the corner cocenter is therefore the number of positive roots with at least one nonzero retained coefficient. This also equals the generic cocenter rank of the retained central extension.

4.3.4 The universal cocenter and its reduced divisor

Theorem 4.19 (The universal multi-curve cocenter)**.** *The module $M_J = HH_0(\mathscr{E}_J/S)$ is free of rank N_J. For $\mathcal{H}_J = HH_0(C_J/R_J)$, there is a free resolution of the form*

$$0 \longrightarrow R_J^b \xrightarrow{\Theta_{\mathrm{tr},J}} R_J^b \longrightarrow \mathcal{H}_J \longrightarrow 0 \tag{4.13}$$

with

$$\mathrm{Fitt}_0^{R_J}(\mathcal{H}_J) = (\delta_J) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta \right). \tag{4.14}$$

Proof. Because M_J is a commutator cokernel, it commutes with scalar extension. Its closed fibre is $HH_0(e_J P_\Delta e_J)$, whose dimension is N_J by Theorem 4.15. On the other hand, (4.7) gives generic rank N_J, since each matrix algebra has a one-dimensional cocenter. Lift a basis of the closed fibre and use Nakayama to obtain $S^{N_J} \twoheadrightarrow M_J$. Its kernel is a submodule of a free module over the domain S; its generic fibre is zero, so the kernel itself is zero. Thus M_J is free over the Cartan parameter ring S.

Since z is central,

$$HH_0(D_J/S) = M_J/zM_J.$$

On the root factor α, z acts on the cocenter by the same scalar as on the algebra. Therefore

$$\det(z \mid M_J) = u \prod_{\pi_I(\alpha) \neq 0} \ell_\alpha = u\, q^* \delta_J, \qquad u \in \mathbb{C}^\times.$$

Its nonvanishing makes multiplication by z injective on the free module M_J, hence gives a square presentation of $HH_0(D_J/S)$. By (4.5), this module is the Cartan pullback of $\mathcal{H}_J$. As in the proof of Theorem 4.13, a first syzygy of a finite free surjection onto $\mathcal{H}_J$ becomes projective after faithfully flat base change. Faithfully flat descent [35, Tag 058S] makes it projective over R_J, hence free because R_J is local. Since $\mathcal{H}_J$ is torsion, the two free ranks agree, proving (4.13). Fitting ideals commute with base change and contract faithfully flatly, which gives (4.14). □

The product defining P_β accounts for all Cartan roots above β. The Fitting exponents of the three presentations are

universal module	exponent of P_β
$e_i C_J e_j$	$\beta_i \beta_j$
C_J	s_β^2
$HH_0(C_J/R_J)$	1

After pullback along the same family of classifying arcs, the closed-arc determinant orders give the ranks over the deformation base.

4.3.5 Expressing a geometric family as a pullback

Let $(B, \mathfrak{m}, \mathbb{C})$ be complete Noetherian local, and let $f_B : X_B \to Y_B$ be a deformation of f_0 as a projective contraction, with source and affine target flat over B. We complete along the central contraction, retaining the full moving support. Write $T_0 = \mathcal{O} \oplus \bigoplus_{j \in J} T_{0,j}$ for the fixed normalized basic generator.

The variable w is the surface-section parameter; B parametrizes the external deformation.

Proposition 4.20 (The multi-curve classifying pullback)**.** *There is a continuous local map $F : R_J \to D_B = B[[w]]$ and a relative tilting generator T_B lifting T_0, such that the relative contraction quotient satisfies*

$$\mathscr{A}_B := \operatorname{End}(T_B)^{\mathrm{op}} / \operatorname{End}(T_B)^{\mathrm{op}} e_0 \operatorname{End}(T_B)^{\mathrm{op}} \cong C_J \otimes_{R_J, F} B[[w]]. \tag{4.15}$$

All labelled projectors are preserved. Any other lift of the same marked generator gives an isomorphic marked quotient, and $\mathscr{A}_B / \mathfrak{m} \mathscr{A}_B \cong A_0$.

Proof. Lift a general surface-section parameter w_0 to w. The smoothness of the central source and the cDV property of the target allow the flatness argument of Proposition 4.6 to apply. That flatness argument gives D_B-flat source and target by the fibrewise criterion [35, Tag 00MP]. The closed surface is the partial resolution with retained set J. The source-deformation and compatible Artin-lifting argument in Proposition 4.6 applies to every nonempty retained subgraph, with base $\mathfrak{h}/W_I$. Its two-stage target and source comparison therefore supplies F and an identification of the actual marked contraction.

Proposition 4.10 fixes the normalized labelled universal generator. For its pullback to the completed family, we first compare endomorphisms on finite Artin quotients of D_B, and then take the formal limit as described below. The labelled summands of the pulled-back universal generator on the central threefold are basic: modulo w, these summands are the distinct indecomposable projectives on the partial surface. A nontrivial decomposition of one such summand would restrict to a nontrivial decomposition on that partial surface, because a summand whose reduction is zero vanishes by Nakayama in the completed neighbourhood. The determinant degrees distinguish the curve labels. Van den Bergh's classification of normalized perverse projectives [37, Proposition 3.5.4 and Theorem 3.5.5] consequently identifies the central generator with T_0.

The higher self-Ext groups of these summands vanish. Karmazyn's lifting lemma [22, Lemma 3.2] identifies their lifts through every external-base thickening and in the formal limit while preserving the chosen projectors. Put $D_n = D_B/(\mathfrak{m}, w)^n$. The universal source, target, and

endomorphism algebra are flat over R_J. The Tor-independence calculation in Proposition 4.10 and derived base change [35, Tag 08IB] therefore identify the actual endomorphism algebra on the n-th pullback with $\Lambda_J \otimes_{R_J} D_n$, preserving multiplication and every projector. Taking the idempotent quotient gives $C_J \otimes_{R_J} D_n$. The endomorphism algebra and its projector ideal are finite modules over the Noetherian completed affine target. Formal functions and exactness of completion on finite modules [35, Tags 02OC and 00MA] identify the geometric quotient with $\varprojlim_n (C_J \otimes_{R_J} D_n)$. Since C_J is finite over R_J, its ordinary pullback is a finite module over the complete ring D_B, and this limit is $C_J \otimes_{R_J} D_B$. This proves (4.15).

The ordinary tensor expression must still be compared with the completed support. The central classifying arc lies in no curve divisor: otherwise the pulled-back family of exceptional curves would give an exceptional surface in X_0. Choose a divisor equation with pullback $g(w) \neq 0$. Then $\mathbb{C}[[w]]$ is finite over $\mathbb{C}[[g(w)]]$, and hence over R_J. Let U_J be the completed universal affine target ring, with maximal ideal $\mathfrak{n}_J$, and set

$$Z_0 = U_J \otimes_{R_J, f} \mathbb{C}[[w]], \qquad f = F \bmod \mathfrak{m}.$$

The finiteness of $\mathbb{C}[[w]]$ over R_J makes Z_0 finite over U_J. Its closed fibre over U_J is

$$Z_0/\mathfrak{n}_J Z_0 \cong \mathbb{C}[[w]]/\mathfrak{m}_{R_J}\mathbb{C}[[w]],$$

which is local Artinian: the ideal on the right contains $g(w) \neq 0$ and is contained in (w). Since a finite algebra is integral, every maximal ideal of Z_0 lies over $\mathfrak{n}_J$. The local closed fibre therefore gives a unique maximal ideal $\mathfrak{n}_0 = \mathfrak{n}_J Z_0 + wZ_0$. For some $c \geq 1$ one has $\mathfrak{n}_0^c \subseteq \mathfrak{n}_J Z_0 \subseteq \mathfrak{n}_0$, so the two adic topologies coincide. Every finite module over the complete Noetherian local ring U_J is complete [35, Tag 00MA]. Consequently Z_0, the central pullback of the completed universal endomorphism algebra, and its idempotent quotient are already $\mathfrak{n}_0$-adically complete. Ordinary scalar extension commutes with the idempotent quotient, since the image of the extended ideal is exactly the ideal generated by the extended projector. Thus the marked identification with the completed central generator established above identifies the ordinary central-arc quotient with A_0, as in Lemma 3.18.

The ordinary pullback on the right-hand side of (4.15) is finite over D_B by Theorem 4.13, and its closed B-fibre is finite-dimensional. Corollary 4.3 applied to the pulled-back square presentation makes this ordinary pullback finite over B. On the resulting finite B-module w is nilpotent modulo $\mathfrak{m}$, so the $(\mathfrak{m}, w)$-adic and $\mathfrak{m}$-adic topologies agree. The ordinary pullback is already complete, which identifies the ordinary tensor expression with the quotient carrying the full relative support. □

The closed classifying arc supplies the divisor orders m_β. They determine the ranks below; the perturbation count of Nabijou–Wemyss identifies them with the GV invariants afterwards.

Proposition 4.21 (Ranks and intersection multiplicities after pullback)**.** *In the preceding setting, $\mathscr{A}_B$ is finite free over B of rank $\dim_{\mathbb{C}} A_0$, and*

$$C_J \otimes^L_{R_J,F} B[[w]] \simeq \mathscr{A}_B[0].$$

Write $f = F \bmod \mathfrak{m}$ and $m_\beta = \operatorname{ord}_w P_\beta(f(w))$. These integers are positive and finite, and

$$\operatorname{rank}_B \mathscr{A}_B = \sum_{\beta \in \mathcal{B}_J} m_\beta s_\beta^2, \tag{4.16}$$

$$\operatorname{rank}_B (e_i \mathscr{A}_B e_j) = \sum_{\beta \in \mathcal{B}_J} m_\beta \beta_i \beta_j. \tag{4.17}$$

Moreover $\operatorname{Spec} B[[w]]/(P_\beta(F))$ *is finite flat over B of degree m_β.*

Proof. Use the square algebra presentation. Its reduction modulo $\mathfrak{m}$ has cokernel A_0, which is finite-dimensional, so its matrix is invertible over $\mathbb{C}((w))$. Lemma 4.2 gives finite freeness over B and injectivity of the pulled-back presentation. The resulting exact sequence proves the derived statement.

Smith normal form on the closed classifying arc and the Fitting ideal give (4.16). Each $P_\beta(f)$ is nonzero because their product, with positive exponents, is nonzero. Since f is local, $P_\beta(f)$ vanishes at the origin, and hence $m_\beta > 0$. Apply Lemma 4.2 to the presentations of the Peirce spaces and to the one-by-one matrices $P_\beta(F)$. This gives the remaining rank formula and the stated finite flat schemes cut out by the divisor equations. Thus the ranks are determined by determinant orders on the closed classifying arc. □

The numerical comparison also requires control of the support of sheaves through a perturbation. The following two lemmas separate this support question and the contribution of a split curve from deformation invariance of the relative sheaf moduli space.

Lemma 4.22 (A proper relative support from the ADE critical locus)**.** *For the classifying family above, the relative surface-critical locus of the affine ADE target is finite over B. Its projective inverse image contains every geometric exceptional curve and is projective over B. For a holomorphic classifying bidisc, one may choose a fixed closed surface-parameter disc and a small external disc so that this critical locus has no points on the surface-parameter boundary, is proper with finite fibres over the external disc, and has proper inverse image under the contraction.*

Proof. Use the uncompleted polynomial marked base H_J^{alg} and write its universal affine equation as $f_{\mathrm{ADE}}(x_1, x_2, x_3; a) = 0$. The algebra

$$\mathcal{J}_{\mathrm{crit}} = \frac{H_J^{\mathrm{alg}}[x_1, x_2, x_3]}{(f_{\mathrm{ADE}}, \partial_{x_1} f_{\mathrm{ADE}}, \partial_{x_2} f_{\mathrm{ADE}}, \partial_{x_3} f_{\mathrm{ADE}})}$$

is finite over H_J^{alg}. Indeed the central Jacobian algebra is finite-dimensional. For the standard ADE equations its relations are, up to nonzero constants,

A_n	$x_1 = x_2 = x_3^n = 0$
D_n	$x_1 = x_2 x_3 = x_2^2 + (n-1)x_3^{n-2} = 0$
E_6	$x_1 = x_2^2 = x_3^3 = 0$
E_7	$x_1 = 3x_2^2 + x_3^3 = x_2 x_3^2 = 0$
E_8	$x_1 = x_2^2 = x_3^4 = 0.$

In type D_n these imply $x_3^{n-1} = x_2^3 = 0$; in type E_7 they imply $x_3^5 = x_2^3 = 0$. Lift a homogeneous basis of the central critical algebra. Reduction modulo the positive parameter ideal and induction on weight show that these lifts generate $\mathcal{J}_{\mathrm{crit}}$ over the parameter ring.

Let Δ_{tot} cut out the full surface discriminant, including both curve and residual-singularity components [29, Remark 2.8 and the following equality]. One may use the invariant square of the product of all positive-root forms. Since the support of the finite critical module is this hypersurface, some power Δ_{tot}^q annihilates it. Its pullback on the central arc is nonzero. Otherwise the critical algebra over $\mathbb{C}[[w]]$ would have a nonzero generic fibre, hence a field quotient finite over $\mathbb{C}((w))$. At the corresponding critical point the central threefold equation G_0 and its three surface partial derivatives vanish. Extend d/dw to this finite separable field extension and differentiate $G_0 = 0$; it follows that $\partial_w G_0 = 0$ as well. This gives a branch of the singular locus dominating the w-disc, contradicting the isolated central threefold singularity. Equivalently, its finite-length four-variable Jacobian quotient cannot map to a field in which w is nonzero.

Put $g = \Delta_{\mathrm{tot}}(F) \in B[[w]]$. Lemma 4.2, applied to g^q, makes $B[[w]]/(g^q)$ finite over B. The pulled-back critical algebra is finite over this ring and therefore finite over B. A partial resolution is an isomorphism over the smooth part of each surface, so all exceptional images lie in its critical locus. The inverse image of this finite scheme under the projective contraction is projective over B. Its finite target support is already complete in the full central topology.

For a holomorphic bidisc, choose the central closed w-disc with no full-discriminant boundary zero and then shrink the external disc to preserve that property. There are finitely many critical points over each of its finitely many w-roots. The affine coordinates on the critical scheme satisfy monic equations over the parameter ring; on a compact set of parameter values these give uniform bounds on the coordinates. Thus the critical incidence over a compact external subdisc is closed in a compact set. The absence of w-boundary points proves properness. Its points specialize only to the central critical point, so the neighbourhood can be chosen inside the original target germ. Projectivity of the contraction then gives properness of the inverse image. This controls escape in the affine target as well as in the w-direction. □

Lemma 4.23 (Stable sheaves on the split local curves)**.** *Let F be a field of characteristic zero. On the split local threefold $Z_{\mathrm{loc}} = \mathrm{Tot}_{\mathbb{P}^1_F}(\mathcal{O}(-1)^{\oplus 2})$, or its formal completion along the zero section C_{loc}, the only pure one-dimensional semistable sheaf with $\chi = 1$ and support on C_{loc} is $\mathcal{O}_{C_{\mathrm{loc}}}$. Its first and second self-Ext groups vanish. It therefore gives one reduced isolated stable-sheaf point, rational over F, with local virtual contribution $+1$ under the usual stable-sheaf obstruction theory. For finitely many disjoint such local curves, there are no additional stable objects supported on higher multiples or on more than one curve.*

Proof. First, a pure one-dimensional sheaf with Hilbert polynomial $dn + 1$ cannot be strictly semistable. Equality of reduced Hilbert polynomials for a proper subsheaf of degree $d' \le d$ would give $d\chi' = d'$, hence $d' = d$ and $\chi' = 1$. The quotient would then have dimension zero and Euler characteristic zero, and would be zero, a contradiction.

Let $p : Z_{\mathrm{loc}} \to \mathbb{P}^1_F$ be the projection, and let $\mathcal{F}$ be such a sheaf of generic multiplicity m. In the algebraic total space, Noetherian coherence and the stated set-theoretic support imply that a power of the zero-section ideal annihilates $\mathcal{F}$. We justify this implication separately in the formal case. The split contraction

$$\mathrm{Tot}_{\mathbb{P}^1_F}(\mathcal{O}(-1)^{\oplus 2}) \longrightarrow \operatorname{Spec} F[a,b,c,d]/(ad - bc)$$

is projective and an isomorphism off the origin: in homogeneous coordinates $[s_0 : s_1]$, it sends the two fibre coordinates x, y to (s_0x, s_1x, s_0y, s_1y), and a nonzero column recovers $[s_0 : s_1]$. On each standard chart the pulled-back origin ideal is the zero-section ideal. Complete the target at the origin, put $\widehat{Q} = F[[a, b, c, d]]/(ad - bc)$, and let $X_{\widehat{Q}}$ be the corresponding projective source. Its closed-fibre formal completion is the formal neighbourhood in the statement.

Coherent formal existence algebraizes $\mathcal{F}$ to a coherent sheaf $\mathcal{F}^{\mathrm{alg}}$ on $X_{\widehat{Q}}$ [35, Remark 30.27.2, Tag 088F]. Formal functions [35, Theorem 30.20.5, Tag 020C] and completeness of finite $\widehat{Q}$-modules identify

$$H^0(X_{\widehat{Q}}, \mathcal{F}^{\mathrm{alg}}) \cong H^0(\widehat{Z}_{\mathrm{loc}}, \mathcal{F}).$$

Here $\widehat{Z}_{\mathrm{loc}}$ denotes the zero-section formal completion. The right-hand side is finite-dimensional over F, as required for the stated Euler characteristic to be defined. The coherent pushforward is therefore a finite-length $\widehat{Q}$-module, so it vanishes off the closed point. The contraction is an isomorphism there; hence $\mathcal{F}^{\mathrm{alg}}$ is supported on its closed fibre. Noetherian coherence now implies annihilation by a power of the closed-fibre ideal, which is the zero-section ideal. The same holds after completion. Thus in both cases $\mathcal{F}$ is a sheaf on a finite zero-section thickening, and $V = p_*\mathcal{F}$ is coherent on $\mathbb{P}^1_F$. Its torsion subsheaf is invariant under the normal-coordinate action, and would give a zero-dimensional subsheaf of $\mathcal{F}$. Purity therefore makes V a vector bundle, of rank m, with $\chi(V) = 1$.

Choose the largest integer a with $\mathrm{Hom}(\mathcal{O}(a), V) \neq 0$. It exists because a rational frame embeds V into $\mathcal{O}(D)^{\oplus m}$ for an effective divisor D, and $a \geq 0$ because $h^0(V) - h^1(V) = 1$. Since $\mathrm{Pic}(\mathbb{P}^1_F) = \mathbb{Z}[\mathcal{O}(1)]$, a nonzero map $\mathcal{O}(a) \to V$ is saturated: otherwise its saturation is a line bundle of larger degree, contradicting maximality. The normal-coordinate algebra acts by $\mathcal{O}(1)^{\oplus 2} \otimes V \to V$. Its action on this subline is zero, since $\mathrm{Hom}(\mathcal{O}(a+1), V) = 0$. Thus $\mathcal{O}(a)$ gives a subsheaf on the reduced zero section. For a polarization of degree $h > 0$ there, its reduced constant Hilbert term is $(a+1)/h$, whereas that of $\mathcal{F}$ is $1/(mh)$. If $m > 1$, this contradicts semistability. Hence $m = 1$, $V = \mathcal{O}$, and the normal action is zero because $H^0(\mathcal{O}(-1)) = 0$. This proves $\mathcal{F} = \mathcal{O}_{C_{\mathrm{loc}}}$, which is stable by the same Hilbert-polynomial criterion.

The zero-section Koszul resolution computes its self-Ext groups as

$$\mathrm{Ext}^k(\mathcal{O}_{C_{\mathrm{loc}}}, \mathcal{O}_{C_{\mathrm{loc}}}) \cong \bigoplus_{q=0}^{2} H^{k-q}\left(\mathbb{P}^1_F, \bigwedge^q (\mathcal{O}(-1)^{\oplus 2})\right).$$

The differentials vanish since the normal coordinates annihilate the zero-section sheaf. The only nonzero groups are F in degrees zero and three, from $H^0(\mathbb{P}^1_F, \mathcal{O}) = F$ and $H^1(\mathbb{P}^1_F, \mathcal{O}(-2)) = F$, respectively. Thus the tangent and obstruction spaces are both zero, and the local virtual point has degree $+1$. This calculation also holds in the formal neighbourhood, since the same finite Koszul resolution computes the proper-support Ext groups. The same argument applies after every field extension of F, so the point is geometrically reduced and isolated.

For disjoint curves, the comaximal support ideals split a sheaf into the direct sum of its pieces. If two pieces are nonzero, one has Euler characteristic at least one and degree strictly smaller than the total degree, and hence destabilizes. Only one curve can support a stable object, and the calculation above applies to it. □

Lemma 4.24 (Central cycles and local Hilbert lengths)**.** *Choose a normal partial ADE surface S_0 in the central smooth threefold, and let p be the exceptional image. For a nonzero effective*

cycle $\beta = \sum_{i\in J} \beta_i[C_i]$*, put*

$$\mathcal{I}_{Z_\beta,S_0} = \bigcap_{\beta_i>0} \mathcal{I}^{(\beta_i)}_{C_i,S_0}.$$

Every pure one-dimensional stable sheaf of Euler characteristic one and cycle β*, supported on the exceptional locus, is* $\mathcal{O}_{Z_\beta}$*. Such a sheaf exists precisely when* $Z_\beta \subset f_0^{-1}(p)$ *scheme-theoretically and* $\chi(\mathcal{O}_{Z_\beta}) = 1$*. It then has* $H^0 = \mathbb{C}$ *and* $H^1 = 0$*, is stable for every ample polarization, and determines an isolated point of the threefold Hilbert space. Denote its local length by* k_β*.*

Consider an actual completed classifying deformation over $R_s = \mathbb{C}[[s]]$*, with flat source and target, and with the retained determinant line bundles fixed. Suppose its geometric generic complete curves are finitely many disjoint split* $(-1,-1)$*-curves. If the central point above exists, its relative local Hilbert algebra is finite free of rank* k_β *over* R_s*. Its geometric generic points are reduced and correspond exactly to the curves of retained degree vector* β*. Conversely, a generic curve of that vector specializes to* Z_β*, so its existence implies the stated central conditions.*

Proof. A stable sheaf F with proper support has only scalar endomorphisms. Indeed a nonzero endomorphism with nonzero kernel would have an image of slope greater than that of F, whereas stability forces the slope of that proper subsheaf to be smaller. An injective endomorphism is surjective, since its cokernel has zero Hilbert polynomial. Thus $\mathrm{End}(F)$ is a finite-dimensional division algebra over $\mathbb{C}$, hence is $\mathbb{C}$. Every function in the maximal ideal of p acts nilpotently on F; being scalar, that action is zero. Consequently F lies on the scheme-theoretic contraction fibre and on S_0.

Rationality of the partial ADE surface gives $H^1(S_0, \mathcal{O}) = 0$. For any closed subscheme Z supported on its exceptional fibre, formal functions and the one-dimensional fibres give $H^2(S_0, \mathcal{I}_Z) = 0$ [35, Tag 02OC]. The cohomology sequence therefore gives $H^1(\mathcal{O}_Z) = 0$, and $\chi(\mathcal{O}_Z) \geq 1$ when $Z \neq \varnothing$. Since $\chi(F) = 1$, F has a nonzero section. Its image is a pure cyclic subsheaf $\mathcal{O}_Z \subset F$. The preceding inequality, together with its degree being at most the degree of F, makes its reduced Hilbert polynomial at least that of F. Stability forces $\mathcal{O}_Z = F$, so $H^0(F) = \mathbb{C}$ and $H^1(F) = 0$.

On a normal surface, purity excludes embedded height-two primes from the quotient by its ideal. At a height-one prime the local ring is a DVR, and the primary ideal is determined by its length. Intersecting these primary ideals gives exactly the displayed symbolic-power ideal. Thus $Z = Z_\beta$. Conversely, a proper quotient of $\mathcal{O}_{Z_\beta}$ is the structure sheaf of a nonempty proper subscheme and has Euler characteristic at least one. Every proper one-dimensional subsheaf therefore has Euler characteristic at most zero, which proves stability for every ample polarization.

For the infinitesimal statement, an Artin flat deformation F_A has $H^0(F_A) \cong A$ and $H^1(F_A) = 0$, by induction over small extensions. A generator of $H^0(F_A)$ gives a surjective evaluation map $\mathcal{O} \to F_A$; changing the generator by a unit preserves its kernel. Its endomorphism algebra is A, so the affine target acts through a unique moving target section. This identifies the pointed sheaf and embedded Hilbert functors, including prescribed ambient deformations and full target completion.

The fixed-cycle locus used below is an actual open-and-closed locus of a relative Hilbert space. Write L_i for the retained determinant line bundles, so that $\deg_{C_j} L_i = \delta_{ij}$. For either the central contraction or the prescribed R_s-family, start with $M = \mathrm{Hilb}_{X/Y}$ and restrict to its one-dimensional Hilbert-polynomial strata. On an étale scheme chart $U \to M$, let $\mathcal{Z}_U \to U$ be

the universal proper flat family. Define

$$d_i(Z) = \chi(\mathcal{O}_Z \otimes L_i) - \chi(\mathcal{O}_Z).$$

The functions $\chi(\mathcal{O}_Z)$ and $d_i(Z)$ are locally constant and commute with base change: apply [35, Lemma 36.32.2, Tag 0B9T] to the proper finitely presented family $\mathcal{Z}_U \to U$, with its flat finitely presented structure sheaf and its twists by L_i. The conditions

$$\chi(\mathcal{O}_Z) = 1, \qquad d_i(Z) = \beta_i \quad (i \in J)$$

therefore define an open-and-closed subspace $M_\beta \subset M$. They do not assert that the ambient contraction is proper over the external base. Properness here is that of the universal Hilbert family. For a one-dimensional subscheme with cycle $\sum_j a_j[C_j]$, additivity modulo zero-dimensional sheaves gives

$$d_i(Z) = \sum_j a_j \deg_{C_j} L_i = a_i.$$

Indeed a zero-dimensional sheaf has the same Euler characteristic after an invertible twist, and the one-dimensional part of its class is the sum of its generic lengths times the integral component classes. Thus these numerical conditions specify the full retained cycle even when the Hilbert subscheme is not initially assumed pure.

Assume first that the central point Z_β exists. Every point of the central M_β lies over p: its Hilbert subscheme has positive one-dimensional degree, whereas the central contraction has zero-dimensional fibres away from p. Every closed subscheme of $f_0^{-1}(p)$ with the prescribed invariants equals Z_β. To see this without assuming purity, let T be the maximal zero-dimensional subsheaf of its structure sheaf. This is an ideal, and its quotient is pure and has cycle β. The symbolic-power argument above identifies that quotient with $\mathcal{O}_{Z_\beta}$. Additivity and $\chi(\mathcal{O}_{Z_\beta}) = 1$ give $1 = \operatorname{length} T + 1$, so $T = 0$.

Here is the finiteness argument for the local ring. By [35, Proposition 99.9.4, Tag 0D01], the relative Hilbert space is locally of finite presentation over the Noetherian target. Choose an affine étale chart near $[Z_\beta]$. Its central restriction is Noetherian and of finite type over the central target. The preceding set-theoretic support assertion says that the target maximal ideal $\mathfrak{m}_p$ is contained in the nilradical of this chart. Consequently some power of $\mathfrak{m}_p$ annihilates its structure sheaf. The chart is therefore of finite type over an Artinian local $\mathbb{C}$-algebra. Its fibre over p has only the Hilbert point $[Z_\beta]$; on an étale chart the points above it form a discrete zero-dimensional fibre. The finite-type $\mathbb{C}$-scheme chart is Jacobson, so it has dimension zero near a chosen lift of this point. Its local ring is Noetherian of dimension zero, hence Artinian. The moving-target identification already proved identifies its pointed Artin functor with the embedded threefold Hilbert functor. It follows that the latter has the same Artinian local ring and intrinsic length k_β. In particular the threefold Hilbert point is isolated.

For the actual R_s-family use the analogous M_β, choose an étale chart and a lift of the central point, and let $\mathscr{H}_\beta$ be its completed local ring. The preceding argument identifies $\mathscr{H}_\beta/(s)$ with the central threefold Hilbert ring. The local Artin functor is unchanged by taking this open-and-closed locus. Its tangent and complete quotient obstruction spaces are

$$T_\beta = \operatorname{Hom}(I_\beta, \mathcal{O}_{Z_\beta}) \cong \operatorname{Ext}^1(\mathcal{O}_{Z_\beta}, \mathcal{O}_{Z_\beta}),$$
$$V_\beta = \operatorname{Ext}^1(I_\beta, \mathcal{O}_{Z_\beta}) \cong \operatorname{Ext}^2(\mathcal{O}_{Z_\beta}, \mathcal{O}_{Z_\beta}).$$

These comparisons follow from the ideal sequence using $H^0 = \mathbb{C}$, $H^1 = H^2 = 0$, and evaluation at 1. For a small extension of R_s-algebras, ambient flatness gives the canonical quotient extension of Tag 0CZU; its splitting is equivalent to a flat quotient lift by Tag 08MQ. This construction commutes with linear pushouts of the small kernel and keeps the prescribed ambient family and s fixed. The lifted quotient remains proper: the central target ideal acts trivially modulo the Artin maximal ideal, so a power of it annihilates the lift. The support consequently lies in the projective inverse image of an Artinian target thickening. Thus this is a complete obstruction space for the stated local Hilbert functor. Proper-support duality on the central smooth crepant threefold gives $V_\beta \cong T_\beta^\vee$. Here the structure sheaf is perfect, has proper exceptional support, and the canonical bundle is trivial near that support. The completion comparison in the proof of Proposition 2.19 transfers these self-Ext groups to the chosen model; no numerical GV conclusion of that proposition is used. Lemma 2.18 now gives

$$\mathscr{H}_\beta \text{ finite free over } R_s, \qquad \operatorname{rank}_{R_s} \mathscr{H}_\beta = k_\beta.$$

Finally, the actual chart local ring injects into its completion by Krull intersection. Its image is an R_s-submodule of the finite module $\mathscr{H}_\beta$, hence is finite and closed in the s-adic topology. It is also dense in that topology, since the closed quotient is Artinian and the maximal-ideal and s-adic topologies of the completion are cofinal. The local ring therefore already equals $\mathscr{H}_\beta$. It carries the actual universal Hilbert family and the actual generic Hilbert tangent spaces.

It remains to identify the generic points. After a finite extension of $\mathbb{C}((s))$, the image of a generic exceptional curve extends to a DVR section by Lemma 4.22. Its projective inverse image contains the unique saturated schematic closure of that curve. The closure is DVR-flat, with central cycle β and Euler characteristic one. Its central fibre lies scheme-theoretically in $f_0^{-1}(p) \subset S_0$. Its pure quotient is $\mathcal{O}_{Z_\beta}$; the rational-surface cohomology calculation above gives its Euler characteristic at least one. Additivity then forces that value to be one and the zero-dimensional kernel to vanish. Thus the central fibre is exactly Z_β. This proves both the existence of the central point and the specialization assertion.

Conversely, a generic Hilbert object has positive one-dimensional degree and lies in one contraction fibre. Every such generic fibre is a reduced integral $\mathbb{P}^1$; its only one-dimensional closed subscheme is itself. The locally constant Euler differences for the retained determinant line bundles preserve the vector β. Uniqueness of saturated closure and of the pointed étale lift prevents duplicate Hilbert points. Finally Lemma 4.23 gives $\operatorname{Ext}^1(\mathcal{O}_E, \mathcal{O}_E) = 0$. Together with $H^0(\mathcal{O}_E)$ equal to the ground field and $H^1(\mathcal{O}_E) = 0$, the ideal sequence makes the embedded Hilbert tangent space zero. The finite geometric generic local rings are therefore fields. Their number is the rank k_β. □

Corollary 4.25 (The Cartan matrix and intersection multiplicities)**.** *The Cartan matrix with respect to the idempotents $(e_i)_{i\in J}$ is*

$$\left(\dim_{\mathbb{C}} e_i A_0 e_j\right)_{i,j\in J} = \sum_{\beta\in\mathcal{B}_J} m_\beta\, \beta\beta^{\mathsf{T}}.$$

It is symmetric positive definite. For the genus-zero invariants of the central flopping contraction, $m_\beta = n_\beta$.

Proof. Restricting (4.17) to the closed fibre gives the displayed matrix formula. Every retained simple root projects to a standard basis vector and every m_β is positive, so the sum is positive definite.

We compare the formal classifying arc f with an analytic marked arc g_{an} for the same original contraction. Fix the same standard partial resolution, its chosen side, and the retained curve basis. Let g be the completion of g_{an}. The convention for n_β is the one fixed in the Introduction: the curve count of [29, Section 3.1, equation (3.A)]. Their contracted subset is I, and their retained subset is our J. Lemma 4.9 identifies their class-β component with our reduced divisor $P_\beta = 0$.

Choose a small disk for g_{an} on whose boundary the full surface discriminant is nonzero. A sufficiently small transverse perturbation avoids the multiple-root strata and satisfies [29, Section 2.6, condition (2.G)]. The argument principle and their curve–intersection correspondence [29, Lemma 2.9 and Corollary 2.10] give

$$n_\beta = \operatorname{ord}_w P_\beta(g(w)) =: m_\beta(g).$$

These are intersections with the reduced downstairs divisor. Distinct proportional restricted roots retain their separate components.

To compare this order with the order along f, form a transverse formal translation of each of the two arcs, using Lemma 5.2 for the full surface discriminant and its root-intersection strata. For each translation, take the ordinary pullback of the fixed algebraic model over $\mathbb{C}[[s,w]]$, followed by full central target completion. Universal parameter flatness and the flatness of this completion make both source and target flat over $\mathbb{C}[[s]]$; their marked central contraction is the prescribed f_0. For an arc $a \in \{f, g\}$, the finite root algebra has degree $m_\beta(a)$, by Lemma 4.2. At an active single-root point, the chosen simultaneous partial resolution has the transverse A_1 model: the full Weyl quotient has normal coordinate $t = \zeta^2$, while the partial quotient is etale at that non-Levi root. The transverse pullback is the split ODP $xy - z^2 = \zeta^2$ after a finite splitting extension, with its selected small resolution and one $(-1,-1)$-curve. At an inactive single-root point no curve is retained. The same local equations underlie [29, Lemma 2.9] and apply over the formal splitting field. Different roots have different w-coordinates, and the active discriminant is the full curve locus. Thus there are exactly $m_\beta(a)$ complete curves with the indicated label and no further complete curves. Lemma 4.22 ensures that their full target coordinates, and hence their projective inverse images, survive central target completion.

The line bundles used on these two families are the pullbacks of the specified labelled bundles in Proposition 4.10. On a chosen local Cartan lift, specialize the projective flat root-divisor family of Lemma 2.8. Its determinant degrees give, for each retained i,

$$\deg_E \det(P_i) = \sum_{j \in J} \beta_j \deg_{C_j} \det(P_i) = \beta_i.$$

Here the central degrees are δ_{ij}; the root specialization uses Lemma 2.8 and the same full-to-partial resolution and labelled bundles as in Remark 2.9. Consequently every retained coordinate, and not only their sum, agrees with the chosen class basis.

Apply Lemma 4.24 to these two actual families. Each contains a class-β generic curve, since its root degree is positive, and hence has central point Z_β. The marked identification of their central completed contractions identifies this unique cyclic sheaf and its embedded threefold Hilbert functor. Its local length k_β is therefore the same for both families. The lemma gives the individual equalities

$$m_\beta(f) = k_\beta = m_\beta(g) = n_\beta.$$

This comparison does not require the two classifying maps to be equal.

Returning to the original formal translation of f, its root algebra has degree $m_\beta(f) = n_\beta$ and the determinant degrees just computed retain its original marking. We form its generic fibre by localizing the full relative family before completing at moving points. For a single retained curve, Proposition 2.19 identifies the same count with the original Katz Hilbert invariant. □

4.3.6 Hua–Toda flatness over general bases

Kawamata proved algebra flatness over a disk [23, Theorem 3.1(2)]. In the present setting, the rank calculation above treats the algebra and its Peirce spaces, the cocenter has a corresponding pulled-back presentation to which Lemma 4.2 applies, and a relative symmetrizing trace identifies the centre. The theorem records the three flatness assertions and their base-change consequences over the complete local bases in its statement.

Theorem 4.26 (Hua–Toda flatness and base change)**.** *Let f_B, T_B, and $\mathscr{A}_B$ be as in Proposition 4.20, with arbitrary complete Noetherian local complex base $(B, \mathfrak{m}, \mathbb{C})$. Let $n_\beta = m_\beta$ be the central curve multiplicities of Corollary 4.25, and put $N = \sum_{\beta \in \mathcal{B}_J} n_\beta$. Then*

$$\mathscr{A}_B \cong B^{\sum_\beta n_\beta s_\beta^2}, \qquad HH_0(\mathscr{A}_B/B) \cong B^N, \qquad Z(\mathscr{A}_B) \cong B^N$$

as B-modules. For every commutative B-algebra B', the canonical maps

$$HH_0(\mathscr{A}_B/B) \otimes_B B' \xrightarrow{\sim} HH_0(\mathscr{A}_B \otimes_B B'/B'),$$
$$Z(\mathscr{A}_B) \otimes_B B' \xrightarrow{\sim} Z(\mathscr{A}_B \otimes_B B')$$

are isomorphisms. Every symmetrizing trace of A_0 lifts to a perfect B-linear symmetrizing trace of $\mathscr{A}_B$. Moreover,

$$C_J \otimes^L_{R_J,F} B[[w]] \simeq \mathscr{A}_B[0],$$
$$\mathcal{H}_J \otimes^L_{R_J,F} B[[w]] \simeq HH_0(\mathscr{A}_B/B)[0].$$

The free bases and lifted trace can be chosen, and the base-change maps are canonical. The auxiliary B-algebra B' need not be Noetherian, local, or complete. These are ordinary tensors of the finite relative algebra; no replacement by $B'[[w]]$ is made.

Proof. Proposition 4.21 gives the algebra assertion and its derived pullback. Put $H_B = HH_0(\mathscr{A}_B/B)$. The B-span of commutators is already a D_B-module, since $d[a,b] = [da,b]$ for a central scalar $d \in D_B$. Hence

$$H_B \cong \mathcal{H}_J \otimes_{R_J,F} D_B \cong \operatorname{coker} \Theta_{\mathrm{tr},J}(F).$$

On the closed classifying arc, the determinant of this presentation is a unit times $\prod_\beta P_\beta(f(w))$. It is nonzero and has order N. Lemma 4.2 gives freeness of rank N and injectivity of the pulled-back presentation. This also proves the second derived statement over B, without assuming that B is reduced.

Finally, the central contraction algebra is symmetric: the stable endomorphism algebra of the entire maximal Cohen–Macaulay generator is covered by [2, Theorem 2.6]. For any symmetrizing trace of A_0, apply Lemma 4.4 to the finite free algebra and cocenter just obtained. It supplies a perfect relative lift. The lift is a choice; it proves centre freeness, whereas the two base-change maps are canonical. □

For the local consequences below, write $d = \dim_{\mathbb{C}} A_0$.

Corollary 4.27 (Local bases, geometric base change, and saturation)**.** *The commutator submodule of $\mathscr{A}_B$ is a B-direct summand. For every ideal $L \subseteq B$,*

$$[\mathscr{A}_B, \mathscr{A}_B] \cap L\mathscr{A}_B = L[\mathscr{A}_B, \mathscr{A}_B].$$

For a continuous local map $B \to B'$ of complete Noetherian local complex bases, $\mathscr{A}_B \otimes_B B'$ is the geometric relative quotient formed from the pulled-back fixed generator and full finite support. Over a Noetherian algebraic or analytic base whose relative quotient is finite over the base, the algebra, centre, and cocenter are locally free near a fibre of the stated kind, with ranks d, N, N, and the centre and cocenter commute with arbitrary base change. Several exceptional image points are treated by their completed pieces.

Proof. The cokernel of the commutator submodule is finite free, so the commutator sequence splits. Intersecting the resulting direct-sum decomposition with $L\mathscr{A}_B$ proves the saturation formula.

By Proposition 4.20, geometric base change reduces to comparing the power-series pullbacks of C_J. Let $W(w)$ be the characteristic polynomial of multiplication by w on the finite free B-module $\mathscr{A}_B$. Cayley–Hamilton makes W annihilate this module. Since w is nilpotent on its closed fibre, W is monic of degree d with reduction w^d. Both $B[w]/(W)$ and $B[[w]]/(W)$ are free of rank d, the latter by Lemma 4.2. Their natural comparison is an isomorphism modulo $\mathfrak{m}$, hence is an isomorphism over B. The same argument after a continuous local map $B \to B'$ gives

$$B'[[w]]/(W') \cong B'[w]/(W') \cong (B[w]/(W)) \otimes_B B'.$$

Tensoring the module killed by W over these finite polynomial quotients therefore identifies its ordinary B-base change with its power-series base change. This retains every point of its finite support.

For the noncomplete local assertion, retain the standing projective, source-flat, target-flat and fixed-generator hypotheses, and write A for the actual relative quotient, assumed finite over the parameter ring $(B, \mathfrak{m}, \mathbb{C})$. First complete at a central target point y, with local ring $(R, \mathfrak{n})$. Proper coherence makes $E = (f_*\mathcal{E}nd(T)^{\mathrm{op}})_y$, its projector ideal, and its quotient finite R-modules. Formal functions is used for this target-point completion: in the algebraic case use [35, Tag 02OC], and in the analytic case use [32, Theorems 19.57 and 19.59]. It identifies $\widehat{E}$ with the endomorphisms of the completed marked bundle. Exact completion on finite target modules then identifies the completed quotient with $\widehat{E}/\widehat{E}e_0\widehat{E}$ [35, Tag 00MA].

Here the analytic completed contraction is a projective scheme over $\widehat{R}$. Choose the given projective embedding and put $D_k = R/\mathfrak{n}^k$. At each level the coherent structure algebra, its defining ideal and the pushed-forward marked bundle are coherent modules on the reduced projective space $\mathbb{P}^r_{\mathbb{C}}$, with finite D_k-actions. Apply projective GAGA on that reduced space: its exact, fully faithful coherent-sheaf equivalence algebraizes both the modules and all their morphisms. Choose a $\mathbb{C}$-basis of D_k; its action is finitely many endomorphisms whose multiplication and unit identities remain valid by faithfulness. The same argument transfers the quotient-algebra action, ideal inclusion, marked inclusions and retractions, and their labelled projectors. Exactness preserves the defining quotient. Compatibility with tensor products and cokernels identifies reduction to D_{k-1} for $k \geq 2$, so the algebraized structure algebras and bundles form one compatible infinitesimal system; no flatness over R is used here. Finally,

algebraic and analytic local rings on $\mathbb{P}^r_{\mathbb{C}}$ have the same completions. This remains true after the finite D_k-extension and the defining quotient, and for the completed bundle modules. Faithfully flat local completion therefore shows that each algebraized marked module is locally free. These are the precise GAGA inputs used below [42, no. 6, Proposition 3, p. 9; no. 9, Proposition 10, p. 17; no. 12, Theorems 1–3, pp. 19–20]. Grothendieck algebraization and existence then give the projective $\widehat{R}$-scheme and marked bundle [35, Tags 089A and 088F]. Their completed local rings and bundle modules along the closed target fibre are those of the original analytic family. Flatness of local maps is preserved by completion [35, Tag 0C4G], so the target and source are flat over $\widehat{B}$: descend flatness from these completed local rings, and use that every point of the projective source specializes to the closed target fibre. The bundle is locally free there by faithfully flat completion, hence everywhere by projectivity and the closed nonfree locus. Formal full faithfulness identifies the central algebraization with the prescribed completed contraction [35, Tag 0A4Z]; all marked summands are retained, so the complete-base theorem applies. Algebraically the corresponding model is the projective base change to $\operatorname{Spec}\widehat{R}$.

The target and parameter completions agree only after taking the finite quotient. When the central support is y, its target maximal ideal acts nilpotently on $A/\mathfrak{m}A$, so for some $c \geq 1$

$$\mathfrak{n}^{ck}A \subseteq \mathfrak{m}^k A \subseteq \mathfrak{n}^k A \qquad (k \geq 1).$$

Thus $\widehat{A}_{\mathfrak{n}} = A \otimes_B \widehat{B}$. For several central support points, let C be the commutative image of the target acting on A. It is finite over B, being a B-submodule of $\operatorname{End}_B(A)$. The finite-algebra completion decomposition [35, Tag 07N9] identifies $A \otimes_B \widehat{B}$ with the product of the quotients completed at all central support points. No splitting or selection of a local branch over the original base is required.

Apply the complete-base theorem to these pieces. The cocenter commutes with scalar extension as the cokernel of the commutator map. Faithfully flat descent of finite projectivity [35, Tag 058S] therefore makes A and its cocenter finite free over B. Lift a central symmetrizing trace through this free cocenter; its pairing is perfect since its determinant is a unit on the closed fibre. The same duality argument gives the centre and its arbitrary scalar extension. These bases and the perfect pairing extend over a neighbourhood of the chosen base point, proving the local assertion. □

4.3.7 The exponents in two examples

For $\Delta = A_3$, with chain $1 - 2 - 3$, take $J = \{1, 3\}$. The restricted roots are $(1, 0), (0, 1), (1, 1)$, and

$$P_{10} = \lambda_1(\lambda_1 + \lambda_2), \qquad P_{01} = \lambda_3(\lambda_2 + \lambda_3), \qquad P_{11} = \lambda_1 + \lambda_2 + \lambda_3.$$

Thus the algebra and cocenter Fitting ideals are respectively

$$(P_{10}P_{01}P_{11}^4), \qquad (P_{10}P_{01}P_{11}).$$

If the three intersection orders are a, b, c, the algebra rank is $a + b + 4c$, its centre and cocenter have rank $a + b + c$, and its Cartan matrix with respect to the two curve idempotents is

$$\begin{pmatrix} a + c & c \\ c & b + c \end{pmatrix}.$$

This example distinguishes the number of Cartan roots above a restricted root from the exponent contributed by its matrix block.

For the coefficient-five vertex of E_8, deleting that vertex leaves $A_3 \sqcup A_4$. Theorem 4.15 gives

$$\dim HH_0(e_p \Pi_0^\rho e_p) = 120 - 6 - 10 = 104.$$

This recovers (3.18) through the full-algebra trace formula of Etingof–Latour–Rains. The explicit rank certificate in Subsection 3.6 gives the direct corner calculation used in the single-curve proof.

The Peirce dimensions and cocenter count computed here are the numerical quantities used in the orbit argument with fixed idempotents in Section 5.

5 Semisimple generic fibres with the curve projectors retained

The universal presentations extend the single-curve construction to every retained subset, with ranks giving the Peirce dimensions and the number of simple factors. For an analytic disk deformation satisfying the hypotheses of Kawamata's Theorem 3.1, the relative contraction algebra is flat and its generic fibre is a product of matrix algebras; the multiplicities of the original summands are recorded by the integers $m_{j,i}$ [23, Theorem 3.1(2)–(3)]. A formal classifying arc keeps the separable algebra of labelled projectors fixed. After a finite extension, the lattice construction produces a polynomially split algebra. The orbit argument identifies the simple matrix factors and the ranks of those projectors in every geometrically semisimple generic fibre.

Throughout this section, $f_0 : X_0 \to Y_0$ is the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold. We use the marked generator, retained set J, restricted roots $\mathcal{B}_J$, and quotient C_J of Subsection 4.3. The original curves are labelled $(C_i)_{i\in J}$. In the positive-dual convention the generator is

$$P_0 = \mathcal{O} \oplus \bigoplus_{i\in J} P_{0,i}, \qquad \deg_{C_j} \det P_{0,i} = \delta_{ij}, \qquad A = \operatorname{End}(P_0)/\operatorname{End}(P_0)e_0 \operatorname{End}(P_0).$$

Dualizing identifies this quotient with the convention $\operatorname{End}(T_0)^{\mathrm{op}}/(e_0)$ used earlier, preserving all projectors. Put

$$E = \bigoplus_{i\in J} \mathbb{C}e_i, \qquad V_\beta = \bigoplus_{i\in J} \mathbb{C}^{\beta_i}, \qquad S_E = \prod_{\beta\in\mathcal{B}_J} \operatorname{End}_{\mathbb{C}}(V_\beta)^{\times n_\beta}. \tag{5.1}$$

On each β-factor, e_i acts as the projection onto $\mathbb{C}^{\beta_i}$. These labelled idempotents are generally noncentral. A marking consists of the fixed embedding $E \hookrightarrow A$, which prescribes the projectors and their left and right actions. Equivalently, the simple module of a β-factor has dimension vector $(\dim_{\mathbb{C}} e_i V_\beta)_{i\in J} = \beta$ relative to the idempotents $(e_i)_{i\in J}$, and this dimension vector occurs n_β times. Write $s_\beta = \sum_i \beta_i$, keeping distinct proportional restricted roots separate.

5.1 Constructing a semisimple generic fibre with fixed projectors

Theorem 5.1 (Semisimple generic fibre with fixed projectors)**.** *There is a finite free unital $\mathbb{C}[[s]]$-algebra $\mathscr{A}$, obtained from the universal marked geometric family, with a marked identification*

$\mathscr{A}/(s) \cong A$. *Its labelled orthogonal idempotents* e_i *sum to its identity. For a finite extension* $K'/\mathbb{C}((s))$, *there is an isomorphism preserving these idempotents*

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong S_E \otimes_{\mathbb{C}} K'.$$

In particular, e_i *has rank* β_i *on each of the* n_β *factors of size* s_β. *After writing* $K' = \mathbb{C}((u))$, $s = u^e$, *there is also a finite free marked* $\mathbb{C}[u]$*-algebra* $\mathscr{P}$ *such that*

$$\mathscr{P}/(u) \cong A, \qquad \mathscr{P}[u^{-1}] \cong S_E \otimes_{\mathbb{C}} \mathbb{C}[u, u^{-1}].$$

Thus the deformation can be chosen polynomially split, with all projector ranks preserved.

Lemma 5.2 (Transverse translation of a formal arc)**.** *Let* $R = \mathbb{C}[[t_1, \dots, t_r]]$, *and let* $f(w) = (f_1(w), \dots, f_r(w))$, *with* $f_\nu(w) \in w\mathbb{C}[[w]]$. *Let* $P_1, \dots, P_q \in R$ *be pairwise nonassociated prime elements such that* $d_b = \operatorname{ord}_w P_b(f(w))$ *is positive and finite. Let* Σ *be a reduced closed formal subscheme of* $\bigcup_b V(P_b)$, *of codimension at least two in* $\operatorname{Spec} R$. *There is a nonempty Zariski open set of vectors* $v \in \mathbb{C}^r$ *for which* $F(w, s) = f(w) + sv$ *has the following properties. Each* $P_b(F)$ *is a unit times a distinguished polynomial in* w *of degree* d_b; *over* $\mathbb{C}((s))$ *these polynomials are separable, pairwise coprime, and have no root mapping into* Σ. *After a finite ramified extension their roots lie in* $u\mathbb{C}[[u]]$.

Proof. Introduce independent parameters $a_1, \dots, a_r$, and put $B_{\mathrm{tr}} = \mathbb{C}[[a_1, \dots, a_r]]$ and $D = B_{\mathrm{tr}}[[w]]$ and $G_b = P_b(f(w) + a)$. Apply Lemma 4.2 to the one-by-one presentation matrix G_b. Its cokernel is free of rank d_b over B_{tr}, with basis $1, w, \dots, w^{d_b-1}$, since these classes are a basis on the closed fibre. Expressing w^{d_b} in this basis gives a monic polynomial $W_b(a, w)$, whose lower coefficients vanish at $a = 0$. It belongs to (G_b). The same finite-slice lemma applied to W_b shows that the surjection $D/(W_b) \to D/(G_b)$ is an isomorphism between free modules with the same displayed basis. Hence $(W_b) = (G_b)$; since D is a domain, G_b is a unit times this monic distinguished polynomial of degree d_b. The coordinate change $t_\nu = a_\nu + f_\nu(w)$ gives

$$D/(G_b) \cong (R/(P_b))[[w]],$$

which is a domain. Its generic fibre over B_{tr} is therefore a finite field extension. In characteristic zero that extension is separable, so the discriminant of W_b is nonzero.

For $b \neq c$, the incidence module $D/(G_b, G_c)$ is finite over B_{tr}. The coordinate change identifies it with $(R/(P_b, P_c))[[w]]$, of dimension at most $r - 1$, whereas $\dim B_{\mathrm{tr}} = r$. It is consequently torsion, so the resultant of W_b, W_c is nonzero. To impose avoidance of Σ, let K_Σ be its radical ideal. Since $\prod_b P_b \in K_\Sigma$, the module

$$D/K_\Sigma(f(w) + a)$$

is finite over B_{tr}. After the coordinate change its dimension is at most $r-1$, whereas $\dim B_{\mathrm{tr}} = r$; it is therefore torsion. Choose a nonzero annihilator, taking 1 when the module is zero. Multiply it by the discriminants and resultants already obtained, and call the resulting nonzero element $b(a)$.

Let b_h be the lowest nonzero homogeneous term of b. For every v with $b_h(v) \neq 0$, $b(sv) = s^h(b_h(v) + O(s)) \neq 0$. This element is invertible over $\mathbb{C}((s))$, proving the three generic assertions. Division by the monic polynomial $\prod_b W_b$ shows that the power-series quotients used to test avoidance of Σ can be computed in $B_{\mathrm{tr}}[w]/(\prod_b W_b)$. Thus ordinary substitution $a = sv$ agrees

with completed substitution and preserves the annihilation relation for $D/K_\Sigma(f(w)+a)$, with the annihilating element $b(a)$ specialized to $b(sv)$. The roots of a distinguished monic polynomial are integral over $\mathbb{C}[[s]]$ and reduce to zero. A finite splitting extension is a complete discretely valued field with residue field $\mathbb{C}$, so its valuation ring is $\mathbb{C}[[u]]$. Absorbing a unit into u gives $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. □

Katz–Morrison's simultaneous partial resolution supplies the geometry at the roots. Nabijou–Wemyss identify its discriminant components with restricted roots and the unique curve class at a general point of each component [29, Theorem 2.6]. Apply this description to the translated slice of Lemma 5.2.

Proposition 5.3 (A formal deformation separating the curve divisors)**.** *Let $f : R_J \to \mathbb{C}[[w]]$ classify f_0. There is a translated arc $F : R_J \to \mathbb{C}[[s,w]]$ reducing to f, such that its geometric generic contraction has exactly n_β disjoint split $(-1,-1)$-curves of each class β. At each such curve the completed target is a split ordinary double point and the source is its chosen small resolution. The generic fibre is formed from the full finite relative family.*

Proof. The marked base R_J is regular, and its curve-divisor equations P_β are distinct primes by Lemma 4.9. Their orders on the central arc are

$$\operatorname{ord}_w P_\beta(f(w)) = n_\beta$$

by Proposition 4.21 and Corollary 4.25. Transversality is imposed on every component of the surface discriminant, including inactive components, because the full threefold slice must be controlled rather than only its active curves.

The central arc is contained in none of these components. Write the central isolated cDV hypersurface as $G(x_1,x_2,x_3,w) = 0$. The relative critical scheme defined by $G, G_{x_1}, G_{x_2}, G_{x_3}$ is finite over $\mathbb{C}[[w]]$, because the central surface of the cDV hypersurface is ADE; this is the topological finiteness argument of Lemma 2.32. If the relative critical scheme had a component dominating the w-axis, the normalization of that component would have a parameter u with $w = u^e$. Differentiating $G(x(u),u^e) = 0$, while using the three vanishing partial derivatives, would give $G_w = 0$ along that curve. This contradicts the isolated singularity of the threefold.

Take Σ to be the reduced image of the intersections of two distinct root hyperplanes on the Cartan cover. The finite quotient preserves the codimension of the Cartan-cover intersection locus, so Σ has codimension at least two; Σ contains the singular and intersection loci of the full discriminant components. Apply Lemma 5.2 to the reduced equations of the full discriminant components. At an active point of the resulting slice, exactly one root α vanishes, with $\pi_I(\alpha) = \beta$. The full Weyl stabilizer is $\{1, s_\alpha\}$: a real vector's stabilizer is generated by reflections in its vanishing roots, as follows by moving it to the closed dominant chamber and expanding $v - wv$ along a reduced reflection word into positive roots with nonnegative coefficients. For a complex vector $a + ib$, choose a generic real c so that $a + cb$ has the same stabilizer and vanishing roots. These are finite linear fixed-subspace conditions, so the assertion also holds over algebraically closed extension fields. Since W_I fixes the retained fundamental weights and $\pi_I(\alpha) \neq 0$, it does not contain s_α. Thus the Cartan-to-marked map is étale here. Completing a finite quotient at its orbit gives invariants under the stabilizer of one lift. A normal Cartan coordinate ζ is negated by s_α, while the tangential coordinates are fixed; hence the unmarked normal parameter is $\tau = \zeta^2$.

The single-root fibre of the actual semiuniversal ADE family has one A_1 singularity. This is the single-coincidence case in the A, D constructions and the single effective-root configuration in the E construction [27, Theorem 1, Sections 4–5 and Table 4]. The same paper's preferred versal equations contain a constant-term base coefficient, so the total unmarked hypersurface is smooth [27, Section 3, Tables 2–3]. We spell out the resulting normal form. Its nondegenerate fibre Hessian first solves for a formal critical section. After translating by that section, eliminate a fibre-homogeneous term r_k of degree $k \geq 3$ by the substitution $\mathbf{x} \mapsto \mathbf{x} - k^{-1}H^{-1}\nabla r_k$, where $\mathbf{x}$ is the column of three fibre variables and H is the quadratic Hessian. Euler's identity removes r_k; the successive substitutions converge. Diagonalizing the quadratic form gives $xy - \xi^2 = g$. The critical-value function g vanishes exactly on the smooth discriminant $\tau = 0$, and total-space smoothness gives $dg \neq 0$. Thus $g = \tau$ times a unit. Absorbing that unit in τ and its square root in ζ gives the actual marked local model

$$xy - \xi^2 = \zeta^2.$$

No Levi root vanishes at this active point, so the full-to-partial resolution contracts nothing near its sole exceptional curve. Formal versality for the resolved A_1 surface identifies its resolved parameter σ, whose affine smoothing parameter is σ^2, with one of $\zeta, -\zeta$ [21, Section 2.3]. These are the two explicit simultaneous A_1 resolutions in [27, Section 4]. The simple root of $P_\beta(F)$ makes ζ a parameter along the slice. After a finite splitting extension the equation is $xy - p_+p_- = 0$, and the chosen source is its small resolution. The charts of Lemma 2.26 give its normal bundle $\mathcal{O}(-1)^{\oplus 2}$. Its retained curve class is β [29, Theorem 2.6 and its proof]. Only finitely many root points and Cartan lifts occur; their residue extensions and the constants needed to split the quadratic forms lie in a single finite extension. The formal coordinate recursions then remain over that field.

At an inactive root, $s_\alpha \in W_I$. Here the marked normal parameter is already τ, so the transverse model $xy - \xi^2 = \tau$ has smooth threefold slice and no contracted curve. Away from the full discriminant the surface family is smooth. Hence the active roots account for every exceptional curve. The distinguished polynomial for $P_\beta(F)$ has degree n_β and simple roots, which gives the stated multiplicity. Forming the finite relative family before inverting s retains every moving root, as in Remark 2.21. □

Proposition 5.4 (The individual projector ranks)**.** *At a split ordinary double point of class β in Proposition 5.3, the completed relative contraction quotient is*

$$\mathrm{End}_{K'}\left(\bigoplus_{i\in J}(K')^{\beta_i}\right),$$

and e_i is the projection onto $(K')^{\beta_i}$. The maximal ideal of the completed target acts trivially on this quotient.

Proof. Fix the actual normalized positive generator and its labelled summands. The comparison in Proposition 4.10 and the completed pullback in Proposition 4.20 preserve those summands separately. All full endomorphisms in this paragraph are formed on the specified completed target model.

Choose a Cartan lift of the moving root after a finite ramified extension. It is a formal arc through the Cartan origin on one root hyperplane H_α, and its generic point lies on no other root hyperplane. Write $\alpha = \sum_k c_k\alpha_k$, so that $\beta_i = c_i$ for $i \in J$. By Lemma 2.8, on

the chosen standard full simultaneous resolution the effective Cartier divisor $\sum_k c_k C_k$ is the scheme-theoretic central fibre of a projective flat family over H_α. Pulling that chosen-root family back to the formal arc gives a proper flat curve family over its complete discrete valuation ring. Its generic fibre is the unique root curve, and its central fundamental cycle is $\sum_k c_k[C_k]$. The full-to-partial morphism contracts the curves indexed by I and is an isomorphism on this generic non-Levi root curve [29, Theorem 2.6 and its proof]. No curve family over the entire quotient divisor is used.

Pull back $\det P_i$ to this proper root-curve family. Its compatible formal restrictions algebraize to a line bundle by projective formal existence [35, Lemma 30.24.3, Tag 0885]; the nonfree locus is empty since its closed image in the local base would otherwise meet the closed fibre. The degree is constant in the flat proper family. More explicitly, the derived pushforwards of this line bundle and of the structure sheaf are perfect and commute with base change [35, Lemmas 36.35.5 and 36.35.10, Tags 0DI4 and 0DJT]; over the local base, alternating sums of the ranks of finite free complexes give their fibrewise Euler characteristics. Their difference is the degree. On the central fundamental cycle it is the sum of the component degrees with multiplicities c_k. The pulled-back line bundle has degree zero on deleted curves and degree δ_{ij} on each retained curve. Consequently, separately for every label,

$$\deg_{C_\beta} \det P_i = \sum_{k \in J} c_k \delta_{ik} = \beta_i.$$

This is the arbitrary-retained-subset form of the chosen-root argument in Lemma 2.10.

The positivity proved on the actual algebraic lift in Proposition 4.10 applies to $P = (T_J^{\mathrm{alg}})^\vee$, with the prescribed labelled summands. Its finite evaluation surjection pulls back to a surjection onto the positive generator here, and its total determinant remains relatively ample under this pullback and the specified full target completion. The structure-sheaf summand is preserved, while $H^1(P^\vee) = 0$ follows from the actual tilting comparison and that summand. Together with $R\pi_*\mathcal{O} = \mathcal{O}$ and the one-dimensional fibre bound, these are the hypotheses of the positive perverse-generator criterion [37, Propositions 3.2.6–3.2.7]. Thus its individual summands are projectives in the fixed positive perverse heart on the completed split ODP resolution.

The split Atiyah calculation of Lemma 2.26 gives exactly two indecomposable positive perverse projectives, $\mathcal{O}$ and $\mathcal{Q}$, with determinant degrees zero and one. Consequently, if $q_i = \operatorname{rank} P_i$, Krull–Schmidt and the displayed degree give, separately for each label,

$$\widehat{P}_i \cong \mathcal{O}^{\oplus(q_i - \beta_i)} \oplus \mathcal{Q}^{\oplus \beta_i}.$$

After pushforward, the generator is a direct sum of free modules and $\bigoplus_i I^{\oplus \beta_i}$, where I is the rank-one Atiyah module. The original e_0 projects onto one free summand. The ideal it generates consists of all maps factoring through finite free modules, so it also kills every additional free summand in this decomposition. Lemma 2.28 identifies $\underline{\mathrm{End}}(I)$ with K', with the target maximal ideal acting by zero. The surviving matrix entries are therefore copies of K', with their usual composition. Grouping them by the original summands gives precisely the asserted projectors and multiplication. □

Lemma 5.5 (Polynomialization with fixed projectors)**.** *Let L be a finite free $\mathbb{C}[[u]]$-algebra with marked special fibre A, and suppose a marked generic isomorphism identifies $L[u^{-1}]$ with $S_E \otimes \mathbb{C}((u))$. There is a finite free marked $\mathbb{C}[u]$-algebra $\mathscr{P}$ whose completion is L and whose localization at u is $S_E \otimes \mathbb{C}[u, u^{-1}]$.*

Proof. The generic marked isomorphism lets us regard L as a lattice in $S_E((u))$. For some $m \geq 0$,

$$u^m S_E[[u]] \subseteq L \subseteq u^{-m} S_E[[u]].$$

Define $\mathscr{P} = L \cap S_E[u, u^{-1}]$. It is a unital subalgebra containing the constant labelled projectors, and

$$u^m S_E[u] \subseteq \mathscr{P} \subseteq u^{-m} S_E[u].$$

The sandwich makes $\mathscr{P}$ finite and torsion-free over the principal ideal domain $\mathbb{C}[u]$, hence free, and its localization is the stated split algebra. To compare completions, take an element of L and truncate its Laurent expansion far enough. The truncation lies in $\mathscr{P}$, because the discarded Laurent tail lies in $u^m S_E[[u]] \subseteq L$, and the truncation approximates the original element to arbitrarily high order. The same sandwich shows that the ambient u-adic topology agrees with the lattice topology. Completion therefore identifies $\mathscr{P} \otimes_{\mathbb{C}[u]} \mathbb{C}[[u]]$ with L, preserving products and projectors. Reduction modulo u gives the prescribed marked algebra A. □

For finite-dimensional algebras without distinguished idempotents, Jelisiejew–Smoktunowicz relate formal, polynomially split, and orbit-closure formulations of deformation [20, Theorems 2.19–2.20]. The lattice argument retains the chosen embedding of E and the constant projectors while passing from the formal family to the polynomial one.

Proof of Theorem 5.1. Let $f : R_J \to \mathbb{C}[[w]]$ classify the given marked contraction. Choose the translation $F : R_J \to \mathbb{C}[[s, w]]$ of Lemma 5.2, applied to the active prime equations P_β. For its excluded locus use the intersection of the active divisors with the reduced image of all intersections of two distinct Cartan root hyperplanes. Each intersection has Cartan codimension two, and the finite Cartan quotient preserves its dimension, so the excluded locus has codimension at least two. The orders of the central root functions are n_β, by Corollary 4.25.

Pull back the finite-type standard simultaneous partial resolution and then complete at the full central target point and its inverse image. Parameter flatness is preserved by this base change and by the flat Noetherian target completion. The prescribed central contraction and all its labelled generator summands are retained. Proposition 4.20 identifies its actual relative quotient with the ordinary tensor

$$\mathscr{A} = C_J \otimes_{R_J, F} \mathbb{C}[[s, w]].$$

The weighted square presentation in Theorem 4.13 and Lemma 4.2 make it finite free over $\mathbb{C}[[s]]$, of rank $\sum_\beta n_\beta s_\beta^2$, with marked special fibre A. The tensor statement here concerns the parameter-finite contraction quotient; full endomorphism algebras are compared on the specified completed geometric model.

By Proposition 4.14, $\delta_J = \prod_\beta P_\beta$ annihilates C_J. Its pullback is a unit times a product $W = \prod_\beta W_\beta$ of distinguished polynomials, where $\deg_w W_\beta = n_\beta$. The finite algebra

$$\mathbb{C}[[s, w]]/(W) = \mathbb{C}[[s]][w]/(W)$$

is formed before inverting s. After a finite splitting extension $K_1/\mathbb{C}((s))$, its distinct roots $w_{\beta,a}$ therefore give, by the Chinese remainder theorem, a product decomposition

$$\mathscr{A} \otimes K_1 = \prod_{\beta,a} C_J \otimes_{R_J,\, F(-, w_{\beta,a})} K_1, \qquad 1 \leq a \leq n_\beta.$$

Each root has positive valuation, so evaluation of the formal classifying map is defined in the valuation ring before localization. This construction retains the whole finite moving support.

Pass temporarily to an algebraic closure of K_1 and choose a Cartan lift of one factor. Avoidance of the excluded locus means that exactly one positive root α vanishes there. Its retained vector is β, because the Cartan pullback of P_β is the product of exactly those root forms. Lemma 4.12, together with the actual labelled comparison in Proposition 4.10, identifies this factor with

$$\operatorname{End}\left(\bigoplus_{i\in J}\overline{K_1}^{\beta_i}\right),$$

with each e_i its indicated projector. Thus the matrix multiplication and every individual label are identified. A matrix for each marked isomorphism and its inverse has finitely many coefficients algebraic over K_1. A single finite extension K'/K_1 contains all these coefficients and splits every factor, giving the stated marked product $S_E \otimes K'$.

The finite-extension argument in Lemma 5.2 writes its valuation ring as $\mathbb{C}[[u]]$, with $s = u^e$. Apply Lemma 5.5 to $L = \mathscr{A} \otimes_{\mathbb{C}[[s]]} \mathbb{C}[[u]]$. It gives the polynomial algebra with its prescribed special fibre, all constant projectors, and completion equal to this entire marked formal family. □

Remark 5.6 (Relation to Kawamata's existence result). Kawamata begins with an analytic disk deformation and proves flatness of the relative contraction algebra; when the noncentral fibres are disjoint $(-1,-1)$-curves, he obtains the generic matrix product [23, Theorem 3.1(2)–(3)]. His proof also decomposes the restriction of each original summand with multiplicities $m_{j,i}$, the geometric source of the projector-rank data. Kawamata's theorem has analytic disk scope. The formal classifying-arc statement needed here is established through the universal quotient with the embedding of E fixed, and is then polynomialized. The comparison after forgetting the idempotents with other deformation formulations is supplied by [20, Theorems 2.19–2.20]; the marked lattice step is written out here.

5.2 Fixed-idempotent laws and the orbit criterion

The existence theorem constructs a product of matrix algebras with fixed idempotents. The unmarked structure-constant and orbit method used here is reviewed by Jelisiejew–Smoktunowicz: semisimple orbits are open, and the formal, polynomially split, and orbit-closure deformation notions agree after the appropriate finite extensions [20, Section 2.1, Theorems 2.14 and 2.20]. With a fixed separable algebra E, the method shows that finite free associative formal deformations of A with geometrically semisimple generic fibre have the same simple matrix factors and the same dimension vectors relative to the lifted idempotents. Its relative tangent formula uses the centralizer of E, generally larger than the centre. Peirce dimensions and the number of simple factors are necessary numerical inputs, but they do not determine the marked factorization by themselves.

For a semisimple algebra whose simple modules have dimension vectors v_a relative to $(e_i)_{i\in J}$, the equalities $C = (c_{ij}) = \sum_a v_a v_a^{\mathsf{T}}$ and $N = \#\{a\}$ do not, by themselves, determine the vectors. For example, the two multisets

$$\{(2,1),(2,1),(2,1),(0,3),(1,0)\}, \qquad \{(3,0),(1,2),(1,2),(1,2),(1,0)\}$$

both have five vectors, Gram matrix $\left(\begin{smallmatrix}13 & 6\\ 6 & 12\end{smallmatrix}\right)$, and unmarked algebra $M_3(\mathbb{C})^4 \times \mathbb{C}$. They are distinct even after exchanging the two labels. Thus the Gram matrix and the number of simple

modules admit several multisets of dimension vectors. The orbit argument in Theorem 5.12 starts from the product constructed geometrically in Theorem 5.1 and proves that every geometrically semisimple generic fibre, together with the lifted idempotents, is isomorphic to it.

Let $V = \bigoplus_{i,j\in J} V_{ij}$ be a fixed E-bimodule, with $c_{ij} = \dim_{\mathbb{C}} V_{ij}$ and distinguished nonzero vectors $e_i \in V_{ii}$. The fixed-idempotent law scheme $X_E(V)$ parametrizes associative E-balanced products $V \otimes_E V \to V$ with these prescribed left and right E-actions. Thus $1 = \sum_i e_i$ is fixed before the multiplication varies, and the associativity equations define the scheme structure. Its change-of-basis group is

$$G_E = \{g \in \mathrm{Aut}_{E\text{-}E}(V) : g(e_i) = e_i \text{ for all } i\}.$$

This group is smooth and irreducible: its factors are $\mathrm{GL}(V_{ij})$ for $i \neq j$ and the stabilizers of the nonzero vectors e_i in $\mathrm{GL}(V_{ii})$. Set

$$b = \sum_i c_{ii}, \qquad g = \dim G_E = \sum_{i,j} c_{ij}^2 - b. \tag{5.2}$$

For a law $B \in X_E(V)$, its centralizer $B^E = \{x : xe = ex \text{ for all } e \in E\}$ is $\bigoplus_i e_i B e_i$, of dimension b.

Relative Hochschild cochains impose the fixed separable algebra data. They give the E-relative tangent dimension needed below, rather than merely the ordinary fixed-unit tangent space.

Lemma 5.7 (Relative cochains and marked tangent dimensions)**.** *For $B \in X_E(V)$, normalized relative Hochschild cochains compute ordinary Hochschild cohomology. If $Z^2_E(B)$ and $B^2_E(B)$ are their degree-two cocycles and coboundaries, then*

$$T_B X_E(V) = Z^2_E(B), \qquad T_B(G_E \cdot B) = B^2_E(B).$$

In particular,

$$\dim T_B X_E(V) = g - b + \dim Z(B) - \dim HH^1(B) + \dim HH^2(B). \tag{5.3}$$

Proof. The normalized relative complex has

$$C^0_E(B) = B^E, \qquad C^n_E(B) = \mathrm{Hom}_{E^e}((B/E)^{\otimes_E n}, B) \quad (n \geq 1).$$

Insertion of the identity gives a contracting homotopy for the augmentation of the relative bar complex, so that complex is exact. Normalization removes its split degenerate subcomplex and leaves the terms $B \otimes_E (B/E)^{\otimes_E n} \otimes_E B$. Each is induced from an E^e-module. Since E^e is a product of copies of $\mathbb{C}$, every such module is projective, and induction makes each term B^e-projective. This is a projective resolution of B, so its cohomology is ordinary $HH^*(B)$.

Linearizing associativity gives the relative two-cocycle equation: fixed E-actions require the cochain to vanish when an input belongs to E. Linearizing a change of E-bimodule basis gives the coboundaries. The group scheme of algebra automorphisms fixing E is a closed subgroup scheme of G_E, hence is smooth in characteristic zero [35, Lemma 39.8.2, Tag 047N]. Its orbit map is smooth [4, Proposition 1.11 and Theorem 1.16], so its orbit has the asserted tangent space and stabilizer Lie algebra $\mathrm{Der}_E(B)$. The degree-zero part of the same complex gives the exact sequence

$$0 \longrightarrow Z(B) \longrightarrow B^E \longrightarrow \mathrm{Der}_E(B) \longrightarrow HH^1(B) \longrightarrow 0.$$

Moreover, $\dim C^1_E(B) = g$, because these cochains are precisely the E-bimodule endomorphisms vanishing on the distinguished vectors. It follows that $\dim B^2_E(B) = g - b + \dim Z(B) - \dim HH^1(B)$. Adding $\dim HH^2(B)$ proves the formula. □

Lemma 5.8 (Semisimple laws with fixed idempotents)**.** *A finite-dimensional semisimple complex algebra with a fixed complete labelled system of orthogonal idempotents is classified, up to marked isomorphism, by a multiset of nonzero vectors $v_a = (v_{a,i})_{i\in J}$ of nonnegative integers. Its algebra and Peirce dimensions are*

$$S(\{v_a\}) = \prod_a \operatorname{End}_{\mathbb{C}}\left(\bigoplus_i \mathbb{C}^{v_{a,i}}\right), \qquad c_{ij} = \sum_a v_{a,i}v_{a,j}.$$

Every such orbit in $X_E(V)$ is open, has dimension $g - b + \dim Z(S)$, and is dense in an irreducible component.

Proof. On a simple matrix factor, simultaneous orthogonal projections are classified by the dimensions of their images. Inner automorphisms change bases in those images, and an algebra automorphism may permute factors carrying the same vector. This proves the classification and the Peirce formula; in particular $\dim Z(S)$ is the number of vectors. A marked isomorphism between laws on V is exactly an element of G_E.

Semisimplicity gives $HH^n(S) = 0$ for $n > 0$, so Lemma 5.7 makes the tangent and orbit dimensions equal to $g - b + \dim Z(S)$. The stabilizer is a product of groups $(\prod_i \mathrm{GL}_{v_{a,i}})/\mathbb{G}_m$, followed by finite permutations of identical factors, and is smooth. Apply Lemma 3.4(i) to the smooth irreducible group G_E. The orbit is open, so its closure is an irreducible component. □

Theorem 5.9 (The fixed-idempotent orbit criterion)**.** *Let $B \in X_E(V)$, and suppose a marked semisimple algebra $S \in X_E(V)$ satisfies*

$$B \in \overline{G_E \cdot S}, \qquad \dim Z(B) = \dim Z(S), \qquad \dim HH^1(B) = \dim HH^2(B).$$

Then B is a smooth point of $X_E(V)$, and $\overline{G_E \cdot S}$ is its unique irreducible component. If another marked semisimple law S' has $B \in \overline{G_E \cdot S'}$, then $S' \cong S$ as marked algebras.

Proof. The hypotheses and the preceding lemmas give

$$\dim T_B X_E(V) = g - b + \dim Z(S) = \dim(G_E \cdot S).$$

The semisimple orbits are open by Lemma 5.8. Applying Lemma 3.4(ii) proves smoothness, the unique-component assertion, and equality of the two marked orbits. □

5.3 Periodicity and general uniqueness

Return to the multi-curve contraction algebra A, and write

$$c_{ij} = \dim e_i A e_j = \sum_\beta n_\beta \beta_i \beta_j, \qquad N = \sum_\beta n_\beta = \dim Z(A) = \dim HH_0(A). \tag{5.4}$$

Corollary 4.25 and Theorem 4.26, restricted to the central fibre, place A and the geometrically constructed product on the same E-bimodule. The Peirce ranks specify the ambient fixed-idempotent law scheme, the cocenter count enters the tangent-dimension comparison, and the formal family supplies a semisimple orbit whose closure contains A.

Proposition 5.10 (The multi-curve Hochschild equality)**.** *The algebra A is symmetric and satisfies*

$$\Omega^4_{A^e}(A) \simeq A \quad \textit{in the stable bimodule category,} \qquad DHH^1(A) \cong HH^2(A).$$

Proof. The target is an isolated cDV hypersurface with a smooth crepant resolution, so August's symmetry theorem applies to the stable endomorphism algebra of the entire nonfree maximal Cohen–Macaulay generator [2, Theorem 2.6]. The four-term projective bimodule extension in [19, Setting 3.2.1 and Theorem 3.2.5] also applies to this generator and has the diagonal bimodule at both ends. Thus the untwisted four-periodicity used in Proposition 3.9 applies here. Lemma 3.11 then gives the asserted duality in positive Hochschild degrees. Each semisimple direct factor has zero positive Hochschild cohomology. □

Lemma 5.11 (Lifting and comparing the fixed idempotents)**.** *Let $(B, \mathfrak{m}, \mathbb{C})$ be complete Noetherian local, and let $\mathscr{D}$ be a finite free B-algebra with identified special fibre A. The system (e_i) lifts to orthogonal idempotents $(\widetilde{e}_i)$ summing to one. Any two such lifts are simultaneously conjugate by a unit congruent to one modulo $\mathfrak{m}\mathscr{D}$. Each $\widetilde{e}_i\mathscr{D}\widetilde{e}_j$ is free of rank c_{ij}, and an adapted trivialization defines a morphism $\operatorname{Spec} B \to X_E(V)$ whose closed point is A.*

Proof. Across a square-zero ideal, if $x^2 - x = q$ lies in that ideal, then $x + (1 - 2x)q$ is idempotent. Lift the first projector and work successively in its complementary corner to preserve orthogonality; take the last projector to be the remaining complement. Applying this construction to successive powers of $\mathfrak{m}$ and using completeness gives the system. For two lifts (p_i) and (q_i), the element

$$h = \sum_i q_i p_i \equiv 1 \pmod{\mathfrak{m}\mathscr{D}}$$

is invertible by completeness and satisfies $hp_i = q_i h$. This proves simultaneous conjugacy.

The maps $x \mapsto p_i x p_j$ are mutually orthogonal B-linear projectors. Their images are direct summands of the free module $\mathscr{D}$, hence free over the local ring, with ranks determined by the special fibre. Choose bases lifting fixed bases of $e_i A e_j$; in each diagonal block include p_i, which is unimodular because its reduction is nonzero. These choices give the claimed E-bimodule trivialization and classifying morphism. □

Theorem 5.12 (Geometric semisimple fibres with and without fixed idempotents)**.** *Let $\mathscr{D}$ be any finite free unital $\mathbb{C}[[t]]$-algebra with special fibre identified with A, and suppose its geometric generic fibre is semisimple. For any lifts of the labelled original idempotents, that fibre is marked-isomorphic to $S_E \otimes_{\mathbb{C}} \overline{\mathbb{C}((t))}$. Equivalently, the multiset of dimension vectors of its simple modules, relative to the lifted idempotents, is exactly*

$$\{\, \beta \text{ repeated } n_\beta \text{ times} : \beta \in \mathcal{B}_J \,\}. \tag{5.5}$$

Thus there is exactly one isomorphism class of semisimple geometric generic fibre together with the lifted idempotents. After forgetting the idempotents, this fibre is obtained by scalar extension from the complex semisimple algebra

$$\prod_{m\geq 1} M_m(\mathbb{C})^{\times N_m}, \qquad N_m = \sum_{s_\beta = m} n_\beta.$$

Moreover A is smooth in $X_E(V)$, and the unique irreducible component through A is $\overline{G_E \cdot S_E}$.

Proof. The Peirce formula in (5.4) places A and S_E on the same E-bimodule V. Theorem 5.1 places A in $\overline{G_E \cdot S_E}$: choose bases in every Peirce summand of the free family, including each projector, and use the generic marked isomorphism. Every equation of the orbit closure vanishes at the generic point after a finite field extension, hence on the whole parameter DVR, and therefore at A. Equation (5.4) gives $\dim Z(A) = N = \dim Z(S_E)$, while Proposition 5.10 gives the Hochschild equality. Theorem 5.9 now proves smoothness and the unique-component assertion.

Let $\mathscr{D}$ be an arbitrary family as in the statement. Lift the projectors and choose the adapted frame of Lemma 5.11. The geometric generic fibre of $\mathscr{D}$ is represented by a constant semisimple complex algebra S' with lifted idempotents, classified by the multiset of dimension vectors of its simple S'-modules. The finitely many coefficients of a marked isomorphism lie in a finite extension of $\mathbb{C}((t))$. The same orbit-closure argument gives $A \in \overline{G_E \cdot S'}$, so the orbit criterion forces $S' \cong S_E$. This proves (5.5). Changing the lifted idempotents conjugates them simultaneously, so the answer does not depend on their choice. Forgetting the projectors combines precisely the factors with the same total size s_β, giving the stated unmarked algebra. □

Corollary 5.13 (Smooth points of $X_E(V)$ and semisimple fibres)**.** *Put $d = \dim A$, and let b, g be as in (5.2). Then*

$$\widehat{\mathcal{O}}_{X_E(V),A} \cong \mathbb{C}[[x_1, \ldots, x_{g-b+N}]],$$
$$\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[y_1, \ldots, y_{d(d-1)-d+N}]].$$

The unique component of X_d through A is the closure of the unmarked orbit of S_E. In every finite free associative deformation of A over a complete Noetherian local complex base $(B, \mathfrak{m}, \mathbb{C})$, every semisimple geometric fibre has the multiset of dimension vectors of simple modules given in (5.5), with respect to any lifts of the original projectors.

Proof. Theorem 5.12 gives smoothness at A in $X_E(V)$. For ordinary smoothness, forget the fixed idempotents in the existence theorem and apply Theorem 3.6, using Proposition 5.10 and $\dim Z(A) = N$. The dimension formulas then follow from the corresponding orbit dimensions

and the power-series description of complete regular local $\mathbb{C}$-algebras with residue field $\mathbb{C}$ [35, Lemma 10.160.10(2), Tag 0C0S]. This description uses the given scalar embedding and any lifts of a cotangent-space basis.

For an arbitrary complete local associative family, choose a smooth irreducible open neighbourhood U_E of A avoiding all other components. Its distinguished semisimple orbit is dense. Every semisimple complex orbit meeting U_E is open, hence meets that dense orbit and equals it.

This conclusion persists over an algebraically closed extension $\Omega/\mathbb{C}$. In fixed dimension d, there are only finitely many multisets of nonzero integral vectors v_a with $\sum_a(\sum_i v_{a,i})^2 = d$; the Peirce equations select a subset. Each gives a constant complex semisimple marked law. By Lemma 5.8, every such law has an open orbit. All these orbits except the distinguished one are disjoint from U_E, and their open subschemes remain disjoint after base change. Every semisimple Ω-law is marked-isomorphic to one of these constant laws after extension: its simple modules give exactly the corresponding integral vectors. The orbit map is smooth and surjective, and its nonempty fibres over the algebraically closed field Ω have Ω-points. Thus those base-changed orbits exhaust the semisimple geometric laws, and only the distinguished orbit meets $(U_E)_\Omega$.

By Lemma 5.11, the given family defines $\operatorname{Spec} B \to X_E(V)$. This map factors through U_E, since an open subset of a local spectrum containing its closed point is the whole spectrum. Every semisimple geometric fibre therefore has the asserted dimension-vector multiset, for any choice of the lifts. □

Geometry constructs the matrix factors and the ranks of the original idempotents on their simple modules; orbit theory makes the resulting multiset of dimension vectors unique. The smooth open neighbourhood of A in X_d is used in Section 6 for arbitrary finite free associative families. It is separate from the geometric relative quotients of Section 4.

Remark 5.14 (What the geometric generic fibre determines). Theorem 5.12 determines the multiset of dimension vectors of the simple modules after a splitting extension. Over $\mathbb{C}((t))$, Galois action may permute simple factors with the same full dimension vector, since every e_i is defined over the ground field. Different integral families can therefore have isomorphic geometric generic fibres. For the basic algebra A, complete primitive idempotent systems are conjugate after a permutation of their labels. Hence the multiset of dimension vectors is intrinsic to A up to such a permutation. The underlying semisimple algebra retains only the aggregated numbers N_m. In particular, keeping the idempotents retains more information than the matrix sizes after they are forgotten.

5.4 Projector ranks in examples

Let A be the normalized basic contraction algebra of a completed smooth-threefold flop whose surface section is the A_3, $J = \{1, 3\}$ configuration in Subsection 4.3.7. Let a, b, c be the finite positive orders of P_{10}, P_{01}, P_{11} on its central marked classifying arc. For any complete local finite free associative family with identified special fibre A and any complete labelled orthogonal lifts of e_1, e_3, Corollary 5.13 identifies every semisimple geometric fibre with the scalar extension, to its geometric fibre field, of the complex marked algebra

$$\mathbb{C}^a \times \mathbb{C}^b \times M_2(\mathbb{C})^c, \qquad e_1 = (1^a, 0^b, E_{11}^c), \quad e_3 = (0^a, 1^b, E_{22}^c).$$

The three vectors $(1,0),(0,1),(1,1)$ recover both the matrix sizes and the Cartan matrix computed from the same universal presentations.

Consider the truncated two-cycle algebra with basis e_1, e_2, a, b, ab, ba, in which every path of length three is zero. It has

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = 4, \quad g = 6, \quad \dim HH^0 = 3, \quad \dim HH^1 = \dim HH^2 = 1.$$

The relative cochain spaces in degrees zero through three have dimensions $4, 6, 12, 24$, and their differentials have ranks $1, 4, 7$. Thus the tangent dimension in $X_E(V)$ is 5, equal to the orbit dimension for the algebra with its fixed idempotents $\mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, with

$$e_1 = (1, 0, E_{11}), \qquad e_2 = (0, 1, E_{22}).$$

The accompanying exact rational calculation records these ranks and the two Gram decompositions. The example shows that the relative tangent formula agrees with the orbit dimension for the dimension vectors supplied by the construction.

6 Consequences for centres and their deformations

Section 5 proves that the contraction algebra A defines a smooth point of X_d. On a smooth open neighbourhood of that point, distinguish two questions. For an already given finite free associative family, how do its centre and cocenter vary? For a prescribed commutative deformation of the centre, does an associative deformation of the whole algebra realize it? The first question has a uniform positive answer near a contraction algebra. The first-order form of the second is governed by the centre tangent map, and the type D family below supplies a direction outside its image.

Conjecture 6.1 (Jelisiejew–Smoktunowicz [20, Conjecture 1.4])**.** *Let A be a contraction algebra and $d = \dim A$.*

(1) *The algebra A defines a smooth point of the moduli space of d-dimensional unital associative algebras; in other words, A is unobstructed.*

(2) *The dimension of the centre is locally constant near this point, and the map sending an algebra to its centre, regarded as a commutative algebra, is smooth.*[1]

For the completed smooth-threefold contractions considered here, the smoothness already proved establishes Conjecture 6.1 (1). The bundle theorem gives local constancy and arbitrary base change for centres and cocenters in already given associative families. The centre tangent map gives the first-order lifting criterion for prescribed commutative centre deformations. Corollary 6.18 gives counterexamples $D_{n,2}$, $n \geq 2$, to the smoothness assertion in Conjecture 6.1(2). Write $c = N = \sum_\beta n_\beta$ when the common rank is used as the dimension of a commutative algebra law. In this section, a marking of a centre deformation means an identification of its special fibre with $Z(A)$; it is distinct from the labelled projectors used in Section 5.

[1] In personal correspondence with the author, Joachim Jelisiejew confirmed that Conjecture 1.4(2) is formulated for the map to commutative algebra structures on the centre (October 2026).

Recall the fixed-unit law scheme X_d and its change-of-basis group G from Subsection 3.2. The points of X_d are unital associative products on a fixed d-dimensional vector space. Write $\mathcal{V} = V \otimes \mathcal{O}_{X_d}$ for the tautological algebra and $\mathcal{V}_U$ for its restriction to an open subscheme U.

The structure-constant space and its Hochschild tangent description come from associative deformation theory [13]; the fixed-unit form used here is reviewed in [20, Section 2.1 and Proposition 2.8]. The preceding description of semisimple fibres gives the local bundle statement, from which the centre morphism and the map χ_B are constructed.

6.1 Unobstructedness

The smooth local lifting argument for Conjecture 6.1 is phrased in the fixed-unit deformation framework of [20].

Corollary 6.2 (Unobstructedness)**.** *Let A be the basic contraction algebra of a completed projective flopping contraction of a smooth complex threefold. Put $d = \dim A$ and $c = \sum_\beta n_\beta$; for one curve this is $c = \sum_j n_j$. Then*

$$\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+c}]]. \tag{6.1}$$

Consequently A is unobstructed: if $R' \to R$ is a surjection of local Artinian $\mathbb{C}$-algebras with residue field $\mathbb{C}$, every finite free unital R-algebra $\mathscr{B}$ with $\mathscr{B} \otimes_R \mathbb{C} \cong A$ lifts to a finite free unital R'-algebra $\mathscr{B}'$ with $\mathscr{B}' \otimes_{R'} R \cong \mathscr{B}$. In particular, Conjecture 6.1(1) holds for these algebras.

Proof. The completed local-ring description follows from Corollary 5.13; in the one-curve case it already follows from Theorem 3.1.

For the lifting statement, interpret the multiplication on $\mathscr{B}$ as a point of X_d specializing to the smooth point A. Choose an R-basis of $\mathscr{B}$ whose first vector is 1 and which lifts the fixed basis of A; the unit is unimodular because it reduces to the unit of A. The structure constants define a homomorphism $\mathcal{O}(X_d) \to R$. It sends the maximal ideal of the point A into the nilpotent maximal ideal of R, and hence factors through $\widehat{\mathcal{O}}_{X_d,A}$. Lift the images of the variables t_i arbitrarily to R'. The resulting homomorphism $\widehat{\mathcal{O}}_{X_d,A} \to R'$ gives structure constants over R' satisfying the associativity and unit equations, hence an algebra $\mathscr{B}'$ reducing to $\mathscr{B}$. □

Remark 6.3. Smoothness of X_d at A gives the lifting property even when $HH^3(A)$ is nonzero [20, Section 1]. The number $d(d-1) - d + c$ is the dimension of the framed space of algebra laws with fixed identity. The deformation functor of A up to isomorphism has tangent space $HH^2(A)$, of dimension $\dim HH^1(A)$ by Lemma 3.11; the second assertion of Corollary 6.2 says that this functor is unobstructed.

6.2 Centres and cocenters in associative families

Section 4 proves freeness for relative contraction quotients arising from geometric families. The result below concerns an arbitrary finite free associative family with special fibre A. The smooth open neighbourhood of A in X_d transfers rank information from the geometric construction to these families. Hua–Toda's generic matrix decomposition and flatness question concern geometric families [17, Theorem 4.2 and Conjecture 4.3]; the smooth neighbourhood in X_d extends the centre and cocenter conclusions to associative families.

Theorem 6.4 (Centre and cocenter near A)**.** *Let A, d, and c be as in Corollary 6.2. There is a smooth irreducible open neighbourhood U of A in X_d with the following properties.*

(i) *The centre $\mathsf{Z}_U = Z(\mathcal{V}_U)$ and the cocenter $\mathsf{M}_U = HH_0(\mathcal{V}_U/\mathcal{O}_U)$ are locally free of rank c, and their formation commutes with arbitrary base change: for every morphism $g\colon U' \to U$, the natural maps $g^*\mathsf{Z}_U \to Z(g^*\mathcal{V}_U)$ and $g^*\mathsf{M}_U \to HH_0(g^*\mathcal{V}_U/\mathcal{O}_{U'})$ are isomorphisms.*

(ii) *Let τ_0 be a symmetrizing trace of A. After shrinking U, there is an $\mathcal{O}_U$-linear map $\tau\colon \mathcal{V}_U \to \mathcal{O}_U$ that vanishes on commutators, restricts to τ_0 at A, and for which $(a,b) \mapsto \tau(ab)$ is a perfect pairing. Every fibre of $\mathcal{V}_U$ is then a symmetric algebra, and*

$$\mathsf{Z}_U \xrightarrow{\sim} \mathsf{M}_U^{\vee}, \qquad z \longmapsto (\bar{a} \mapsto \tau(za)).$$

In particular, the dimension of the centre is constant on U, as the first half of Conjecture 6.1(2) predicts.

Proof. By Corollary 5.13, A is smooth in X_d, and the unique component of X_d through A is the closure of the unmarked orbit of S_E. Choose a smooth irreducible open neighbourhood U meeting no other component. Its semisimple locus is dense, and each semisimple geometric fibre has c simple factors.

For the centre, consider the commutator operator

$$\kappa\colon \mathcal{V}_U \longrightarrow \mathcal{E}nd_{\mathcal{O}_U}(\mathcal{V}_U), \qquad a \longmapsto (b \mapsto ab - ba),$$

a map of free modules of ranks d and d^2 with kernel Z_U. At a point B, the rank of κ is $d - \dim Z(B)$. This rank is $d-c$ both on the semisimple locus, because S_E has c simple factors, and at A, by (5.4). All minors of size $d-c+1$ consequently vanish on the dense semisimple locus and hence on the reduced scheme U. Some minor of size $d-c$ is nonzero at A; shrink U so that it is a unit. A matrix with an invertible $r \times r$ minor and vanishing $(r+1)$-minors is equivalent to $\left(\begin{smallmatrix} 1_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Thus $\ker\kappa$ is a direct summand of rank c, and $\ker(g^*\kappa) = g^*\ker\kappa$ for every g. Since the centre of $g^*\mathcal{V}_U$ is $\ker(g^*\kappa)$, this proves the assertion for Z_U.

For the cocenter, use the parallel cokernel calculation with

$$b\colon \mathcal{V}_U \otimes_{\mathcal{O}_U} \mathcal{V}_U \longrightarrow \mathcal{V}_U, \qquad a \otimes a' \longmapsto aa' - a'a,$$

whose cokernel is M_U. The rank of b at B is $d - \dim HH_0(B)$, which equals $d-c$ on the semisimple locus and, by (5.4), at A. Thus M_U is locally free of rank c. Formation of a cokernel commutes with base change by right exactness.

For (ii), choose an extension of the functional τ_0 to a local section of $\mathsf{M}_U^{\vee}$, and compose it with $\mathcal{V}_U \to \mathsf{M}_U$. At the local ring of A, Lemma 4.4 makes the pairing defined by the extended trace perfect. Shrink U so that the determinant remains invertible. The same trace formula gives $\mathsf{Z}_U \cong \mathsf{M}_U^{\vee}$ throughout U and after base change. Every fibre is therefore symmetric. □

Corollary 6.5 (Invariants of arbitrary local associative families)**.** *Let $(B, \mathfrak{m}, \mathbb{C})$ be a Noetherian local complex algebra, and let $\mathscr{D}$ be a finite free unital B-algebra with special fibre A. Then $Z(\mathscr{D})$ and $HH_0(\mathscr{D}/B)$ are free of rank c, commute with every commutative scalar extension $B \to B'$, and every symmetrizing trace of A lifts to a perfect trace on $\mathscr{D}$. The assertion applies to every local associative family specified here, including all complete local bases in the preceding sections.*

Proof. Choose a basis beginning with the unit and lifting a fixed basis of A. The structure constants of $\mathscr{D}$ in this basis define $\operatorname{Spec} B \to X_d$. This map factors through U, because an open subset of a local spectrum containing its closed point is the whole spectrum. Pull back Theorem 6.4; for a prescribed closed trace, use Lemma 4.4. □

In particular, these conclusions apply when R is a local Artinian $\mathbb{C}$-algebra with residue field $\mathbb{C}$ and $\mathscr{B}$ is a finite free unital R-algebra with $\mathscr{B} \otimes_R \mathbb{C} \cong A$, including every local Artinian scalar extension $R \to R''$. The base-change isomorphism $Z(\mathscr{B}) \otimes_R \mathbb{C} \cong Z(A)$ shows that every central element of A lifts to a central element of $\mathscr{B}$.

Lifting central elements inside a fixed associative family is weaker than lifting an arbitrarily prescribed multiplication on the centre. The latter problem begins with the centre morphism below. The bundle theorem also determines the difference of the first two positive Hochschild dimensions throughout the smooth neighbourhood.

Corollary 6.6. *For every $B \in U(\mathbb{C})$, $\dim HH^1(B) = \dim HH^2(B)$.*

Proof. The scheme U is smooth and irreducible of dimension $d(d-1) - d + c$, so $\dim T_B X_d = d(d-1)-d+c$. As in the proof of Corollary 3.7, $\dim T_B X_d = d(d-1) - \dim \operatorname{Der}(B) + \dim HH^2(B)$ and $\dim \operatorname{Der}(B) = d - \dim Z(B) + \dim HH^1(B)$. Since $\dim Z(B) = c$ by Theorem 6.4, the claim follows. □

Remark 6.7 (Relation to geometric flatness). The universal presentations in Theorem 4.26 give cocenter freeness on geometric slices and the numerical input for marked uniqueness. The resulting orbit geometry supplies the smooth neighbourhood used here. The bundle theorem on this neighbourhood extends the centre and cocenter properties to arbitrary associative families.

6.3 The centre morphism

The centre bundle has constant rank but no preferred basis. A unit-preserving frame turns its multiplication into a point of a fixed scheme of commutative algebra laws. The fixed-unit scheme X_d used above is the structure-constant scheme of [20, Section 2.1 and Proposition 2.8]; see also [13, 12]. The commutative analogue of X_d is as follows.

For $c \geq 1$, let Com_c be the affine scheme of commutative associative multiplications on $\mathbb{C}^c$ with identity e_1, and let $G_c = \mathrm{Stab}_{\mathrm{GL}_c}(e_1)$, a smooth connected group of dimension $c(c-1)$. Poonen constructs the scheme of based finite-rank commutative algebras and its fixed-identity locus [31, Definition 1.5 and Proposition 3.1]; Com_c is the corresponding complex fixed-unit law scheme used here. For a commutative algebra Z, write $Z^2_{\mathrm{com}}(Z)$ for the symmetric normalized Hochschild 2-cocycles of Z with values in Z, and $B^2_{\mathrm{com}}(Z) = \delta(C^1_{\mathrm{norm}}(Z, Z))$, where $(\delta f)(a, b) = af(b) - f(ab) + f(a)b$. Put

$$T^1_{\mathrm{com}}(Z) = Z^2_{\mathrm{com}}(Z)/B^2_{\mathrm{com}}(Z).$$

This is Harrison's cohomology in cohomological degree two, which governs first-order commutative deformations of Z [15, Section 1, p. 192]. To compare its original unnormalized definition, a symmetric cocycle f satisfies $f(1, a) = af(1, 1)$; subtracting the coboundary of $a \mapsto af(1, 1)$ normalizes it. If a coboundary δh is normalized, then $h(1) = \delta h(1, 1) = 0$, so the normalized quotient is unchanged. The following tangent calculation identifies this commutative deformation space with the quotient of the tangent space to the framed law scheme by infinitesimal changes of frame.

Lemma 6.8 (Commutative tangent spaces)**.** *Let $Z \in \mathrm{Com}_c(\mathbb{C})$. Then*

$$T_Z\mathrm{Com}_c = Z^2_{\mathrm{com}}(Z), \qquad T_Z(G_c \cdot Z) = B^2_{\mathrm{com}}(Z),$$

and $T^1_{\mathrm{com}}(Z)$ is a subspace of $HH^2(Z)$. The G_c-orbits are the isomorphism classes. The orbit $G_c \cdot \mathbb{C}^c$ of the split étale algebra is open in Com_c and has dimension $c(c-1)$.

Proof. The fixed-unit calculation is the commutative specialization of Lemma 3.2. A dual-number multiplication $m + \varepsilon\psi$ remains commutative exactly when ψ is symmetric. Since Z is commutative, every normalized coboundary is symmetric, so $B^2_{\mathrm{com}}(Z)$ is the space of all normalized coboundaries and $T^1_{\mathrm{com}}(Z) \subset HH^2(Z)$. The algebra $\mathbb{C}^c$ is separable, so $HH^2(\mathbb{C}^c) = 0$ and its tangent space equals the orbit tangent space. Its stabilizer is the finite group permuting the primitive idempotents. The proof of Proposition 3.5 then shows that the orbit is open, of dimension $\dim G_c = c(c-1)$. □

We call the closure of $G_c \cdot \mathbb{C}^c$, with its reduced structure, the *smoothable component* $\mathrm{Sm}_c \subset \mathrm{Com}_c$. It is the component reached by degenerating the split c-point algebra, and it is irreducible of dimension $c(c-1)$.

A unit-preserving frame places a commutative deformation of the centre in Com_c. The following construction gives a local model of the map in Conjecture 6.1(2), relating algebra deformations to commutative deformations of $Z(A)$ [20, Section 1]. Smoothness of the centre morphism would lift a prescribed commutative centre deformation. We test smoothness on Com_c and on its smoothable component. Corollary 6.18 gives counterexamples even after restriction to the scheme-theoretic image.

Definition 6.9 (The centre morphism)**.** Let $U \subset X_d$ be an open subscheme on which the centre $\mathsf{Z}_U = Z(\mathcal{V}_U)$ is a locally free $\mathcal{O}_U$-module of rank c whose formation commutes with arbitrary base change. Let $\mathscr{F}_U \to U$ be the scheme of unit-preserving frames: for a U-scheme U', the U'-points of $\mathscr{F}_U$ are the isomorphisms $\mathcal{O}^c_{U'} \to \mathsf{Z}_U \otimes \mathcal{O}_{U'}$ sending e_1 to 1. The unit section spans a subbundle of Z_U, so $\mathscr{F}_U$ is a G_c-torsor that is locally trivial for the Zariski topology. Transporting the multiplication of Z_U along the tautological frame defines a G_c-equivariant morphism

$$\mathrm{cent}\colon \mathscr{F}_U \longrightarrow \mathrm{Com}_c.$$

We say that the *centre morphism is smooth at* $B \in U(\mathbb{C})$ if cent is smooth at a point of $\mathscr{F}_U$ over B. By equivariance, this does not depend on the choice of the point.

The just-defined smoothness condition is equivalently the smoothness at B of the morphism $U \to [\mathrm{Com}_c/G_c]$ classifying Z_U: the quotient stack classifies locally free rank-c commutative algebras [31, Definition 3.2], and the frame bundle $\mathscr{F}_U$ is its base change along the smooth atlas $\mathrm{Com}_c \to [\mathrm{Com}_c/G_c]$. An object of this fibre product is a law on $\mathcal{O}^c$ and a unit-preserving algebra isomorphism to the centre; forgetting the law leaves exactly a centre frame. Smoothness and its open locus are detected by this smooth surjective base change [35, Lemmas 101.33.4 and 101.33.6, Tags `0DN7` and `0DZR`]. The unit is unimodular in every local trivialization of Z_U, so its span is a local direct summand; this is what makes the unit-preserving frame bundle a G_c-torsor.

6.4 The centre tangent map and the smooth locus

Let U be as in Theorem 6.4, so that the centre morphism $\mathrm{cent}\colon \mathscr{F}_U \to \mathrm{Com}_c$ of Definition 6.9 is defined. Fix $B \in U(\mathbb{C})$ and put $Z_B = Z(B)$. For $\phi \in Z^2_{\mathrm{norm}}(B,B)$, let B_ϕ be $B[\varepsilon]$ with multiplication $aa' + \varepsilon\phi(a,a')$, where $\varepsilon^2 = 0$. By Lemma 3.2, these are the tangent vectors to X_d at B. The centre bundle lifts central elements through B_ϕ; their induced first-order multiplication, modulo changes of the chosen frame, defines χ_B.

Lemma 6.10 (The centre tangent map)**.** *Let $\phi \in Z^2_{\mathrm{norm}}(B,B)$.*

(i) *There is a linear map $\tau\colon Z_B \to B$ with $\tau(1) = 0$ such that $z + \varepsilon\tau(z)$ is central in B_ϕ for all $z \in Z_B$; equivalently,*

$$[\tau(z), a] = \phi(a,z) - \phi(z,a) \qquad (a \in B). \tag{6.2}$$

Any two such maps differ by a linear map $Z_B \to Z_B$ vanishing at 1.

(ii) *For such τ, the bilinear map*

$$\psi(z,z') = \phi(z,z') + \tau(z)z' + z\tau(z') - \tau(zz') \tag{6.3}$$

takes values in Z_B and lies in $Z^2_{\mathrm{com}}(Z_B)$. The map $z + \varepsilon z'' \mapsto z + \varepsilon(\tau(z) + z'')$ is an isomorphism from $Z_B[\varepsilon]$, with multiplication $zz' + \varepsilon\psi(z,z')$, onto $Z(B_\phi)$.

(iii) *The class of ψ in $T^1_{\mathrm{com}}(Z_B)$ depends only on the class of ϕ in $HH^2(B)$. This defines a linear map*

$$\chi_B\colon HH^2(B) \longrightarrow T^1_{\mathrm{com}}(Z_B),$$

the centre tangent map *of B.*

Proof. (i) The algebra B_ϕ gives a dual-number point of X_d with closed point in the open subscheme U, so the corresponding morphism factors through U. By Theorem 6.4(i), $Z(B_\phi)$ is free of rank c over $\mathbb{C}[\varepsilon]$ and reduces to Z_B. Every z thus lifts to a central element $z + \varepsilon b$; choose such lifts on a basis containing 1 to define τ. Comparing $(z + \varepsilon\tau(z))a$ with $a(z + \varepsilon\tau(z))$ in B_ϕ gives (6.2). An element εb is central in B_ϕ exactly when $b \in Z_B$, which proves the last assertion.

(ii) Multiply these chosen central lifts. The product of $z + \varepsilon\tau(z)$ and $z' + \varepsilon\tau(z')$ is

$$zz' + \varepsilon(\phi(z,z') + \tau(z)z' + z\tau(z')) = (zz' + \varepsilon\tau(zz')) + \varepsilon\psi(z,z').$$

It is central, so $\varepsilon\psi(z,z')$ is central and $\psi(z,z') \in Z_B$. Central elements commute, so ψ is symmetric, and $\psi(1,z') = 0$ because ϕ is normalized and $\tau(1) = 0$. The displayed map is $\mathbb{C}[\varepsilon]$-linear and bijective, since both sides are free of rank c and it reduces to the identity of Z_B. By the computation above it is multiplicative for the stated product. That product is therefore associative, so ψ is a Hochschild cocycle.

(iii) Finally, changing τ by $f\colon Z_B \to Z_B$ with $f(1) = 0$ changes ψ by δf. If $\phi = \delta g$ with $g \in C^1_{\mathrm{norm}}(B,B)$, then $1 + \varepsilon g$ is an algebra isomorphism $B_\phi \to B[\varepsilon]$. Hence $z - \varepsilon g(z)$ is central in B_ϕ, and the choice $\tau = -g|_{Z_B}$ in (6.3) gives $\psi = 0$. For linearity, choose a fixed vector-space complement to Z_B in B. Projection to this complement selects the unique complement-valued solution τ of (6.2); since that equation has a right-hand side linear in ϕ, this choice and the induced class of ψ are linear in ϕ. □

Proposition 6.11 (The differential of the centre morphism)**.** *Let $(B, \mathfrak{f})$ be a $\mathbb{C}$-point of $\mathscr{F}_U$, and identify $\mathbb{C}^c$ with Z_B by the frame $\mathfrak{f}$. The image of $d\mathrm{cent}\colon T_{(B,\mathfrak{f})}\mathscr{F}_U \to T_{Z_B}\mathrm{Com}_c = Z^2_{\mathrm{com}}(Z_B)$ is spanned by $B^2_{\mathrm{com}}(Z_B)$ and the cocycles (6.3). Consequently*

$$\operatorname{rank} d\mathrm{cent}_{(B,\mathfrak{f})} = \dim B^2_{\mathrm{com}}(Z_B) + \operatorname{rank} \chi_B, \tag{6.4}$$

and $d\mathrm{cent}_{(B,\mathfrak{f})}$ is surjective if and only if χ_B is surjective.

Proof. A tangent vector to $\mathscr{F}_U$ at $(B, \mathfrak{f})$ is a first-order deformation B_ϕ together with a unit-preserving frame of $Z(B_\phi)$ lifting $\mathfrak{f}$. By Lemma 6.10(i), these frames are $z \mapsto z + \varepsilon\tau(z)$, where τ runs through a torsor under $C^1_{\mathrm{norm}}(Z_B, Z_B) = \operatorname{Lie} G_c$. By Lemma 6.10(ii), $d\mathrm{cent}$ sends this tangent vector to ψ. Varying τ gives all of $\psi + B^2_{\mathrm{com}}(Z_B)$, which proves the description of the image. The image of $d\mathrm{cent}$ contains $B^2_{\mathrm{com}}(Z_B)$. Modulo $B^2_{\mathrm{com}}(Z_B)$, the image of $d\mathrm{cent}$ is the image of χ_B in $T^1_{\mathrm{com}}(Z_B)$. □

The rank formula separates changes of frame from the induced commutative deformation class. After quotienting the centre tangent space by frame directions, the remaining first-order obstruction is the cokernel of χ_B.

To infer smoothness of cent from this criterion, its source must be smooth. The frame bundle is smooth because it lies over the smooth neighbourhood U. The following lemma applies even when the target is singular.

Lemma 6.12 (A tangent criterion for smoothness)**.** *Let $g\colon X \to Y$ be a morphism of schemes of finite type over $\mathbb{C}$, let $x \in X(\mathbb{C})$ and $y = g(x)$, and suppose that X is smooth at x. Then g is smooth at x if and only if $dg_x\colon T_xX \to T_yY$ is surjective. The target Y may be singular.*

Proof. Apply the smooth-source Jacobian criterion of [14, Section 8.1, Proposition 8.1] with base $\operatorname{Spec}\mathbb{C}$. That statement concerns quasi-separated algebraic stacks of finite presentation and their tangent categories. Here we may restrict to affine neighbourhoods of x and y, so its finite-presentation and quasi-separatedness hypotheses hold. Both residue fields are $\mathbb{C}$. For schemes, the tangent category is the discrete category of maps from $\operatorname{Spec}\mathbb{C}[\varepsilon]/(\varepsilon^2)$ with the prescribed closed point, so essential surjectivity of the tangent functor is exactly surjectivity of dg_x. The source-smoothness hypothesis is precisely the assumption on X. The cited criterion gives the claimed equivalence, without any reducedness or smoothness assumption on Y; it also implies that Y is smooth at y whenever these equivalent conditions hold. □

The smooth-source hypothesis cannot be omitted. For example, the closed immersion

$$\operatorname{Spec}\mathbb{C}[t]/(t^2) \longrightarrow \operatorname{Spec}\mathbb{C}[t]/(t^3)$$

induces an isomorphism on tangent spaces and is not flat.

Theorem 6.13 (The smooth locus of the centre morphism)**.** *Let U be as in Theorem 6.4 and $B \in U(\mathbb{C})$. The centre morphism is smooth at B if and only if the centre tangent map*

$$\chi_B\colon HH^2(B) \longrightarrow T^1_{\mathrm{com}}(Z(B))$$

is surjective. These points form the $\mathbb{C}$-points of a dense open subscheme $U^{\mathrm{cent}} \subset U$, which contains every semisimple algebra in U.

Proof. The torsor $\mathscr{F}_U$ is locally trivial with smooth fibre G_c over the smooth scheme U, so it is smooth. Lemma 6.12 and Proposition 6.11 give the criterion. The smooth locus of cent is open and, by equivariance, G_c-stable. The torsor projection is smooth and surjective, hence open, so the image of this locus in U is open; call it U^{cent}. If B is semisimple, then $HH^2(B) = 0$, and $Z(B) \cong \mathbb{C}^c$ has $T^1_{\mathrm{com}} = 0$ by Lemma 6.8; so χ_B is surjective. The semisimple locus is dense in U. □

The cokernel of χ_B is the first-order obstruction to smoothness of the centre morphism. The type D family below gives an explicit nonzero class at the original contraction algebra.

6.5 Type D counterexamples to Conjecture 1.4(2)

For $n \geq 2$, consider the algebra $D_{n,2}$ of [20, Definition 3.2 and Proposition 3.3],

$$A_n := D_{n,2} = \frac{\mathbb{C}\langle x, y\rangle}{(xy + yx,\ y^2 + x^2 + x^{2n-1},\ x^{2n+1})}. \tag{6.5}$$

It is the completed Jacobi algebra of the potential $\frac{1}{2n}x^{2n} + \frac{1}{3}x^3 + xy^2$; the ideal displayed in (6.5) contains a power of (x, y), so completion changes nothing [20, Proposition 3.3]. By Brown–Wemyss, A_n is the contraction algebra of a simple flop of a smooth threefold, of type D and length two [5, Theorems 1.5 and 8.11]. It has dimension $4n$ by Lemma 6.14. Lemma 6.15 below constructs a split associative family and proves that its only semisimple geometric generic type is $M_2(\mathbb{C})^{\times(n-1)} \times \mathbb{C}^4$. This type is also recorded in [20, Theorems 3.7 and 3.13]; our construction and uniqueness argument here are independent. Theorem 3.1 therefore identifies its GV invariants as $n_1 = 4$ and $n_2 = n - 1$, with $c = n + 3$.

These cited results provide the presentation and geometric realization of A_n, and determine the possible geometrically semisimple generic fibre of an associative deformation of A_n. We use the presentation to construct a polynomial family of centres and prove the non-lifting assertion directly.

Throughout this subsection we write

$$\mathsf{u} = x^2, \qquad \mathsf{v} = x^{2n-1}, \qquad \mathsf{w} = x^{2n-2}y,$$

using sans-serif letters for elements of the centre. The symbols u and w continue to denote geometric parameters.

The obstruction uses the contrast between v and w. The former is a power of the noncentral element x, whereas the latter lies outside the subalgebra $\mathbb{C}[x]$. A commutative centre family makes v^2 acquire a w-component. The calculation below excludes this symbolically for every $n \geq 2$; the exact $n = 2$ calculation later addresses the stronger unmarked statement.

Lemma 6.14 (Structure of $D_{n,2}$)**.** *Let $n \geq 2$.*

(i) *In A_n, $x^{2n-1}y = 0$, and the elements $1, x, \dots, x^{2n}$ and $y, xy, \dots, x^{2n-2}y$ form a basis. With $x^i = 0$ for $i > 2n$ and $x^i y = 0$ for $i > 2n - 2$,*

$$x^i y \cdot x^j = (-1)^j x^{i+j} y, \qquad x^i y \cdot x^j y = (-1)^{j+1}(x^{i+j+2} + x^{i+j+2n-1}).$$

(ii) *The centralizer of x is $\mathbb{C}[x] \oplus \mathbb{C}\mathsf{w}$, where $\mathbb{C}[x] = \operatorname{span}(1, x, \dots, x^{2n})$.*

(iii) *The centre is*

$$Z_n := Z(A_n) = \operatorname{span}(1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}) \cong \frac{\mathbb{C}[\mathsf{u}, \mathsf{v}, \mathsf{w}]}{(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{uw}, \mathsf{v}^2, \mathsf{vw}, \mathsf{w}^2)},$$

of dimension $n + 3$.

(iv) $\mathsf{w}x = x\mathsf{w} = 0$, *and* $\mathsf{v}A_n + A_n\mathsf{v} \subseteq \mathbb{C}[x]$.

Proof. (i) Since $yx^i = (-1)^i x^i y$, multiplying $y^2 = -x^2 - x^{2n-1}$ by y on the left and on the right gives $-x^2 y + x^{2n-1} y = -x^2 y - x^{2n-1} y$, so $x^{2n-1} y = 0$. The listed elements span. To prove independence directly, take a vector space with basis $E_0, \dots, E_{2n}, F_0, \dots, F_{2n-2}$, interpreting larger indices as zero, and define operators

$$XE_i = E_{i+1}, \quad XF_j = F_{j+1}, \quad YE_i = (-1)^i F_i, \quad YF_j = (-1)^{j+1}(E_{j+2} + E_{j+2n-1}).$$

On each basis vector, $XY + YX = 0$ and $X^{2n+1} = 0$. Moreover $Y^2 E_i = -E_{i+2} - E_{i+2n-1}$ and $Y^2 F_j = -F_{j+2}$; since $X^{2n-1} F_j = 0$, these identities give $Y^2 + X^2 + X^{2n-1} = 0$. Thus $x \mapsto X$, $y \mapsto Y$ defines a representation of A_n. Applying the spanning monomials to E_0 gives the distinct basis vectors E_i and F_j, so the monomials are independent. The products follow from $yx^j = (-1)^j x^j y$ and the relation for y^2.

(ii) Write $a = f(x) + g(x)y$. Then $[a, x] = [g(x)y, x] = -2xg(x)y$. By (i), this vanishes exactly when g is a multiple of x^{2n-2}.

(iii) By (ii), a central element has the form $f(x) + \lambda\mathsf{w}$. Since $2n - 2$ is even, w commutes with y. Moreover $[f(x), y] = 2\sum_{i \text{ odd}} f_i x^i y$, which vanishes exactly when $f_i = 0$ for all odd $i \le 2n - 3$. This gives the displayed basis. The relations hold because $x^{2n+1} = 0$, $x^{2n-1} y = 0$ and $\mathsf{w}^2 = x^{4n-4} y^2 = -x^{4n-2} - x^{6n-5} = 0$; comparing dimensions gives the presentation.

(iv) This follows from the products in (i). □

Lemma 6.15 (An explicit split family for $D_{n,2}$)**.** *For every $n \ge 2$, there is a finite free unital $\mathbb{C}[[h]]$-algebra $\mathscr{B}_n$, with special fibre A_n, whose generic fibre is*

$$\mathscr{B}_n \otimes_{\mathbb{C}[[h]]} \mathbb{C}((h)) \cong \left(M_2(\mathbb{C})^{\times(n-1)} \times \mathbb{C}^4\right) \otimes_{\mathbb{C}} \mathbb{C}((h)).$$

This is the only semisimple geometric generic type of a finite free formal associative deformation of A_n. The point $A_n \in X_{4n}$ is smooth, of dimension $16n^2 - 7n + 3$. The unique irreducible component of X_{4n} through A_n is the closure of this semisimple orbit.

Proof. Put $R_h = \mathbb{C}[[h]]$, $D_h = R_h[[x]]$, and

$$p_h = x \prod_{i=1}^{n-1} (x^2 - ih^2), \qquad q_h = x^2 - nh^2, \qquad g_h = p_h + q_h,$$
$$T_h = D_h/(p_h g_h), \qquad M_h = D_h/(p_h).$$

Let $\pi : T_h \to M_h$ be the quotient and let σ be the involution of M_h induced by $x \mapsto -x$. For $k, l \in M_h$, the expression $\mathsf{b}(k, l) = -\widetilde{k}\,\sigma(\widetilde{l}) g_h$ is well defined in T_h, since changing either lift changes it by a multiple of $p_h g_h$. On $T_h \oplus M_h y$ set

$$(f, k)(f', k') = \left(ff' + \mathsf{b}(k, k'),\ \pi(f)k' + k\sigma(\pi(f'))\right).$$

The unit is $(1,0)$. Associativity with at most one M_h-factor follows from the commuting left and right actions $\pi, \sigma\pi$. With two such factors it follows from

$$\mathsf{b}(\pi(f)k,l) = f\mathsf{b}(k,l), \quad \mathsf{b}(k,l\sigma(\pi(f))) = \mathsf{b}(k,l)f, \quad \mathsf{b}(k\sigma(\pi(f)),l) = \mathsf{b}(k,\pi(f)l).$$

With three factors, the two products are $-k\sigma(l)\pi(g_h)m$ and $-k\sigma(l)m\sigma(\pi(g_h))$, equal because $\pi(g_h) = \pi(q_h)$ is fixed by σ.

Here is the freeness calculation. The equation $g_h(hX) = 0$ becomes

$$X^2 - n + h^{2n-3}X\prod_{i=1}^{n-1}(X^2 - i) = 0.$$

At $X = \pm\sqrt{n}$, each successive coefficient in a formal solution is determined by division by the nonzero number $\pm 2\sqrt{n}$. It has solutions $\rho_\pm/h \in R_h$, with $\rho_\pm = \pm\sqrt{n}\,h + O(h^2)$. Division by $x - \rho_+$ is valid in $R_h[[x]]$, since $\rho_+ \in hR_h$ makes evaluation and the quotient converge h-adically. The resulting quotient vanishes at ρ_-: its value times $\rho_- - \rho_+$ is zero, and R_h is a domain. Dividing once more gives $g_h = (x - \rho_+)(x - \rho_-)v_h$, where $v_h(0,x) = 1 + x^{2n-3}$, so v_h is a unit. Consequently T_h is the quotient by the monic distinguished polynomial $p_h(x - \rho_+)(x - \rho_-)$ of degree $2n+1$. For a monic distinguished polynomial of degree a, division modulo h is division by x^a; successive corrections modulo h^r give a unique remainder of degree below a over R_h. Thus T_h and M_h are free of ranks $2n+1$ and $2n-1$. Our algebra $\mathscr{B}_n = T_h \oplus M_h y$ is free of rank $4n$. Its reduction has the basis of Lemma 6.14 and relations $xy = -yx$, $y^2 = -x^2 - x^{2n-1}$, $x^{2n+1} = 0$; hence its special fibre is A_n.

For each $1 \le i \le n-1$, the matrices

$$X_i = h\sqrt{i}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad Y_i = h\sqrt{n-i}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

and the four scalar pairs

$$(0, h\sqrt{n}), \quad (0, -h\sqrt{n}), \quad (\rho_+, 0), \quad (\rho_-, 0)$$

give representations of $\mathscr{B}_n$. Indeed, in every case $XY = -YX$, $Y^2 = -g_h(X)$, $p_h(X)Y = 0$ and $p_h(X)g_h(X) = 0$; evaluation of the formal series converges since X is divisible by h. Over $\mathbb{C}((h))$, the values of x^2 are $ih^2, 0, \rho_+^2, \rho_-^2$. They are distinct except for the two zero-x characters. For the last two values, equality would imply $\rho_- = -\rho_+$, hence $p_h(\rho_+) = q_h(\rho_+) = 0$; but $q_h(\rho_+) = 0$ gives

$$p_h(\rho_+) = \rho_+ h^{2n-2}\prod_{i=1}^{n-1}(n-i) \neq 0.$$

Interpolation in x^2 separates these factors; y separates the two zero-x characters. On each matrix factor, X_i gives the two diagonal projectors and Y_i gives both nonzero off-diagonal entries. The resulting map onto $M_2(\mathbb{C}((h)))^{\times(n-1)} \times \mathbb{C}((h))^4$ is surjective. Both sides have dimension $4n$, so it is an isomorphism.

Finally apply Theorem 5.9 with $E = \mathbb{C}1$. The constructed family places A_n in this semisimple orbit closure; both centres have dimension $n+3$ by Lemma 6.14. The Hochschild equality is Proposition 5.10, applicable to the geometric realization of A_n. The criterion proves smoothness, the unique component and uniqueness for any other semisimple orbit closure through A_n. A

formal deformation with geometrically semisimple generic fibre places its special point in that generic type's orbit closure: each defining equation vanishes after the generic field extension, hence in the parameter domain and on its special fibre. This gives the stated uniqueness for every such formal family. The orbit dimension is $(4n)(4n-1)-4n+(n+3) = 16n^2-7n+3$. □

Lemma 6.16 (A polynomial family of centres)**.** *Let*

$$\mathscr{Z}_n = \frac{\mathbb{C}[t, \mathsf{u}, \mathsf{v}, \mathsf{w}]}{(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{uw}, \mathsf{vw}, \mathsf{w}^2,\ \mathsf{v}^2 - t\mathsf{w})}. \tag{6.6}$$

(i) *The algebra $\mathscr{Z}_n$ is free over $\mathbb{C}[t]$ with basis $1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}$, and $\mathscr{Z}_n/(t) \cong Z_n$.*

(ii) *For $t \neq 0$, the fibre is isomorphic to $\mathbb{C}[\mathsf{u}, \mathsf{v}]/(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{v}^3)$, which is smoothable.*

(iii) *In the basis of (i), $\mathscr{Z}_n$ defines a morphism $\mathbb{A}^1 \to \mathrm{Com}_{n+3}$ that factors through Sm_{n+3}. Its tangent vector at $t = 0$ is the cocycle $\psi_0 \in Z^2_{\mathrm{com}}(Z_n)$ with $\psi_0(\mathsf{v}, \mathsf{v}) = \mathsf{w}$ and $\psi_0 = 0$ on all other pairs of basis vectors.*

Proof. (i) Let the powers of u multiply as in $\mathbb{C}[\mathsf{u}]/(\mathsf{u}^{n+1})$, put $\mathsf{v}^2 = t\mathsf{w}$, and let all other products of two basis vectors other than 1 vanish. The element w annihilates $\mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}$, and v annihilates the powers of u. Hence every product of three such basis vectors is zero in either order, unless all three are powers of u. The multiplication is therefore associative and commutative. It satisfies the relations of (6.6), and those relations reduce every monomial to the stated basis; this gives inverse isomorphisms. Setting $t = 0$ recovers Lemma 6.14(iii).

(ii) For $t \neq 0$, eliminate $\mathsf{w} = t^{-1}\mathsf{v}^2$. The family

$$\frac{\mathbb{C}[h, \mathsf{u}, \mathsf{v}]}{(\mathsf{uv},\ \mathsf{u}^{n+1} - h^n\mathsf{u},\ \mathsf{v}^3 - h^2\mathsf{v})}$$

is free over $\mathbb{C}[h]$ with basis $1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{v}^2$. Indeed, it maps onto the fibre product over $\mathbb{C}[h]$ of $\mathbb{C}[h][\mathsf{u}]/(\mathsf{u}^{n+1} - h^n\mathsf{u})$ and $\mathbb{C}[h][\mathsf{v}]/(\mathsf{v}^3 - h^2\mathsf{v})$ along $\mathsf{u} \mapsto 0$ and $\mathsf{v} \mapsto 0$; that fibre product is free with the images of these elements as a basis, and they span the source. For $h \neq 0$, its fibre has the $n+3$ distinct points $(\mathsf{u}, \mathsf{v}) = (h\xi, 0)$ with $\xi^n = 1$, and $(0,0)$, $(0,h)$, $(0,-h)$, so it is isomorphic to $\mathbb{C}^{n+3}$. In this basis the family is a morphism $\mathbb{A}^1 \to \mathrm{Com}_{n+3}$ mapping $h \neq 0$ into the irreducible set $G_{n+3} \cdot \mathbb{C}^{n+3}$. Its fibre at $h = 0$, which is the algebra in (ii), therefore lies in the closure Sm_{n+3}.

(iii) By (ii), the points with $t \neq 0$ map to Sm_{n+3}, which is closed. Since $\mathbb{A}^1$ is reduced, the whole morphism factors through Sm_{n+3}. The structure constants are affine-linear in t, and the only t-dependent one is that of $\mathsf{v} \cdot \mathsf{v}$. □

Theorem 6.17 (A centre deformation that does not lift)**.** *Let $n \geq 2$, let $R_\varepsilon = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, and let $\mathscr{Z}_{n,\varepsilon} = \mathscr{Z}_n \otimes_{\mathbb{C}[t]} R_\varepsilon$ with $t \mapsto \varepsilon$. There is no finite free R_ε-algebra $\mathscr{B}$ with $\mathscr{B}/\varepsilon\mathscr{B} \cong A_n$ together with a homomorphism of R_ε-algebras $\iota\colon \mathscr{Z}_{n,\varepsilon} \to \mathscr{B}$ whose reduction modulo ε is the inclusion $Z_n \subset A_n$.*

The statement rules out even a lift of the central inclusion $Z_n \subset A_n$. If the displayed commutative family were the centre of an associative deformation with that marking, the inclusion of the deformed centre into the total algebra would give this homomorphism. This stronger algebraic statement therefore yields the centre-lifting obstruction.

Proof. Suppose that $(\mathscr{B}, \iota)$ exists. The argument compares two descriptions of the square of a lift of v: the relation on x forces it into $\varepsilon\mathbb{C}[x]$, whereas the proposed centre family forces it to equal $\varepsilon\mathsf{w}$. Since $\mathscr{B}$ is free over R_ε, multiplication by ε induces an isomorphism of A_n-bimodules

$$A_n \xrightarrow{\sim} \varepsilon\mathscr{B}, \qquad a \longmapsto \varepsilon\tilde{a}, \tag{6.7}$$

where $\tilde{a}$ is any lift of a. We write εa for the image of a. Choose a lift $\tilde{x} \in \mathscr{B}$ of x, and put $\tilde{\mathsf{v}} = \iota(\mathsf{v})$ and $\tilde{\mathsf{w}} = \iota(\mathsf{w})$.

Since $x^{2n+1} = 0$, we have $\tilde{x}^{2n+1} = \varepsilon b$ for some $b \in A_n$. As $\tilde{x}$ commutes with its own powers, $0 = [\tilde{x}^{2n+1}, \tilde{x}] = \varepsilon[b, x]$, so $[b, x] = 0$ by (6.7). Lemma 6.14(ii) gives $b \in \mathbb{C}[x] \oplus \mathbb{C}\mathsf{w}$.

Now $\tilde{\mathsf{v}}$ lifts $\mathsf{v} = x^{2n-1}$, so $\tilde{\mathsf{v}} = \tilde{x}^{2n-1} + \varepsilon g$ for some $g \in A_n$. Since $\varepsilon^2 = 0$ and $4n - 2 \geq 2n + 1$,

$$\tilde{\mathsf{v}}^{\,2} = \tilde{x}^{2n+1}\tilde{x}^{2n-3} + \varepsilon(\mathsf{v}g + g\mathsf{v}) = \varepsilon(bx^{2n-3} + \mathsf{v}g + g\mathsf{v}).$$

Here $2n - 3 \geq 1$ and $\mathsf{w}x = 0$, so $bx^{2n-3} \in \mathbb{C}[x]$, and $\mathsf{v}g + g\mathsf{v} \in \mathbb{C}[x]$ by Lemma 6.14(iv). Hence $\tilde{\mathsf{v}}^{\,2} \in \varepsilon\,\mathbb{C}[x]$. On the other hand, $\tilde{\mathsf{v}}^{\,2} = \iota(\mathsf{v}^2) = \iota(\varepsilon\mathsf{w}) = \varepsilon\tilde{\mathsf{w}} = \varepsilon\mathsf{w}$. By (6.7), $\mathsf{w} \in \mathbb{C}[x]$, which contradicts Lemma 6.14(i). □

Corollary 6.18 (The centre morphism is not smooth at $D_{n,2}$)**.** *Let $n \geq 2$, $c = n + 3$, and let U be a neighbourhood of A_n as in Theorem 6.4.*

(i) *The class $[\psi_0]$ is not in the image of χ_{A_n}. Hence the centre morphism is not smooth at A_n, and $A_n \notin U^{\mathrm{cent}}$.*

(ii) *The centre morphism factors through Sm_c, which is the scheme-theoretic image of* cent$\colon \mathscr{F}_U \to \mathrm{Com}_c$*. The morphism* cent$\colon \mathscr{F}_U \to \mathrm{Sm}_c$ *is not smooth at the points over A_n either.*

(iii) *The family $\mathscr{Z}_n$ does not lift near $t = 0$: there is no finite free algebra over $\mathbb{C}[t]_{(t)}$, over the convergent power series in t, or over $\mathbb{C}[[t]]$, with special fibre A_n and centre identified with $\mathscr{Z}_n$ compatibly with $Z_n \subset A_n$.*

Thus $A_n = D_{n,2}$, $n \geq 2$, gives a counterexample to the smoothness assertion in Conjecture 6.1(2). The local-constancy assertion holds for A_n by Theorem 6.4.

Proof. (i) Suppose that $[\psi_0] = \chi_{A_n}([\phi])$. By Lemma 6.10, there is a map τ for which (6.3) equals ψ_0, after changing τ by an element of $C^1_{\mathrm{norm}}(Z_n, Z_n)$. The map $z \mapsto z + \varepsilon\tau(z)$ is then an algebra homomorphism $\mathscr{Z}_{n,\varepsilon} \to (A_n)_\phi$ lifting $Z_n \subset A_n$, contradicting Theorem 6.17. The smoothness conclusion then follows from Theorem 6.13.

(ii) Over the dense semisimple locus of U, the centres are isomorphic to $\mathbb{C}^c$, so cent maps a dense open subscheme of the reduced scheme $\mathscr{F}_U$ into $G_c \cdot \mathbb{C}^c$. Hence cent factors through Sm_c. By equivariance its image contains the open orbit. Moreover, the kernel of $\mathcal{O}(\mathrm{Com}_c) \to \Gamma(\mathscr{F}_U, \mathcal{O})$ is radical, because $\mathscr{F}_U$ is reduced. Thus the scheme-theoretic image is exactly the reduced orbit closure Sm_c. Let $\mathfrak{f}$ be the frame of Z_n given by the basis of Lemma 6.16(i). By part (iii) of that lemma, ψ_0 is tangent to Sm_c at $\mathrm{cent}(A_n, \mathfrak{f})$. By (i) and Proposition 6.11, it is not in the image of $d\mathrm{cent}_{(A_n,\mathfrak{f})}$. So the differential of cent$\colon \mathscr{F}_U \to \mathrm{Sm}_c$ at $(A_n, \mathfrak{f})$ is not surjective, and cent is not smooth there by Lemma 6.12; by equivariance, the same holds at every point over A_n.

(iii) Reducing such a lift modulo t^2 would give a pair $(\mathscr{B}, \iota)$ as excluded by Theorem 6.17. □

Remark 6.19. The socle element $\mathsf{v} = x^{2n-1}$ of Z_n is a power of the noncentral element x. In every first-order deformation, the square of a lift of v stays in the subalgebra generated by a chosen lift of x, because of the relation $x^{2n+1} = 0$ and the shape of the centralizer of x. The commutative centre deformation $\mathsf{v}^2 = t\mathsf{w}$ instead gives the square a component along $\mathsf{w} = x^{2n-2}y$, outside that subalgebra.

The obstruction is first order. A ramified base change $t = h^r$ with $r \geq 2$ has zero first-order class, so lifts of that pullback are a higher-order problem.

6.6 Exact tangent data for $D_{2,2}$

The preceding symbolic obstruction applies to every $n \geq 2$. For $n = 2$, exact linear algebra computes the full first-order image of χ_A. This image calculation also gives an independent certificate for the stronger conclusion without a chosen marking. Here $A = A_2$ has basis $1, x, x^2, x^3, x^4, y, xy, x^2y$, and Z_2 has basis $1, \mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q}$ with $\mathsf{u} = x^2$, $\mathsf{v} = x^3$, $\mathsf{w} = x^2y$ and $\mathsf{q} = \mathsf{u}^2 = x^4$. The only nonzero product of two of $\mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q}$ is $\mathsf{u}^2 = \mathsf{q}$.

Proposition 6.20 (Tangent data of $D_{2,2}$)**.** *For $A = D_{2,2}$ and $Z_2 = Z(A)$, the following hold.*

(i) $\dim \mathrm{Der}(A) = 6$, $\dim HH^1(A) = \dim HH^2(A) = 3$, *and* $\dim Z^2_{\mathrm{norm}}(A, A) = 53 = \dim_A X_8$.

(ii) $\dim B^2_{\mathrm{com}}(Z_2) = 10$, $\dim Z^2_{\mathrm{com}}(Z_2) = 26$, *and* $\dim T^1_{\mathrm{com}}(Z_2) = 16$.

(iii) *The centre tangent map χ_A has rank* 3. *Its image is spanned by three explicitly given cocycle classes $[\psi_1], [\psi_2], [\psi_3]$, recorded in Appendix C. The representatives satisfy $\psi_i(s, s) \in \mathbb{C}\mathsf{q}$ for every $s \in \mathrm{Soc}(Z_2)$.*

(iv) *At every point of $\mathscr{F}_U$ over A,* $\operatorname{rank} d\mathrm{cent} = 10+3 = 13$, *whereas* $\dim T_{Z_2}\mathrm{Sm}_5 \geq \dim \mathrm{Sm}_5 = 20$ *and* $\dim T_{Z_2}\mathrm{Com}_5 = 26$.

Proof. The cochain dimensions and ranks in (i)–(iii) are exact linear algebra over $\mathbb{Q}$, described in Appendix C. The equality with the local dimension in (i) uses Lemma 6.15 with $n = 2$: its split family places A in the orbit closure of $M_2(\mathbb{C}) \times \mathbb{C}^4$, of dimension $8 \cdot 7 - 8 + 5 = 53$. Hence $53 \leq \dim_A X_8 \leq \dim T_A X_8 = 53$, so A is smooth and that orbit closure is the unique component through A. Part (iv) follows from (ii), (iii), Proposition 6.11, and Lemma 6.8. □

At $D_{2,2}$, the differential of the centre morphism misses at least seven dimensions of the tangent space of the scheme-theoretic image of the centre morphism. Theorem 6.17 fixes the special-fibre identification $\mathscr{Z}_{n,\varepsilon}/(\varepsilon) \cong Z_n$. For $n = 2$, the socle condition below is invariant under every automorphism of the centre, and the symbolic socle argument below removes that marking from the conclusion.

Proposition 6.21 (Non-lifting without a marking)**.** *No first-order deformation of $A = D_{2,2}$ over R_ε has centre isomorphic to $\mathscr{Z}_{2,\varepsilon}$ as an R_ε-algebra. Equivalently, the* $\mathrm{Aut}(Z_2)$*-orbit of $[\psi_0]$ does not meet the image of χ_A.*

Proof. We first prove the socle constraint directly for any finite free commutative R_ε-algebra $\mathscr{D}$ with special fibre Z_2 admitting a homomorphism to a finite free associative deformation $\mathscr{B}$ of

A that reduces to $Z_2 \subset A$. Write the multiplication of $\mathscr{D}$, in a unit-preserving lifted basis, as $ab + \varepsilon\psi(a,b)$. Put $\mathfrak{m} = (\mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q})$. Then $\mathfrak{m}^2 = \mathbb{C}\mathsf{q}$ and $\operatorname{Soc}(Z_2) = \mathbb{C}\mathsf{v} \oplus \mathbb{C}\mathsf{w} \oplus \mathbb{C}\mathsf{q}$. We will show

$$\psi(s,s) \in \mathbb{C}s + \mathfrak{m}^2 \qquad \text{for every } s \in \operatorname{Soc}(Z_2). \tag{6.8}$$

Associativity on (u, s, s) gives $\mathsf{u}\psi(s,s) = s\psi(\mathsf{u}, s)$. If the v- or w-coordinate of s is nonzero, the left side lies in $\mathbb{C}\mathsf{u} + \mathbb{C}\mathsf{q}$, while the right side is a scalar multiple of s. Comparing coefficients forces the 1- and u-coordinates of $\psi(s,s)$ to vanish. These coordinates are polynomials in s, so $\psi(s,s) \in \operatorname{Soc}(Z_2)$ for every socle element.

For $a, b, c \in \mathbb{C}$ with $d = a^2 - b^2 \neq 0$, put $z = ax + by$ and $s = a\mathsf{v} + b\mathsf{w} + c\mathsf{q}$. Lemma 6.14 gives

$$z^2 = d\mathsf{u} - b^2\mathsf{v}, \qquad z^3 = d(a\mathsf{v} + b\mathsf{w}) - ab^2\mathsf{q}, \qquad z^4 = d^2\mathsf{q}, \qquad z^5 = 0.$$

For $r \in A$, let r_x, r_y, r_{xy} be its coefficients on x, y, xy; the other basis elements are central. Then

$$[r, z] = 2(br_x - ar_y)xy - 2r_{xy}(a\mathsf{w} + b\mathsf{v} + b\mathsf{q}).$$

It follows that

$$C_A(z) = \mathbb{C}[z] \oplus \mathbb{C}(b\mathsf{v} + a\mathsf{w}), \qquad (b\mathsf{v} + a\mathsf{w})z = 0, \qquad \mathbb{C}[z] \cap \operatorname{Soc}(Z_2) = \mathbb{C}s + \mathbb{C}\mathsf{q}.$$

Indeed, the centralizer is $Z_2 + \mathbb{C}z$, and the displayed powers of z, together with $b\mathsf{v} + a\mathsf{w}$, are independent because $d \neq 0$. Also

$$s = P(z), \qquad P(T) = d^{-1}T^3 + (c + ab^2/d)d^{-2}T^4.$$

Choose a lift X of z in $\mathscr{B}$. The isomorphism (6.7) writes $X^5 = \varepsilon r$ with $r \in C_A(z)$, since X commutes with its powers. For every $j \geq 6$, therefore, $X^j = \varepsilon r z^{j-5}$ lies in $\varepsilon\mathbb{C}[z]$. If S is the image in $\mathscr{B}$ of the chosen lift of s in $\mathscr{D}$, then $S = P(X) + \varepsilon g$ for some $g \in A$. The products of s with $1, x, y$ are $s, a\mathsf{q}, -b\mathsf{q}$, respectively, and its products with every other basis element are zero. Since s is central,

$$S^2 = P(X)^2 + \varepsilon(sg + gs) \in \varepsilon\mathbb{C}[z].$$

On the other hand $S^2 = \varepsilon\psi(s,s)$, because $s^2 = 0$. Thus $\psi(s,s) \in \mathbb{C}[z] \cap \operatorname{Soc}(Z_2) = \mathbb{C}s + \mathbb{C}\mathsf{q}$ when $d \neq 0$.

To include the remaining socle directions, write

$$\psi(a\mathsf{v} + b\mathsf{w} + c\mathsf{q}, a\mathsf{v} + b\mathsf{w} + c\mathsf{q}) = F(a,b,c)\mathsf{v} + G(a,b,c)\mathsf{w} + H(a,b,c)\mathsf{q}.$$

The quadratic polynomials F, G satisfy $bF - aG = 0$ on $a^2 - b^2 \neq 0$, hence identically. Coprimality of a, b in $\mathbb{C}[a,b,c]$ gives $F = aL$ and $G = bL$ for a polynomial L. The displayed diagonal is $Ls + (H - cL)\mathsf{q}$, proving (6.8) also when $s \in \mathbb{C}\mathsf{q}$.

This condition depends only on the cohomology class: for a normalized linear map $f : Z_2 \to Z_2$, $(\delta f)(s,s) = 2sf(s) \in \mathbb{C}s$. It is invariant under $\operatorname{Aut}(Z_2)$, which preserves $\operatorname{Soc}(Z_2)$ and $\mathfrak{m}^2$. The class $[\psi_0]$ violates it, since $\psi_0(\mathsf{v}, \mathsf{v}) = \mathsf{w} \notin \mathbb{C}\mathsf{v} + \mathbb{C}\mathsf{q}$. The centre tangent map is constructed from precisely such lifted multiplications, so its image satisfies the condition; no automorphic transport of $[\psi_0]$ belongs to that image.

Finally suppose $Z(\mathscr{B}) \cong \mathscr{Z}_{2,\varepsilon}$ as an R_ε-algebra. Then $Z(\mathscr{B})$ is free of rank five. Its reduction maps injectively into Z_2: if a central element lies in $\varepsilon\mathscr{B}$, it is annihilated by ε, and freeness of $Z(\mathscr{B})$ puts it in $\varepsilon Z(\mathscr{B})$. Dimension five makes the reduced map an isomorphism. The putative centre isomorphism consequently transports $[\psi_0]$ by an automorphism of Z_2, up to a coboundary, to a class satisfying (6.8). The invariance and failure just proved give a contradiction. $\square$

Centres in families and the scheme of algebra laws. The smoothness proved in Section 5 gives an open neighbourhood of A in X_d on which centre and cocenter have constant rank. Corollary 6.2 and Theorem 6.4 control deformations that are already associative. A prescribed commutative centre deformation has the additional first-order condition that its class lie in the image of χ. Corollary 6.18 gives counterexamples $D_{n,2}$, $n \geq 2$, to the smoothness assertion in Conjecture 6.1(2), even when the target is restricted to the smoothable component or to the scheme-theoretic image.

The type A and D algebras studied in [20] belong to this geometric class [5, Theorem 1.5]. The type D examples show that base-change compatibility of centres in associative families does not guarantee that a prescribed commutative centre deformation lifts.

Acknowledgements

Large language models, including ChatGPT, and Math Exploration Agents, including Danus and Rethlas, were used extensively in developing this manuscript. Alongside language editing and computational assistance, these tools supported mathematical exploration, the proposal of proof strategies and intermediate arguments, and the checking and reorganization of calculations. I selected, revised, and assembled the arguments presented here, have made every effort to verify them against the cited literature and the accompanying exact certificates, and take full responsibility for the contents of the manuscript and the decision to make it public. At my present level of expertise, I cannot regard every part of the geometric argument as something I independently rediscovered without such assistance.

For this reason, I do not wish the existence or priority date of this manuscript to discourage a genuinely independent human derivation of these results. If a subsequent work obtains the same results through a mathematical argument developed independently and entirely by humans, I make no claim to special priority or intellectual credit for the corresponding proof ideas beyond the ordinary bibliographic acknowledgement of prior or concurrent results.

If, however, a subsequent argument makes substantive use of the results, constructions, or proof strategies contained in this manuscript, ordinary scholarly citation remains appropriate regardless of the tools used. In particular, I would appreciate explicit attribution when closely related arguments appear in AI-assisted work, since the provenance of ideas generated through overlapping language-model systems can otherwise be difficult to determine.

A Root enumeration for the primitive marked pairs

The root enumeration verifies the assertion used in Lemma 2.5: each nonzero marked-coefficient stratum is one Levi Weyl-group orbit. Karmazyn's primitive presentations determine the six marked pairs [21, Examples 4.22–4.27]. This appendix records the exact root-and-orbit calculation and the diagram-numbering convention it uses. Number the vertices in the diagrams of Figure 2 from left to right along the horizontal chain, and number the lower vertex last. Thus the retained vertices are $1, 2, 3, 3, 4, 3$ for lengths $1, \ldots, 6$, respectively. The enumeration uses this diagram numbering. The table in the proof of Theorem 3.13 instead uses Karmazyn's numbering; the two conventions must therefore not be mixed.

Represent a root by its vector $v = (v_1, \ldots, v_r)$ in the simple-root basis. If $i \sim h$ denotes

adjacency in the diagram, the simple reflection acts by

$$s_i(v)_h = v_h \quad (h \neq i), \qquad s_i(v)_i = -v_i + \sum_{h \sim i} v_h.$$

Starting from the simple roots, repeatedly apply every reflection s_i and retain each new vector once. Since the ADE root system is finite, this terminates and gives the full Weyl orbits of the simple roots, hence the entire root system. Keeping the vectors with nonnegative coordinates produces Φ^+.

For each marked coefficient $j > 0$, choose a positive root with $v_p = j$ and repeat the procedure using only reflections with $i \neq p$. The resulting orbit agrees, in all six primitive cases, with the complete set of positive roots of coefficient j. The numbers of positive roots are

$$1, \quad 12, \quad 36, \quad 63, \quad 120, \quad 120,$$

and the non-Levi totals are

$$N_p = 1, \quad 9, \quad 29, \quad 53, \quad 104, \quad 106.$$

The individual orbit cardinalities are those of Lemma 2.5. The same calculation checks that the largest marked coefficient is the stated length.

The integer implementation is included in `tools/check_mathematics.py` in the source distribution. At every step it verifies

$$2\sum_i v_i^2 - 2 \sum_{\{i,h\} \text{ an edge}} v_i v_h = 2.$$

Integer vector operations make the enumeration exact. Distinct coefficient orbits give disjoint sets of hyperplanes, because proportional roots in a reduced root system differ only by sign. The curve divisors descended from these disjoint sets of hyperplanes are therefore distinct.

B The coefficient-five E_8 calculation

B.1 The ρ-line relations

This appendix supplies the direct coefficient-five E_8 calculation. It derives the homogeneous relations and the relation matrices whose modular ranks bound the cocenter. Together with the tail and generic-rank arguments in Subsection 3.6, these bounds give cocenter dimension 104 for the remaining closed corner. The central extension is that of [10, Theorem 3.2], and the form of the nodal-corner relations comes from [11, Section 3]. The coefficient-five specialization, relation reduction, and exact row-rank certificate are carried out here. We use the normalization $\rho_i = (\rho, \alpha_i) = 1$ from Subsection 3.5. The algebra Π_0^ρ is the path algebra over $\mathbb{C}[z]$ of the double of a finite E_8 quiver Q, with z central, modulo

$$\sum_{a \in Q_1} [a, a^*] = z \sum_i e_i.$$

Different orientations of Q give isomorphic algebras [11, Section 2.1]. We compose paths from left to right. For an edge joining v and v', let $\ell_{v,v'}$ denote the summand of the left-hand side that is a loop at v through v'. It is aa^* or $-a^*a$, according to the orientation of the arrow a of

Q joining them. The relation at v reads $\sum_{v'\sim v}\ell_{v,v'}=z\,e_v$. If $p\colon v\to v'$ and $q\colon v'\to v$ are the two arrows of the double quiver, then $\ell_{v,v'}=\pm pq$ and $\ell_{v',v}=\mp qp$; hence

$$\ell_{v,v'}\,p=-p\,\ell_{v',v},\qquad q\,\ell_{v,v'}=-\ell_{v',v}\,q. \tag{B.1}$$

Lemma B.1 (Arms)**.** *Let $v_0,v_1,\dots,v_r$ be a chain in the Dynkin graph such that $v_1,\dots,v_r$ have no neighbours other than those in the chain. Then $X=\ell_{v_0,v_1}$ satisfies*

$$X(X+z)(X+2z)\cdots(X+rz)=0.$$

Proof. Put $Y=\ell_{v_1,v_0}$. By (B.1), $qf(X)=f(-Y)q$ for every polynomial f over $\mathbb{C}[z]$; hence $Xf(X)=\pm pf(-Y)q$. It is therefore enough to find a degree-r polynomial f with $f(-Y)=0$. If $r=1$, the relation at v_1 is $Y=z$, and $f(T)=T+z$ has the required property. If $r>1$, the relation at v_1 gives $\ell_{v_1,v_2}=z-Y$, and the induction hypothesis gives $\prod_{m=0}^{r-1}(\ell_{v_1,v_2}+mz)=0$. Thus $\prod_{m=1}^{r}(mz-Y)=0$, and $f(T)=\prod_{m=1}^{r}(T+mz)$ works. □

The nodal corners of Etingof–Latour–Rains are presented by relations of the same form [11, equation (3.1)]. We number the vertices of E_8 as in [21, Example 4.26]: the finite diagram is the chain $1,2,3,4,8,7,6$ with 5 attached to the branching vertex 8, and the retained vertex is $p=4$. Choose Q so that its arrow a between 4 and 8 points from 4 to 8, and put

$$d=-\ell_{4,3},\qquad b=-\ell_{8,5},\qquad c=-\ell_{8,7}.$$

Apply Lemma B.1 to the three arms, of lengths 3, 1, and 2, and then use the relations at vertices 4 and 8. This gives

$$\begin{gathered}d(d-z)(d-2z)(d-3z)=0,\qquad b(b-z)=0,\qquad c(c-z)(c-2z)=0,\\ aa^*=d+z\,e_4,\qquad a^*a+b+c+z\,e_8=0.\end{gathered} \tag{B.2}$$

These are the relations of Karmazyn's intermediate presentation [21, Example 4.26] at $t_i=z$, after the free vertex has been removed. The following change of generators is the form used for the closed-corner calculation. Put

$$B=b-2z\,e_8,\qquad C=c,\qquad H=a^*a=-B-C-3z\,e_8,$$

so that

$$B^2=-3zB-2z^2,\qquad C^3=3zC^2-2z^2C. \tag{B.3}$$

In the corner $B_5=e_4\Pi_0^{\rho}e_4$, define

$$\beta=aa^*+3z\,e_4=d+4z\,e_4,\qquad \gamma=aBa^*-2z^2e_4.$$

Lemma B.2 (Generators)**.** *The elements z,β,γ generate the algebra B_5.*

Proof. We reduce closed paths alternately at vertices 4 and 8. A closed path at 4 is a product of paths meeting 4 only at their ends. Those entering the arm $\{1,2,3\}$ are polynomials in d and z, by the arm relations and (B.1); the remaining paths have the form aqa^* with q closed at 8. Similarly, a closed path at 8 is a product of polynomials in b,z, polynomials in c,z, and elements a^*ra with r closed at 4. Since $da=a(H-z)$ by (B.2), we have $a^*d^ka=H(H-z)^k$. Induction on path length now shows that every closed path at 8 is a polynomial in B, C, and z. Substitute $C=-a^*a-B-3z$ in an element $aq(B,C)a^*$. Each occurrence of a^*a splits the expression into a product of two elements of the same form, while powers of B reduce by (B.3). Thus B_5 is generated by z, aa^*, and aBa^*, equivalently by z, β, and γ. □

Lemma B.3 (Relations)**.** *In B_5, the elements*

$$\begin{aligned}\mathsf{P} &= (\beta-4z)(\beta-5z)(\beta-6z)(\beta-7z),\\ \mathsf{R}_4 &= (\gamma+\beta^2)^2+\beta\gamma\beta-5z^2(\gamma+\beta^2)+4z^4,\\ \mathsf{R}_5 &= \gamma\beta\gamma+\gamma^2\beta+\gamma\beta^3-5z^2\gamma\beta\end{aligned}$$

vanish.

Proof. Since $\beta = d+4z$, the element P is $d(d-z)(d-2z)(d-3z)=0$. For the remaining two relations, multiply elements of the form $s\,e_4 + aPa^*$, with $s\in\mathbb{C}[z]$ and P a noncommutative polynomial in B and C, using the rule

$$(s\,e_4+aPa^*)(s'e_4+aP'a^*) = ss'e_4 + a(sP'+s'P+PHP')a^*.$$

Only (B.3) and $H=-B-C-3z$ enter the reduction. They give $HB = 2z^2-CB$, $BH = 2z^2-BC$ and

$$\gamma+\beta^2 = 7z^2+a(3z-C)a^*,\qquad \beta\gamma = -6z^3+a(3z-C)Ba^*,\qquad \gamma\beta = -6z^3+aB(3z-C)a^*.$$

For R_5, they give $\beta\gamma+(\gamma+\beta^2)\beta-5z^2\beta = a(C-z)(C-2z)a^*$. Hence

$$\mathsf{R}_5 = \gamma(\beta\gamma+(\gamma+\beta^2)\beta-5z^2\beta) = a(BH-2z^2)(C-z)(C-2z)a^* = -aBC(C-z)(C-2z)a^*,$$

which vanishes by (B.3). For R_4, put $g=\gamma+\beta^2$, so that $\mathsf{R}_4 = (g-z^2)(g-4z^2)+\beta\gamma\beta$. The same rules give

$$\begin{aligned}(g-z^2)(g-4z^2) &= 18z^4+a(27z^3-9z^2C+(3z-C)H(3z-C))a^*,\\ (3z-C)H(3z-C) &= -27z^3-9z^2B+11z^2C+3zBC+3zCB-CBC,\\ \beta\gamma\beta &= -18z^4+a(9z^2B-2z^2C-3zBC-3zCB+CBC)a^*,\end{aligned}$$

where the second line uses $C^3 = 3zC^2-2z^2C$. The sum is zero. □

The program `tools/check_e8_rho_relations_formal.py` independently verifies these reductions over the polynomial ring $\mathbb{Z}[z]$, with z retained as a variable.

Remark B.4 (Sign conventions in Karmazyn's arXiv version)*.* The comparison in this remark concerns arXiv:1709.02720v1 (8 September 2017), Example 4.26, pp. 23–25 of [21]. The invariant generators are listed on p. 23, the intermediate presentation is on pp. 24–25, and the final display is on p. 25. The relations used here are derived independently in Lemma B.3. This convention check matters because the rank matrices below use $\mathsf{P},\mathsf{R}_4,\mathsf{R}_5$ exactly as displayed in Lemma B.3. Those relations agree with the last two relations of Karmazyn's intermediate presentation, written in the invariants R_1,R_2,R_3,S_2 of the five-arm parameters $\tau_1,\dots,\tau_5$. On the ρ-line, $(\tau_1,\dots,\tau_5)=(2,1,0,-1,-2)z$ and $(R_1,R_2,R_3,S_2)=(3z,2z^2,0,2z^2)$; here $\beta = a_4a_4^*+R_1$ and $\gamma = a_4Ba_4^*-S_2$. For the quadratic and cubic relations displayed there, the shifts must be $B=b+\tau_5$ and $C=c+\tau_3$. The final display on p. 25 of that version rewrites these relations through $T_3=R_2+S_2+R_1S_1$, $T_2=R_3+R_2S_1+R_1S_2$, $T_1=-R_3R_1+R_2S_2$ and $T_0=R_3S_2$. The expressions T_3,T_2,T_1,T_0 are $+\sigma_2,\dots,+\sigma_5$ in the parameters τ_i, whereas Karmazyn's list of invariant generators sets $T_i=-\sigma(\tau)$. Substituting the sign convention $T_i=-\sigma(\tau)$ into Karmazyn's final display reverses the signs of the parameter terms and gives

$$\mathsf{R}_4' = (\gamma+\beta^2)^2+\beta\gamma\beta+5z^2(\gamma+\beta^2)-4z^4,\qquad \mathsf{R}_5' = \gamma\beta\gamma+\gamma^2\beta+\gamma\beta^3+5z^2\gamma\beta.$$

These are not relations of B_5. This can be checked independently of the large rank table. Put $\mathcal{F} = \mathbb{C}[z]\langle\beta,\gamma\rangle$. The matrix Hilbert polynomial in [10, Theorem 3.2(iv)] gives

$$(\dim(B_5)_m)_{m=0}^{5} = (1,2,4,7,10,14), \qquad (\dim \mathcal{F}_m)_{m=0}^{5} = (1,2,4,7,12,20).$$

For the first calculation, let C_Q be the adjacency matrix of the numbered E_8 tree. The diagonal coefficients at vertex 4 of $(1 - C_Q t + t^2)^{-1}$ in arrow degrees $0,2,4,6,8,10$ are $1,1,2,3,3,4$, obtained from the recurrence $v_d = C_Q v_{d-1} - v_{d-2}$, with $v_{-1} = 0$ and v_0 the fourth standard basis vector. Multiplying by $1 + t^2 + \cdots + t^{58}$ gives the displayed partial sums. The second calculation counts words of weights 1 and 2, together with powers of the central variable.

Thus the degree-four kernel of $\mathcal{F} \to B_5$ has dimension two. The independent relations P, R_4 span it. In degree five the kernel has dimension six and is spanned by

$$z\mathsf{P}, \quad \beta\mathsf{P}, \quad z\mathsf{R}_4, \quad \beta\mathsf{R}_4, \quad \mathsf{R}_4\beta, \quad \mathsf{R}_5.$$

Their independence follows by comparing successively the coefficients of $\gamma\beta\gamma$, $\beta\gamma^2$, $\gamma^2\beta$, $z\gamma^2$, β^5, and $z\beta^4$. Consequently $Q_\rho \to B_5$ and the induced cocenter maps are isomorphisms through degree five. In degree four, the cyclic relations are therefore exactly the span of $[\mathsf{P}]$ and $[\mathsf{R}_4]$. In cyclic words,

$$[\mathsf{P}] = [\beta^4] - 22z[\beta^3] + 179z^2[\beta^2] - 638z^3[\beta] + 840z^4,$$
$$[\mathsf{R}_4] = [\gamma^2] + 3[\beta^2\gamma] + [\beta^4] - 5z^2[\gamma] - 5z^2[\beta^2] + 4z^4,$$

and $[\mathsf{R}_4'] - [\mathsf{R}_4] = 10z^2[\gamma] + 10z^2[\beta^2] - 8z^4$. If $[\mathsf{R}_4']$ were in the span, comparing the coefficients of $[\gamma^2]$ and of $z[\beta^3]$ would force $[\mathsf{R}_4'] = [\mathsf{R}_4]$, which is false. Hence $\mathsf{R}_4' \neq 0$ in B_5.

For the second expression, $[\mathsf{R}_5'] - [\mathsf{R}_5] = 10z^2[\gamma\beta]$. The degree-five cyclic relation space is spanned by $[z\mathsf{P}], [\beta\mathsf{P}], [z\mathsf{R}_4], [\beta\mathsf{R}_4], [\mathsf{R}_5]$. The coefficient of $z[\gamma^2]$ first eliminates $[z\mathsf{R}_4]$ from a possible expression for $z^2[\gamma\beta]$. The coefficients of $[\gamma^2\beta]$ and $[\gamma\beta^3]$ in $[\beta\mathsf{R}_4]$ and $[\mathsf{R}_5]$ are respectively $(1,3)$ and $(2,1)$; their determinant is -5. They therefore eliminate these two classes as well. The remaining classes $[z\mathsf{P}]$ and $[\beta\mathsf{P}]$ contain no γ, and cannot produce $z^2[\gamma\beta]$. Thus $[\mathsf{R}_5'] \neq 0$, and hence $\mathsf{R}_5' \neq 0$ in B_5.

The rank table below is computed from $\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5$.

B.2 Relation matrices and modular rank bounds

The C++17 program `tools/check_e8_corrected_rho.cpp` constructs the finite relation matrices used in Proposition 3.15. In each half-path degree $m \leq 23$, it enumerates the cyclic-word columns $z^a[w]_{\mathrm{cyc}}$, with $a \geq 0$ and $a + \mathrm{wt}(w) = m$, taking one word from each cyclic class. For each $\mathsf{R} \in \{\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5\}$, $a \geq 0$, and word w in β, γ satisfying $a + \deg \mathsf{R} + \mathrm{wt}(w) = m$, it includes the row $z^a[\mathsf{R}w]_{\mathrm{cyc}}$. The empty word is included and has weight zero; the weights of z, β, γ are $1, 1, 2$, respectively. The program then performs exact Gaussian elimination over $\mathbb{F}_{1000003}$. The command `make math` compiles the program, repeats the calculation over $\mathbb{F}_{1000033}$ and compares both outputs with the table below. Agreement of the two ranks is a reproducibility check. For the characteristic-zero estimate, a nonzero minor modulo either prime remains nonzero over $\mathbb{Q}$. Hence r_m is a lower bound for the characteristic-zero relation rank and $u_m = c_m - r_m$ is an upper bound for the characteristic-zero cocenter nullity. The tail and generic-rank arguments in Subsection 3.6 establish the exact total dimension. The table lists m, c_m, r_m, and u_m.

Degree m	Columns c_m	Rank r_m over $\mathbb{F}_{1000003}$, $\mathbb{F}_{1000033}$	Nullity bound u_m
0	1	0	1
1	2	0	2
2	4	0	4
3	6	0	6
4	9	2	7
5	12	5	7
6	17	10	7
7	22	16	6
8	30	24	6
9	40	34	6
10	55	49	6
11	74	69	5
12	105	100	5
13	146	142	4
14	210	206	4
15	304	300	4
16	447	443	4
17	658	655	3
18	987	984	3
19	1480	1478	2
20	2246	2244	2
21	3416	3414	2
22	5227	5225	2
23	8014	8013	1

The SHA-256 checksums of the two E_8 programs are:

File	SHA-256
`check_e8_corrected_rho.cpp`	`2545d84e337ea8b4ac650c428b90880a36ab47cd168e8935e0d4598ac1dbabd9`
`check_e8_rho_relations_formal.py`	`208305e7e5a7613b75b66959b4ac7022c31644b54c988cbd642baad7382d5f1a`

C Exact linear algebra for $D_{2,2}$

Theorem 6.17 proves the centre-lifting obstruction symbolically for every $D_{n,2}$. This appendix gives an independent exact-rational calculation for $D_{2,2}$: it computes the tangent data in Proposition 6.20 and the automorphism-invariant socle condition (6.8) used in Proposition 6.21. The program `tools/verify_conjecture_1_4_counterexample.py` computes the cochain ranks, and `tools/check_d22_tangent_image.py` verifies the tangent image and socle condition. Both use exact rational arithmetic (`fractions.Fraction` in Python) and are run by `make math`. They provide the finite $n = 2$ certificate; Theorem 6.17 and Corollary 6.18 remain the uniform symbolic arguments for every $n \geq 2$.

Cochains and ranks. Fix the bases and normalized cochain complexes. The algebra $A = D_{2,2}$ is given in the basis $(1, x, x^2, x^3, x^4, y, xy, x^2y)$ by the multiplication table of Lemma 6.14(i), and the first program checks associativity and the unit on all triples of basis vectors. The centre Z_2 has the basis $(1, \mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q})$, embedded as $(1, x^2, x^3, x^2y, x^4)$, and the same program checks that the embedding is multiplicative. Normalized cochains are multilinear maps on the span of the basis vectors other than 1; for Z_2, the second cochains are symmetric. The coboundary maps are sparse rational matrices. Exact row reduction gives the following dimensions.

	A, associative	Z_2, commutative
normalized 1-cochains	56	20
rank of δ on 1-cochains	50	10
normalized 2-cochains	392	50 (symmetric)
rank of δ on 2-cochains	339	24
2-cocycles	53	26
cocycles modulo coboundaries	$HH^2(A)$: 3	$T^1_{\mathrm{com}}(Z_2)$: 16

The derivations of A form the kernel on 1-cochains, of dimension $56-50=6$; inner derivations have dimension $8-5=3$, so $\dim HH^1(A)=3$.

The centre tangent map. For each of the 53 basis cocycles ϕ of $Z^2_{\mathrm{norm}}(A,A)$, the first program solves the linear system (6.2) for τ with values in the chosen complement $\mathrm{span}(x,y,xy)$ of Z_2. This complement fixes a representative for the calculation. The first program checks that the solution exists and is unique, forms ψ by (6.3), and checks that ψ is symmetric, takes values in Z_2, and is a cocycle. It also verifies that the coboundaries of A map to commutative coboundaries and that the 53 cocycles ψ have rank 3 modulo $B^2_{\mathrm{com}}(Z_2)$. Finally, the coefficient of w in $\psi(\mathsf{v},\mathsf{v})$ vanishes on all these cocycles and on $B^2_{\mathrm{com}}(Z_2)$, but equals 1 on ψ_0. This is an independent exact verification of Corollary 6.18(i) for $n=2$.

Representatives and the socle condition. The representatives in Proposition 6.20(iii) are normalized and vanish on every pair not displayed below, apart from the values imposed by symmetry:

$$\begin{aligned}
&\psi_1(\mathsf{u},\mathsf{w})=-\mathsf{w}, \quad \psi_1(\mathsf{u},\mathsf{q})=-2\mathsf{q}, \quad \psi_1(\mathsf{v},\mathsf{v})=\mathsf{q};\\
&\psi_2(\mathsf{u},\mathsf{v})=-\tfrac{1}{2}\mathsf{w}, \quad \psi_2(\mathsf{u},\mathsf{w})=\tfrac{1}{2}\mathsf{v}, \quad \psi_2(\mathsf{v},\mathsf{w})=\tfrac{1}{2}\mathsf{q};\\
&\psi_3(\mathsf{u},\mathsf{w})=-2\mathsf{w}, \quad \psi_3(\mathsf{u},\mathsf{q})=-2\mathsf{q}, \quad \psi_3(\mathsf{w},\mathsf{w})=2\mathsf{q}.
\end{aligned}$$

These formulas also give $\psi_i(s,s)\in\mathbb{C}\mathsf{q}$ for every $s\in\mathrm{Soc}(Z_2)$.

The second program checks that ψ_1,ψ_2,ψ_3 of Proposition 6.20(iii) are commutative cocycles whose classes lie in the image of χ_A and span it. Reducing them modulo $B^2_{\mathrm{com}}(Z_2)$ by the normalized row-echelon basis of the coboundaries leaves the coefficients of w in $\psi(\mathsf{u},\mathsf{v})$, of w in $\psi(\mathsf{u},\mathsf{w})$, and of q in $\psi(\mathsf{u},\mathsf{q})$ as non-pivot coordinates. The coefficient vectors of ψ_1,ψ_2,ψ_3 in these three coordinates form the matrix

$$\begin{pmatrix} 0 & -1 & -2\\ -\frac{1}{2} & 0 & 0\\ 0 & -2 & -2 \end{pmatrix}, \qquad \det=-1,$$

whose rows correspond to ψ_1,ψ_2,ψ_3. The program then verifies (6.8) for all 53 image cocycles and for the 20 generators of $B^2_{\mathrm{com}}(Z_2)$, as a polynomial identity in the coordinates of $s=\alpha\mathsf{v}+\beta\mathsf{w}+\gamma\mathsf{q}$, and checks that ψ_0 violates (6.8).

Uniform checks. For $n=2,\dots,6$, the first program also checks the identities of Lemma 6.14 used in the proof of Theorem 6.17: $\dim A_n=4n$, $\dim Z(A_n)=n+3$, the centralizer of x has dimension $2n+2$, $\mathsf{w}x=x\mathsf{w}=0$, and $\mathsf{v}A_n+A_n\mathsf{v}\subseteq\mathbb{C}[x]$. It also checks that the multiplication of (6.6) for $n=2$ is associative over $\mathbb{Q}[t]$. These are finite checks of identities whose general proof is already given in Lemma 6.14.

The SHA-256 checksums of the two programs are:

File	SHA-256
`verify_conjecture_1_4_counterexample.py`	`e51dd6b0816f019f3cd2f38c9fb12312ec4b9f49d18323a18748055bdbb7ccef`
`check_d22_tangent_image.py`	`2140afdf251c2fb1f123a02cdb92fac87b4d029e822f336a4a2ca1e9e92dc9ae`

References

[1] C. Amiot, *Cluster categories for algebras of global dimension 2 and quivers with potential*, Ann. Inst. Fourier **59** (2009), 2525–2590. arXiv:0805.1035v2.

[2] J. August, *The tilting theory of contraction algebras*, Adv. Math. **374** (2020), Article 107372. doi:10.1016/j.aim.2020.107372. arXiv:1802.10366v2.

[3] D. Bowman, D. Puljić, and A. Smoktunowicz, *A construction of deformations to general algebras*, Int. Math. Res. Not. IMRN (2024), no. 12, 9827–9848. doi:10.1093/imrn/rnae077.

[4] M. Brion, *Introduction to actions of algebraic groups*, Les cours du C.I.R.M. **1** (2010), no. 1, 1–22.

[5] G. Brown and M. Wemyss, *Local normal forms of noncommutative functions*, Forum Math. Pi **13** (2025), Paper e8. doi:10.1017/fmp.2025.2.

[6] J. Bryan, S. Katz, and N. C. Leung, *Multiple covers and the integrality conjecture for rational curves in Calabi–Yau threefolds*, J. Algebraic Geom. **10** (2001), no. 3, 549–568.

[7] C. Chevalley, *Invariants of finite groups generated by reflections*, Amer. J. Math. **77** (1955), no. 4, 778–782. doi:10.2307/2372597.

[8] W. Donovan and M. Wemyss, *Noncommutative deformations and flops*, Duke Math. J. **165** (2016), no. 8, 1397–1474.

[9] A. Dugas, *Periodicity of d-cluster-tilted algebras*, J. Algebra **368** (2012), 40–52. arXiv:1007.2811v2.

[10] P. Etingof and E. Rains, *Central extensions of preprojective algebras, the quantum Heisenberg algebra, and 2-dimensional complex reflection groups*, J. Algebra **299** (2006), no. 2, 570–588. arXiv:math/0503393v3.

[11] P. Etingof, F. Latour, and E. Rains, *On central extensions of preprojective algebras*, J. Algebra **313** (2007), no. 1, 165–175. arXiv:math/0606403v1.

[12] P. Gabriel, *Finite representation type is open*, in V. Dlab and P. Gabriel (eds.), *Representations of Algebras*, Lecture Notes in Math. **488**, Springer, 1975, pp. 132–155. doi:10.1007/BFb0081219.

[13] M. Gerstenhaber, *On the deformation of rings and algebras*, Ann. of Math. (2) **79** (1964), 59–103.

[14] P. Gille, R. Parimala, and V. Suresh, *Local triviality for G-torsors*, arXiv:1901.04722v5 (2021).

[15] D. K. Harrison, *Commutative algebras and cohomology*, Trans. Amer. Math. Soc. **104** (1962), 191–204.

[16] M. Herschend and O. Iyama, *Selfinjective quivers with potential and 2-representation-finite algebras*, Compos. Math. **147** (2011), no. 6, 1885–1920. doi:10.1112/S0010437X11005367. Preprint: arXiv:1006.1917v2.

[17] Z. Hua and Y. Toda, *Contraction algebra and invariants of singularities*, Int. Math. Res. Not. IMRN (2018), no. 10, 3173–3198. Preprint: arXiv:1601.04881v2. The theorem numbering used here follows this preprint.

[18] O. Iyama, M. Kalck, M. Wemyss, and D. Yang, *Frobenius categories, Gorenstein algebras and rational surface singularities*, Compos. Math. **151** (2015), no. 3, 502–534. doi:10.1112/S0010437X14007647. Preprint: arXiv:1209.4215v3.

[19] G. Jasso, B. Keller, and F. Muro, *The Donovan–Wemyss conjecture via the derived Auslander–Iyama correspondence*, in *Triangulated Categories in Representation Theory and Beyond*, Abel Symposia **17**, Springer, 2024, 105–140. doi:10.1007/978-3-031-57789-5_4. Preprint: arXiv:2301.11593v3. Theorem numbering follows this preprint.

[20] J. Jelisiejew and A. Smoktunowicz, *On a conjecture by Michael Wemyss regarding the calculation of GV invariants*, arXiv:2602.05919v1 (2026).

[21] J. Karmazyn, *The length classification of threefold flops via noncommutative algebras*, Adv. Math. **343** (2019), 393–447. The convention comparison in Remark B.4 uses arXiv:1709.02720v1 (2017).

[22] J. Karmazyn, *Deformations of algebras defined by tilting bundles*, J. Algebra **513** (2018), 388–434.

[23] Y. Kawamata, *Deformations over non-commutative base*, C. R. Math. Acad. Sci. Paris **362** (2024), special issue S1, 159–169. doi:10.5802/crmath.622. Preprint: arXiv:2303.06323v7. Theorem numbering follows this preprint; its Theorem 3.1 is Theorem 4 in the published version.

[24] B. Keller, with an appendix by M. Van den Bergh, *Deformed Calabi–Yau completions*, J. Reine Angew. Math. **654** (2011), 125–180. arXiv:0908.3499v5.

[25] B. Keller, *Erratum to "Deformed Calabi–Yau completions"*, arXiv:1809.01126 (2018).

[26] B. Keller, *On differential graded categories*, arXiv:math/0601185v5 (2006).

[27] S. Katz and D. R. Morrison, *Gorenstein threefold singularities with small resolutions via invariant theory for Weyl groups*, J. Algebraic Geom. **1** (1992), no. 3, 449–530. Preprint: arXiv:alg-geom/9202002v1.

[28] M. Linckelmann, *Tate duality and transfer in Hochschild cohomology*, J. Pure Appl. Algebra **217** (2013), no. 12, 2387–2399. arXiv:1211.5999v1.

[29] N. Nabijou and M. Wemyss, *GV and GW invariants via the enhanced movable cone*, Moduli **1** (2024), Paper e8, 1–38. doi:10.1112/mod.2024.4.

[30] Y. Namikawa, *Flops and Poisson deformations of symplectic varieties*, Publ. Res. Inst. Math. Sci. **44** (2008), no. 2, 259–314. doi:10.2977/prims/1210167328.

[31] B. Poonen, *The moduli space of commutative algebras of finite rank*, J. Eur. Math. Soc. **10** (2008), no. 3, 817–836. doi:10.4171/JEMS/131.

[32] O. Riemenschneider, *Supplement: A short introduction to complex analytic spaces*, Chapter 19 of online lecture notes, Section 19.22, Theorems 19.57 and 19.59. Author's online notes.

[33] A. Sard, *The measure of the critical values of differentiable maps*, Bull. Amer. Math. Soc. **48** (1942), 883–890. Original article.

[34] A. Smoktunowicz, *On flat deformations and their applications*, arXiv:2509.10121v2 (2025).

[35] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu, 2026; Tags 00MP and 08IB.

[36] Y. Toda, *Non-commutative width and Gopakumar–Vafa invariants*, Manuscripta Math. **148** (2015), no. 3–4, 521–533.

[37] M. Van den Bergh, *Three-dimensional flops and non-commutative rings*, Duke Math. J. **122** (2004), no. 3, 423–455.

[38] S. Katz, *Genus zero Gopakumar–Vafa invariants of contractible curves*, J. Differential Geom. **79** (2008), no. 2, 185–195. arXiv:math/0601193.

[39] O. Iyama and S. Oppermann, *Stable categories of higher preprojective algebras*, Adv. Math. **244** (2013), 23–68. arXiv:0912.3412v2.

[40] W. Crawley-Boevey, *Geometry of the moment map for representations of quivers*, Compos. Math. **126** (2001), 257–293. Author's version.

[41] M. Pérez Rodríguez, *Deformation of formal schemes through local homology*, J. Algebra **445** (2016), 78–102. doi:10.1016/j.jalgebra.2015.07.041. Version checked: arXiv:2402.02879v1. The cited locators follow this preprint.

[42] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier **6** (1956), 1–42, nos. 6, 9 and 12. doi:10.5802/aif.59.